



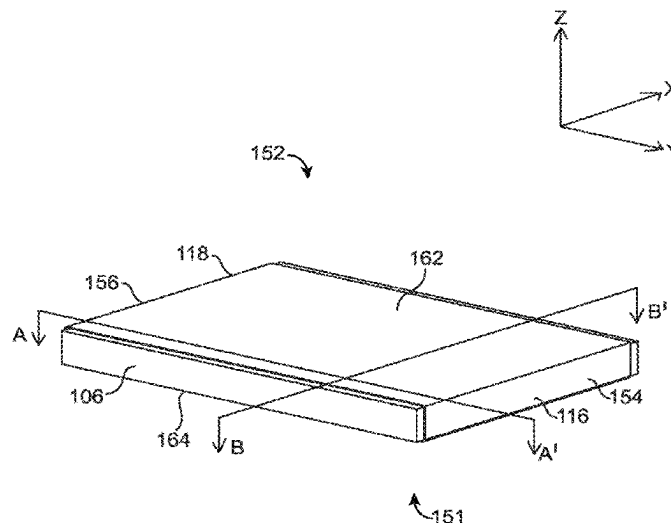
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(19) **United States**(12) **Patent Application Publication**
Busacca et al.(10) **Pub. No.: US 2025/0286099 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **DIMENSIONAL CONSTRAINTS FOR
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Sep. 6, 2022, now Pat. No. 11,961,952, which is a
continuation of application No. 17/335,725, filed on
Jun. 1, 2021, now Pat. No. 11,444,310, which is a
continuation of application No. 16/241,159, filed on
Jan. 7, 2019, now Pat. No. 11,081,718, which is a
continuation of application No. 15/889,338, filed on
Feb. 6, 2018, now Pat. No. 10,177,400, which is a
continuation of application No. PCT/US17/32355,
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H01M 10/058 (2010.01)
H01M 10/0585 (2010.01)
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2004/021 (2013.01)

(57)

ABSTRACT

A secondary battery is provided for cycling between a charged and a discharged state, the secondary battery including a battery enclosure, an electrode assembly, carrier ions, a non-aqueous liquid electrolyte within the battery enclosure, and a set of electrode constraints. The set of electrode constraints includes a primary constraint system having first and second primary growth constraints and at least one primary connecting member, the first and second primary growth constraints separated from each other in the longitudinal direction, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20%. The set of electrode constraints further includes a secondary constraint system having first and second secondary growth constraints connected by at least one secondary connecting member, wherein the secondary constraint system at least partially restrains growth of the electrode assembly in a second direction upon cycling of the secondary battery.



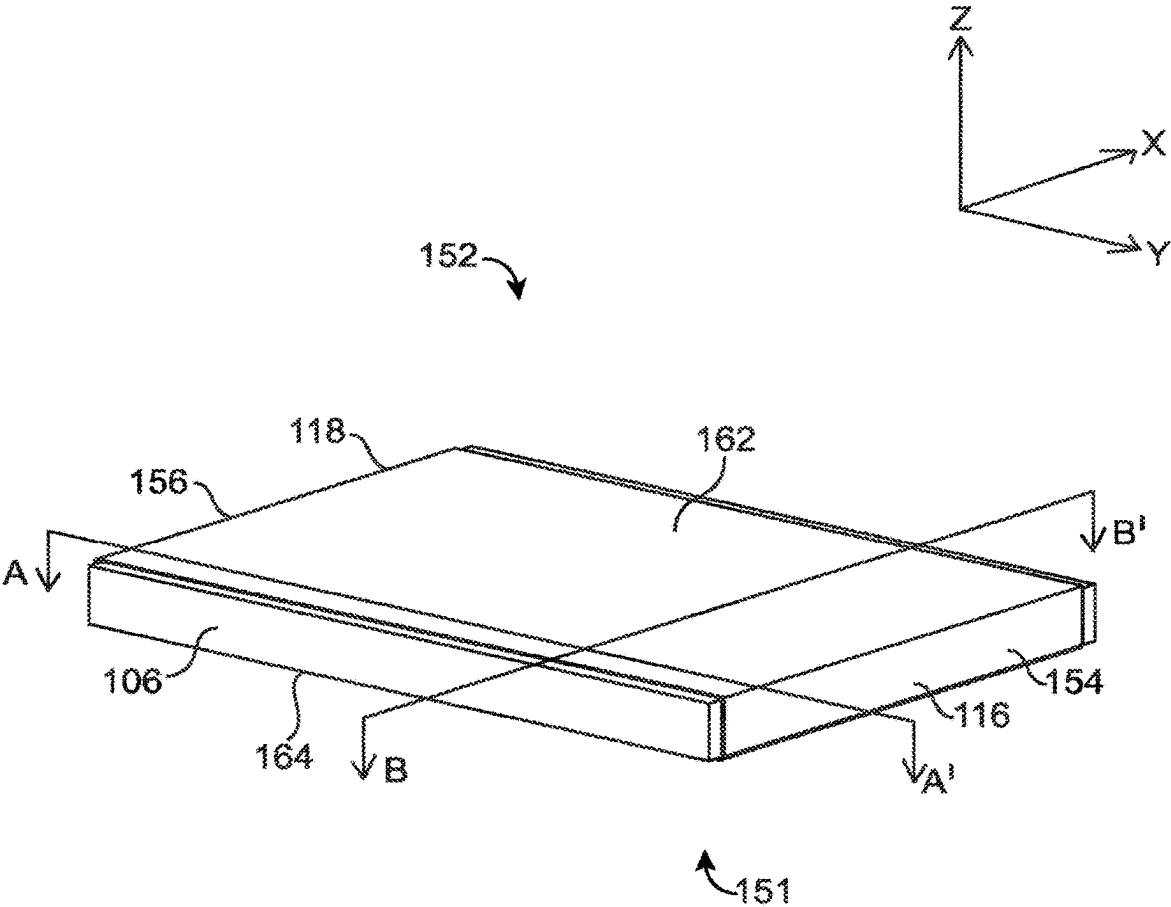


FIG. 1

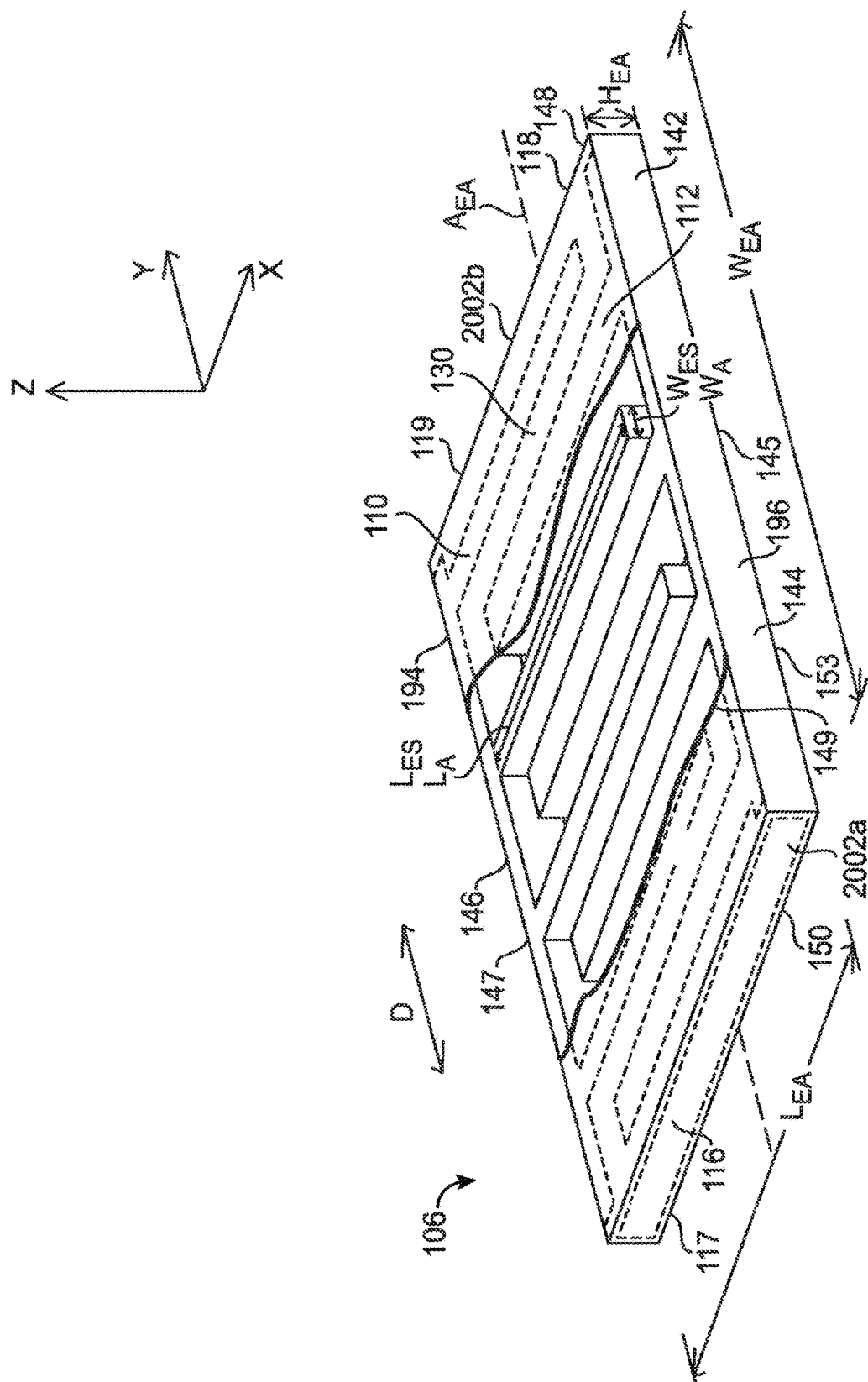


FIG. 2A

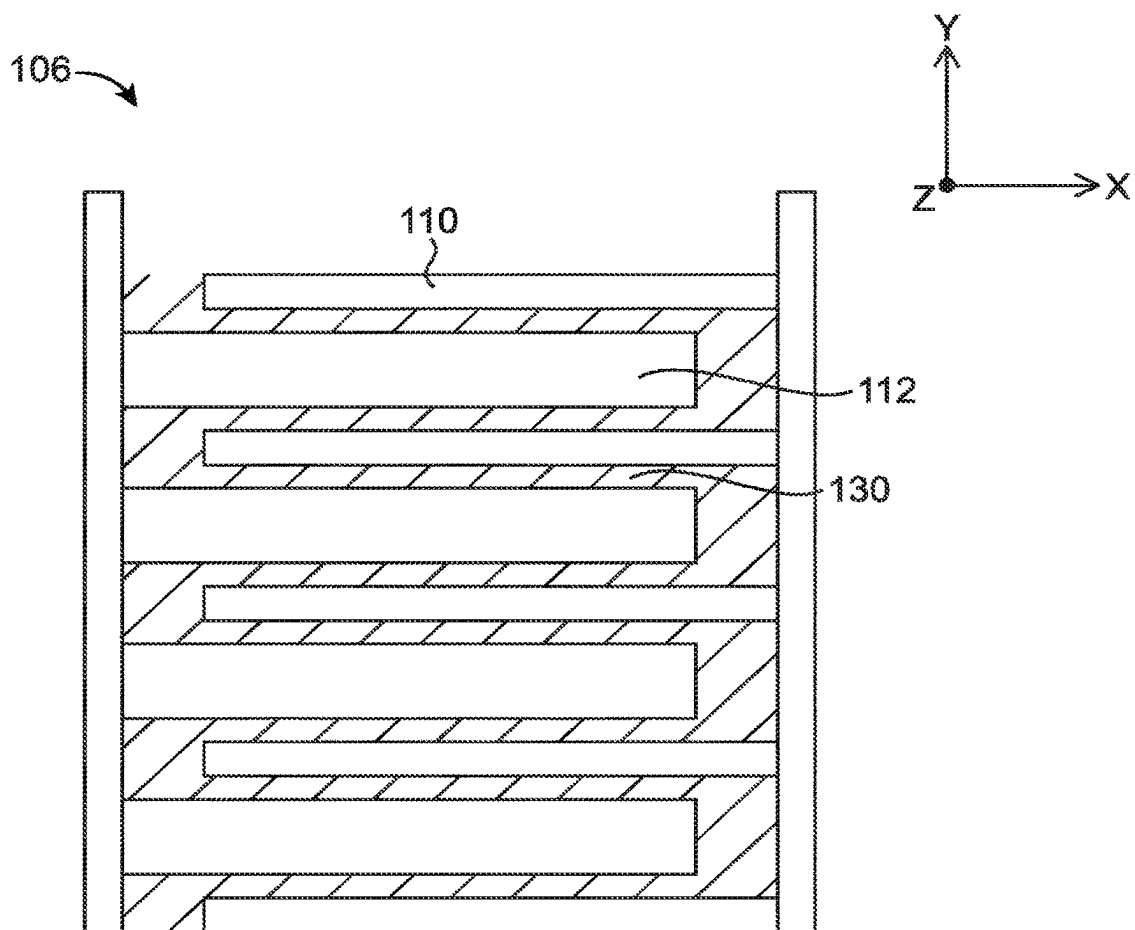


FIG. 2B

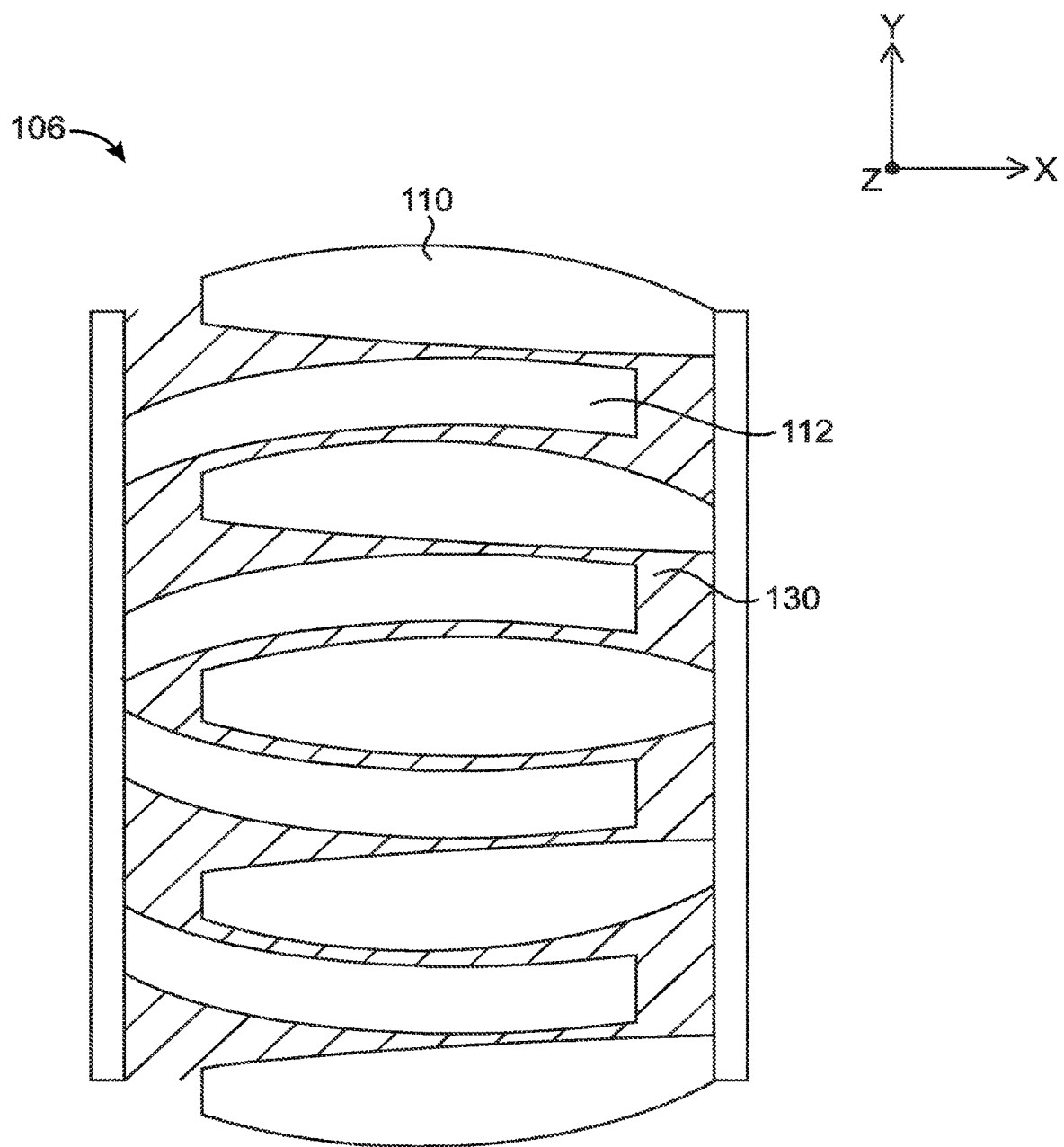
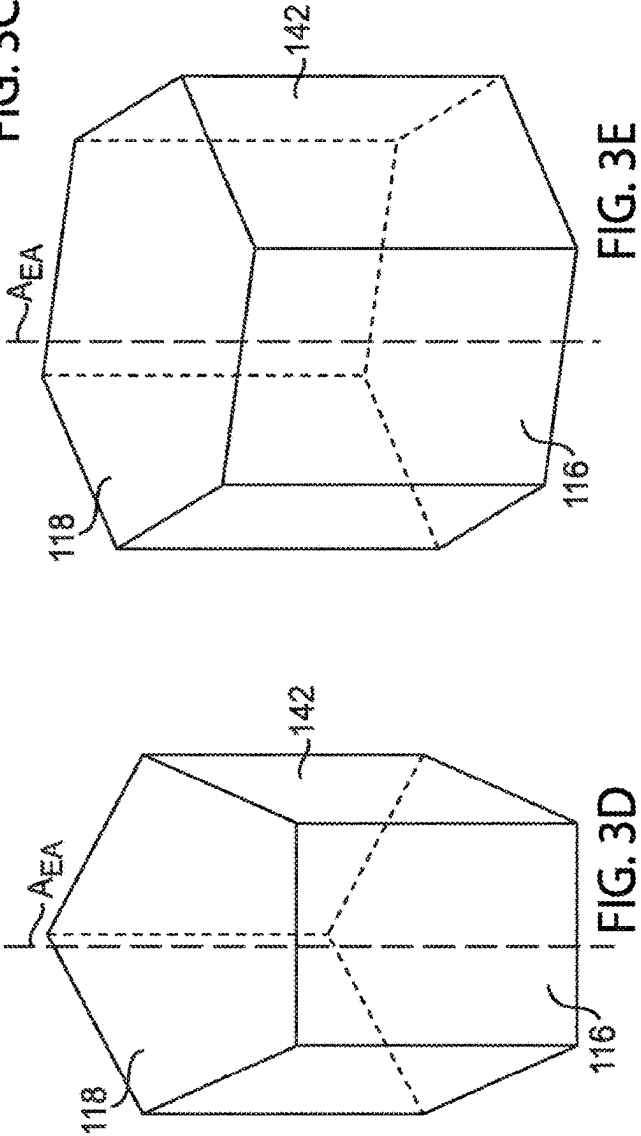
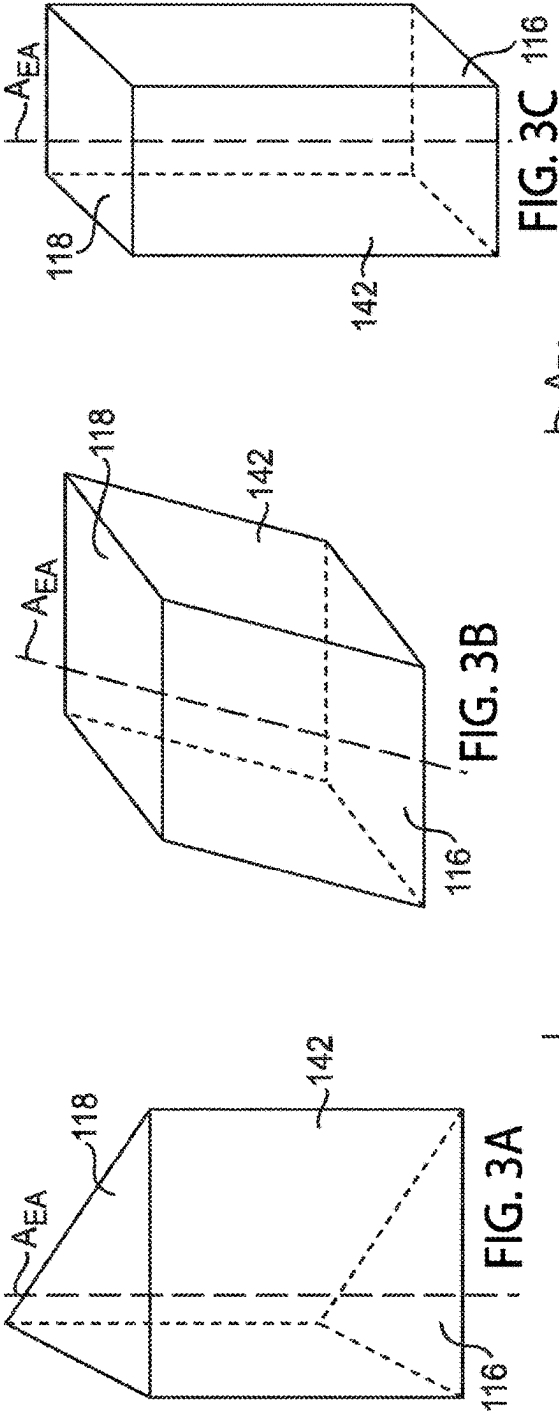


FIG. 2C



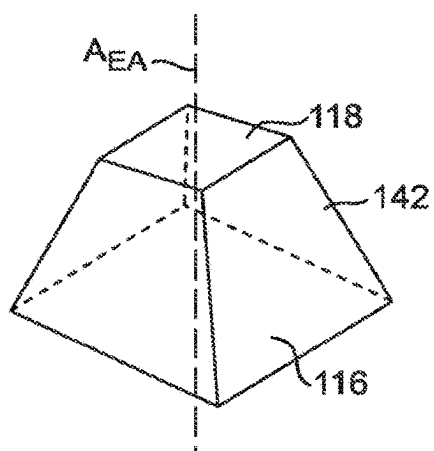


FIG. 3F

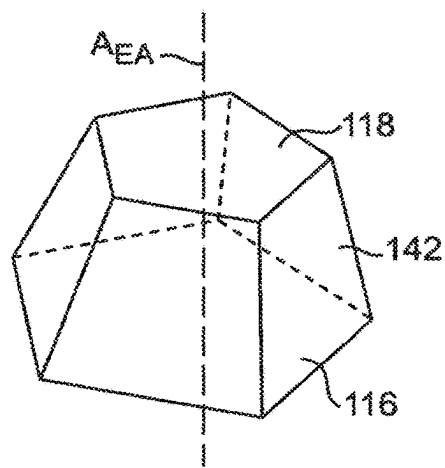


FIG. 3G

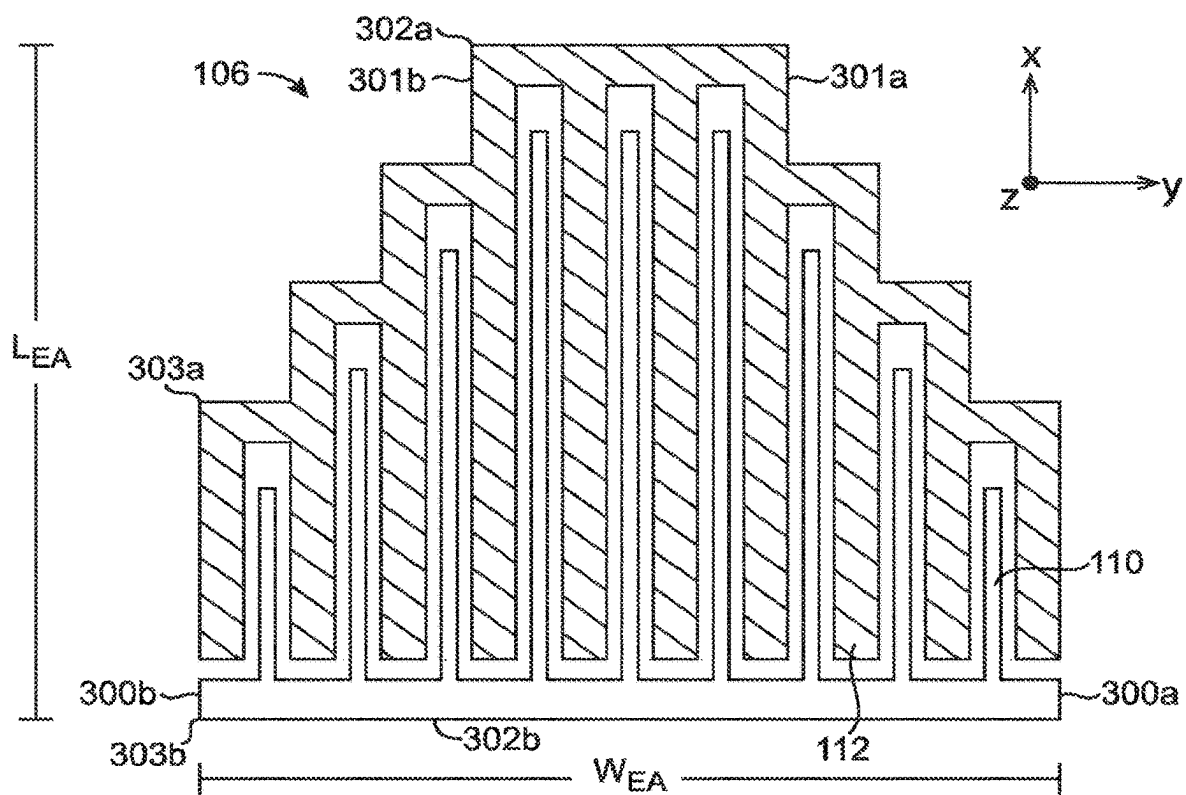


FIG. 3H

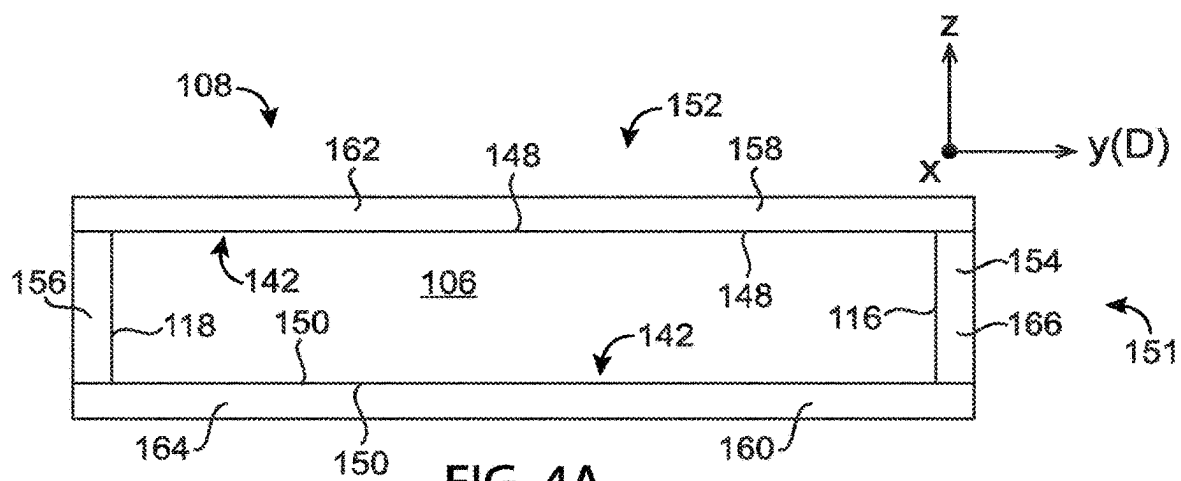


FIG. 4A

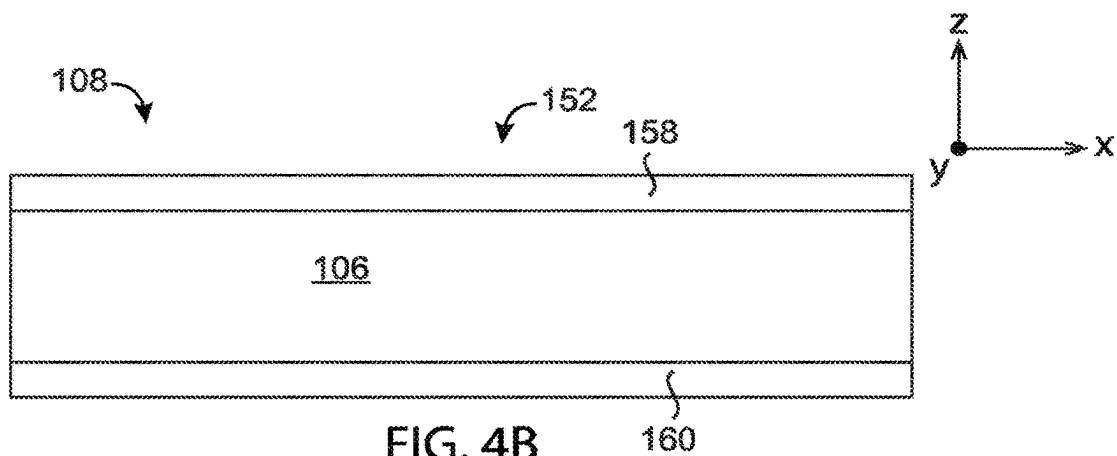


FIG. 4B

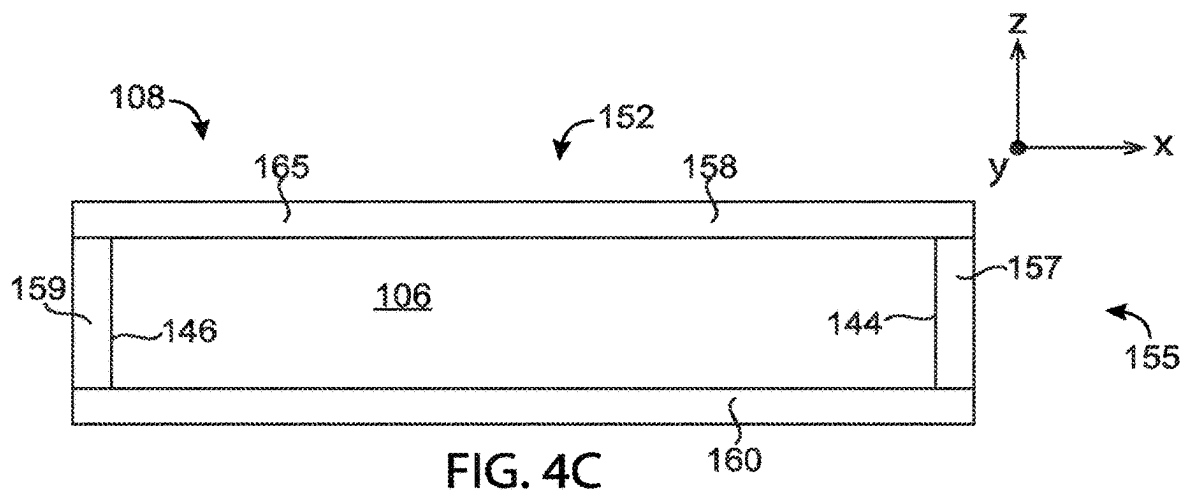


FIG. 4C

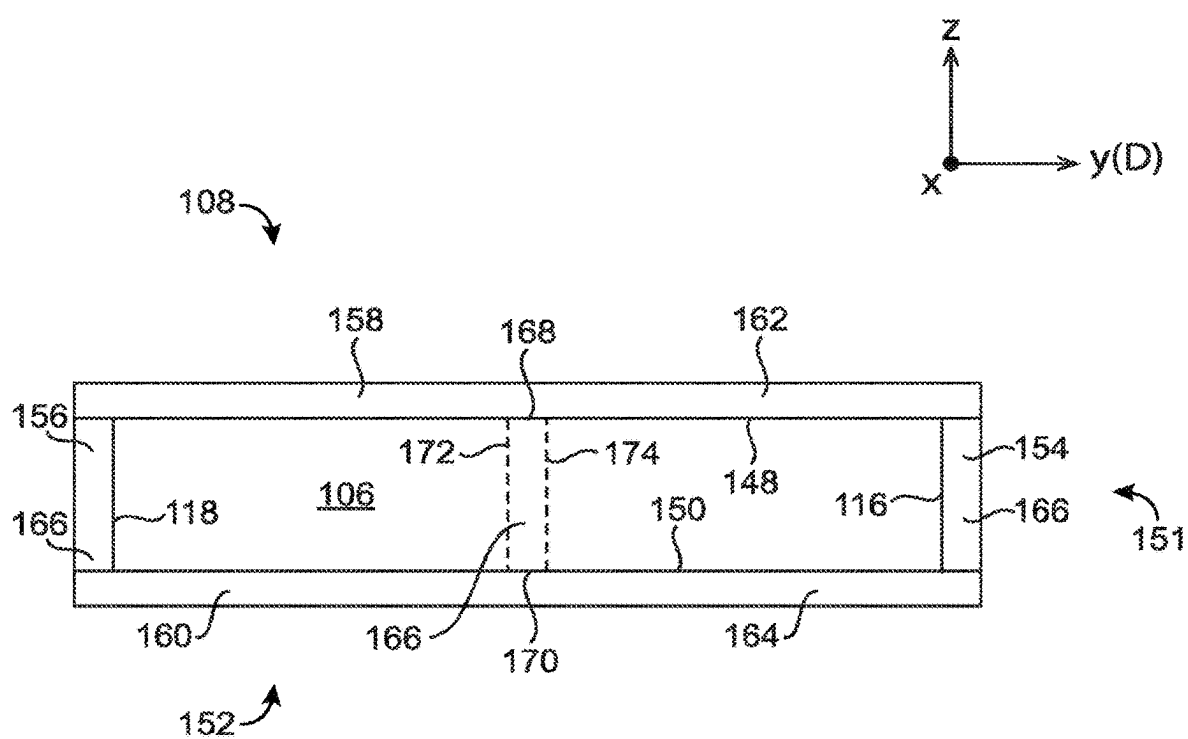


FIG. 5

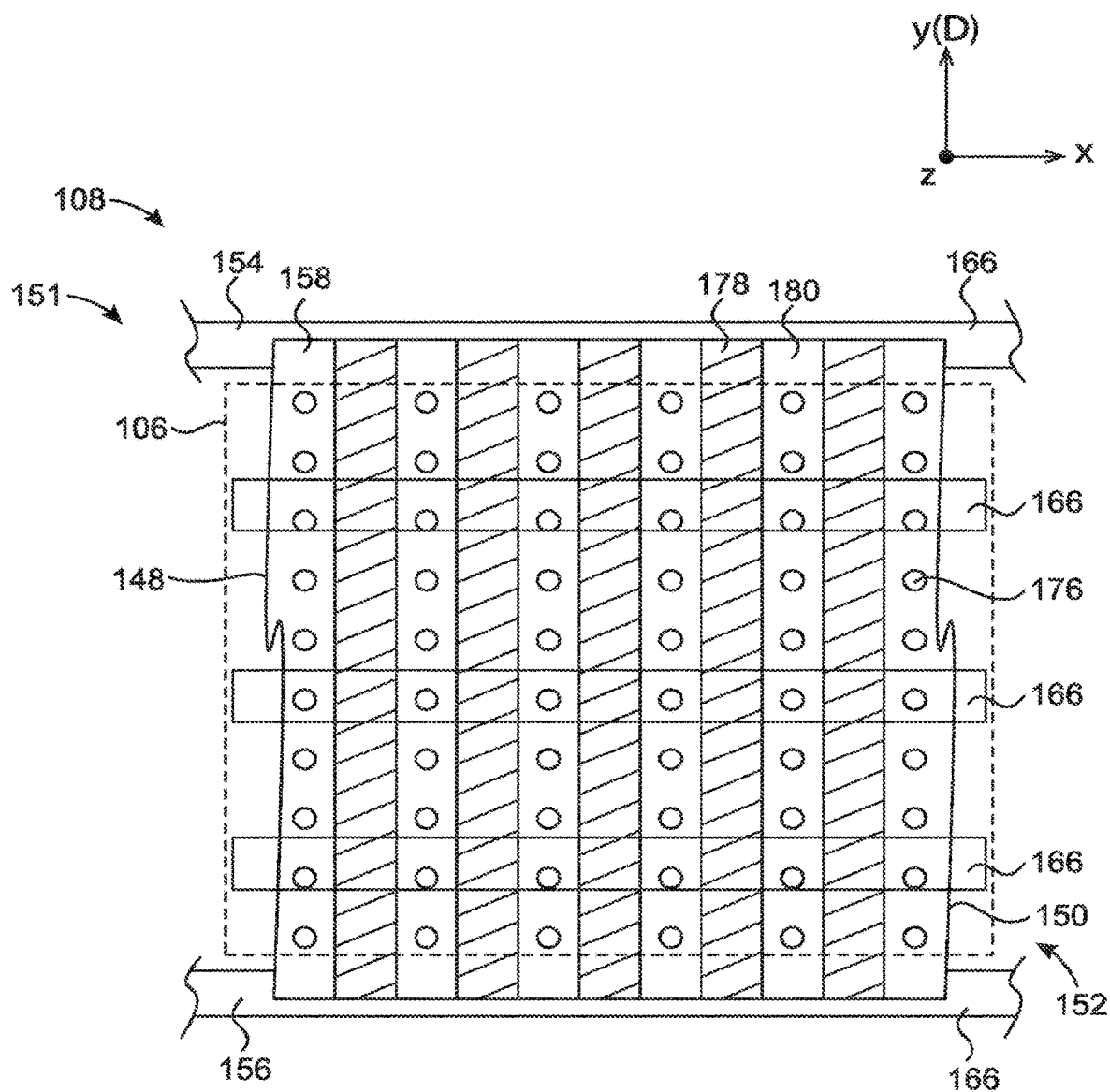


FIG. 6A

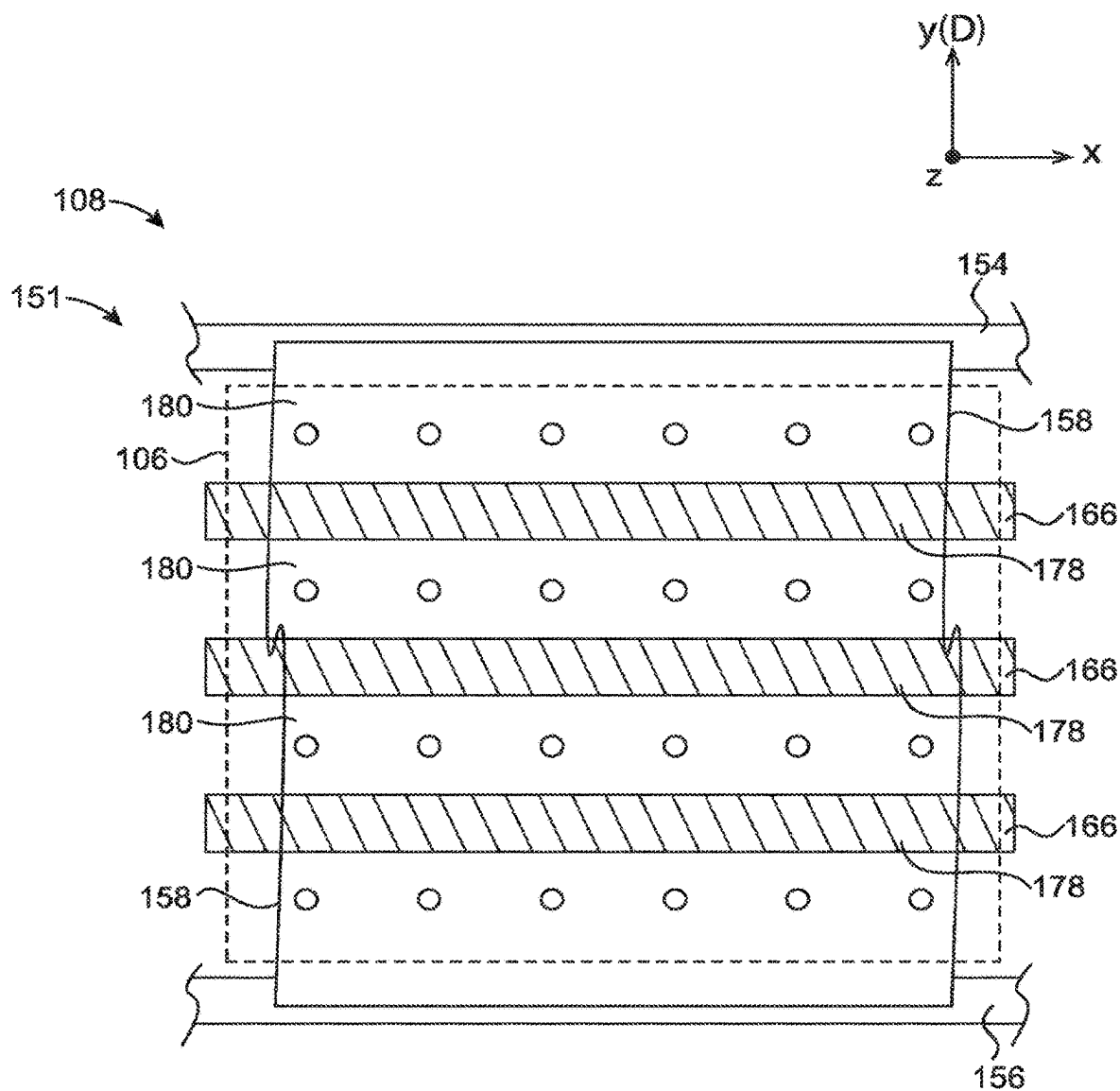


FIG. 6B

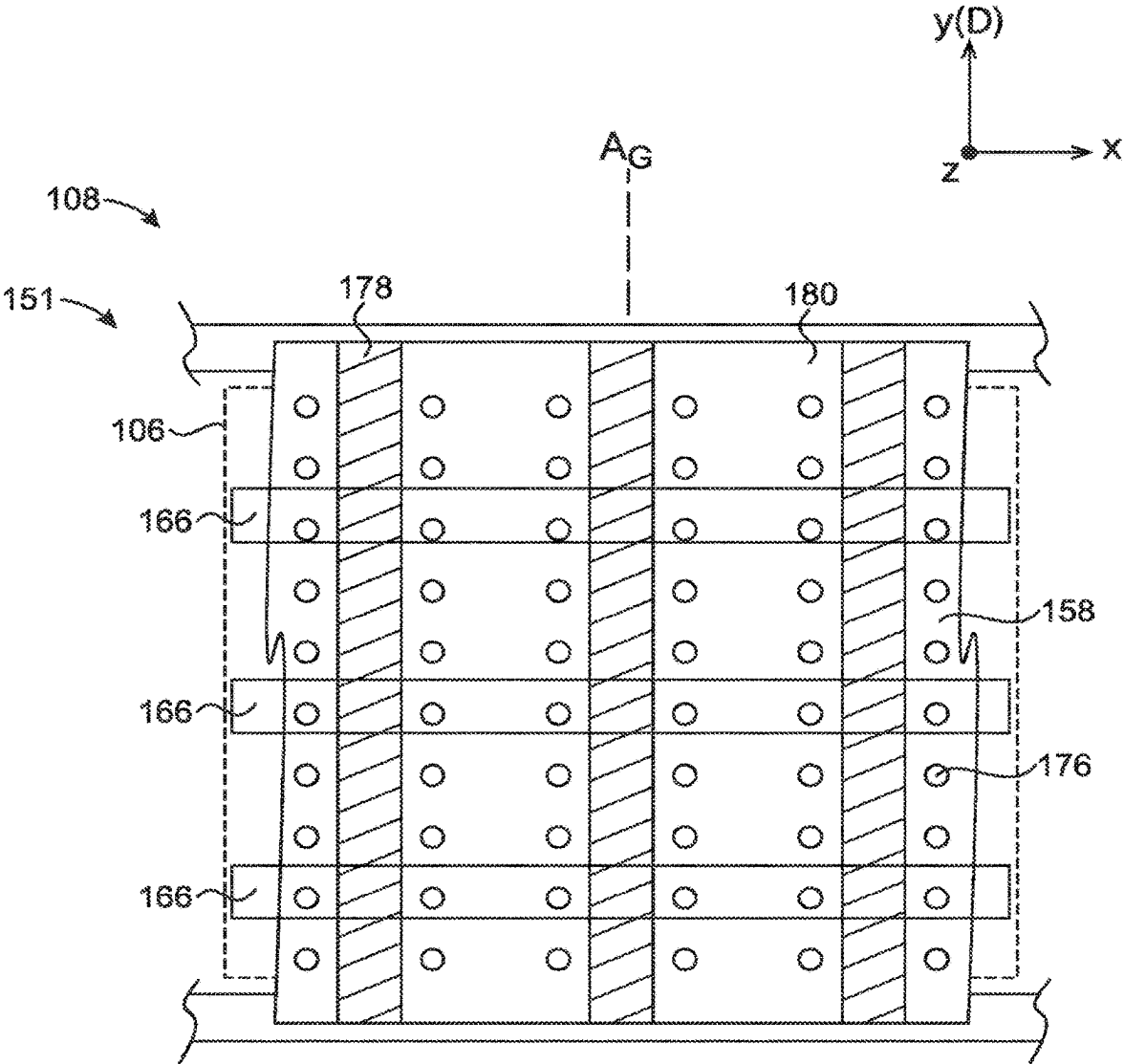


FIG. 6C

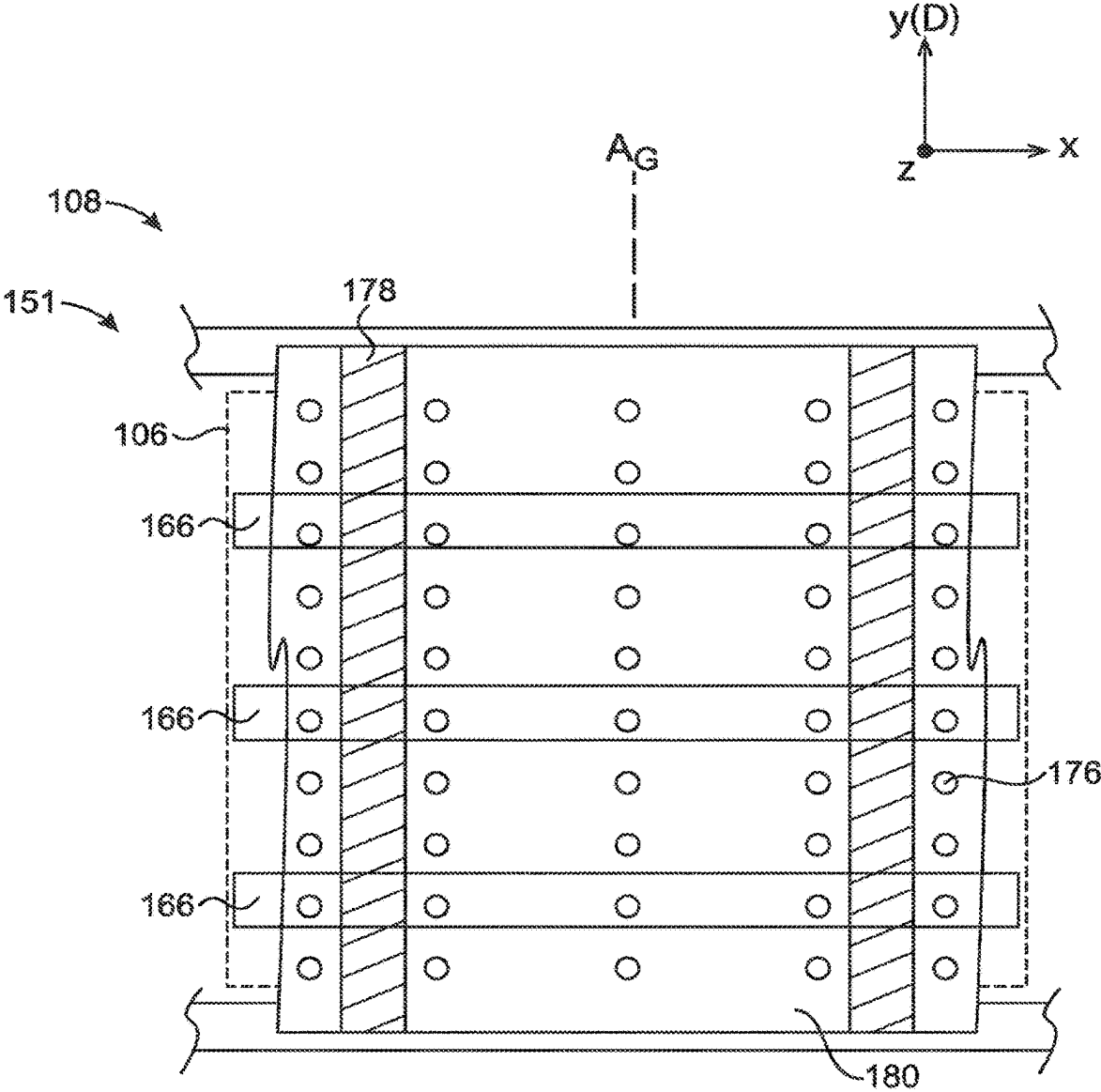


FIG. 6D

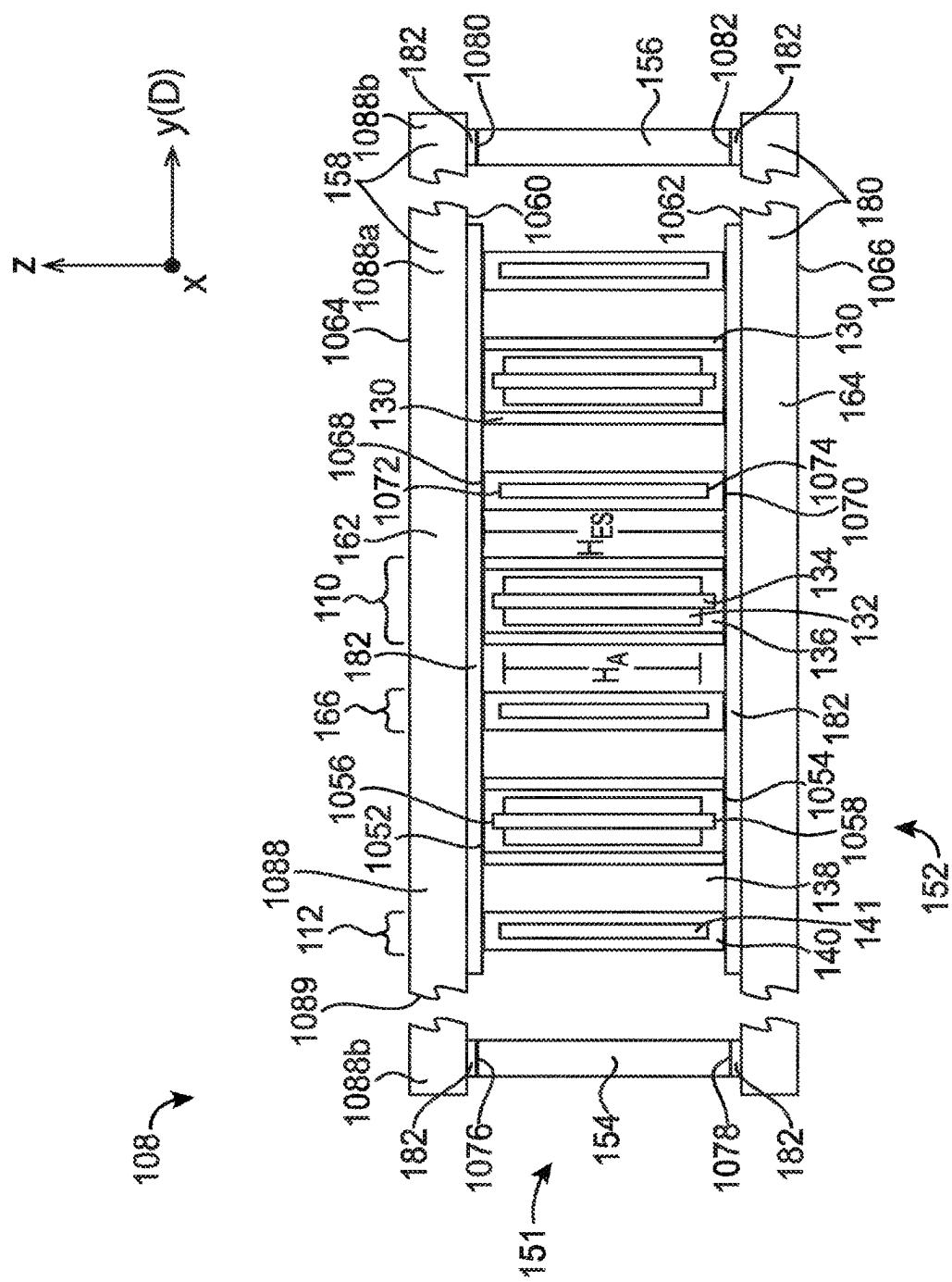


FIG. 7

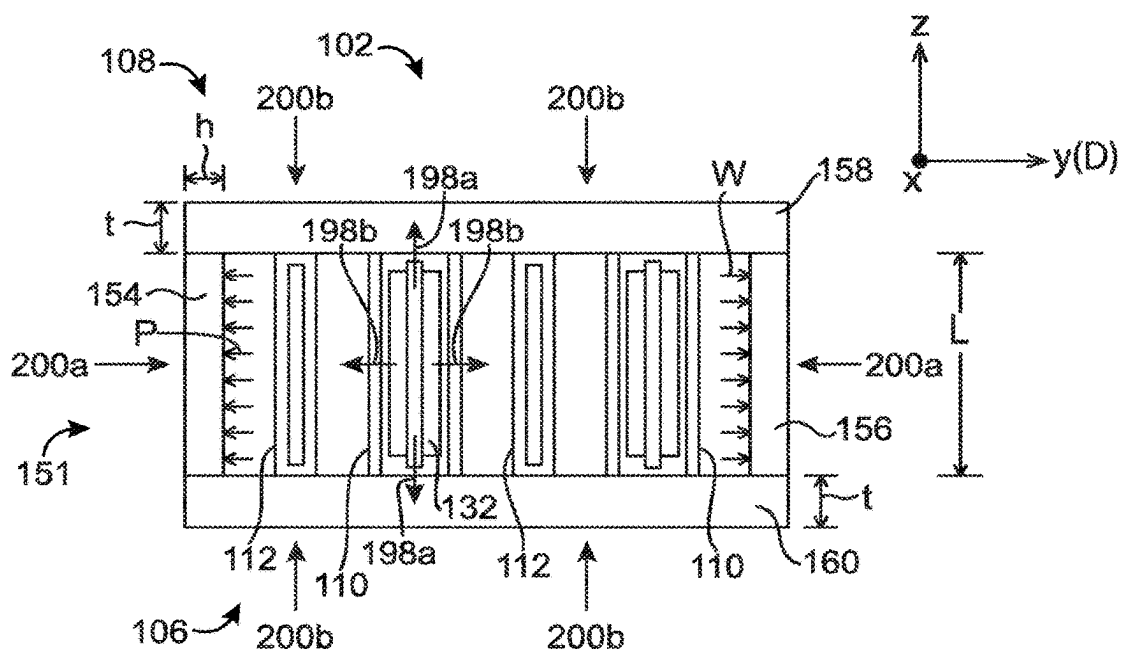


FIG. 8A

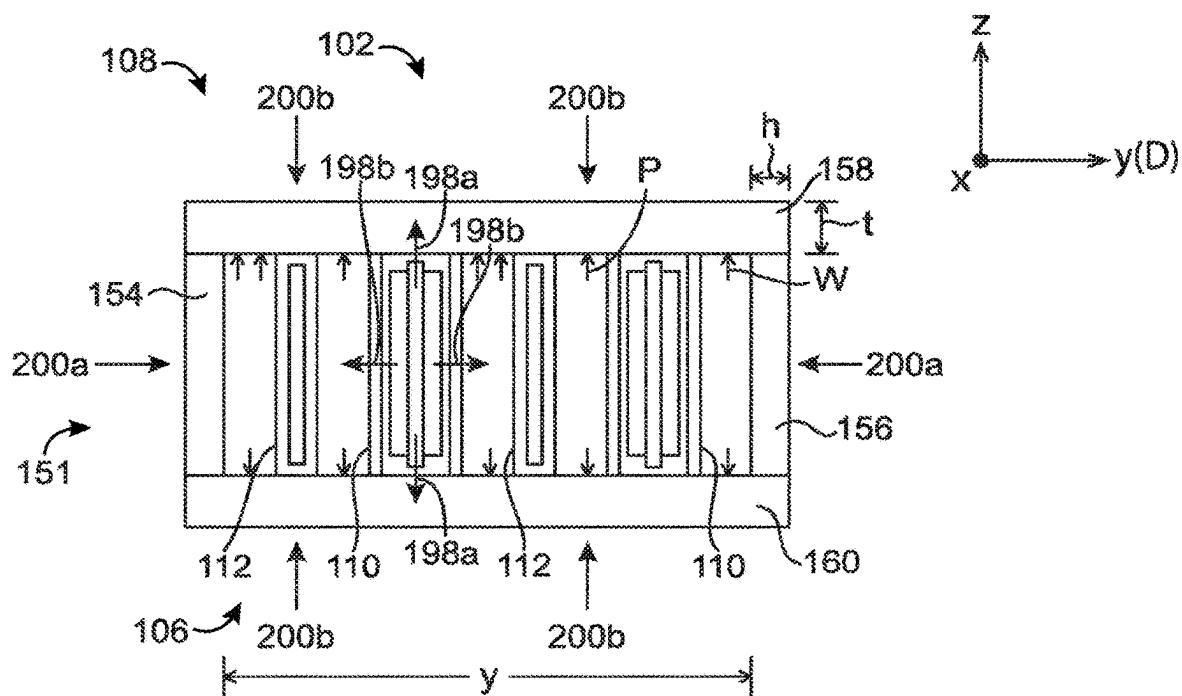


FIG. 8B

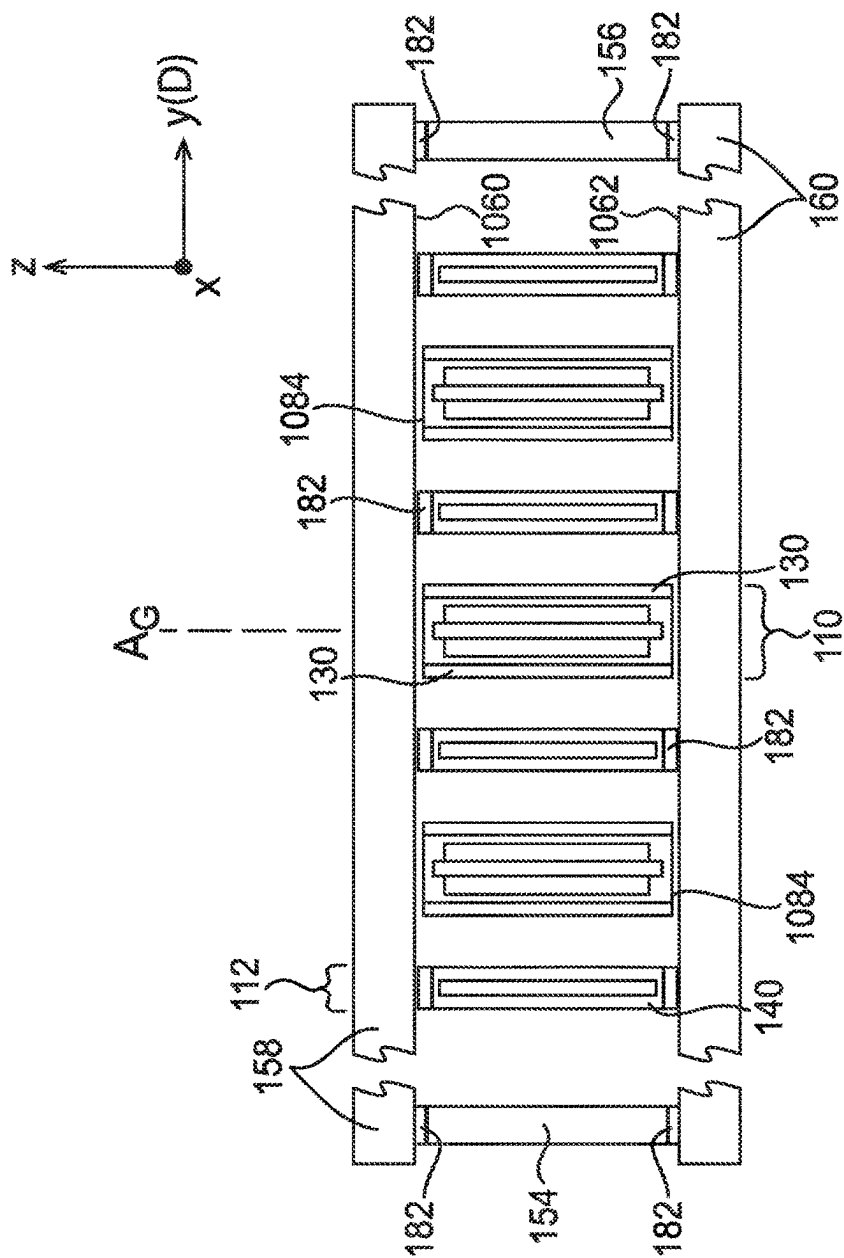


FIG. 9B

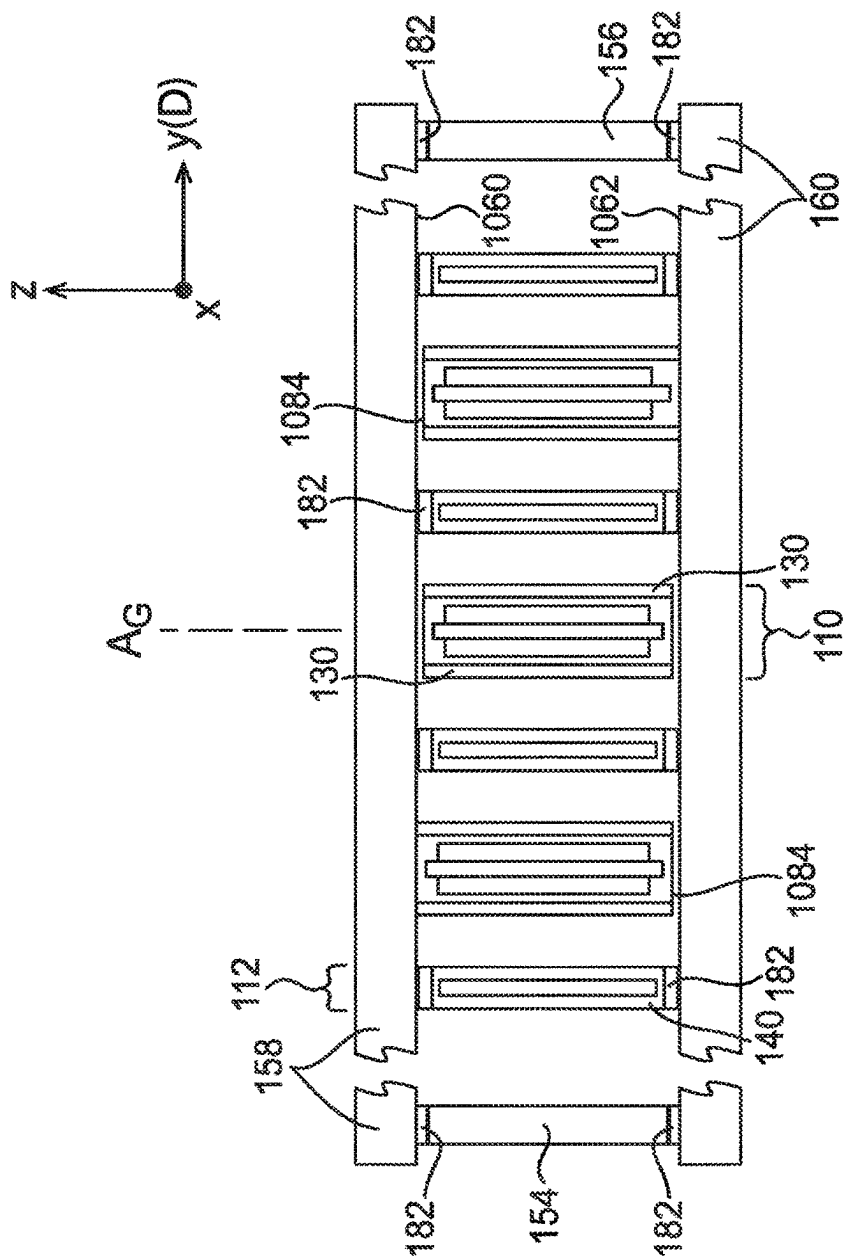


FIG. 9C

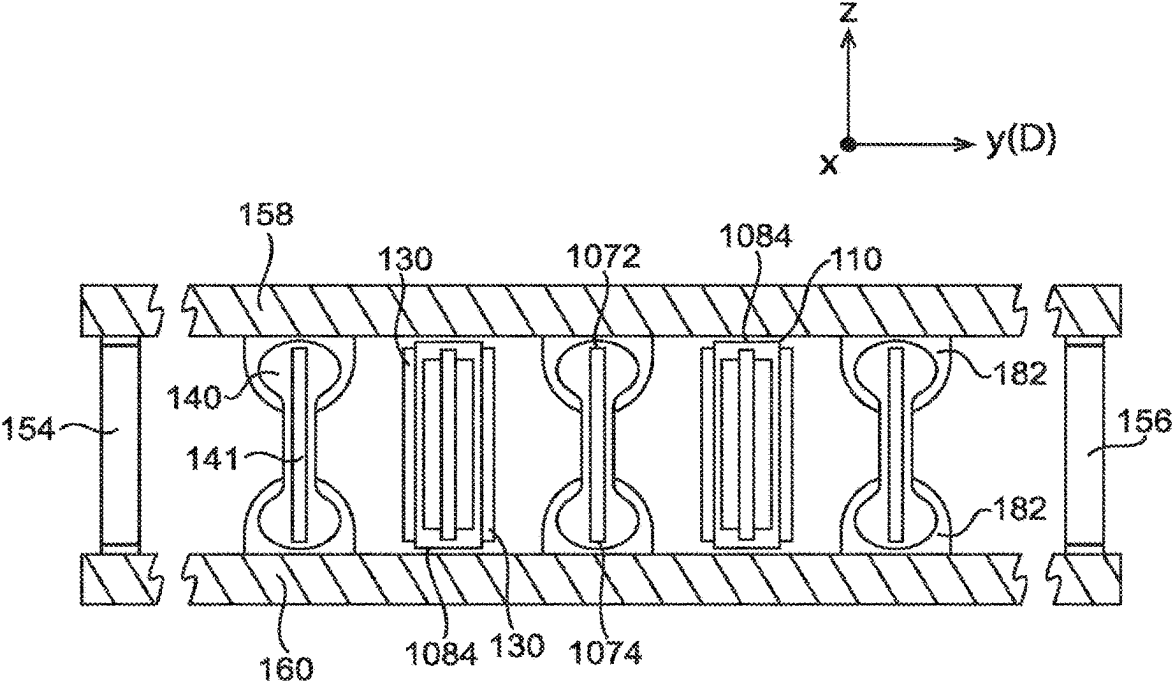


FIG. 10

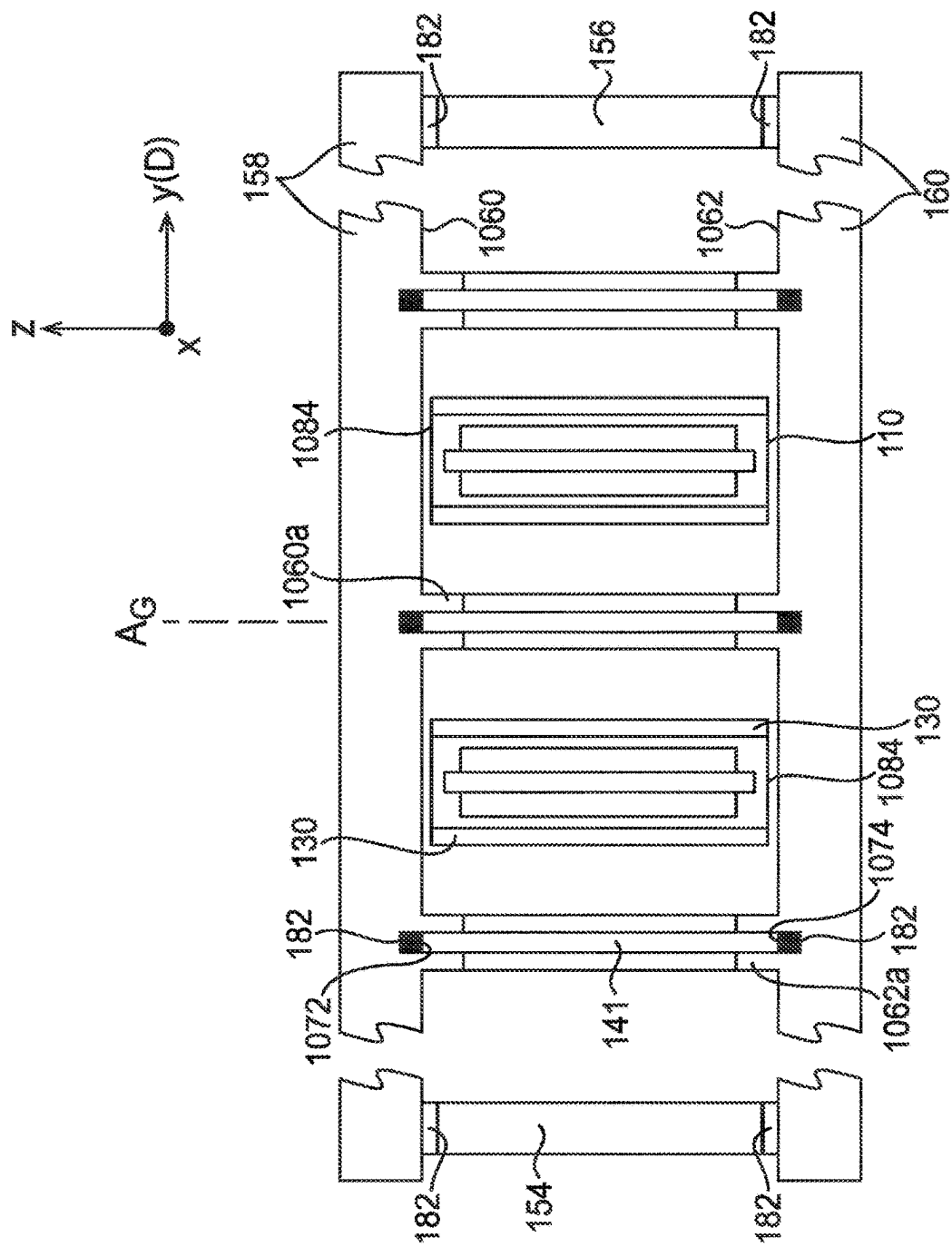
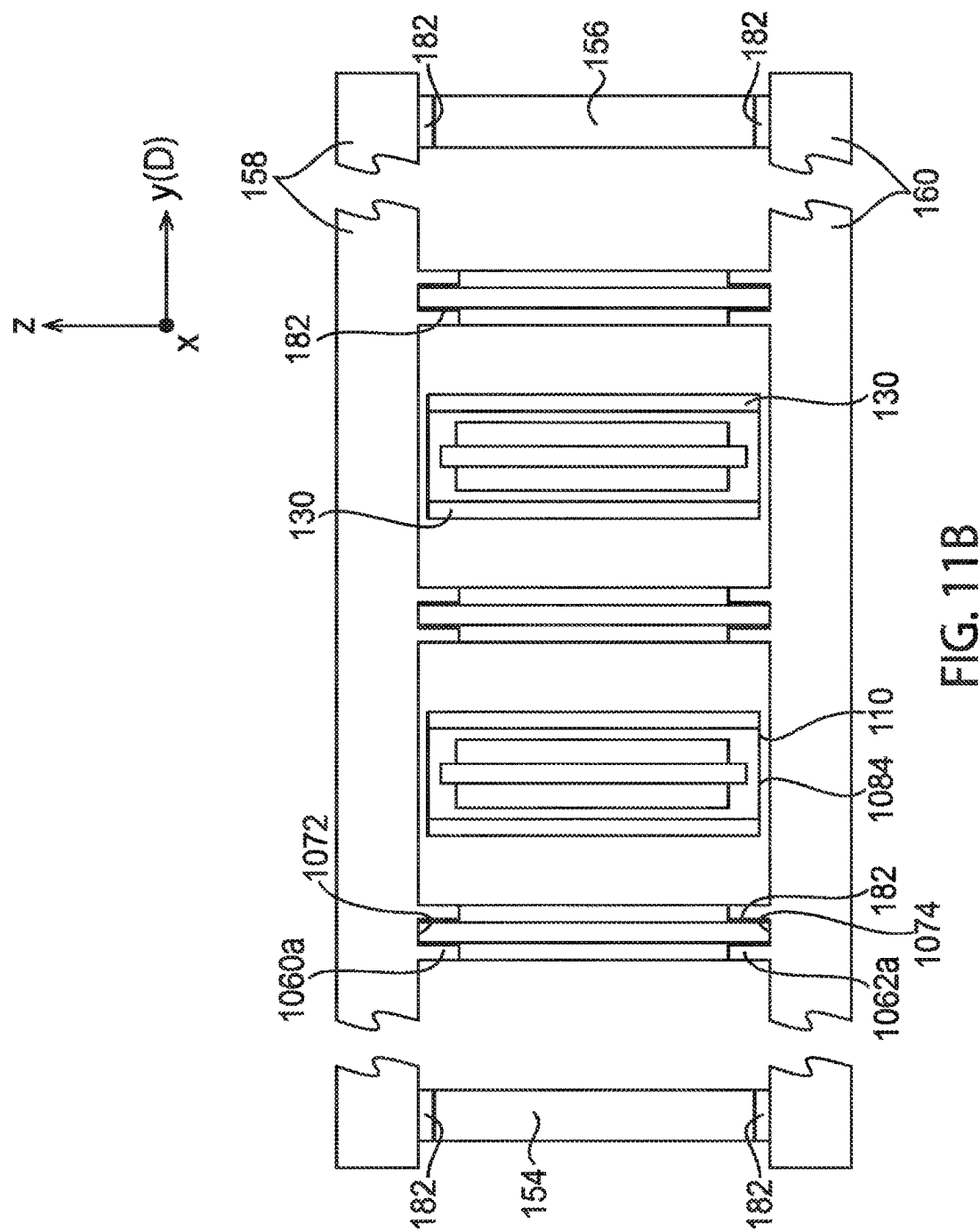


FIG. 11A



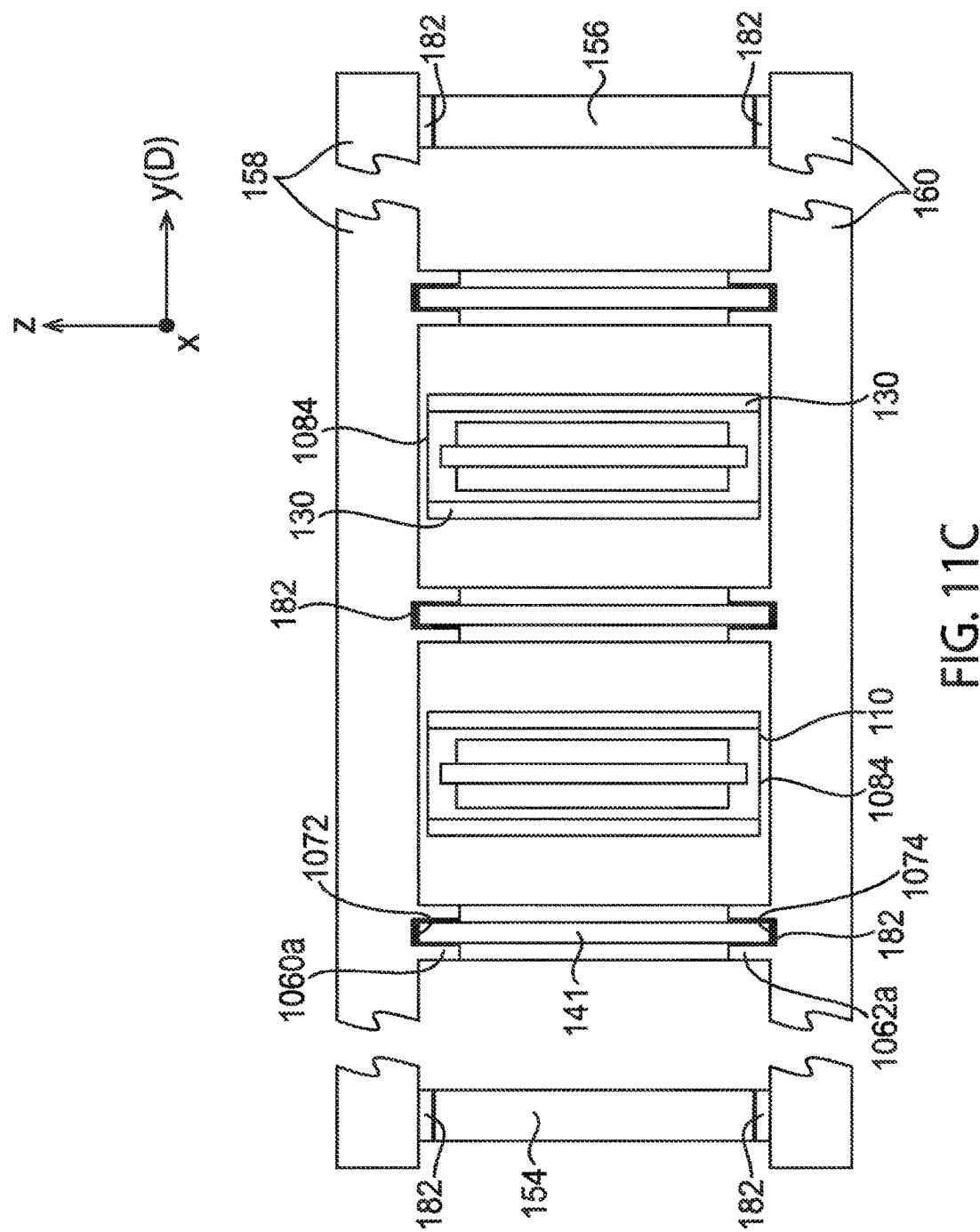
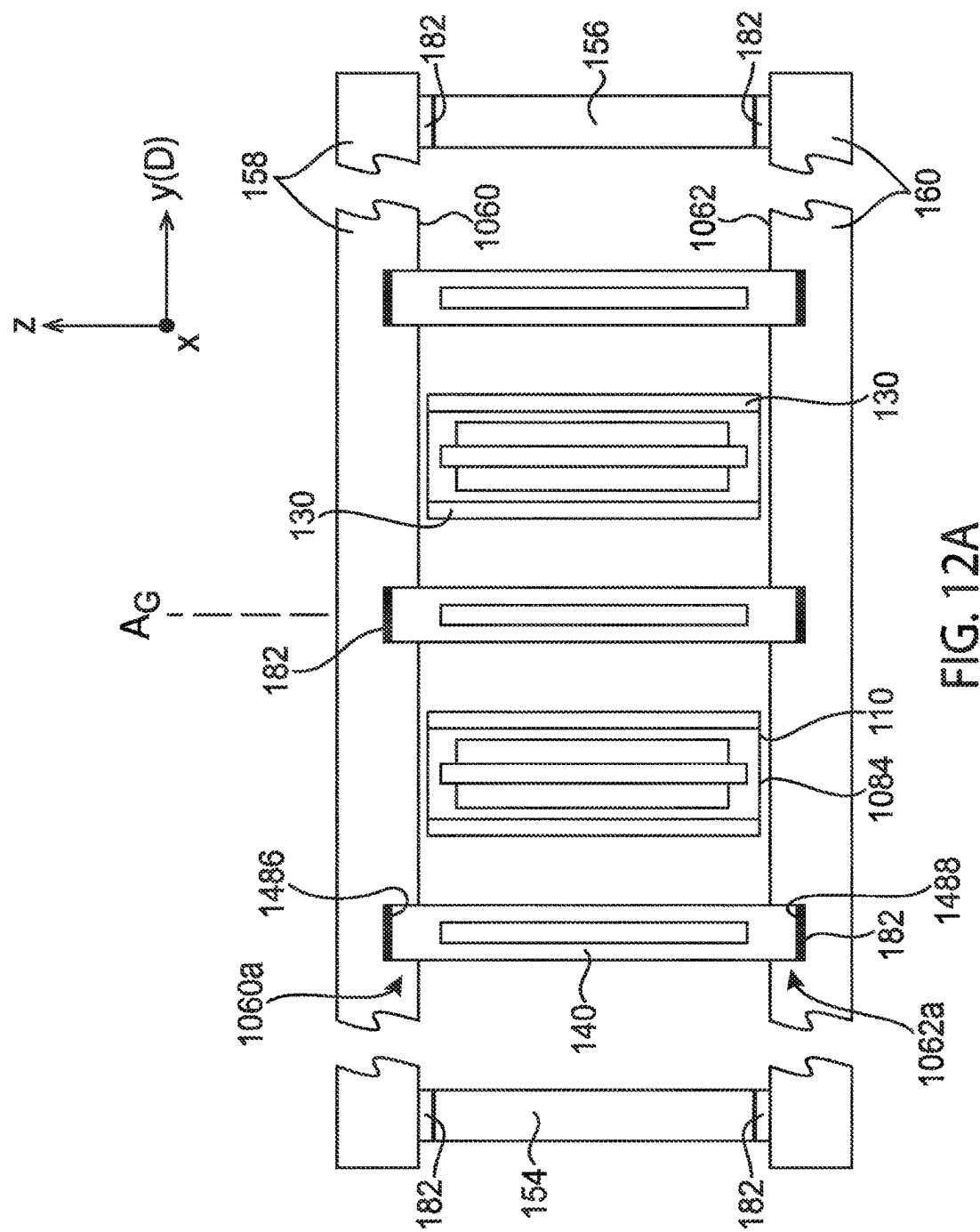
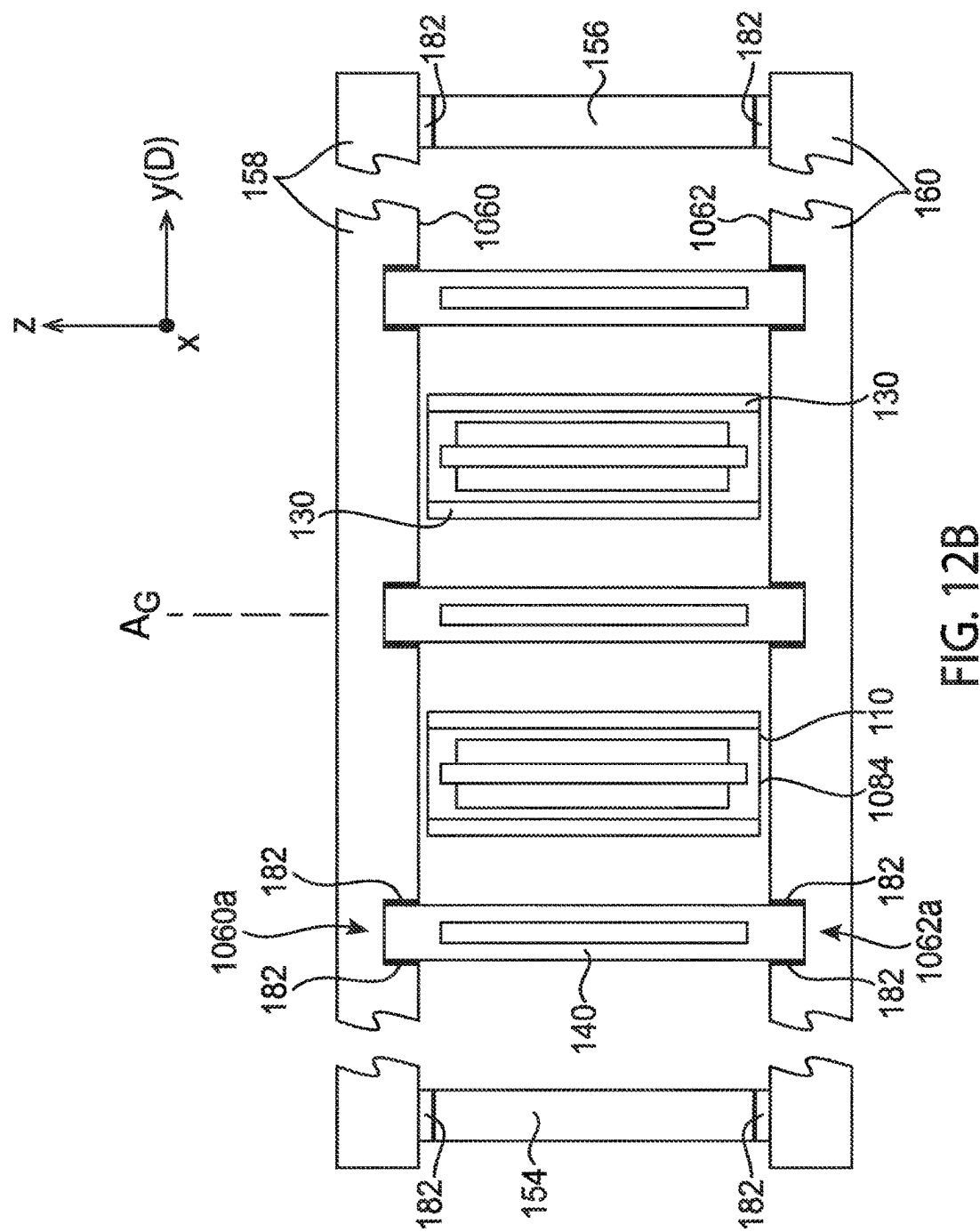
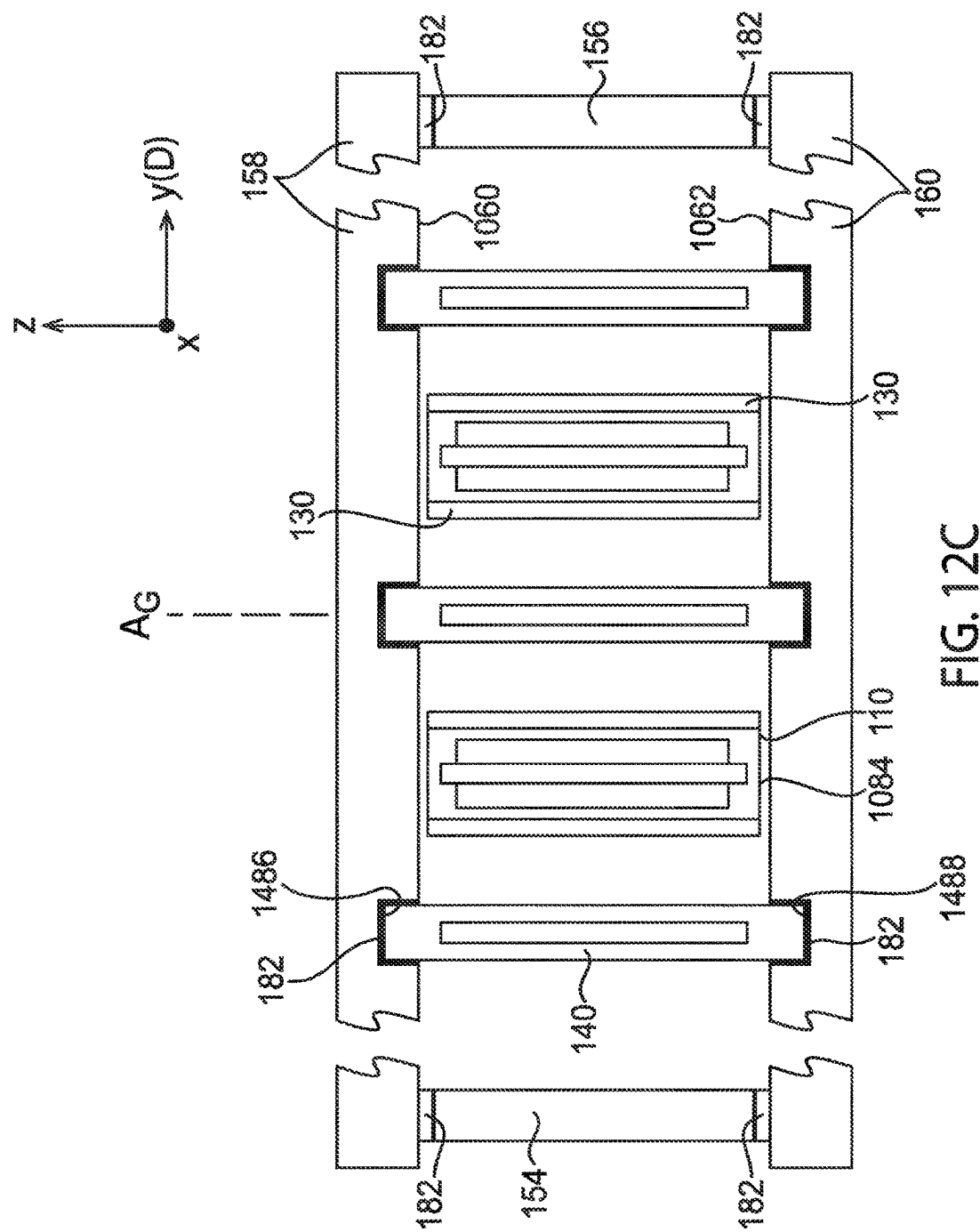


FIG. 11C







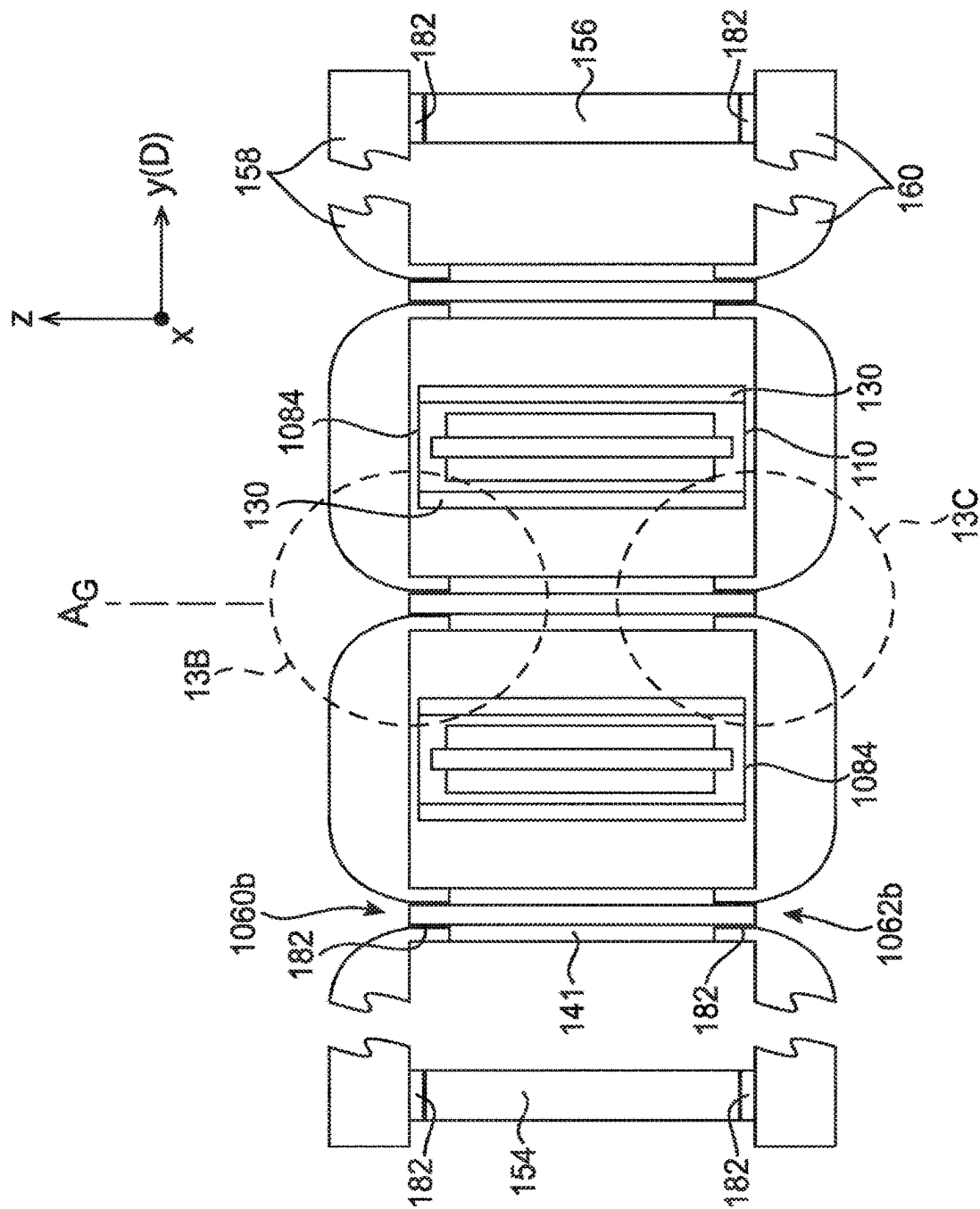


FIG. 13A

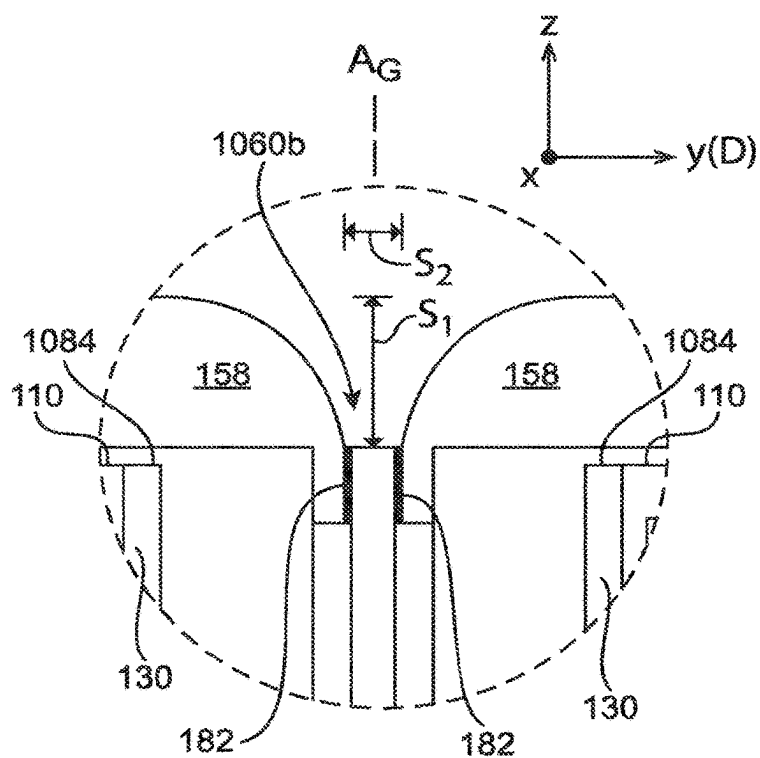


FIG. 13B

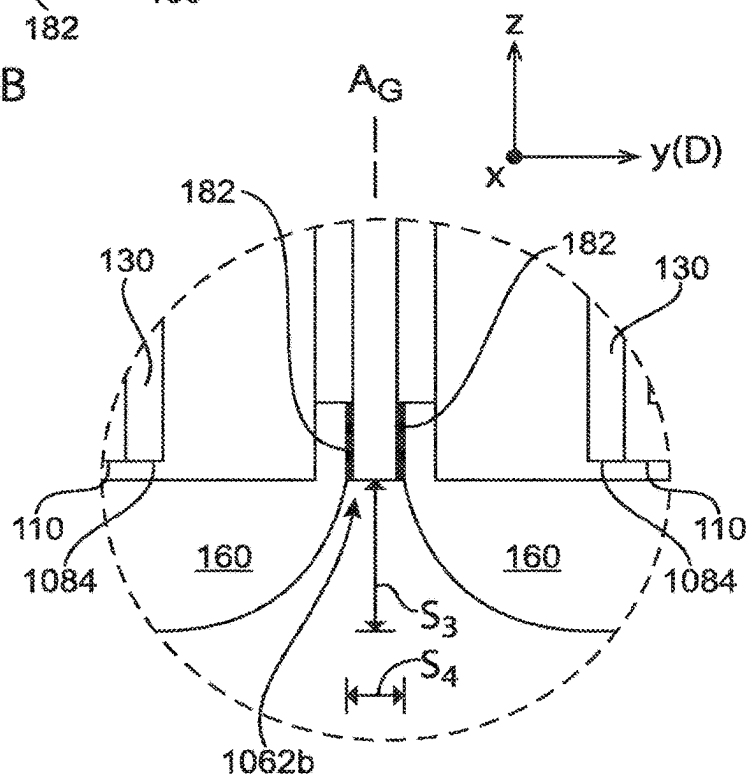
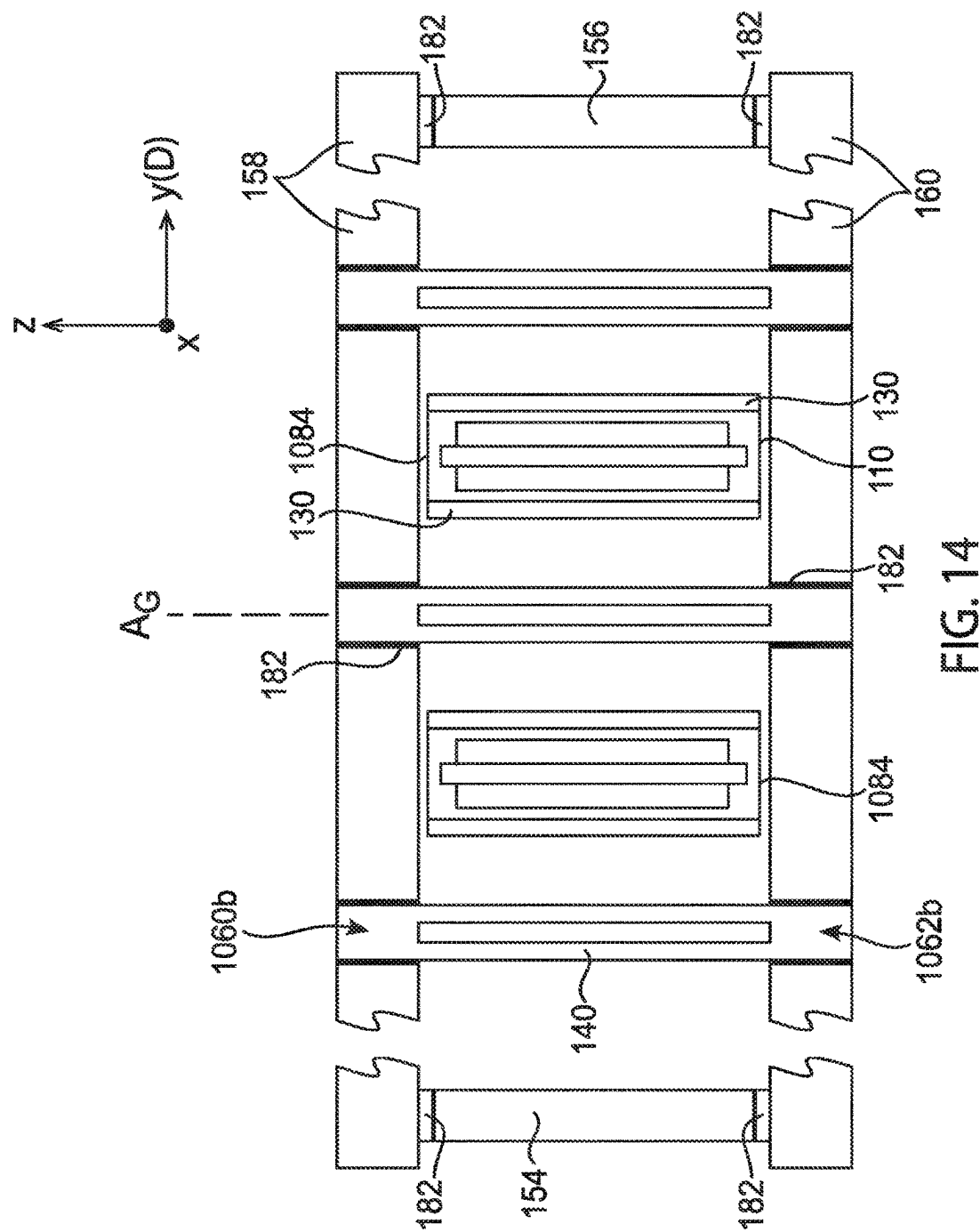


FIG. 13C



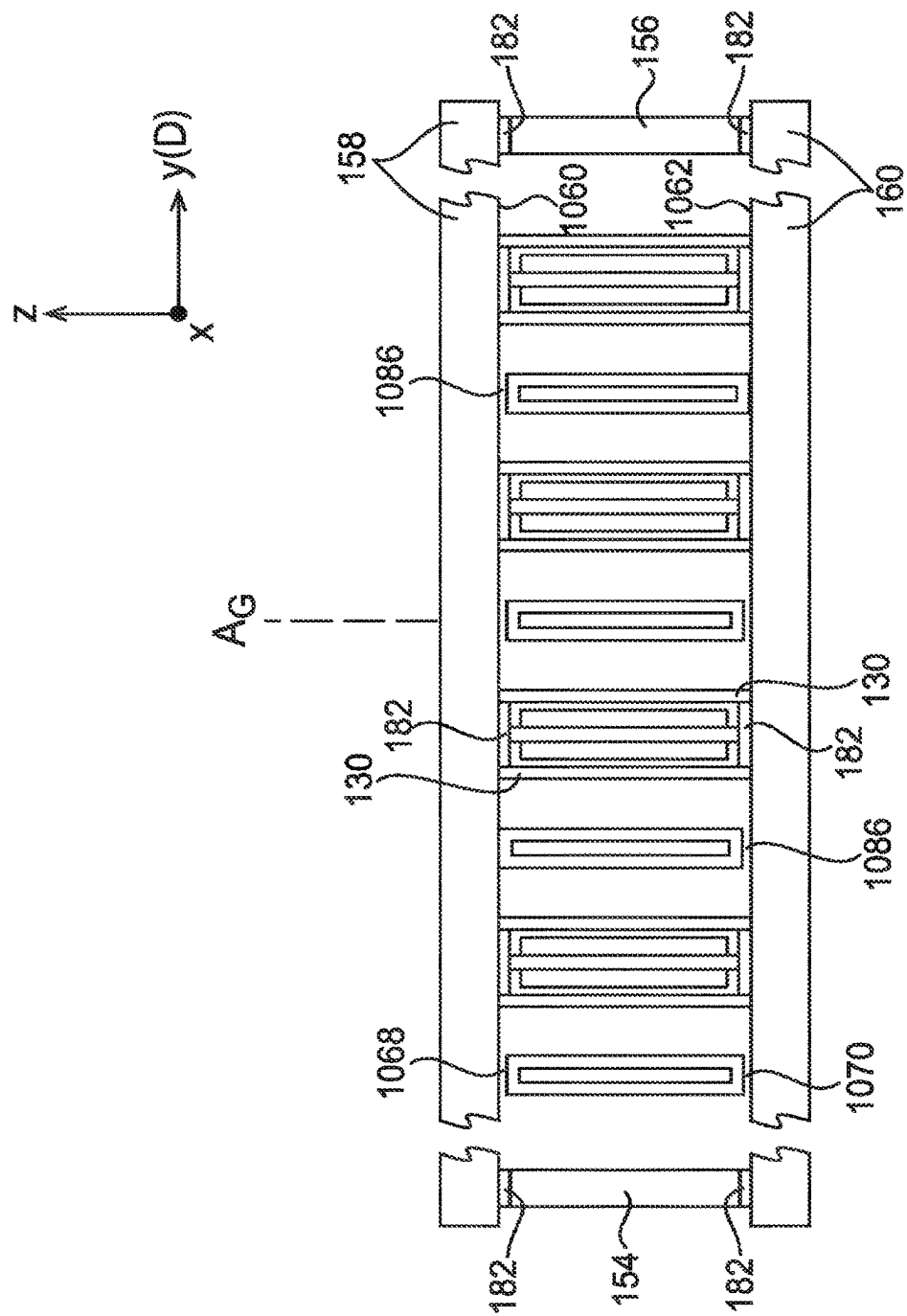


FIG. 15A

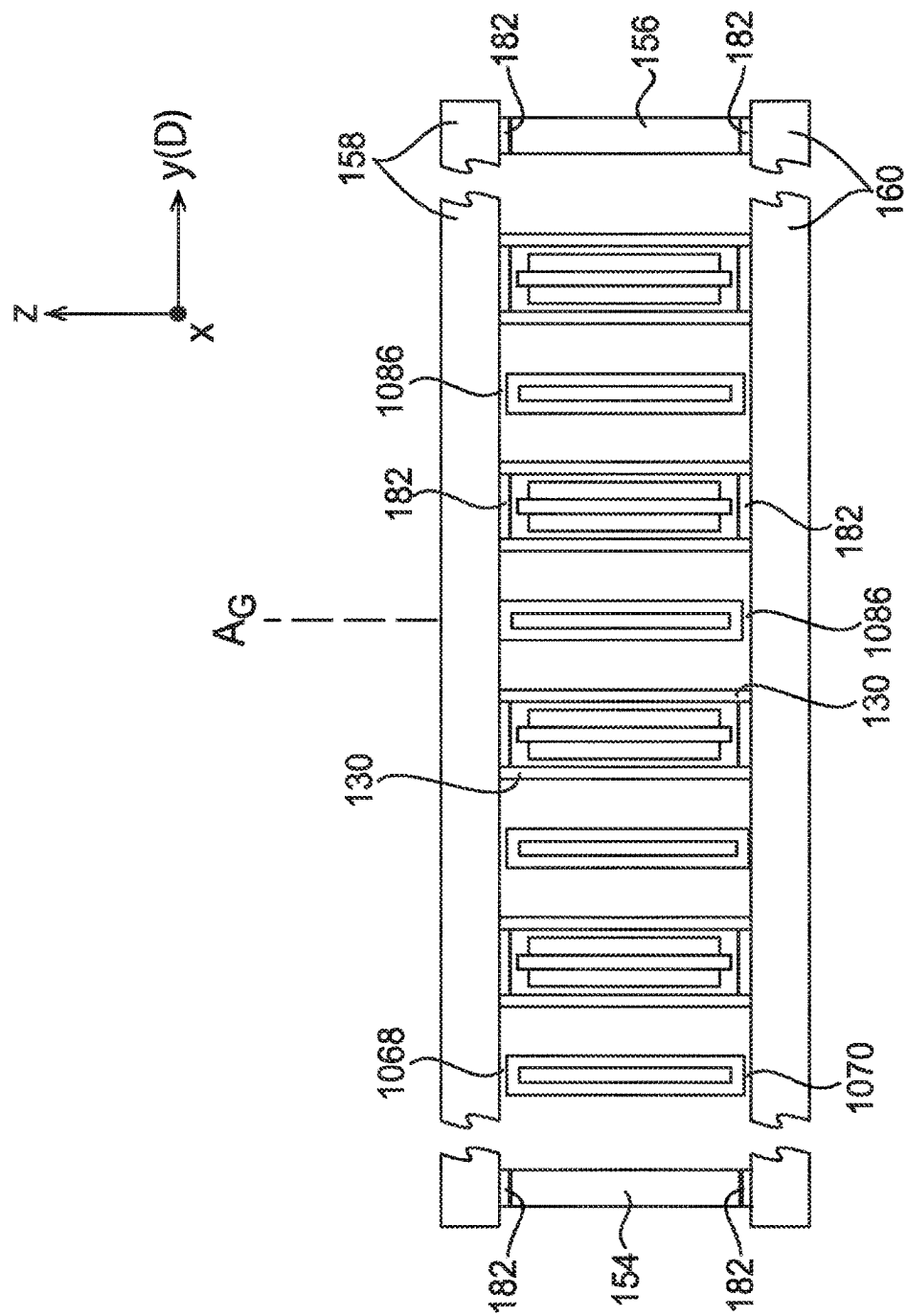


FIG. 15B

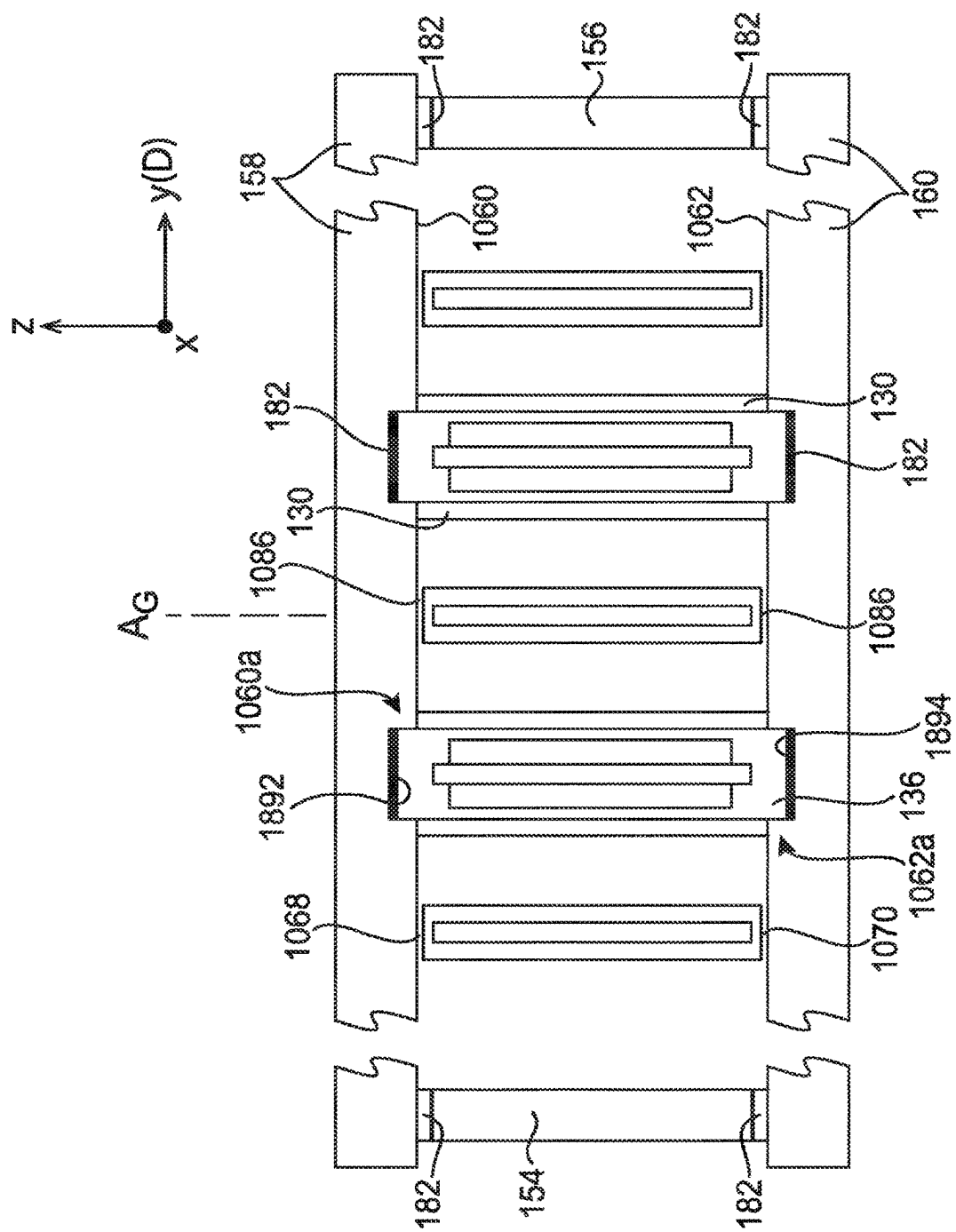
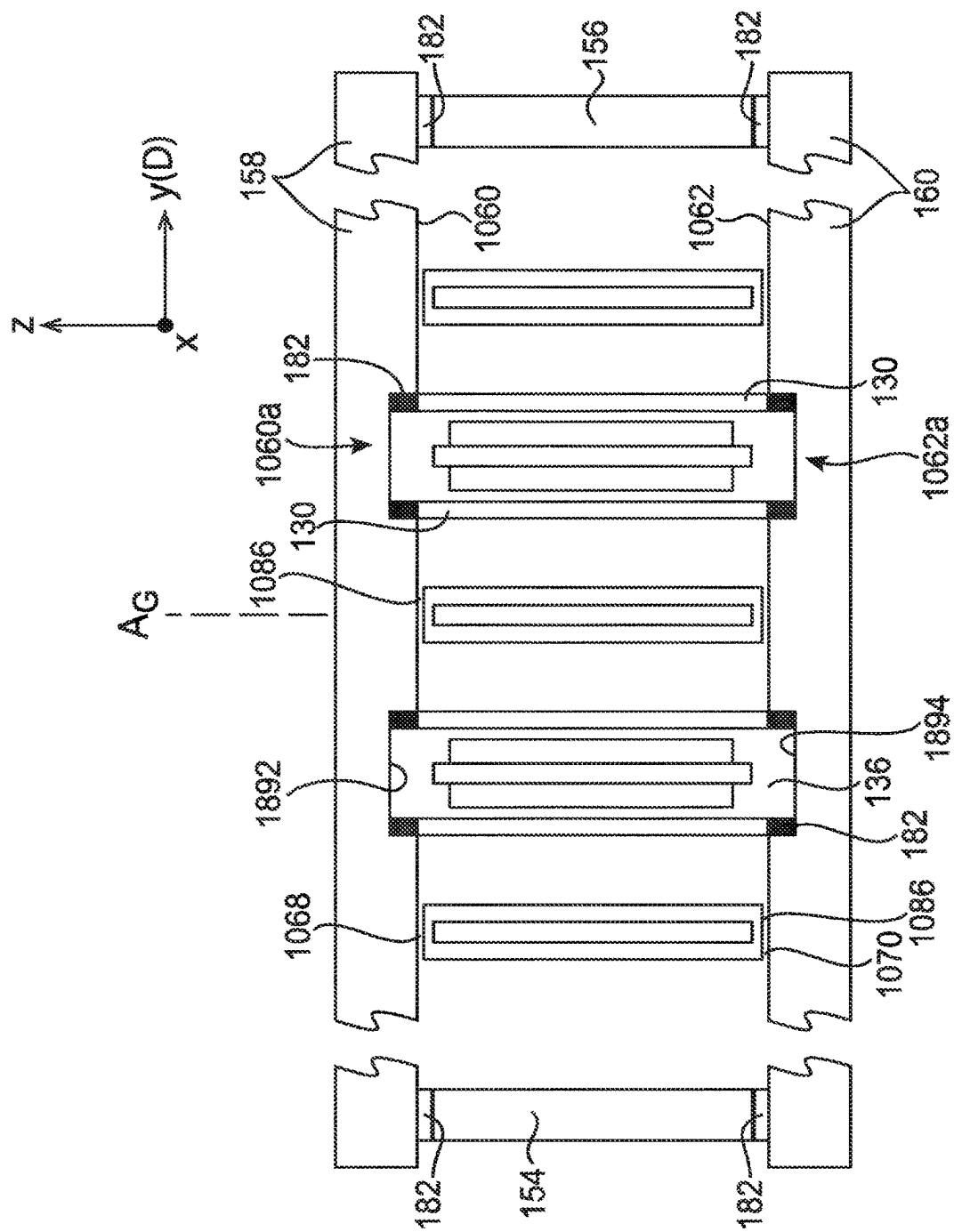


Fig. 16A



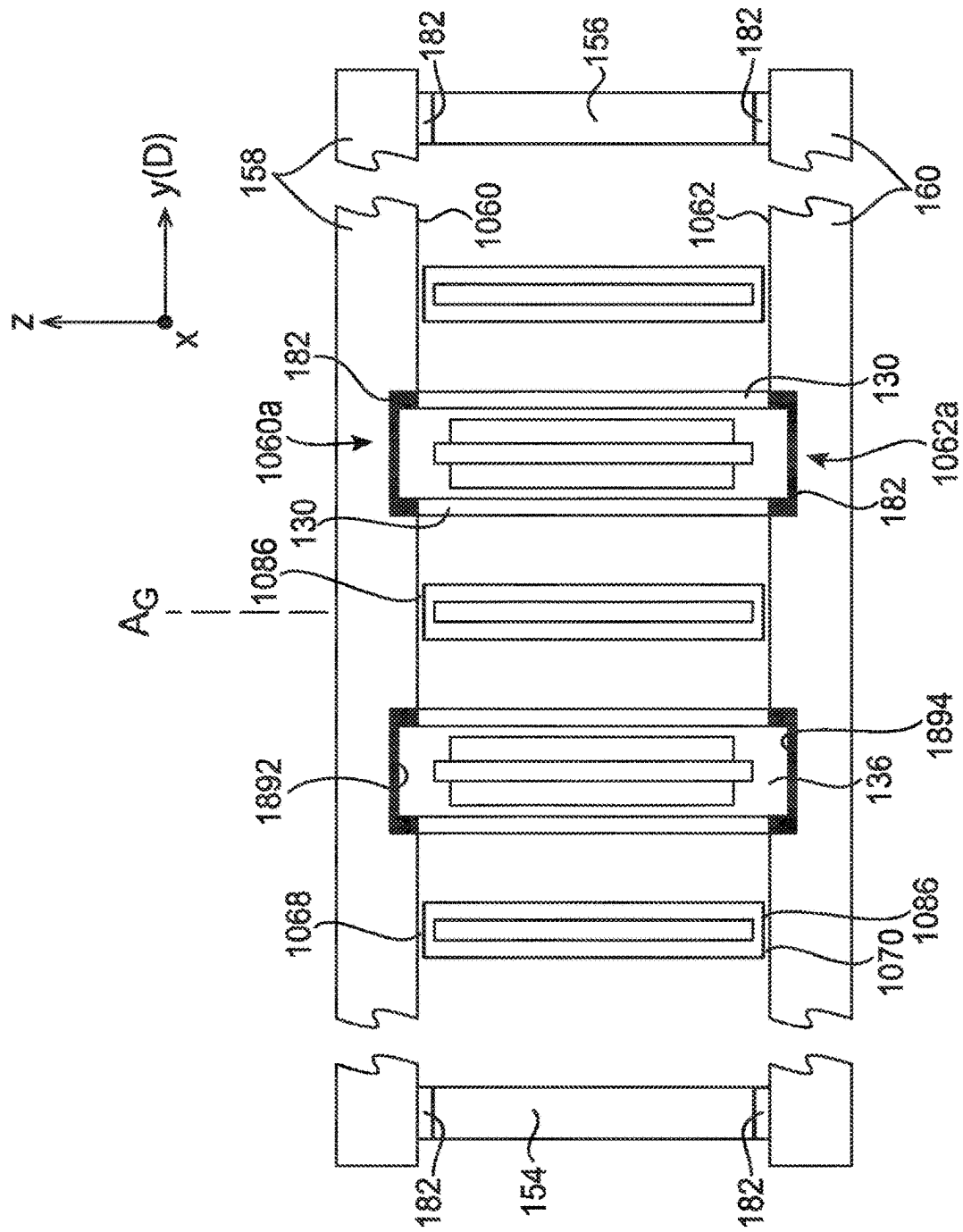


FIG. 16C

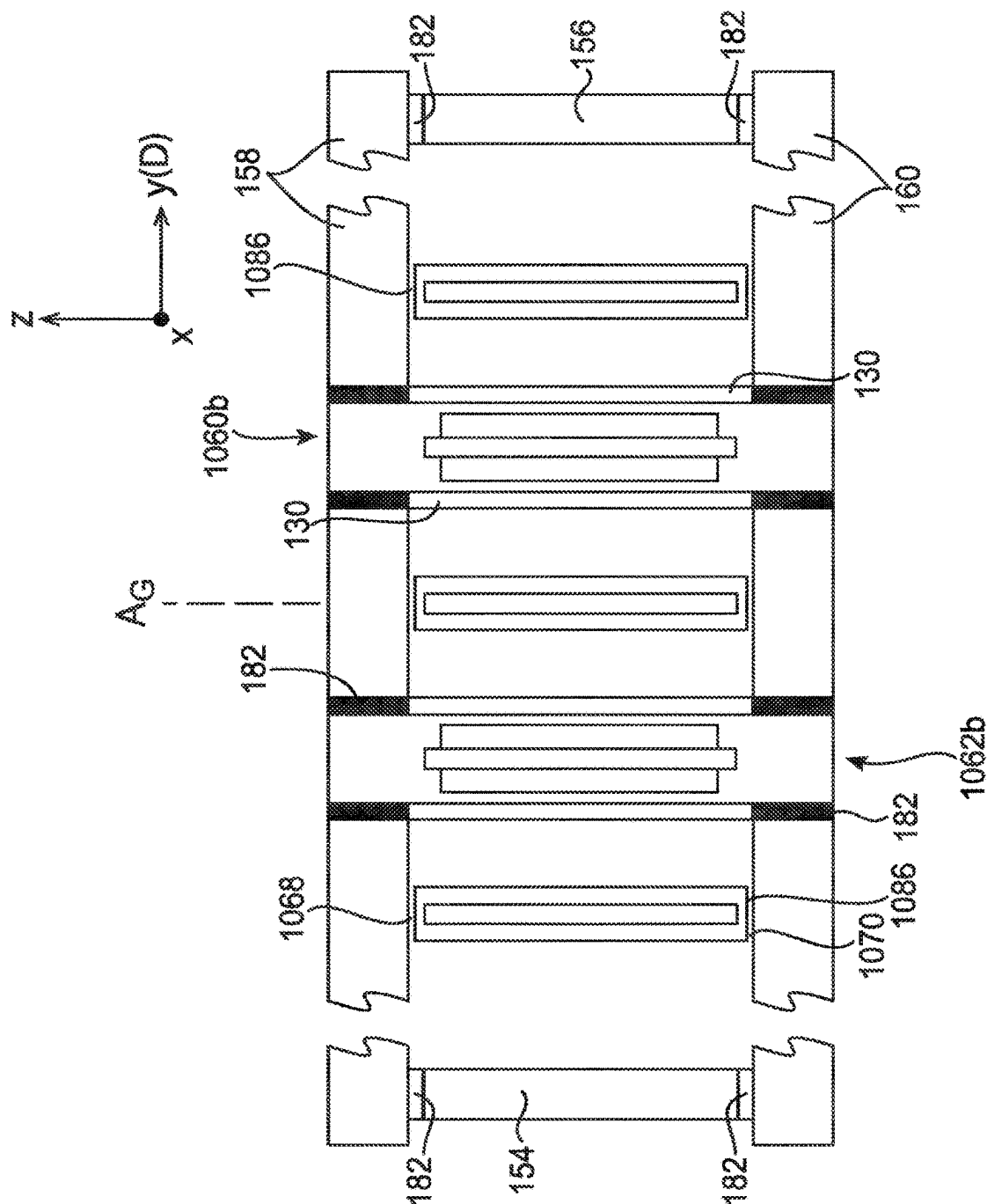


FIG. 17

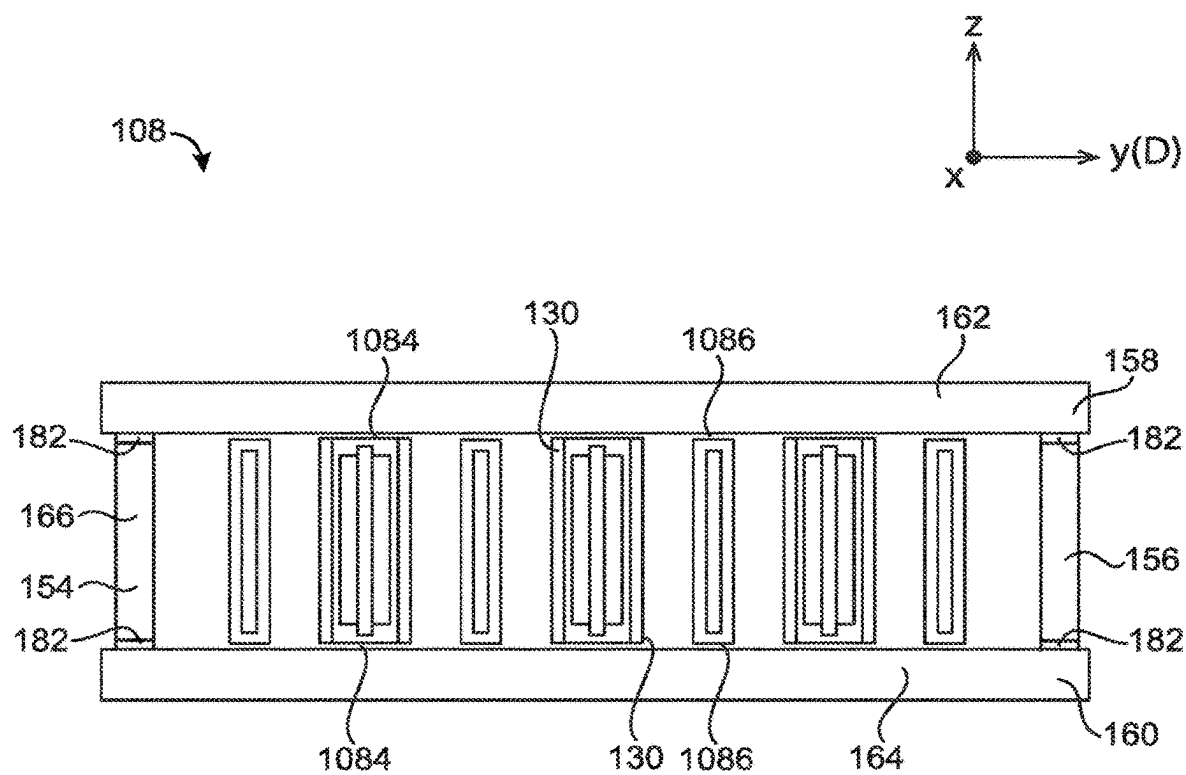


FIG. 18A

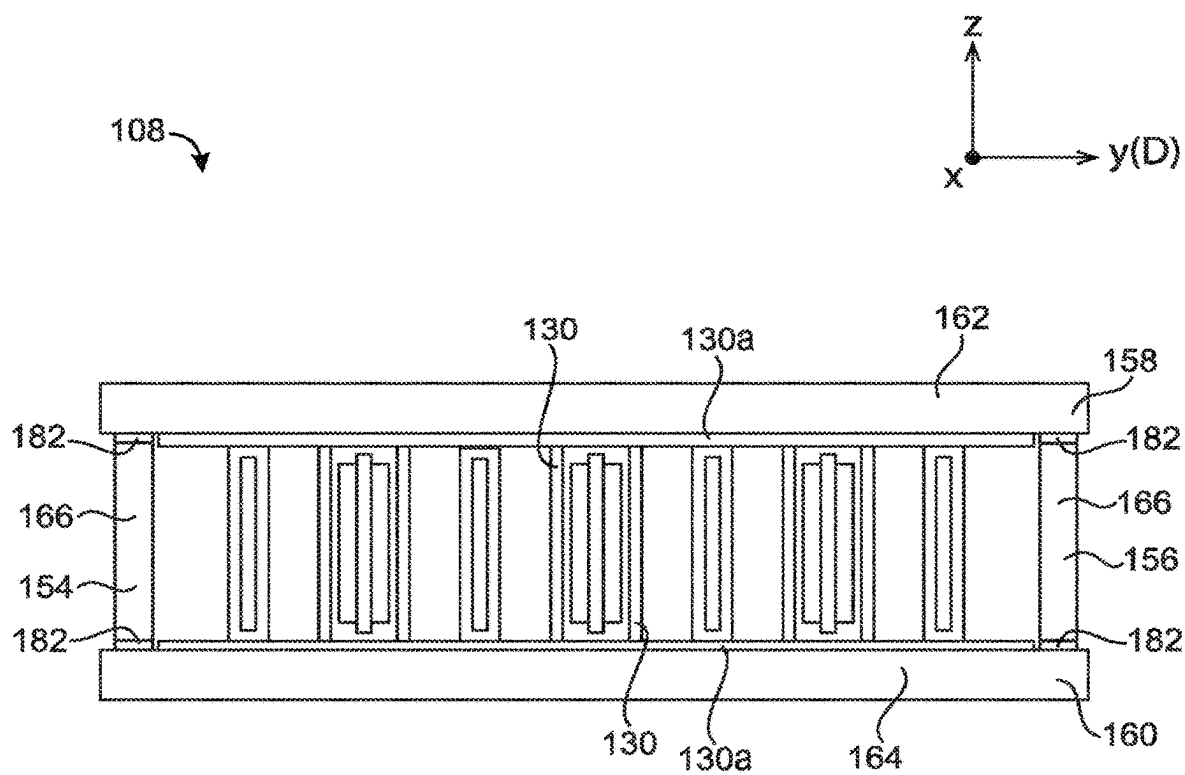


FIG. 18B

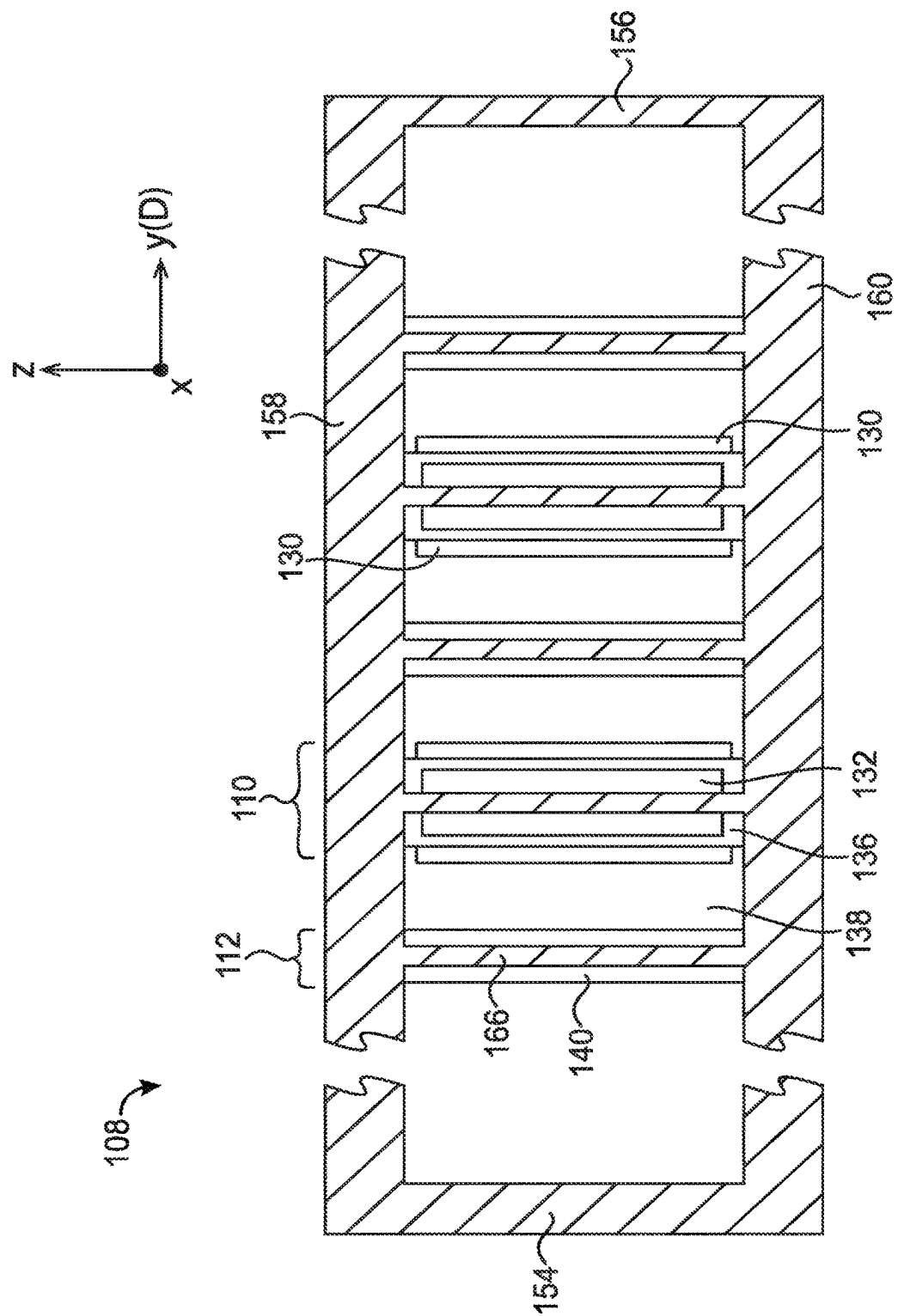


FIG. 19

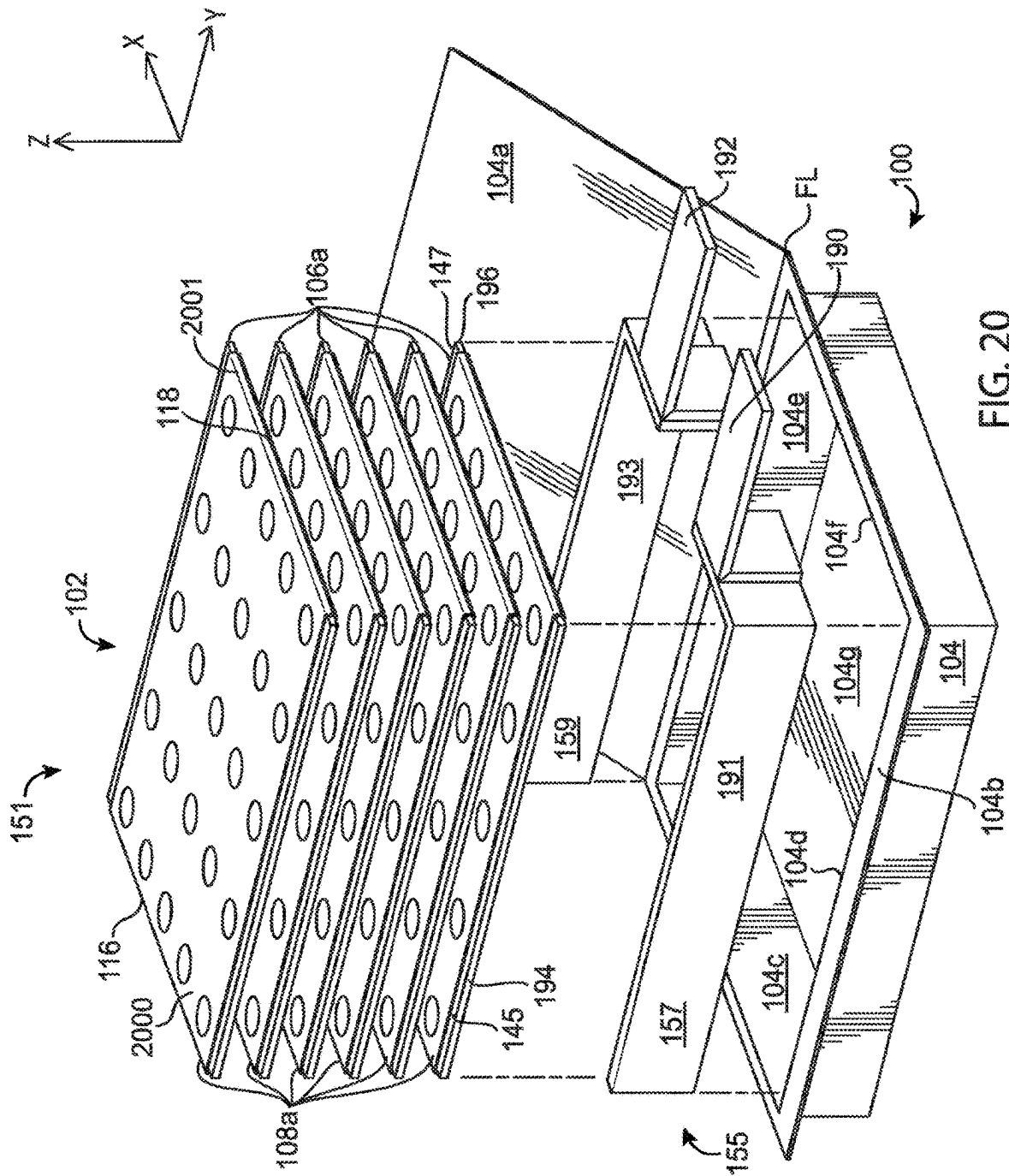


FIG. 20

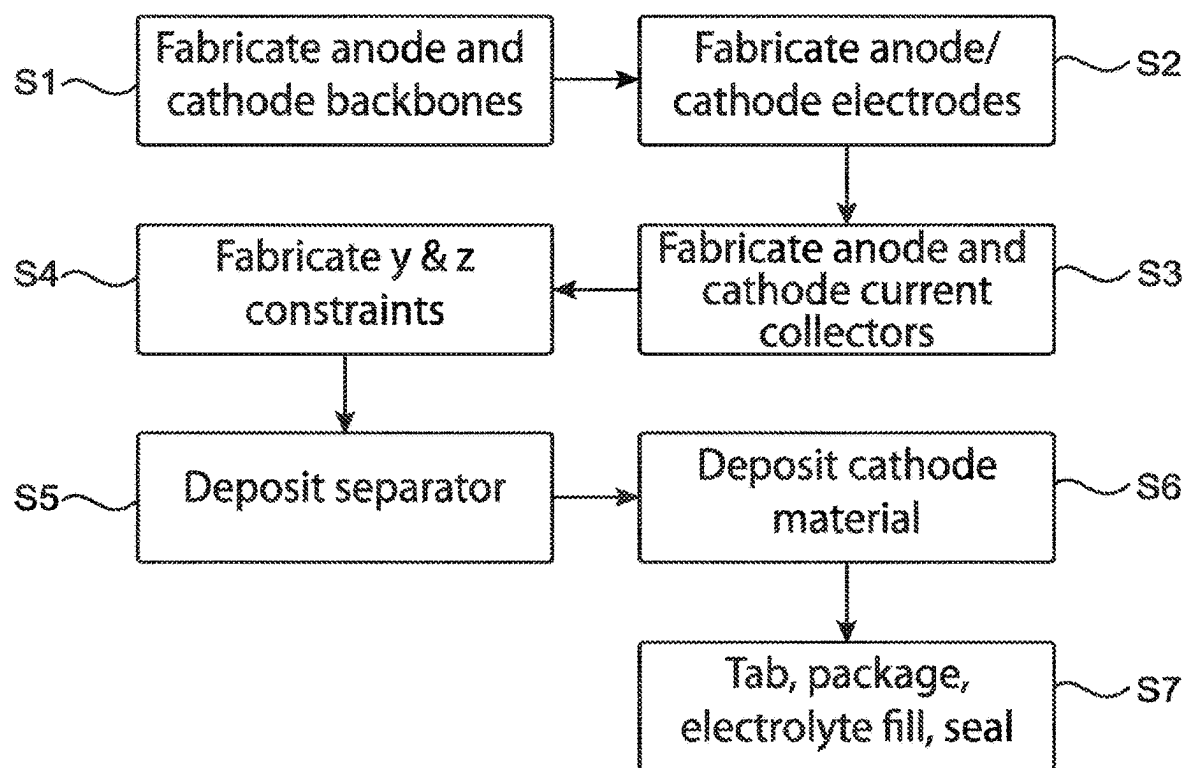


FIG. 21

DIMENSIONAL CONSTRAINTS FOR THREE-DIMENSIONAL BATTERIES

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 18/599,952, filed on Mar. 8, 2024, which is a continuation of U.S. patent application Ser. No. 17/903,250 filed on Sep. 6, 2022, now U.S. Pat. No. 11,961,952, which is a continuation of U.S. patent application Ser. No. 17/335,725 filed on Jun. 1, 2021, now U.S. Pat. No. 11,444,310, which is a continuation of U.S. patent application Ser. No. 16/241,159 filed on Jan. 7, 2019, now U.S. Pat. No. 11,081,718, which is a continuation of U.S. patent application Ser. No. 15/889,338 filed on Feb. 6, 2018, now U.S. Pat. No. 10,177,400, which is a continuation of International Application No. PCT/US2017/032355 filed May 12, 2017, which claims the benefit of priority from U.S. Patent Application No. 62/335,912 filed on May 13, 2016 and claims the benefit of priority from U.S. Patent Application No. 62/422,958 filed on Nov. 16, 2016, all of which are hereby incorporated by reference herein in their entireties.

FIELD OF THE INVENTION

[0002] This disclosure generally relates to structures for use in energy storage devices, to energy storage devices employing such structures, and to methods for producing such structures and energy devices.

BACKGROUND

[0003] Rocking chair or insertion secondary batteries are a type of energy storage device in which carrier ions, such as lithium, sodium, potassium, calcium or magnesium ions, move between a positive electrode and a negative electrode through an electrolyte. The secondary battery may comprise a single battery cell, or two or more battery cells that have been electrically coupled to form the battery, with each battery cell comprising a positive electrode, a negative electrode, a microporous separator, and an electrolyte.

[0004] In rocking chair battery cells, both the positive and negative electrodes comprise materials into which a carrier ion inserts and extracts. As a cell is discharged, carrier ions are extracted from the negative electrode and inserted into the positive electrode. As a cell is charged, the reverse process occurs: the carrier ion is extracted from the positive and inserted into the negative electrode.

[0005] When the carrier ions move between electrodes, one of the persistent challenges resides in the fact that the electrodes tend to expand and contract as the battery is repeatedly charged and discharged. The expansion and contraction during cycling tends to be problematic for reliability and cycle life of the battery because when the electrodes expand, electrical shorts and battery failures occur.

[0006] Therefore, there remains a need for controlling the expansion and contraction of electrodes during battery cycling to improve reliability and cycle life of the battery.

SUMMARY

[0007] Briefly, therefore, one aspect of this disclosure relates to the implementation of constraint structures to

mitigate or prevent the macroscopic expansion of electrodes, thereby improving the energy density, reliability, and cycle life of batteries.

[0008] According to one aspect, a secondary battery is provided for cycling between a charged and a discharged state, the secondary battery having a battery enclosure, an electrode assembly, carrier ions, a non-aqueous liquid electrolyte within the battery enclosure, and a set of electrode constraints. The electrode assembly has mutually perpendicular longitudinal, transverse, and vertical axes, a first longitudinal end surface and a second longitudinal end surface separated from each other in the longitudinal direction, and a lateral surface surrounding an electrode assembly longitudinal axis A_{EA} and connecting the first and second longitudinal end surfaces, the lateral surface having opposing first and second regions on opposite sides of the longitudinal axis and separated in a first direction that is orthogonal to the longitudinal axis, the electrode assembly having a maximum width W_{EA} measured in the longitudinal direction, a maximum length L_{EA} bounded by the lateral surface and measured in the transverse direction, and a maximum height H_{EA} bounded by the lateral surface and measured in the vertical direction, the ratio of each of L_{EA} and W_{EA} to H_{EA} being at least 2:1, respectively. The electrode assembly has a population of electrode structures, a population of counter-electrode structures, and an electrically insulating microporous separator material electrically separating members of the electrode and counter-electrode populations, members of the electrode and counter-electrode structure populations being arranged in an alternating sequence in the longitudinal direction. Each member of the population of electrode structures has a layer of an electrode active material and each member of the population of counter-electrode structures comprises a layer of a counter-electrode active material, wherein the electrode active material has the capacity to accept more than one mole of carrier ion per mole of electrode active material when the secondary battery is charged from a discharged state to a charged state. The set of electrode constraints includes a primary constraint system comprising first and second primary growth constraints and at least one primary connecting member, the first and second primary growth constraints separated from each other in the longitudinal direction, and the at least one primary connecting member connecting the first and second primary growth constraints, wherein the primary constraint system restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20%. The set of electrode constraints further includes a secondary constraint system comprising first and second secondary growth constraints separated in a second direction and connected by at least one secondary connecting member, wherein the secondary constraint system at least partially restrains growth of the electrode assembly in the second direction upon cycling of the secondary battery, the second direction being orthogonal to the longitudinal direction. The charged state is at least 75% of a rated capacity of the secondary battery, and the discharged state is less than 25% of the rated capacity of the secondary battery.

[0009] According to another aspect, a secondary battery for cycling between a charged and a discharged state is provided, the secondary battery having a battery enclosure, an electrode assembly, carrier ions, a non-aqueous liquid

electrolyte within the battery enclosure, and a set of electrode constraints. The electrode assembly has mutually perpendicular longitudinal, transverse, and vertical axes, a first longitudinal end surface and a second longitudinal end surface separated from each other in the longitudinal direction, and a lateral surface surrounding an electrode assembly longitudinal axis A_{EA} and connecting the first and second longitudinal end surfaces, the lateral surface having opposing first and second regions on opposite sides of the longitudinal axis and separated in a first direction that is orthogonal to the longitudinal axis, the electrode assembly having a maximum width W_{EA} measured in the longitudinal direction, a maximum length L_{EA} bounded by the lateral surface and measured in the transverse direction, and a maximum height H_{EA} bounded by the lateral surface and measured in the vertical direction, the ratio of each of L_{EA} and W_{EA} to H_{EA} being at least 2:1, respectively. The electrode assembly further includes a population of electrode structures, a population of counter-electrode structures, and an electrically insulating microporous separator material electrically separating members of the electrode and counter-electrode populations, members of the electrode and counter-electrode structure populations being arranged in an alternating sequence in the longitudinal direction. Each member of the population of electrode structures has a layer of an electrode active material and each member of the population of counter-electrode structures comprises a layer of a counter-electrode active material, wherein the electrode active material has the capacity to accept more than one mole of carrier ion per mole of electrode active material when the secondary battery is charged from a discharged state to a charged state. The set of electrode constraints includes a primary constraint system comprising first and second primary growth constraints and at least one primary connecting member, the first and second primary growth constraints separated from each other in the longitudinal direction, and the at least one primary connecting member connecting the first and second primary growth constraints, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20%. The charged state is at least 75% of a rated capacity of the secondary battery, and the discharged state is less than 25% of the rated capacity of the secondary battery.

[0010] Other aspects, features and embodiments of the present disclosure will be, in part, discussed and, in part, apparent in the following description and drawing.

BRIEF DESCRIPTION OF THE DRAWING

[0011] FIG. 1 is a perspective view of one embodiment of a constraint system employed with an electrode assembly.

[0012] FIG. 2A is a schematic of one embodiment of a three-dimensional electrode assembly.

[0013] FIGS. 2B-2C are schematics of one embodiment of a three-dimensional electrode assembly, depicting anode structure population members in constrained and expanded configurations.

[0014] FIGS. 3A-3H show exemplary embodiments of different shapes and sizes for an electrode assembly.

[0015] FIG. 4A illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, and further illustrates elements of the primary and secondary growth constraint systems.

[0016] FIG. 4B illustrates a cross-section of an embodiment of the electrode assembly taken along the line B-B' as shown in FIG. 1, and further illustrates elements of the primary and secondary growth constraint systems.

[0017] FIG. 4C illustrates a cross-section of an embodiment of the electrode assembly taken along the line B-B' as shown in FIG. 1, and further illustrates elements of the primary and secondary growth constraint systems.

[0018] FIG. 5 illustrates a cross section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1.

[0019] FIG. 6A illustrates one embodiment of a top view of a porous secondary growth constraint over an electrode assembly, and one embodiment for adhering the secondary growth constraint to the electrode assembly.

[0020] FIG. 6B illustrates one embodiment of a top view of a porous secondary growth constraint over an electrode assembly, and another embodiment for adhering the secondary growth constraint to the electrode assembly.

[0021] FIG. 6C illustrates one embodiment of a top view of a porous secondary growth constraint over an electrode assembly, and yet another embodiment for adhering the secondary growth constraint to the electrode assembly.

[0022] FIG. 6D illustrates one embodiment of a top view of a porous secondary growth constraint over an electrode assembly, and still yet another embodiment for adhering the secondary growth constraint to the electrode assembly.

[0023] FIG. 7 illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including one embodiment of a primary constraint system and one embodiment of a secondary constraint system.

[0024] FIGS. 8A-8B illustrate a force schematics, according to one embodiment, showing the forces exerted on the electrode assembly by the set of electrode constraints, as well as the forces being exerted by electrode structures upon repeated cycling of a battery containing the electrode assembly.

[0025] FIG. 9A illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including one embodiment of a primary growth constraint system and one embodiment of a secondary growth constraint system where the counter-electrode backbones are used for assembling the set of electrode constraints.

[0026] FIG. 9B illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including another embodiment of a primary growth constraint system and another embodiment of a secondary growth constraint system where the counter-electrode current collectors are used for assembling the set of electrode constraints.

[0027] FIG. 9C illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including yet another embodiment of a primary growth constraint system and yet another embodiment of a secondary growth constraint system where the counter-electrode current collectors are used for assembling the set of electrode constraints.

[0028] FIG. 10 illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as

straints, including one embodiment of a primary growth constraint system and one embodiment of a secondary growth constraint system where the electrode current collectors are used for assembling the set of electrode constraints via slots.

[0045] FIG. 18A illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including one embodiment of a primary growth constraint system and one embodiment of a secondary growth constraint system where the primary growth constraint system is hybridized with the secondary growth constraint system and used for assembling the set of electrode constraints.

[0046] FIG. 18B illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including another embodiment of a primary growth constraint system and another embodiment of a secondary growth constraint system where the primary growth constraint system is hybridized with the secondary growth constraint system and used for assembling the set of electrode constraints.

[0047] FIG. 19 illustrates a cross-section of an embodiment of the electrode assembly taken along the line A-A' as shown in FIG. 1, further including a set of electrode constraints, including one embodiment of a primary growth constraint system and one embodiment of a secondary growth constraint system where the primary growth constraint system is fused with the secondary growth constraint system and used for assembling the set of electrode constraints.

[0048] FIG. 20 illustrates an exploded view of an embodiment of an energy storage device or a secondary battery utilizing one embodiment of a set of growth constraints.

[0049] FIG. 21 illustrates an embodiment of a flowchart for the general assembly of an energy storage device or a secondary battery utilizing one embodiment of a set of growth constraints.

[0050] Other aspects, embodiments and features of the inventive subject matter will become apparent from the following detailed description when considered in conjunction with the accompanying drawing. The accompanying figures are schematic and are not intended to be drawn to scale. For purposes of clarity, not every element or component is labeled in every figure, nor is every element or component of each embodiment of the inventive subject matter shown where illustration is not necessary to allow those of ordinary skill in the art to understand the inventive subject matter.

Definitions

[0051] “A,” “an,” and “the” (i.e., singular forms) as used herein refer to plural referents unless the context clearly dictates otherwise. For example, in one instance, reference to “an electrode” includes both a single electrode and a plurality of similar electrodes.

[0052] “About” and “approximately” as used herein refers to plus or minus 10%, 5%, or 1% of the value stated. For example, in one instance, about 250 μm would include 225 μm to 275 μm . By way of further example, in one instance, about 1,000 μm would include 900 μm to 1,100 μm . Unless otherwise indicated, all numbers expressing quantities (e.g., measurements, and the like) and so forth used in the spec-

ification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the following specification and attached claims are approximations. Each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques.

[0053] “Charged state” as used herein in the context of the state of a secondary battery refers to a state where the secondary battery is charged to at least 75% of its rated capacity. For example, the battery may be charged to at least 80% of its rated capacity, at least 90% of its rated capacity, and even at least 95% of its rated capacity, such as 100% of its rated capacity.

[0054] “C-rate” as used herein refers to a measure of the rate at which a secondary battery is discharged, and is defined as the discharge current divided by the theoretical current draw under which the battery would deliver its nominal rated capacity in one hour. For example, a C-rate of 1C indicates the discharge current that discharges the battery in one hour, a rate of 2C indicates the discharge current that discharges the battery in $\frac{1}{2}$ hours, a rate of C/2 indicates the discharge current that discharges the battery in 2 hours, etc.

[0055] “Discharged state” as used herein in the context of the state of a secondary battery refers to a state where the secondary battery is discharged to less than 25% of its rated capacity. For example, the battery may be discharged to less than 20% of its rated capacity, such as less than 10% of its rated capacity, and even less than 5% of its rated capacity, such as 0% of its rated capacity.

[0056] A “cycle” as used herein in the context of cycling of a secondary battery between charged and discharged states refers to charging and/or discharging a battery to move the battery in a cycle from a first state that is either a charged or discharged state, to a second state that is the opposite of the first state (i.e., a charged state if the first state was discharged, or a discharged state if the first state was charged), and then moving the battery back to the first state to complete the cycle. For example, a single cycle of the secondary battery between charged and discharged states can include, as in a charge cycle, charging the battery from a discharged state to a charged state, and then discharging back to the discharged state, to complete the cycle. The single cycle can also include, as in a discharge cycle, discharging the battery from the charged state to the discharged state, and then charging back to a charged state, to complete the cycle.

[0057] “Feret diameter” as referred to herein with respect to the electrode assembly is defined as the distance between two parallel planes restricting the electrode assembly measured in a direction perpendicular to the two planes. For example, a Feret diameter of the electrode assembly in the longitudinal direction is the distance as measured in the longitudinal direction between two parallel planes restricting the electrode assembly that are perpendicular to the longitudinal direction. As another example, a Feret diameter of the electrode assembly in the transverse direction is the distance as measured in the transverse direction between two parallel planes restricting the electrode assembly that are perpendicular to the transverse direction. As yet another example, a Feret diameter of the electrode assembly in the vertical direction is the distance as measured in the vertical

direction between two parallel planes restricting the electrode assembly that are perpendicular to the vertical direction.

[0058] “Longitudinal axis,” “transverse axis,” and “vertical axis,” as used herein refer to mutually perpendicular axes (i.e., each are orthogonal to one another). For example, the “longitudinal axis,” “transverse axis,” and the “vertical axis” as used herein are akin to a Cartesian coordinate system used to define three-dimensional aspects or orientations. As such, the descriptions of elements of the inventive subject matter herein are not limited to the particular axis or axes used to describe three-dimensional orientations of the elements. Alternatively stated, the axes may be interchangeable when referring to three-dimensional aspects of the inventive subject matter.

[0059] “Longitudinal direction,” “transverse direction,” and “vertical direction,” as used herein, refer to mutually perpendicular directions (i.e., each are orthogonal to one another). For example, the “longitudinal direction,” “transverse direction,” and the “vertical direction” as used herein may be generally parallel to the longitudinal axis, transverse axis and vertical axis, respectively, of a Cartesian coordinate system used to define three-dimensional aspects or orientations.

[0060] “Repeated cycling” as used herein in the context of cycling between charged and discharged states of the secondary battery refers to cycling more than once from a discharged state to a charged state, or from a charged state to a discharged state. For example, repeated cycling between charged and discharged states can include cycling at least 2 times from a discharged to a charged state, such as in charging from a discharged state to a charged state, discharging back to a discharged state, charging again to a charged state and finally discharging back to the discharged state. As yet another example, repeated cycling between charged and discharged states at least 2 times can include discharging from a charged state to a discharged state, charging back up to a charged state, discharging again to a discharged state and finally charging back up to the charged state. By way of further example, repeated cycling between charged and discharged states can include cycling at least 5 times, and even cycling at least 10 times from a discharged to a charged state. By way of further example, the repeated cycling between charged and discharged states can include cycling at least 25, 50, 100, 300, 500 and even 1000 times from a discharged to a charged state.

[0061] “Rated capacity” as used herein in the context of a secondary battery refers to the capacity of the secondary battery to deliver a specified current over a period of time, as measured under standard temperature conditions (25° C.). For example, the rated capacity may be measured in units of Amp-hour, either by determining a current output for a specified time, or by determining for a specified current, the time the current can be output, and taking the product of the current and time. For example, for a battery rated 20 Amp-hr, if the current is specified at 2 amperes for the rating, then the battery can be understood to be one that will provide that current output for 10 hours, and conversely if the time is specified at 10 hours for the rating, then the battery can be understood to be one that will output 2 amperes during the 10 hours. In particular, the rated capacity for a secondary battery may be given as the rated capacity at a specified discharge current, such as the C-rate, where the C-rate is a measure of the rate at which the battery is discharged

relative to its capacity. For example, a C-rate of 1C indicates the discharge current that discharges the battery in one hour, 2C indicates the discharge current that discharges the battery in ½ hours, C/2 indicates the discharge current that discharges the battery in 2 hours, etc. Thus, for example, a battery rated at 20 Amp-hr at a C-rate of 1C would give a discharge current of 20 Amp for 1 hour, whereas a battery rated at 20 Amp-hr at a C-rate of 2C would give a discharge current of 40 Amps for ½ hour, and a battery rated at 20 Amp-hr at a C-rate of C/2 would give a discharge current of 10 Amps over 2 hours.

[0062] “Maximum width” (W_{EA}) as used herein in the context of a dimension of an electrode assembly corresponds to the greatest width of the electrode assembly as measured from opposing points of longitudinal end surfaces of the electrode assembly in the longitudinal direction.

[0063] “Maximum length” (L_{EA}) as used herein in the context of a dimension of an electrode assembly corresponds to the greatest length of the electrode assembly as measured from opposing points of a lateral surface of the electrode assembly in the transverse direction.

[0064] “Maximum height” (H_{EA}) as used herein in the context of a dimension of an electrode assembly corresponds to the greatest height of the electrode assembly as measured from opposing points of the lateral surface of the electrode assembly in the transverse direction.

DETAILED DESCRIPTION

[0065] In general, the present disclosure is directed to an energy storage device **100**, such as a secondary battery **102**, as shown for example in FIG. 2A and/or FIG. 20, that cycles between a charged and a discharged state. The secondary battery **102** includes a battery enclosure **104**, an electrode assembly **106**, carrier ions, and a non-aqueous liquid electrolyte within the battery enclosure. The secondary battery **102** also includes a set of electrode constraints **108** that restrain growth of the electrode assembly **106**. The growth of the electrode assembly **106** that is being constrained may be a macroscopic increase in one or more dimensions of the electrode assembly **106**.

[0066] Further, in general, various aspects of the present disclosure include three-dimensional constraint structures offering particular advantages when incorporated into energy storage devices **100** such as batteries, capacitors, fuel cells, and the like. In one embodiment, the constraint structures have a configuration and/or structure that is selected to resist at least one of growth, swelling, and/or expansion of an electrode assembly **106** that can otherwise occur when a secondary battery **102** is repeatedly cycled between charged and discharged states. In particular, in moving from a discharged state to a charged state, carrier ions such as, for example, one or more of lithium, sodium, potassium, calcium and magnesium, move between the positive and negative electrodes in the battery. Upon reaching the electrode, the carrier ions may then intercalate or alloy into the electrode material, thus increasing the size and volume of that electrode. Conversely, reversing to move from the charged state to the discharged state can cause the ions to de-intercalate or de-alloy, thus contracting the electrode. This alloying and/or intercalation and de-alloying and/or de-intercalation can cause significant volume change in the electrode. Thus, the repeated expansion and contraction of the electrodes upon charging and discharging can create

strain in the electrode assembly 106, which can lead to reduced performance and ultimately even failure of the secondary battery.

[0067] Referring to FIGS. 2A-2C, the effects of the repeated expansion and/or contraction of the electrode assembly 106, according to an embodiment of the disclosure, can be described. FIG. 2A shows an embodiment of a three-dimensional electrode assembly 106, with a population of electrode structures 110 and a population of counter-electrode structures 112 (e.g., population of anode and cathode structures, respectively). The three-dimensional electrode assembly 106 in this embodiment provides an alternating set of the electrodes structures 110 and counter-electrode structures 112 that are interdigitated with one another and, in the embodiment shown in FIG. 2A, has a longitudinal axis A_{EA} parallel to the Y axis, a transverse axis (not shown) parallel to the X axis, and a vertical axis (not shown) parallel to the Z axis. The X, Y and Z axes shown herein are arbitrary axes intended only to show a basis set where the axes are mutually perpendicular to one another in a reference space, and are not intended in any way to limit the structures herein to a specific orientation. Upon charge and discharge cycling of a secondary battery 102 having the electrode assembly 106, the carrier ions travel between the electrode and counter-electrode structures 110 and 112, respectively, such as generally in a direction that is parallel to the Y axis as shown in the embodiment depicted in FIG. 2A, and can intercalate into electrode material of one or more of the electrode structures 110 and counter-electrode structures 112 that is located within the direction of travel. The effect of intercalation and/or alloying of carrier ions into the electrode material can be seen in the embodiments illustrated in FIGS. 2B-2C. In particular, FIG. 2B depicts an embodiment of the electrode assembly 106 with electrode structures 110 in a relatively unexpanded state, such as prior to repeated cycling of the secondary battery 106 between charged and discharged states. By comparison, FIG. 2C depicts an embodiment of the electrode assembly 106 with electrode structures 110 after repeated cycling of the secondary battery for a predetermined number of cycles. As shown in this figure, the dimensions of the electrode structures 110 can increase significantly in the stacking direction (e.g., Y-direction), due to the intercalation and/or alloying of carrier ions into the electrode material. The dimensions of the electrode structures 110 can also significantly increase in another direction, such as in the Z-direction (not shown in FIG. 2C). Furthermore, the increase in size of the electrode structures 110 can result in the deformation of the structures inside the electrode assembly, such as deformation of the counter-electrode structures 112 and separator 130 in the assembly, to accommodate the expansion in the electrode structures 110. The expansion of the electrode structures 110 can ultimately result in the bulging and/or warping of the electrode assembly 106 at the longitudinal ends thereof, as depicted in the embodiment shown in FIG. 2C (as well as in other directions such as at the top and bottom surfaces in the Z-direction). Accordingly, the electrode assembly 106 according to one embodiment can exhibit significant expansion and contraction along the longitudinal (Y axis) of the assembly 106, as well as other axis, due to the intercalation and de-intercalation of the carrier ions during the charging and discharging process.

[0068] Thus, in one embodiment, a primary growth constraint system 151 is provided to mitigate and/or reduce at

least one of growth, expansion, and/or swelling of the electrode assembly 106 in the longitudinal direction (i.e., in a direction that parallels the Y axis), as shown for example in FIG. 1. For example, the primary growth constraint system 151 can include structures configured to constrain growth by opposing expansion at longitudinal end surfaces 116, 118 of the electrode assembly 106. In one embodiment, the primary growth constraint system 151 comprises first and second primary growth constraints 154, 156, that are separated from each other in the longitudinal direction, and that operate in conjunction with at least one primary connecting member 162 that connects the first and second primary growth constraints 154, 156 together to restrain growth in the electrode assembly 106. For example, the first and second primary growth constraints 154, 156 may at least partially cover first and second longitudinal end surfaces 116, 118 of the electrode assembly 106, and may operate in conjunction with connecting members 162, 164 connecting the primary growth constraints 154, 156 to one another to oppose and restrain any growth in the electrode assembly 106 that occurs during repeated cycles of charging and/or discharging. Further discussion of embodiments and operation of the primary growth constraint system 151 is provided in more detail below.

[0069] In addition, repeated cycling through charge and discharge processes in a secondary battery 102 can induce growth and strain not only in a longitudinal direction of the electrode assembly 106 (e.g., Y-axis in FIG. 2A), but can also induce growth and strain in directions orthogonal to the longitudinal direction, as discussed above, such as the transverse and vertical directions (e.g., X and Z axes, respectively, in FIG. 2A). Furthermore, in certain embodiments, the incorporation of a primary growth constraint system 151 to inhibit growth in one direction can even exacerbate growth and/or swelling in one or more other directions. For example, in a case where the primary growth constraint system 151 is provided to restrain growth of the electrode assembly 106 in the longitudinal direction, the intercalation of carrier ions during cycles of charging and discharging and the resulting swelling of electrode structures can induce strain in one or more other directions. In particular, in one embodiment, the strain generated by the combination of electrode growth/swelling and longitudinal growth constraints can result in buckling or other failure(s) of the electrode assembly 106 in the vertical direction (e.g., the Z axis as shown in FIG. 2A), or even in the transverse direction (e.g., the X axis as shown in FIG. 2A).

[0070] Accordingly, in one embodiment of the present disclosure, the secondary battery 102 includes not only a primary growth constraint system 151, but also at least one secondary growth constraint system 152 that may operate in conjunction with the primary growth constraint system 151 to restrain growth of the electrode assembly 106 along multiple axes of the electrode assembly 106. For example, in one embodiment, the secondary growth constraint system 152 may be configured to interlock with, or otherwise synergistically operate with, the primary growth constraint system 151, such that overall growth of the electrode assembly 106 can be restrained to impart improved performance and reduced incidence of failure of the secondary battery having the electrode assembly 106 and primary and secondary growth constraint systems 151 and 152, respectively. Further discussion of embodiments of the interrelationship between the primary and secondary growth constraint sys-

tems **151** and **152**, respectively, and their operation to restrain growth of the electrode assembly **106**, is provided in more detail below.

[0071] By constraining the growth of the electrode assembly **106**, it is meant that, as discussed above, an overall macroscopic increase in one or more dimensions of the electrode assembly **106** is being constrained. That is, the overall growth of the electrode assembly **106** may be constrained such that an increase in one or more dimensions of the electrode assembly **106** along (the X, Y, and Z axes) is controlled, even though a change in volume of one or more electrodes within the electrode assembly **106** may nonetheless occur on a smaller (e.g., microscopic) scale during charge and discharge cycles. The microscopic change in electrode volume may be observable, for example, via scanning electron microscopy (SEM). While the set of electrode constraints **108** may be capable of inhibiting some individual electrode growth on the microscopic level, some growth may still occur, although the growth may at least be restrained. The volume change in the individual electrodes upon charge/discharge, while it may be a small change on the microscopic level for each individual electrode, can nonetheless have an additive effect that results in a relatively larger volume change on the macroscopic level for the overall electrode assembly **106** in cycling between charged and discharged states, thereby potentially causing strain in the electrode assembly **106**.

[0072] According to one embodiment, an electrode active material used in an electrode structure **110** corresponding to an anode of the electrode assembly **106** comprises a material that expands upon insertion of carrier ions into the electrode active material during charge of the secondary battery **102**. For example, the electrode active materials may comprise anodically active materials that accept carrier ions during charging of the secondary battery, such as by intercalating with or alloying with the carrier ions, in an amount that is sufficient to generate an increase in the volume of the electrode active material. For example, in one embodiment the electrode active material may comprise a material that has the capacity to accept more than one mole of carrier ion per mole of electrode active material, when the secondary battery **102** is charged from a discharged to a charged state. By way of further example, the electrode active material may comprise a material that has the capacity to accept 1.5 or more moles of carrier ion per mole of electrode active material, such as 2.0 or more moles of carrier ion per mole of electrode active material, and even 2.5 or more moles of carrier ion per mole of electrode active material, such as 3.5 moles or more of carrier ion per mole of electrode active material. The carrier ion accepted by the electrode active material may be at least one of lithium, potassium, sodium, calcium, and magnesium. Examples of electrode active materials that expand to provide such a volume change include one or more of silicon, aluminum, tin, zinc, silver, antimony, bismuth, gold, platinum, germanium, palladium, and alloys thereof.

Electrode Assembly

[0073] Referring again to FIG. 2A, in one embodiment, an interdigitated electrode assembly **106** includes a population of electrode structures **110**, a population of counter-electrode structures **112**, and an electrically insulating microporous separator **130** electrically insulating the electrode structures **110** from the counter-electrode structures **112**. In one

embodiment, the electrode structures **110** comprise an electrode active material layer **132**, an electrode backbone **134** that supports the electrode active material layer **132**, and an electrode current collector **136**, which may be an ionically porous current collector to allow ions to pass therethrough, as shown in the embodiment depicted in FIG. 7. For example, the electrode structure **110**, in one embodiment, can comprise an anode structure, with an anodically active material layer, an anode backbone, and an anode current collector. Similarly, in one embodiment, the counter-electrode structures **112** comprise a counter-electrode active material layer **138**, a counter-electrode current collector **140**, and a counter-electrode backbone **141** that supports one or more of the counter-electrode current collector **140** and/or the counter-electrode active material layer **138**, as shown for example in the embodiment depicted in FIG. 7. For example, the counter-electrode structure **112** can comprise, in one embodiment, a cathode structure comprising a cathodically active material layer, a cathode current collector, and a cathode backbone. The electrically insulating microporous separator **130** allows carrier ions to pass therethrough during charge and/or discharge processes, to travel between the electrode structures **110** and counter-electrode structures **112** in the electrode assembly **106**. Furthermore, it should be understood that the electrode and counter electrode structures **110** and **112**, respectively, are not limited to the specific embodiments and structures described herein, and other configurations, structures, and/or materials other than those specifically described herein can also be provided to form the electrode structures **110** and counter-electrode structures **112**. For example, the electrode and counter electrode structures **110**, **112** can be provided in a form where the structures are substantially absent any electrode and/or counter-electrode backbones **134**, **141**, such as in a case where the region of the electrode and/or counter-electrode structures **110**, **112** that would contain the backbones is instead made up of electrode active material and/or counter-electrode active material.

[0074] According to the embodiment as shown in FIG. 2A, the members of the electrode and counter-electrode structure populations **110** and **112**, respectively, are arranged in alternating sequence, with a direction of the alternating sequence corresponding to the stacking direction D. The electrode assembly **106** according to this embodiment further comprises mutually perpendicular longitudinal, transverse, and vertical axes, with the longitudinal axis A_{EA} generally corresponding or parallel to the stacking direction D of the members of the electrode and counter-electrode structure populations. As shown in the embodiment in FIG. 2A, the longitudinal axis A_{EA} is depicted as corresponding to the Y axis, the transverse axis is depicted as corresponding to the X axis, and the vertical axis is depicted as corresponding to the Z axis.

[0075] Further, the electrode assembly **106** has a maximum width W_{EA} measured in the longitudinal direction (i.e., along the y-axis), a maximum length L_{EA} bounded by the lateral surface and measured in the transverse direction (i.e., along the x-axis), and a maximum height H_{EA} also bounded by the lateral surface and measured in the vertical direction (i.e., along the z-axis). The maximum width W_{EA} can be understood as corresponding to the greatest width of the electrode assembly **106** as measured from opposing points of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** where the electrode assembly is widest in the

longitudinal direction. For example, referring to the embodiment of the electrode assembly **106** in FIG. 2, the maximum width W_{EA} can be understood as corresponding simply to the width of the assembly **106** as measured in the longitudinal direction. However, referring to the embodiment of the electrode assembly **106** shown in FIG. 3H, it can be seen that the maximum width W_{EA} corresponds to the width of the electrode assembly as measured from the two opposing points **300a**, **300b**, where the electrode assembly is widest in the longitudinal direction, as opposed to a width as measured from opposing points **301a**, **301b** where the electrode assembly **106** is more narrow. Similarly, the maximum length L_{EA} can be understood as corresponding to the greatest length of the electrode assembly as measured from opposing points of the lateral surface **142** of the electrode assembly **106** where the electrode assembly is longest in the transverse direction. Referring again to the embodiment in FIG. 2A, the maximum length L_{EA} can be understood as simply the length of the electrode assembly **106**, whereas in the embodiment shown in FIG. 3H, the maximum length L_{EA} corresponds to the length of the electrode assembly as measured from two opposing points **302a**, **302b**, where the electrode assembly is longest in the transverse direction, as opposed to a length as measured from opposing points **303a**, **303b** where the electrode assembly is shorter. Along similar lines, the maximum height H_{EA} can be understood as corresponding to the greatest height of the electrode assembly as measured from opposing points of the lateral surface **143** of the electrode assembly where the electrode assembly is highest in the vertical direction. That is, in the embodiment shown in FIG. 2A, the maximum height H_{EA} is simply the height of the electrode assembly. While not specifically depicted in the embodiment shown in FIG. 3H, if the electrode assembly had different heights at points across one or more of the longitudinal and transverse directions, then the maximum height H_{EA} of the electrode assembly would be understood to correspond to the height of the electrode assembly as measured from two opposing points where the electrode assembly is highest in the vertical direction, as opposed to a height as measured from opposing points where the electrode assembly is shorter, as analogously described for the maximum width W_{EA} and maximum length L_{EA} . The maximum length L_{EA} , maximum width W_{EA} , and maximum height H_{EA} of the electrode assembly **106** may vary depending upon the energy storage device **100** and the intended use thereof. For example, in one embodiment, the electrode assembly **106** may include maximum lengths L_{EA} , widths W_{EA} , and heights H_{EA} typical of conventional secondary battery dimensions. By way of further example, in one embodiment, the electrode assembly **106** may include maximum lengths L_{EA} , widths W_{EA} , and heights H_{EA} typical of thin-film battery dimensions.

[0076] In some embodiments, the dimensions L_{EA} , W_{EA} , and H_{EA} are selected to provide an electrode assembly **106** having a maximum length L_{EA} along the transverse axis (X axis) and/or a maximum width W_{EA} along the longitudinal axis (Y axis) that is longer than the maximum height H_{EA} along the vertical axis (Z axis). For example, in the embodiment shown in FIG. 2A, the dimensions L_{EA} , W_{EA} , and H_{EA} are selected to provide an electrode assembly **106** having the greatest dimension along the transverse axis (X axis) that is orthogonal with electrode structure stacking direction D, as well as along the longitudinal axis (Y axis) coinciding with the electrode structure stacking direction D. That is, the

maximum length L_{EA} and/or maximum width W_{EA} may be greater than the maximum height H_{EA} . For example, in one embodiment, a ratio of the maximum length L_{EA} to the maximum height H_{EA} may be at least 2:1. By way of further example, in one embodiment a ratio of the maximum length L_{EA} to the maximum height H_{EA} may be at least 5:1. By way of further example, in one embodiment, the ratio of the maximum length L_{EA} to the maximum height H_{EA} may be at least 10:1. By way of further example, in one embodiment, the ratio of the maximum length L_{EA} to the maximum height H_{EA} may be at least 15:1. By way of further example, in one embodiment, the ratio of the maximum length L_{EA} to the maximum height H_{EA} may be at least 20:1. The ratios of the different dimensions may allow for optimal configurations within an energy storage device to maximize the amount of active materials, thereby increasing energy density.

[0077] In some embodiments, the maximum width W_{EA} may be selected to provide a width of the electrode assembly **106** that is greater than the maximum height H_{EA} . For example, in one embodiment, a ratio of the maximum width W_{EA} to the maximum height H_{EA} may be at least 2:1. By way of further example, in one embodiment, the ratio of the maximum width W_{EA} to the maximum height H_{EA} may be at least 5:1. By way of further example, in one embodiment, the ratio of the maximum width W_{EA} to the maximum height H_{EA} may be at least 10:1. By way of further example, in one embodiment, the ratio of the maximum width W_{EA} to the maximum height H_{EA} may be at least 15:1. By way of further example, in one embodiment, the ratio of the maximum width W_{EA} to the maximum height H_{EA} may be at least 20:1.

[0078] According to one embodiment, a ratio of the maximum width W_{EA} to the maximum length L_{EA} may be selected to be within a predetermined range that provides for an optimal configuration. For example, in one embodiment, a ratio of the maximum width W_{EA} to the maximum length L_{EA} may be in the range of from 1:5 to 5:1. By way of further example, in one embodiment a ratio of the maximum width W_{EA} to the maximum length L_{EA} may be in the range of from 1:3 to 3:1. By way of yet a further example, in one embodiment a ratio of the maximum width W_{EA} to the maximum length L_{EA} may be in the range of from 1:2 to 2:1.

[0079] In the embodiment as shown in FIG. 2A, the electrode assembly **106** has the first longitudinal end surface **116** and the opposing second longitudinal end surface **118** that is separated from the first longitudinal end surface **116** along the longitudinal axis A_{EA} . The electrode assembly **106** further comprises a lateral surface **142** that at least partially surrounds the longitudinal axis A_{EA} , and that connects the first and second longitudinal end surfaces **116**, **118**. In one embodiment, the maximum width W_{EA} is the dimension along the longitudinal axis A_{EA} as measured from the first longitudinal end surface **116** to the second longitudinal end surface **118**. Similarly, the maximum length L_{EA} may be bounded by the lateral surface **142**, and in one embodiment, may be the dimension as measured from opposing first and second regions **144**, **146** of the lateral surface **142** along the transverse axis that is orthogonal to the longitudinal axis. The maximum height H_{EA} , in one embodiment, may be bounded by the lateral surface **142** and may be measured from opposing first and second regions **148**, **150** of the lateral surface **142** along the vertical axis that is orthogonal to the longitudinal axis.

[0080] For the purposes of clarity, only four electrode structures **110** and four counter-electrode structures **112** are illustrated in the embodiment shown in FIG. 2A. For example, the alternating sequence of members of the electrode and counter-electrode structure populations **110** and **112**, respectively, may include any number of members for each population, depending on the energy storage device **100** and the intended use thereof, and the alternating sequence of members of the electrode and counter-electrode structure populations **110** and **112** may be interdigitated, for example, as shown in FIG. 2A. By way of further example, in one embodiment, each member of the population of electrode structures **110** may reside between two members of the population of counter-electrode structures **112**, with the exception of when the alternating sequence terminates along the stacking direction, D. By way of further example, in one embodiment, each member of the population of counter-electrode structures **112** may reside between two members of the population of electrode structures **110**, with the exception of when the alternating sequence terminates along the stacking direction, D. By way of further example, in one embodiment, and stated more generally, the population of electrode structures **110** and the population of counter-electrode structures **112** each have N members, each of N-1 electrode structure members **110** is between two counter-electrode structure members **112**, each of N-1 counter-electrode structure members **112** is between two electrode structure members **110**, and N is at least 2. By way of further example, in one embodiment, N is at least 4. By way of further example, in one embodiment, N is at least 5. By way of further example, in one embodiment, N is at least 10. By way of further example, in one embodiment, N is at least 25. By way of further example, in one embodiment, N is at least 50. By way of further example, in one embodiment, N is at least 100 or more. In one embodiment, members of the electrode and/or counter-electrode populations extend sufficiently from an imaginary backplane (e.g., a plane substantially coincident with a surface of the electrode assembly) to have a surface area (ignoring porosity) that is greater than twice the geometrical footprint (i.e., projection) of the members in the backplane. In certain embodiments, the ratio of the surface area of a non-laminar (i.e., three-dimensional) electrode and/or counter-electrode structure to its geometric footprint in the imaginary backplane may be at least about 5, at least about 10, at least about 50, at least about 100, or even at least about 500. In general, however, the ratio will be between about 2 and about 1000. In one such embodiment, members of the electrode population are non-laminar in nature. By way of further example, in one such embodiment, members of the counter-electrode population are non-laminar in nature. By way of further example, in one such embodiment, members of the electrode population and members of the counter-electrode population are non-laminar in nature.

[0081] According to one embodiment, the electrode assembly **106** has longitudinal ends **117**, **119** at which the electrode assembly **106** terminates. According to one embodiment, the alternating sequence of electrode and counter-electrode structures **110**, **112**, respectively, in the electrode assembly **106** terminates in a symmetric fashion along the longitudinal direction, such as with electrode structures **110** at each end **117**, **119** of the electrode assembly **106** in the longitudinal direction, or with counter-electrode structures **112** at each end **117**, **119** of the electrode assembly

106, in the longitudinal direction. In another embodiment, the alternating sequence of electrode **110** and counter-electrode structures **112** may terminate in an asymmetric fashion along the longitudinal direction, such as with an electrode structure **110** at one end **117** of the longitudinal axis AHA, and a counter-electrode structure **112** at the other end **119** of the longitudinal axis AHA. According to yet another embodiment, the electrode assembly **106** may terminate with a substructure of one or more of an electrode structure **110** and/or counter-electrode structure **112** at one or more ends **117**, **119** of the electrode assembly **106**. By way of example, according to one embodiment, the alternating sequence of the electrode **110** and counter-electrode structures **112** can terminate at one or more substructures of the electrode **110** and counter-electrode structures **112**, including an electrode backbone **134**, counter-electrode backbone **141**, electrode current collector **136**, counter-electrode current collector **140**, electrode active material layer **132**, counter-electrode active material layer **138**, and the like, and may also terminate with a structure such as the separator **130**, and the structure at each longitudinal end **117**, **119** of the electrode assembly **106** may be the same (symmetric) or different (asymmetric). The longitudinal terminal ends **117**, **119** of the electrode assembly **106** can comprise the first and second longitudinal end surfaces **116**, **118** that are contacted by the first and second primary growth constraints **154**, **156** to constrain overall growth of the electrode assembly **106**.

[0082] According to yet another embodiment, the electrode assembly **106** has first and second transverse ends **145**, **147** (see, e.g., FIG. 2A) that may contact one or more electrode and/or counter electrode tabs **190**, **192** (see, e.g., FIG. 20) that may be used to electrically connect the electrode and/or counter-electrode structures **110**, **112** to a load and/or a voltage supply (not shown). For example, the electrode assembly **106** can comprise an electrode bus **194** (see, e.g., FIG. 2A), to which each electrode structure **110** can be connected, and that pools current from each member of the population of electrode structures **110**. Similarly, the electrode assembly **106** can comprise a counter-electrode bus **196** to which each counter-electrode structure **112** may be connected, and that pools current from each member of the population of counter-electrode structures **112**. The electrode and/or counter-electrode buses **194**, **196** each have a length measured in direction D, and extending substantially the entire length of the interdigitated series of electrode structures **110**, **112**. In the embodiment illustrated in FIG. 20, the electrode tab **190** and/or counter electrode tab **192** includes electrode tab extensions **191**, **193** which electrically connect with, and run substantially the entire length of electrode and/or counter-electrode bus **194**, **196**. Alternatively, the electrode and/or counter electrode tabs **190**, **192** may directly connect to the electrode and/or counter-electrode bus **194**, **196**, for example, an end or position intermediate thereof along the length of the buses **194**, **196**, without requiring the tab extensions **191**, **193**. Accordingly, in one embodiment, the electrode and/or counter-electrode buses **194**, **196** can form at least a portion of the terminal ends **145**, **147** of the electrode assembly **106** in the transverse direction, and connect the electrode assembly to the tabs **190**, **192** for electrical connection to a load and/or voltage supply (not shown). Furthermore, in yet another embodiment, the electrode assembly **106** comprises first and second terminal ends **149**, **153** disposed along the vertical

(Z) axis. For example, according to one embodiment, each electrode **110** and/or counter-electrode structure **112**, is provided with a top and bottom coating of separator material, as shown in FIG. 2A, where the coatings form the terminal ends **149**, **153** of the electrode assembly **106** in the vertical direction. The terminal ends **149**, **153** that may be formed of the coating of separator material can comprise first and second surface regions **148**, **150** of the lateral surface **142** along the vertical axis that can be placed in contact with the first and second secondary growth constraints **158**, **160** to constrain growth in the vertical direction.

[0083] In general, the electrode assembly **106** can comprise longitudinal end surfaces **116**, **118** that are planar, co-planar, or non-planar. For example, in one embodiment the opposing longitudinal end surfaces **116**, **118** may be convex. By way of further example, in one embodiment the opposing longitudinal end surfaces **116**, **118** may be concave. By way of further example, in one embodiment the opposing longitudinal end surfaces **116**, **118** are substantially planar. In certain embodiments, electrode assembly **106** may include opposing longitudinal end surfaces **116**, **118** having any range of two-dimensional shapes when projected onto a plane. For example, the longitudinal end surfaces **116**, **118** may independently have a smooth curved shape (e.g., round, elliptical, hyperbolic, or parabolic), they may independently include a series of lines and vertices (e.g., polygonal), or they may independently include a smooth curved shape and include one or more lines and vertices. Similarly, the lateral surface **142** of the electrode assembly **106** may be a smooth curved shape (e.g., the electrode assembly **106** may have a round, elliptical, hyperbolic, or parabolic cross-sectional shape) or the lateral surface **142** may include two or more lines connected at vertices (e.g., the electrode assembly **106** may have a polygonal cross-section). For example, in one embodiment, the electrode assembly **106** has a cylindrical, elliptic cylindrical, parabolic cylindrical, or hyperbolic cylindrical shape. By way of further example, in one such embodiment, the electrode assembly **106** may have a prismatic shape, having opposing longitudinal end surfaces **116**, **118** of the same size and shape and a lateral surface **142** (i.e., the faces extending between the opposing longitudinal end surfaces **116** and **118**) being parallelogram-shaped. By way of further example, in one such embodiment, the electrode assembly **106** has a shape that corresponds to a triangular prism, the electrode assembly **106** having two opposing triangular longitudinal end surfaces **116** and **118** and a lateral surface **142** consisting of three parallelograms (e.g., rectangles) extending between the two longitudinal ends. By way of further example, in one such embodiment, the electrode assembly **106** has a shape that corresponds to a rectangular prism, the electrode assembly **106** having two opposing rectangular longitudinal end surfaces **116** and **118**, and a lateral surface **142** comprising four parallelogram (e.g., rectangular) faces. By way of further example, in one such embodiment, the electrode assembly **106** has a shape that corresponds to a pentagonal prism, hexagonal prism, etc., wherein the electrode assembly **106** has two pentagonal, hexagonal, etc., respectively, opposing longitudinal end surfaces **116** and **118**, and a lateral surface comprising five, six, etc., respectively, parallelograms (e.g., rectangular) faces.

[0084] Referring now to FIGS. 3A-3H, several exemplary geometric shapes are schematically illustrated for electrode

assembly **106**. More specifically, in FIG. 3A, electrode assembly **106** has a triangular prismatic shape with opposing first and second longitudinal end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including the three rectangular faces connecting the longitudinal end surfaces **116**, **118**, that are about the longitudinal axis A_{HA} . In FIG. 3B, electrode assembly **106** has a parallelepiped shape with opposing first and second parallelogram longitudinal end surfaces **116**, **118** separated along longitudinal axis A_{HA} , and a lateral surface **142** including the four parallelogram-shaped faces connecting the two longitudinal end surfaces **116**, **118**, and surrounding longitudinal axis A_{EA} . In FIG. 3C, electrode assembly **106** has a rectangular prism shape with opposing first and second rectangular longitudinal end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including the four rectangular faces connecting the two longitudinal end surfaces **116**, **118** and surrounding longitudinal axis A_{EA} . In FIG. 3D, electrode assembly **106** has a pentagonal prismatic shape with opposing first and second pentagonal longitudinal end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including the five rectangular faces connecting the two longitudinal end surfaces **116**, **118** and surrounding longitudinal axis A_{EA} . In FIG. 3E, electrode assembly **106** has a hexagonal prismatic shape with opposing first and second hexagonal longitudinal end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including the six rectangular faces connecting the two longitudinal end surfaces **116**, **118** and surrounding longitudinal axis A_{EA} . In FIG. 3F, the electrode assembly has a square pyramidal frustum shape with opposing first and second square end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including four trapezoidal faces connecting the two longitudinal end surfaces **116**, **118** and surrounding longitudinal axis A_{HA} , with the trapezoidal faces tapering in dimension along the longitudinal axis from a greater dimension at the first surface **116** to a smaller dimension at the second surface **118**, and the size of the second surface being smaller than that of the first surface. In FIG. 3G, the electrode assembly has a pentagonal pyramidal frustum shape with opposing first and second square end surfaces **116**, **118** separated along longitudinal axis A_{EA} , and a lateral surface **142** including five trapezoidal faces connecting the two longitudinal end surfaces **116**, **118** and surrounding longitudinal axis A_{EA} , with the trapezoidal faces tapering in dimension along the longitudinal axis from a greater dimension at the first surface **116** to a smaller dimension at the second surface **118**, and the size of the second surface being smaller than that of the first surface. In FIG. 3H, the electrode assembly **106** has a pyramidal shape in the longitudinal direction, by virtue of electrode and counter-electrode structures **110**, **112** having lengths that decrease from a first length towards the middle of the electrode assembly **106** on the longitudinal axis, to second lengths at the longitudinal ends **117**, **119** of the electrode assembly **106**.

Electrode Constraints

[0085] In one embodiment, a set of electrode constraints **108** is provided that that restrains overall macroscopic growth of the electrode assembly **106**, as illustrated for example in FIG. 1. The set of electrode constraints **108** may be capable of restraining growth of the electrode assembly

106 along one or more dimensions, such as to reduce swelling and deformation of the electrode assembly 106, and thereby improve the reliability and cycling lifetime of an energy storage device 100 having the set of electrode constraints 108. As discussed above, without being limited to any one particular theory, it is believed that carrier ions traveling between the electrode structures 110 and counter electrode structures 112 during charging and/or discharging of a secondary battery 102 can become inserted into electrode active material, causing the electrode active material and/or the electrode structure 110 to expand. This expansion of the electrode structure 110 can cause the electrodes and/or electrode assembly 106 to deform and swell, thereby compromising the structural integrity of the electrode assembly 106, and/or increasing the likelihood of electrical shorting or other failures. In one example, excessive swelling and/or expansion and contraction of the electrode active material layer 132 during cycling of an energy storage device 100 can cause fragments of electrode active material to break away and/or delaminate from the electrode active material layer 132, thereby compromising the efficiency and cycling lifetime of the energy storage device 100. In yet another example, excessive swelling and/or expansion and contraction of the electrode active material layer 132 can cause electrode active material to breach the electrically insulating microporous separator 130, thereby causing electrical shorting and other failures of the electrode assembly 106. Accordingly, the set of electrode constraints 108 inhibit this swelling or growth that can otherwise occur with cycling between charged and discharged states to improve the reliability, efficiency, and/or cycling lifetime of the energy storage device 100.

[0086] According to one embodiment, the set of electrode constraints 108 comprises a primary growth constraint system 151 to restrain growth and/or swelling along the longitudinal axis (e.g., Y-axis in FIG. 1) of the electrode assembly 106. In another embodiment, the set of electrode constraints 108 may include a secondary growth constraint system 152 that restrains growth along the vertical axis (e.g., Z-axis in FIG. 1). In yet another embodiment, the set of electrode constraints 108 may include a tertiary growth constraint system 155 that restrains growth along the transverse axis (e.g., X-axis in FIG. 4C). In one embodiment, the set of electrode constraints 108 comprises primary growth and secondary growth constraint systems 151, 152, respectively, and even tertiary growth constraint systems 155 that operate cooperatively to simultaneously restrain growth in one or more directions, such as along the longitudinal and vertical axes (e.g., Y axis and Z axis), and even simultaneously along all of the longitudinal, vertical, and transverse axes (e.g., Y, Z, and X axes). For example, the primary growth constraint system 151 may restrain growth that can otherwise occur along the stacking direction D of the electrode assembly 106 during cycling between charged and discharged states, while the secondary growth constraint system 152 may restrain swelling and growth that can occur along the vertical axis, to prevent buckling or other deformation of the electrode assembly 106 in the vertical direction. By way of further example, in one embodiment, the secondary growth constraint system 152 can reduce swelling and/or expansion along the vertical axis that would otherwise be exacerbated by the restraint on growth imposed by the primary growth constraint system 151. The tertiary growth constraint system 155 can also optionally reduce

swelling and/or expansion along the transverse axis that could occur during cycling processes. That is, according to one embodiment, the primary growth and secondary growth constraint systems 151, 152, respectively, and optionally the tertiary growth constraint system 155, may operate together to cooperatively restrain multi-dimensional growth of the electrode assembly 106.

[0087] Referring to FIGS. 4A-4B, an embodiment of a set of electrode constraints 108 is shown having a primary growth constraint system 151 and a secondary growth constraint system 152 for an electrode assembly 106. FIG. 4A shows a cross-section of the electrode assembly 106 in FIG. 1 taken along the longitudinal axis (Y axis), such that the resulting 2-D cross-section is illustrated with the vertical axis (Z axis) and longitudinal axis (Y axis). FIG. 4B shows a cross-section of the electrode assembly 106 in FIG. 1 taken along the transverse axis (X axis), such that the resulting 2-D cross-section is illustrated with the vertical axis (Z axis) and transverse axis (X axis). As shown in FIG. 4A, the primary growth constraint system 151 can generally comprise first and second primary growth constraints 154, 156, respectively, that are separated from one another along the longitudinal direction (Y axis). For example, in one embodiment, the first and second primary growth constraints 154, 156, respectively, comprise a first primary growth constraint 154 that at least partially or even entirely covers a first longitudinal end surface 116 of the electrode assembly 106, and a second primary growth constraint 156 that at least partially or even entirely covers a second longitudinal end surface 118 of the electrode assembly 106. In yet another version, one or more of the first and second primary growth constraints 154, 156 may be interior to a longitudinal end 117, 119 of the electrode assembly 106, such as when one or more of the primary growth constraints comprise an internal structure of the electrode assembly 106. The primary growth constraint system 151 can further comprise at least one primary connecting member 162 that connects the first and second primary growth constraints 154, 156, and that may have a principal axis that is parallel to the longitudinal direction. For example, the primary growth constraint system 151 can comprise first and second primary connecting members 162, 164, respectively, that are separated from each other along an axis that is orthogonal to the longitudinal axis, such as along the vertical axis (Z axis) as depicted in the embodiment. The first and second primary connecting members 162, 164, respectively, can serve to connect the first and second primary growth constraints 154, 156, respectively, to one another, and to maintain the first and second primary growth constraints 154, 156, respectively, in tension with one another, so as to restrain growth along the longitudinal axis of the electrode assembly 106.

[0088] According to one embodiment, the set of electrode constraints 108 including the primary growth constraint system 151 may be capable of restraining growth of the electrode assembly 106 in the longitudinal direction (i.e., electrode stacking direction, D) such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20% between charged and discharged states. By way of further example, in one embodiment the primary growth constraint system 151 may be capable of restraining growth of the electrode assembly 106 in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30

[illegible]

constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 8000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10,000 consecutive cycles of the secondary battery is less than 20%.

[0089] In yet another embodiment, the set of electrode constraints **108** including the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 50 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 80 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 100 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 200 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode

assembly in the longitudinal direction over 300 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 500 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 800 consecutive cycles of the secondary battery is less than 10% between charged and discharged states. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 1000 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 2000 consecutive cycles is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 3000 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5000 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 8000 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10,000 consecutive cycles of the secondary battery is less than 10%.

[0090] In yet another embodiment, the set of electrode constraints **108** including the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10 consecutive

[illegible]

assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 2000 consecutive cycles of the secondary battery is less than 5% between charged and discharged states. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 3000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 8000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10,000 consecutive cycles of the secondary battery is less than 5%.

[0091] In yet another embodiment, the set of electrode constraints **108** including the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction per cycle of the secondary battery is less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode

[illegible]

151 may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 8000 consecutive cycles of the secondary battery to less than 1%. By way of further example, in one embodiment the primary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10,000 consecutive cycles of the secondary battery to less than 1%.

[0092] By charged state it is meant that the secondary battery **102** is charged to at least 75% of its rated capacity, such as at least 80% of its rated capacity, and even at least 90% of its rated capacity, such as at least 95% of its rated capacity, and even 100% of its rated capacity. By discharged state it is meant that the secondary battery is discharged to less than 25% of its rated capacity, such as less than 20% of its rated capacity, and even less than 10%, such as less than 5%, and even 0% of its rated capacity. Furthermore, it is noted that the actual capacity of the secondary battery **102** may vary over time and with the number of cycles the battery has gone through. That is, while the secondary battery **102** may initially exhibit an actual measured capacity that is close to its rated capacity, the actual capacity of the battery will decrease over time, with the secondary battery **102** being considered to be at the end of its life when the actual capacity drops below 80% of the rated capacity as measured in going from a charged to a discharged state.

[0093] Further shown in FIGS. 4A and 4B, the set of electrode constraints **108** can further comprise the secondary growth constraint system **152**, that can generally comprise first and second secondary growth constraints **158**, **160**, respectively, that are separated from one another along a second direction orthogonal to the longitudinal direction, such as along the vertical axis (Z axis) in the embodiment as shown. For example, in one embodiment, the first secondary growth constraint **158** at least partially extends across a first region **148** of the lateral surface **142** of the electrode assembly **106**, and the second secondary growth constraint **160** at least partially extends across a second region **150** of the lateral surface **142** of the electrode assembly **106** that opposes the first region **148**. In yet another version, one or more of the first and second secondary growth constraints **154**, **156** may be interior to the lateral surface **142** of the electrode assembly **106**, such as when one or more of the secondary growth constraints comprise an internal structure of the electrode assembly **106**. In one embodiment, the first and second secondary growth constraints **158**, **160**, respectively, are connected by at least one secondary connecting member **166**, which may have a principal axis that is parallel to the second direction, such as the vertical axis. The secondary connecting member **166** may serve to connect and hold the first and second secondary growth constraints **158**, **160**, respectively, in tension with one another, so as to restrain growth of the electrode assembly **106** along a direction orthogonal to the longitudinal direction, such as for example to restrain growth in the vertical direction (e.g., along the Z axis). In the embodiment depicted in FIG. 4A, the at least one secondary connecting member **166** can correspond to at least one of the first and second primary growth constraints **154**, **156**. However, the secondary con-

necting member **166** is not limited thereto, and can alternatively and/or in addition comprise other structures and/or configurations.

[0094] According to one embodiment, the set of constraints including the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in a second direction orthogonal to the longitudinal direction, such as the vertical direction (Z axis), such that any increase in the Feret diameter of the electrode assembly in the second direction over 20 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 30 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 50 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 80 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 100 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 200 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 300 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 500 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 800 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 1000 consecutive cycles of the

secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 2000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 3000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 5000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 8000 consecutive cycles of the secondary battery is less than 20%. By way of further example, in one embodiment the secondary growth constraint system **151** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 10,000 consecutive cycles of the secondary battery is less than 20% between charged and discharged states.

[0095] In embodiment, the set of constraints including the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 10 consecutive cycles of the secondary battery is less than 10% between charged and discharged states. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 20 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 30 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 50 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the secondary growth constraint system **152** may be capable of restraining growth of the electrode assembly **106** in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 80 consecutive cycles of the secondary battery is less than 10%. By way of further example, in one embodiment the secondary growth con-

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secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 30 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 50 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 80 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 100 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 200 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 300 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 500 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 800 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 1000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 151 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 2000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that

any increase in the Feret diameter of the electrode assembly in the second direction over 3000 consecutive cycles of the secondary battery is less than 1% between charged and discharged states. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 5000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 152 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 8000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the secondary growth constraint system 151 may be capable of restraining growth of the electrode assembly 106 in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 10,000 consecutive cycles of the secondary battery is less than 1%.

[0098] FIG. 4C shows an embodiment of a set of electrode constraints 108 that further includes a tertiary growth constraint system 155 to constrain growth of the electrode assembly in a third direction that is orthogonal to the longitudinal and second directions, such as the transverse direction (X) direction. The tertiary growth constraint system 155 can be provided in addition to the primary and secondary growth constraint systems 151, 152, respectively, to constrain overall growth of the electrode assembly 106 in three dimensions, and/or may be provided in combination with one of the primary or secondary growth constraint systems 151, 152, respectively, to constrain overall growth of the electrode assembly 106 in two dimensions. FIG. 4C shows a cross-section of the electrode assembly 106 in FIG. 1 taken along the transverse axis (X axis), such that the resulting 2-D cross-section is illustrated with the vertical axis (Z axis) and transverse axis (X axis). As shown in FIG. 4C, the tertiary growth constraint system 155 can generally comprise first and second tertiary growth constraints 157, 159, respectively, that are separated from one another along the third direction such as the transverse direction (X axis). For example, in one embodiment, the first tertiary growth constraint 157 at least partially extends across a first region 144 of the lateral surface 142 of the electrode assembly 106, and the second tertiary growth constraint 159 at least partially extends across a second region 146 of the lateral surface 142 of the electrode assembly 106 that opposes the first region 144 in the transverse direction. In yet another version, one or more of the first and second tertiary growth constraints 157, 159 may be interior to the lateral surface 142 of the electrode assembly 106, such as when one or more of the tertiary growth constraints comprise an internal structure of the electrode assembly 106. In one embodiment, the first and second tertiary growth constraints 157, 159, respectively, are connected by at least one tertiary connecting member 165, which may have a principal axis that is parallel to the third direction. The tertiary connecting member 165 may serve to connect and hold the first and second tertiary growth constraints 157, 159, respectively, in tension with one another, so as to restrain growth of the electrode assembly 106 along a direction orthogonal to the longitudinal direction, for example, to restrain growth in the

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restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 10,000 consecutive cycles of the secondary battery is less than 10%.

[0101] In one embodiment, the set of constraints having the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 5 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 10 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 20 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 30 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 50 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 80 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 100 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 200 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 300 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 500 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint

system **152** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 800 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 1000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 2000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 3000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 5000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 8000 consecutive cycles of the secondary battery is less than 5%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 10,000 consecutive cycles of the secondary battery is less than 5%.

[0102] In one embodiment, the set of constraints having the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction per cycle of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 5 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 10 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system **155** may be capable of restraining growth of the electrode assembly **106** in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 20 consecutive cycles of the secondary battery is less than

[illegible]

tive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 5000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 8000 consecutive cycles of the secondary battery is less than 1%. By way of further example, in one embodiment the tertiary growth constraint system 155 may be capable of restraining growth of the electrode assembly 106 in the third direction such that any increase in the Feret diameter of the electrode assembly in the third direction over 10,000 consecutive cycles of the secondary battery is less than 1%.

[0103] According to one embodiment, the primary and secondary growth constraint systems 151, 152, respectively, and optionally the tertiary growth constraint system 155, are configured to cooperatively operate such that portions of the primary growth constraint system 151 cooperatively act as a part of the secondary growth constraint system 152, and/or portions of the secondary growth constraint system 152 cooperatively act as a part of the primary growth constraint system 151, and the portions of any of the primary and/or secondary constraint systems 151, 152, respectively, may also cooperatively act as a part of the tertiary growth constraint system, and vice versa. For example, in the embodiment shown in in FIGS. 4A and 4B, the first and second primary connecting members 162, 164, respectively, of the primary growth constraint system 151 can serve as at least a portion of, or even the entire structure of, the first and second secondary growth constraints 158, 160 that constrain growth in the second direction orthogonal to the longitudinal direction. In yet another embodiment, as mentioned above, one or more of the first and second primary growth constraints 154, 156, respectively, can serve as one or more secondary connecting members 166 to connect the first and second secondary growth constraints 158, 160, respectively. Conversely, at least a portion of the first and second secondary growth constraints 158, 160, respectively, can act as first and second primary connecting members 162, 164, respectively, of the primary growth constraint system 151, and the at least one secondary connecting member 166 of the secondary growth constraint system 152 can, in one embodiment, act as one or more of the first and second primary growth constraints 154, 156, respectively. In yet another embodiment, at least a portion of the first and second primary connecting members 162, 164, respectively, of the primary growth constraint system 151, and/or the at least one secondary connecting member 166 of the secondary growth constraint system 152 can serve as at least a portion of, or even the entire structure of, the first and second tertiary growth constraints 157, 159, respectively, that constrain growth in the transverse direction orthogonal to the longitudinal direction. In yet another embodiment, one or more of the first and second primary growth constraints 154, 156, respectively, and/or the first and second secondary growth constraints 158, 160, respectively, can serve as one or more tertiary connecting members 166 to connect the first and second tertiary growth constraints 157, 159, respectively.

Conversely, at least a portion of the first and second tertiary growth constraints 157, 159, respectively, can act as first and second primary connecting members 162, 164, respectively, of the primary growth constraint system 151, and/or the at least one secondary connecting member 166 of the secondary growth constraint system 152, and the at least one tertiary connecting member 165 of the tertiary growth constraint system 155 can in one embodiment act as one or more of the first and second primary growth constraints 154, 156, respectively, and/or one or more of the first and second secondary growth constraints 158, 160, respectively. Alternatively and/or additionally, the primary and/or secondary and/or tertiary growth constraints can comprise other structures that cooperate to restrain growth of the electrode assembly 106. Accordingly, the primary and secondary growth constraint systems 151, 152, respectively, and optionally the tertiary growth constraint system 155, can share components and/or structures to exert restraint on the growth of the electrode assembly 106.

[0104] In one embodiment, the set of electrode constraints 108 can comprise structures such as the primary and secondary growth constraints, and primary and secondary connecting members, that are structures that are external to and/or internal to the battery enclosure 104, or may be a part of the battery enclosure 104 itself. For example, the set of electrode constraints 108 can comprise a combination of structures that includes the battery enclosure 104 as well as other structural components. In one such embodiment, the battery enclosure 104 may be a component of the primary growth constraint system 151 and/or the secondary growth constraint system 152; stated differently, in one embodiment, the battery enclosure 104, alone or in combination with one or more other structures (within and/or outside the battery enclosure 104, for example, the primary growth constraint system 151 and/or a secondary growth constraint system 152) restrains growth of the electrode assembly 106 in the electrode stacking direction D and/or in the second direction orthogonal to the stacking direction, D. For example, one or more of the primary growth constraints 154, 156 and secondary growth constraints 158, 160 can comprise a structure that is internal to the electrode assembly. In another embodiment, the primary growth constraint system 151 and/or secondary growth constraint system 152 does not include the battery enclosure 104, and instead one or more discrete structures (within and/or outside the battery enclosure 104) other than the battery enclosure 104 restrains growth of the electrode assembly 106 in the electrode stacking direction, D, and/or in the second direction orthogonal to the stacking direction, D. The electrode assembly 106 may be restrained by the set of electrode constraints 108 at a pressure that is greater than the pressure exerted by growth and/or swelling of the electrode assembly 106 during repeated cycling of an energy storage device 100 or a secondary battery having the electrode assembly 106.

[0105] In one exemplary embodiment, the primary growth constraint system 151 includes one or more discrete structure(s) within the battery enclosure 104 that restrains growth of the electrode structure 110 in the stacking direction D by exerting a pressure that exceeds the pressure generated by the electrode structure 110 in the stacking direction D upon repeated cycling of a secondary battery 102 having the electrode structure 110 as a part of the electrode assembly 106. In another exemplary embodiment, the primary growth constraint system 151 includes one or more discrete struc-

tures within the battery enclosure 104 that restrains growth of the counter-electrode structure 112 in the stacking direction D by exerting a pressure in the stacking direction D that exceeds the pressure generated by the counter-electrode structure 112 in the stacking direction D upon repeated cycling of a secondary battery 102 having the counter-electrode structure 112 as a part of the electrode assembly 106. The secondary growth constraint system 152 can similarly include one or more discrete structures within the battery enclosure 104 that restrain growth of at least one of the electrode structures 110 and counter-electrode structures 112 in the second direction orthogonal to the stacking direction D, such as along the vertical axis (Z axis), by exerting a pressure in the second direction that exceeds the pressure generated by the electrode or counter-electrode structure 110, 112, respectively, in the second direction upon repeated cycling of a secondary battery 102 having the electrode or counter electrode structures 110, 112, respectively.

[0106] In yet another embodiment, the first and second primary growth constraints 154, 156, respectively, of the primary growth constraint system 151 restrain growth of the electrode assembly 106 by exerting a pressure on the first and second longitudinal end surfaces 116, 118 of the electrode assembly 106, meaning, in a longitudinal direction, that exceeds a pressure exerted by the first and second primary growth constraints 154, 156 on other surfaces of the electrode assembly 106 that would be in a direction orthogonal to the longitudinal direction, such as opposing first and second regions of the lateral surface 142 of the electrode assembly 106 along the transverse axis and/or vertical axis. That is, the first and second primary growth constraints 154, 156 may exert a pressure in a longitudinal direction (Y axis) that exceeds a pressure generated thereby in directions orthogonal thereto, such as the transverse (X axis) and vertical (Z axis) directions. For example, in one such embodiment, the primary growth constraint system 151 restrains growth of the electrode assembly 106 with a pressure on first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 by the primary growth constraint system 151 in at least one, or even both, of the two directions that are perpendicular to the stacking direction D, by a factor of at least 3. By way of further example, in one such embodiment, the primary growth constraint system 151 restrains growth of the electrode assembly 106 with a pressure on first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 in at least one, or even both, of the two directions that are perpendicular to the stacking direction D, by a factor of at least 4. By way of further example, in one such embodiment, the primary growth constraint system 151 restrains growth of the electrode assembly 106 with a pressure on first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 in at least one, or even both, of the two directions that are perpendicular to the stacking direction D, by a factor of at least 5.

[0107] Similarly, in one embodiment, the first and second secondary growth constraints 158, 160, respectively, of the primary growth constraint system 151 restrain growth of the

electrode assembly 106 by exerting a pressure on first and second opposing regions of the lateral surface 142 of the electrode assembly 106 in a second direction orthogonal to the longitudinal direction, such as first and second opposing surface regions along the vertical axis 148, 150, respectively (i.e., in a vertical direction), that exceeds a pressure exerted by the first and second secondary growth constraints 158, 160, respectively, on other surfaces of the electrode assembly 106 that would be in a direction orthogonal to the second direction. That is, the first and second secondary growth constraints 158, 160, respectively, may exert a pressure in a vertical direction (Z axis) that exceeds a pressure generated thereby in directions orthogonal thereto, such as the transverse (X axis) and longitudinal (Y axis) directions. For example, in one such embodiment, the secondary growth constraint system 152 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 148, 150, respectively (i.e., in the vertical direction), that exceeds the pressure maintained on the electrode assembly 106 by the secondary growth constraint system 152 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 3. By way of further example, in one such embodiment, the secondary growth constraint system 152 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 148, 150, respectively (i.e., in the vertical direction), that exceeds the pressure maintained on the electrode assembly 106 by the secondary growth constraint system 152 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 4. By way of further example, in one such embodiment, the secondary growth constraint system 152 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 148, 150, respectively (i.e., in the vertical direction), that exceeds the pressure maintained on the electrode assembly 106 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 5.

[0108] In yet another embodiment, the first and second tertiary growth constraints 157, 159, respectively, of the tertiary growth constraint system 155 restrain growth of the electrode assembly 106 by exerting a pressure on first and second opposing regions of the lateral surface 142 of the electrode assembly 106 in a direction orthogonal to the longitudinal direction and the second direction, such as first and second opposing surface regions along the transverse axis 161, 163, respectively (i.e., in a transverse direction), that exceeds a pressure exerted by the tertiary growth constraint system 155 on other surfaces of the electrode assembly 106 that would be in a direction orthogonal to the transverse direction. That is, the first and second tertiary growth constraints 157, 159, respectively, may exert a pressure in a transverse direction (X axis) that exceeds a pressure generated thereby in directions orthogonal thereto, such as the vertical (Z axis) and longitudinal (Y axis) directions. For example, in one such embodiment, the tertiary growth constraint system 155 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 144, 146 (i.e., in the transverse direction) that exceeds the pressure maintained on the electrode assembly 106 by the tertiary growth constraint system 155 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 3. By way of further example, in one such embodiment, the tertiary

growth constraint system 155 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 144, 146, respectively (i.e., in the transverse direction), that exceeds the pressure maintained on the electrode assembly 106 by the tertiary growth constraint system 155 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 4. By way of further example, in one such embodiment, the tertiary growth constraint system 155 restrains growth of the electrode assembly 106 with a pressure on first and second opposing surface regions 144, 146, respectively (i.e., in the transverse direction), that exceeds the pressure maintained on the electrode assembly 106 in at least one, or even both, of the two directions that are perpendicular thereto, by a factor of at least 5.

[0109] In one embodiment, the set of electrode constraints 108, which may include the primary growth constraint system 151, the secondary growth constraint system 152, and optionally the tertiary growth constraint system 155, is configured to exert pressure on the electrode assembly 106 along two or more dimensions thereof (e.g., along the longitudinal and vertical directions, and optionally along the transverse direction), with a pressure being exerted along the longitudinal direction by the set of electrode constraints 108 being greater than any pressure(s) exerted by the set of electrode constraints 108 in any of the directions orthogonal to the longitudinal direction (e.g., the Z and X directions). That is, when the pressure(s) exerted by the primary, secondary, and optionally tertiary growth constraint systems 151, 152, 155, respectively, making up the set of electrode constraints 108 are summed together, the pressure exerted on the electrode assembly 106 along the longitudinal axis exceeds the pressure(s) exerted on the electrode assembly 106 in the directions orthogonal thereto. For example, in one such embodiment, the set of electrode constraints 108 exerts a pressure on the first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 by the set of electrode constraints 108 in at least one or even both of the two directions that are perpendicular to the stacking direction D, by a factor of at least 3. By way of further example, in one such embodiment, the set of electrode constraints 108 exerts a pressure on first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 by the set of electrode constraints 108 in at least one, or even both, of the two directions that are perpendicular to the stacking direction D by a factor of at least 4. By way of further example, in one such embodiment, the set of electrode constraints 108 exerts a pressure on first and second longitudinal end surfaces 116, 118 (i.e., in the stacking direction D) that exceeds the pressure maintained on the electrode assembly 106 in at least one, or even both, of the two directions that are perpendicular to the stacking direction D, by a factor of at least 5.

[0110] According to one embodiment, the first and second longitudinal end surfaces 116, 118, respectively, have a combined surface area that is less than a predetermined amount of the overall surface area of the entire electrode assembly 106. For example, in one embodiment, the electrode assembly 106 may have a geometric shape corresponding to that of a rectangular prism with first and second longitudinal end surfaces 116, 118, respectively, and a lateral surface 142 extending between the end surfaces 116, 118,

respectively, that makes up the remaining surface of the electrode assembly 106, and that has opposing surface regions 144, 146 in the X direction (i.e., the side surfaces of the rectangular prism) and opposing surface regions 148, 150 in the Z direction (i.e., the top and bottom surfaces of the rectangular prism, wherein X, Y and Z are dimensions measured in directions corresponding to the X, Y, and Z axes, respectively). The overall surface area is thus the sum of the surface area covered by the lateral surface 142 (i.e., the surface area of the opposing surfaces 144, 146, 148, and 150 in X and Z), added to the surface area of the first and second longitudinal end surfaces 116, 118, respectively. In accordance with one aspect of the present disclosure, the sum of the surface areas of the first and second longitudinal end surfaces 116, 118, respectively, is less than 33% of the surface area of the total surface of the electrode assembly 106. For example, in one such embodiment, the sum of the surface areas of the first and second longitudinal end surfaces 116, 118, respectively, is less than 25% of the surface area of the total surface of the electrode assembly 106. By way of further example, in one embodiment, the sum of the surface areas of the first and second longitudinal end surfaces 116, 118, respectively, is less than 20% of the surface area of the total surface of the electrode assembly. By way of further example, in one embodiment, the sum of the surface areas of the first and second longitudinal end surfaces 116, 118, respectively, is less than 15% of the surface area of the total surface of the electrode assembly. By way of further example, in one embodiment, the sum of the surface areas of the first and second longitudinal end surfaces 116, 118, respectively, is less than 10% of the surface area of the total surface of the electrode assembly.

[0111] In yet another embodiment, the electrode assembly 106 is configured such that a surface area of a projection of the electrode assembly 106 in a plane orthogonal to the stacking direction (i.e., the longitudinal direction), is smaller than the surface areas of projections of the electrode assembly 106 onto other orthogonal planes. For example, referring to the electrode assembly 106 embodiment shown in FIG. 2A (e.g., a rectangular prism), it can be seen that surface area of a projection of the electrode assembly 106 into a plane orthogonal to the stacking direction (i.e., the X-Z plane) corresponds to $L_{EA} \times H_{EA}$. Similarly, a projection of the electrode assembly 106 into the Z-Y plane corresponds to $W_{EA} \times H_{EA}$, and a projection of the electrode assembly 106 into the X-Y plane corresponds to $L_{EA} \times W_{EA}$. Accordingly, the electrode assembly 106 is configured such that the stacking direction intersects the plane in which the projection having the smallest surface area lies. Accordingly, in the embodiment in FIG. 2A, the electrode assembly 106 is positioned such that the stacking direction intersects the X-Z plane in which the smallest surface area projection corresponding to $H_{EA} \times L_{EA}$ lies. That is, the electrode assembly is positioned such that the projection having the smallest surface area (e.g., $H_{EA} \times L_{EA}$) is orthogonal to the stacking direction.

[0112] In yet another embodiment, the secondary battery 102 can comprise a plurality of electrode assemblies 106 that are stacked together to form an electrode stack, and can be constrained by one or more shared electrode constraints. For example, in one embodiment, at least a portion of one or more of the primary growth constraint system 151 and the secondary growth constraint system 152 can be shared by a plurality of electrode assemblies 106 forming the electrode

assembly stack. By way of further example, in one embodiment, a plurality of electrode assemblies forming an electrode assembly stack may be constrained in a vertical direction by a secondary growth constraint system 152 having a first secondary growth constraint 158 at a top electrode assembly 106 of the stack, and a second secondary growth constraint 160 at a bottom electrode assembly 106 of the stack, such that the plurality of electrode assemblies 106 forming the stack are constrained in the vertical direction by the shared secondary growth constraint system. Similarly, portions of the primary growth constraint system 151 could also be shared. Accordingly, in one embodiment, similarly to the single electrode assembly described above, a surface area of a projection of the stack of electrode assemblies 106 in a plane orthogonal to the stacking direction (i.e., the longitudinal direction), is smaller than the surface areas of projections of the stack of electrode assemblies 106 onto other orthogonal planes. That is, the plurality of electrode assemblies 106 may be configured such that the stacking direction (i.e., longitudinal direction) intersects and is orthogonal to a plane that has a projection of the stack of electrode assemblies 106 that is the smallest of all the other orthogonal projections of the electrode assembly stack.

[0113] According to one embodiment, the electrode assembly 106 further comprises electrode structures 110 that are configured such that a surface area of a projection of the electrode structures 110 into a plane orthogonal to the stacking direction (i.e., the longitudinal direction), is larger than the surface areas of projections of the electrode structures 100 onto other orthogonal planes. For example, referring to the embodiments as shown in FIGS. 2 and 7, the electrodes 110 can each be understood to have a length L_{ES} measured in the transverse direction, a width W_{ES} measured in the longitudinal direction, and a height H_{ES} measured in the vertical direction. The projection into the X-Z plane as shown in FIGS. 2 and 7 thus has a surface area $L_{ES} \times H_{ES}$, the projection into the Y-Z plane has a surface area $W_{ES} \times H_{ES}$, and the projection into the XY plane has a surface area $L_{ES} \times W_{ES}$. Of these, the plane corresponding to the projection having the largest surface area is the one that is selected to be orthogonal to the stacking direction. Similarly, the electrodes 110 may also be configured such that a surface area of a projection of the electrode active material layer 132 into a plane orthogonal to the stacking direction is larger than the surface areas of projections of the electrode active material layer onto other orthogonal planes. For example, in the embodiments shown in FIGS. 2 and 7, the electrode active material layer may have a length L_A measured in the transverse direction, a width W_A measured in the longitudinal direction, and a height H_A measured in the vertical direction, from the surface areas of projections can be calculated (L_{ES} , L_A , W_{ES} , W_A , H_{ES} and H_A may also correspond to the maximum of these dimensions, in a case where the dimensions of the electrode structure and/or electrode active material layer 132 vary along one or more axes). In one embodiment, by positioning the electrode structures 110 such that the plane having the highest projection surface area of the electrode structure 100 and/or electrode active material layer 132 is orthogonal to the stacking direction, a configuration can be achieved whereby the surface of the electrode structure 110 having the greatest surface area of electrode active material faces the direction of travel of the

carrier ions, and thus experiences the greatest growth during cycling between charged and discharged states due to intercalation and/or alloying.

[0114] In one embodiment, the electrode structure 110 and electrode assembly 106 can be configured such that the largest surface area projection of the electrode structure 110 and/or electrode active material layer 132, and the smallest surface area projection of the electrode assembly 106 are simultaneously in a plane that is orthogonal to the stacking direction. For example, in a case as shown in FIGS. 2 and 7, where the projection of the electrode active material layer 132 in the X-Z plane ($L_A \times H_A$) of the electrode active material layer 132 is the highest, the electrode structure 110 and/or electrode active material layer 132 is positioned with respect to the smallest surface area projection of the electrode assembly ($L_{EA} \times H_{EA}$) such the projection plane for both projections is orthogonal to the stacking direction. That is, the plane having the greatest surface area projection of the electrode structure 110 and/or electrode active material is parallel to (and/or in the same plane with) the plane having the smallest surface area projection of the electrode assembly 106. In this way, according to one embodiment, the surfaces of the electrode structures that are most likely to experience the highest volume growth, i.e., the surfaces having the highest content of electrode active material layer, and/or surfaces that intersect (e.g., are orthogonal to) a direction of travel of carrier ions during charge/discharge of a secondary battery, face the surfaces of the electrode assembly 106 having the lowest surface area. An advantage of providing such a configuration may be that the growth constraint system used to constrain in this greatest direction of growth, e.g. along the longitudinal axis, can be implemented with growth constraints that themselves have a relatively small surface area, as compared to the area of other surfaces of the electrode assembly 106, thereby reducing the volume required for implementing a constraint system to restrain growth of the electrode assembly.

[0115] In one embodiment, the constraint system 108 occupies a relatively low volume % of the combined volume of the electrode assembly 106 and constraint system 108. That is, the electrode assembly 106 can be understood as having a volume bounded by its exterior surfaces (i.e., the displacement volume), namely the volume enclosed by the first and second longitudinal end surfaces 116, 118 and the lateral surface 42 connecting the end surfaces. Portions of the constraint system 108 that are external to the electrode assembly 106 (i.e., external to the longitudinal end surfaces 116, 118 and the lateral surface), such as where first and second primary growth constraints 154, 156 are located at the longitudinal ends 117, 119 of the electrode assembly 106, and first and second secondary growth constraints 158, 160 are at the opposing ends of the lateral surface 142, the portions of the constrain system 108 similarly occupy a volume corresponding to the displacement volume of the constraint system portions. Accordingly, in one embodiment, the external portions of the set of electrode constraints 108, which can include external portions of the primary growth constraint system 151 (i.e., any of the first and second primary growth constraints 154, 156 and at least one primary connecting member that are external, or external portions thereof), as well as external portions of the secondary growth constraint system 152 (i.e., any of the first and second secondary growth constraints 158, 160 and at least one secondary connecting member that are external, or

external portions thereof) occupies no more than 80% of the total combined volume of the electrode assembly 106 and external portion of the set of electrode constraints 108. By way of further example, in one embodiment the external portions of the set of electrode constraints occupies no more than 60% of the total combined volume of the electrode assembly 106 and the external portion of the set of electrode constraints. By way of yet a further example, in one embodiment the external portion of the set of electrode constraints 106 occupies no more than 40% of the total combined volume of the electrode assembly 106 and the external portion of the set of electrode constraints. By way of yet a further example, in one embodiment the external portion of the set of electrode constraints 106 occupies no more than 20% of the total combined volume of the electrode assembly 106 and the external portion of the set of electrode constraints. In yet another embodiment, the external portion of the primary growth constraint system 151 (i.e., any of the first and second primary growth constraints 154, 156 and at least one primary connecting member that are external, or external portions thereof) occupies no more than 40% of the total combined volume of the electrode assembly 106 and the external portion of the primary growth constraint system 151. By way of further example, in one embodiment the external portion of the primary growth constraint system 151 occupies no more than 30% of the total combined volume of the electrode assembly 106 and the external portion of the primary growth constraint system 151. By way of yet a further example, in one embodiment the external portion of the primary growth constraint system 151 occupies no more than 20% of the total combined volume of the electrode assembly 106 and the external portion of the primary growth constraint system 151. In yet another embodiment, the external portion of the secondary growth constraint system 152 (i.e., any of the first and second secondary growth constraints 158, 160 and at least one secondary connecting member that are external, or external portions thereof) occupies no more than 40% of the total combined volume of the electrode assembly 106 and the external portion of the secondary growth constraint system 152. By way of further example, in one embodiment, the external portion of the secondary growth constraint system 152 occupies no more than 30% of the total combined volume of the electrode assembly 106 and the external portion of the secondary growth constraint system 152. By way of yet another example, in one embodiment, the external portion of the secondary growth constraint system 152 occupies no more than 20% of the total combined volume of the electrode assembly 106 and the external portion of the secondary growth constraint system 152. By way of yet another example, in one embodiment, the external portion of the secondary growth constraint system 152 occupies no more than 10% of the total combined volume of the electrode assembly 106 and the external portion of the secondary growth constraint system 152.

[0116] According to one embodiment, a rationale for the relatively low volume occupied by portions of the set of electrode constraints 108 can be understood by referring to the force schematics shown in FIGS. 8A and 8B. FIG. 8A depicts an embodiment showing the forces exerted on the

first and second primary growth constraints 154, 156 upon cycling of the secondary battery 102, due to the increase in volume of the electrode active material layers 132. The arrows 198b depict the forces exerted by the electrode active material layers 132 upon expansion thereof, where w shows the load applied to the first and second primary growth constraints 154, 156, due to the growth of the electrode active material layers 132, and P shows the pressure applied to the first and second primary growth constraints 154, 156 as a result of the increase in volume of the electrode active material layers 132. Similarly, FIG. 8B depicts an embodiment showing the forces exerted on the first and second secondary growth constraints 158, 160 upon cycling of the secondary battery 102, due to the increase in volume of the electrode active material layers 132. The arrows 198a depict the forces exerted by the electrode active material layers 132 upon expansion thereof, where w shows the load applied to the first and second secondary growth constraints 158, 160, due to the growth of the electrode active material layers 132, and P shows the pressure applied to the first and second secondary growth constraints 158, 160 as a result of the increase in volume of the electrode active material layers 132. While the electrode active material expands isotropically (i.e., in all directions), during cycling of the secondary battery, and thus the pressure P in each direction is the same, the load w exerted in each direction is different. By way of explanation, referring to the embodiment depicted in FIGS. 8A and 8B, it can be understood that the load in the X-Z plane on a first or secondary primary growth constraint 154, 156 is proportional to $P \times L_{ES} \times H_{ES}$, where P is the pressure exerted due to the expansion of the electrode active material layers 132 on the primary growth constraints 154, 156, L_{ES} is length of the electrode structures 110 in the transverse direction, and H_{ES} is the height of the electrode structures 110 in the vertical direction. Similarly, the load in the X-Y plane on a first or second secondary growth constraint 158, 160 is proportional to $P \times L_{ES} \times W_{ES}$, where P is the pressure exerted due to the expansion of the electrode active material layers 132 on the secondary growth constraints 158, 160, L_{ES} is length of the electrode structures 110 in the transverse direction, and W_{ES} is the width of the electrode structures 110 in the longitudinal direction. In a case where a tertiary constraint system is provided, the load in the Y-Z plane on a first or secondary tertiary growth constraint 157, 159 is proportional to $P \times H_{ES} \times W_{ES}$, where P is the pressure exerted due to the expansion of the electrode active material layers 132 on the tertiary growth constraints 157, 159, H_{ES} is height of the electrode structures 110 in the vertical direction, and W_{ES} is the width of the electrode structures 110 in the longitudinal direction. Accordingly, in a case where L_{ES} is greater than both W_{ES} and H_{ES} , the load in the Y-Z plane will be the least, and in a case where $H_{ES} > W_{ES}$, the load in the X-Y plane will be less than the load in the X-Z plane, meaning that the X-Z plane has the highest load to be accommodated among the orthogonal planes.

[0117] Furthermore, according to one embodiment, if a primary constraint is provided in the X-Z plane in a case where the load in that plane is the greatest, as opposed to providing a primary constraint in the X-Y plane, then the primary constraint in the X-Z plane may require a much lower volume that the primary constraint would be required to have if it were in the X-Y plane. This is because if the primary constraint were in the X-Y plane instead of the X-Z plane, then the constraint would be required to be much

thicker in order to have the stiffness against growth that would be required. In particular, as is described herein in further detail below, as the distance between primary connecting members increases, the buckling deflection can also increase, and the stress also increases. For example, the equation governing the deflection due to bending of the primary growth constraints **154**, **156** can be written as:

$$\delta = 60wL^4/Eh^3$$

[0118] where w =total distributed load applied on the primary growth constraint **154**, **156** due to the electrode expansion; L =distance between the primary connecting members **158**, **160** along the vertical direction; E =elastic modulus of the primary growth constraints **154**, **156**, and h =thickness (width) of the primary growth constraints **154**, **156**. The stress on the primary growth constraints **154**, **156** due to the expansion of the electrode active material **132** can be calculated using the following equation:

$$\sigma = 3wL^2/4h^2$$

[0119] where w =total distributed load applied on the primary growth constraints **154**, **156** due to the expansion of the electrode active material layers **132**; L =distance between primary connecting members **158**, **160** along the vertical direction; and h =thickness (width) of the primary growth constraints **154**, **156**. Thus, if the primary growth constraints were in the X-Y plane, and if the primary connecting members were much further apart (e.g., at longitudinal ends) than they would otherwise be if the primary constraint were in the X-Z plane, this can mean that the primary growth constraints would be required to be thicker and thus occupy a larger volume that they otherwise would if they were in the X-Z plane.

[0120] According to one embodiment, a projection of the members of the electrode and counter-electrode populations onto first and second longitudinal end surfaces **116**, **118** circumscribes a first and second projected areas **2002a**, **2002b**. In general, first and second projected areas **2002a**, **2002b** will typically comprise a significant fraction of the surface area of the first and second longitudinal end surfaces **122**, **124**, respectively. For example, in one embodiment the first and second projected areas each comprise at least 50% of the surface area of the first and second longitudinal end surfaces, respectively. By way of further example, in one such embodiment the first and second projected areas each comprise at least 75% of the surface area of the first and second longitudinal end surfaces, respectively. By way of further example, in one such embodiment the first and second projected areas each comprise at least 90% of the surface area of the first and second longitudinal end surfaces, respectively.

[0121] In certain embodiments, the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a significant compressive load. For example, in some embodiments, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 0.7 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). For example, in one embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 1.75 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). By way of further example, in one

such embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 2.8 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). By way of further example, in one such embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 3.5 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). By way of further example, in one such embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 5.25 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). By way of further example, in one such embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 7 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). By way of further example, in one such embodiment, each of the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of at least 8.75 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). In general, however, the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will be under a compressive load of no more than about 10 kPa (e.g., averaged over the total surface area of each of the longitudinal end surfaces, respectively). The regions of the longitudinal end surface of the electrode assembly that are coincident with the projection of members of the electrode and counter-electrode populations onto the longitudinal end surfaces (i.e., the projected surface regions) may also be under the above compressive loads (as averaged over the total surface area of each projected surface region, respectively). In each of the foregoing exemplary embodiments, the longitudinal end surfaces **116**, **118** of the electrode assembly **106** will experience such compressive loads when an energy storage device **100** having the electrode assembly **106** is charged to at least about 80% of its rated capacity.

[0122] According to one embodiment, the secondary growth constraint system **152** is capable of restraining growth of the electrode assembly **106** in the vertical direction (Z direction) by applying a restraining force at a predetermined value, and without excessive skew of the growth restraints. For example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restraining force to opposing vertical regions **148**, **150** of greater than 1000 psi and a skew of less than 0.2 mm/m. By way of further example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restraining force to opposing vertical regions **148**, **150** with less than 5% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m. By way of further example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restraining force to opposing vertical regions **148**, **150** with less than 3% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m. By way of further example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restrain-

ing force to opposing vertical regions **148**, **150** with less than 1% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m. By way of further example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restraining force to opposing vertical regions **148**, **150** in the vertical direction with less than 15% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m after 50 battery cycles. By way of further example, in one embodiment, the secondary growth constraint system **152** may restrain growth of the electrode assembly **106** in the vertical direction by applying a restraining force to opposing vertical regions **148**, **150** with less than 5% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m after 150 battery cycles.

[0123] Referring now to FIG. 5, an embodiment of an electrode assembly **106** with a set of electrode constraints **108** is shown, with a cross-section taken along the line A-A' as shown in FIG. 1. In the embodiment shown in FIG. 5, the primary growth constraint system **151** can comprise first and second primary growth constraints **154**, **156**, respectively, at the longitudinal end surfaces **116**, **118** of the electrode assembly **106**, and the secondary growth constraint system **152** comprises first and second secondary growth constraints **158**, **160** at the opposing first and second surface regions **148**, **150** of the lateral surface **142** of the electrode assembly **106**. According to this embodiment, the first and second primary growth constraints **154**, **156** can serve as the at least one secondary connecting member **166** to connect the first and second secondary growth constraints **158**, **160** and maintain the growth constraints in tension with one another in the second direction (e.g., vertical direction) that is orthogonal to the longitudinal direction. However, additionally and/or alternatively, the secondary growth constraint system **152** can comprise at least one secondary connecting member **166** that is located at a region other than the longitudinal end surfaces **116**, **118** of the electrode assembly **106**. Also, the at least one secondary connecting member **166** can be understood to act as at least one of a first and second primary growth constraint **154**, **156** that is internal to the longitudinal ends **116**, **118** of the electrode assembly, and that can act in conjunction with either another internal primary growth restraint and/or a primary growth restraint at a longitudinal end **116**, **118** of the electrode assembly **106** to restrain growth. Referring to the embodiment shown in FIG. 5, a secondary connecting member **166** can be provided that is spaced apart along the longitudinal axis away from the first and second longitudinal end surfaces **116**, **118**, respectively, of the electrode assembly **106**, such as toward a central region of the electrode assembly **106**. The secondary connecting member **166** can connect the first and second secondary growth constraints **158**, **160**, respectively, at an interior position from the electrode assembly end surfaces **116**, **118**, and may be under tension between the secondary growth constraints **158**, **160** at that position. In one embodiment, the secondary connecting member **166** that connects the secondary growth constraints **158**, **160** at an interior position from the end surfaces **116**, **118** is provided in addition to one or more secondary connecting members **166** provided at the electrode assembly end surfaces **116**, **118**, such as the secondary connecting members **166** that also serve as primary growth constraints **154**, **156** at the longitudinal end surfaces **116**, **118**. In another embodiment, the secondary growth constraint system **152** comprises one or

more secondary connecting members **166** that connect with first and second secondary growth constraints **158**, **160**, respectively, at interior positions that are spaced apart from the longitudinal end surfaces **116**, **118**, with or without secondary connecting members **166** at the longitudinal end surfaces **116**, **118**. The interior secondary connecting members **166** can also be understood to act as first and second primary growth constraints **154**, **156**, according to one embodiment. For example, in one embodiment, at least one of the interior secondary connecting members **166** can comprise at least a portion of an electrode or counter electrode structure **110**, **112**, as described in further detail below.

[0124] More specifically, with respect to the embodiment shown in FIG. 5, secondary growth constraint system **152** may include a first secondary growth constraint **158** that overlies an upper region **148** of the lateral surface **142** of electrode assembly **106**, and an opposing second secondary growth constraint **160** that overlies a lower region **150** of the lateral surface **142** of electrode assembly **106**, the first and second secondary growth constraints **158**, **160** being separated from each other in the vertical direction (i.e., along the Z-axis). Additionally, secondary growth constraint system **152** may further include at least one interior secondary connecting member **166** that is spaced apart from the longitudinal end surfaces **116**, **118** of the electrode assembly **106**. The interior secondary connecting member **166** may be aligned parallel to the Z axis and connects the first and second secondary growth constraints **158**, **160**, respectively, to maintain the growth constraints in tension with one another, and to form at least a portion of the secondary constraint system **152**. In one embodiment, the at least one interior secondary connecting member **166**, either alone or with secondary connecting members **166** located at the longitudinal end surfaces **116**, **118** of the electrode assembly **106**, may be under tension between the first and second growth constraints **158**, **160** in the vertical direction (i.e., along the Z axis), during repeated charge and/or discharge of an energy storage device **100** or a secondary battery **102** having the electrode assembly **106**, to reduce growth of the electrode assembly **106** in the vertical direction. Furthermore, in the embodiment as shown in FIG. 5, the set of electrode constraints **108** further comprises a primary growth constraint system **151** having first and second primary growth constraints **154**, **156**, respectively, at the longitudinal ends **117**, **119** of the electrode assembly **106**, that are connected by first and second primary connecting members **162**, **164**, respectively, at the upper and lower lateral surface regions **148**, **150**, respectively, of the electrode assembly **106**. In one embodiment, the secondary interior connecting member **166** can itself be understood as acting in concert with one or more of the first and second primary growth constraints **154**, **156**, respectively, to exert a constraining pressure on each portion of the electrode assembly **106** lying in the longitudinal direction between the secondary interior connecting member **166** and the longitudinal ends **117**, **119** of the electrode assembly **106** where the first and second primary growth constraints **154**, **156**, respectively, can be located.

[0125] In one embodiment, one or more of the primary growth constraint system **151** and secondary growth constraint system **152** includes first and secondary primary growth constraints **154**, **156**, respectively, and/or first and second secondary growth constraints **158**, **160**, respectively,

that include a plurality of constraint members. That is, each of the primary growth constraints **154**, **156** and/or secondary growth constraints **158**, **160** may be a single unitary member, or a plurality of members may be used to make up one or more of the growth constraints. For example, in one embodiment, the first and second secondary growth constraints **158**, **160**, respectively, can comprise single constraint members extending along the upper and lower surface regions **148**, **150**, respectively, of the electrode assembly lateral surface **142**. In another embodiment, the first and second secondary growth constraints **158**, **160**, respectively, comprise a plurality of members extending across the opposing surface regions **148**, **150**, of the lateral surface. Similarly, the primary growth constraints **154**, **156** may also be made of a plurality of members, or can each comprise a single unitary member at each electrode assembly longitudinal end **117**, **119**. To maintain tension between each of the primary growth constraints **154**, **156** and secondary growth constraints **158**, **160**, the connecting members (e.g., **162**, **164**, **165**, **166**) are provided to connect the one or plurality of members comprising the growth constraints to the opposing growth constraint members in a manner that exerts pressure on the electrode assembly **106** between the growth constraints.

[0126] In one embodiment, the at least one secondary connecting member **166** of the secondary growth constraint system **152** forms areas of contact **168**, **170** with the first and second secondary growth constraints **158**, **160**, respectively, to maintain the growth constraints in tension with one another. The areas of contact **168**, **170** are those areas where the surfaces at the ends **172**, **174** of the at least one secondary connecting member **166** touches and/or contacts the first and second secondary growth constraints **158**, **160**, respectively, such as where a surface of an end of the at least one secondary connecting member **166** is adhered or glued to the first and second secondary growth constraints **158**, **160**, respectively. The areas of contact **168**, **170** may be at each end **172**, **174** and may extend across a surface area of the first and second secondary growth constraints **158**, **160**, to provide good contact therebetween. The areas of contact **168**, **170** provide contact in the longitudinal direction (Y axis) between the second connecting member **166** and the growth constraints **158**, **160**, and the areas of contact **168**, **170** can also extend into the transverse direction (X-axis) to provide good contact and connection to maintain the first and second secondary growth constraints **158**, **160** in tension with one another. In one embodiment, the areas of contact **168**, **170** provide a ratio of the total area of contact (e.g., the sum of all areas **168**, and the sum of all areas **170**) of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction that is at least 1%. For example, in one embodiment, a ratio of the total area of contact of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction is at least 2%. By way of further example, in one embodiment, a ratio of the total area of contact of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction, is at least 5%. By way of further example, in one embodiment, a ratio of the total area of contact of the one or

more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction, is at least 10%. By way of further example, in one embodiment, a ratio of the total area of contact of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction, is at least 25%. By way of further example, in one embodiment, a ratio of the total area of contact of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction, is at least 50%. In general, a ratio of the total area of contact of the one or more secondary connecting members **166** in the longitudinal direction (Y axis) with the growth constraints **158**, **160**, per W_{EA} of the electrode assembly **106** in the longitudinal direction, will be less than 100%, such as less than 90%, and even less than 75%, as the one or more connecting members **166** typically do not have an area of contact **168**, **170** that extends across the entire longitudinal axis. However, in one embodiment, an area of contact **168**, **170** of the secondary connecting members **166** with the growth constraints **158**, **160**, may extend across a significant portion of the transverse axis (X axis), and may even extend across the entire L_{EA} of the electrode assembly **106** in the transverse direction. For example, a ratio of the total area of contact (e.g., the sum of all areas **168**, and the sum of all areas **170**) of the one or more secondary connecting members **166** in the transverse direction (X axis) with the growth constraints **158**, **160**, per L_{EA} of the electrode assembly **106** in the transverse direction, may be at least about 50%. By way of further example, a ratio of the total area of contact of the one or more secondary connecting members **166** in the transverse direction (X axis) with the growth constraints **158**, **160**, per L_{EA} of the electrode assembly **106** in the transverse direction (X-axis), may be at least about 75%. By way of further example, a ratio of the total area of contact of the one or more secondary connecting members **166** in the transverse direction (X axis) with the growth constraints **158**, **160**, per L_{EA} of the electrode assembly **106** in the transverse direction (X axis), may be at least about 90%. By way of further example, a ratio of the total area of contact of the one or more secondary connecting members **166** in the transverse direction (X axis) with the growth constraints **158**, **160**, per L_{EA} of the electrode assembly **106** in the transverse direction (X axis), may be at least about 95%.

[0127] According to one embodiment, the areas of contact **168**, **170** between the one or more secondary connecting members **166** and the first and second secondary growth constraints **158**, **160**, respectively, are sufficiently large to provide for adequate hold and tension between the growth constraints **158**, **160** during cycling of an energy storage device **100** or a secondary battery **102** having the electrode assembly **106**. For example, the areas of contact **168**, **170** may form an area of contact with each growth constraint **158**, **160** that makes up at least 2% of the surface area of the lateral surface **142** of the electrode assembly **106**, such as at least 10% of the surface area of the lateral surface **142** of the electrode assembly **106**, and even at least 20% of the surface area of the lateral surface **142** of the electrode assembly **106**. By way of further example, the areas of contact **168**, **170** may form an area of contact with each growth constraint

158, 160 that makes up at least 35% of the surface area of the lateral surface **142** of the electrode assembly **106**, and even at least 40% of the surface area of the lateral surface **142** of the electrode assembly **106**. For example, for an electrode assembly **106** having upper and lower opposing surface regions **148, 150**, respectively, the at least one secondary connecting member **166** may form areas of contact **168, 170** with the growth constraints **158, 160** along at least 5% of the surface area of the upper and lower opposing surface regions **148, 150**, respectively, such as along at least 10% of the surface area of the upper and lower opposing surface regions **148, 150**, respectively, and even at least 20% of the surface area of the upper and lower opposing surface regions **148, 150**, respectively. By way of further example, an electrode assembly **106** having upper and lower opposing surface regions **148, 150**, respectively, the at least one secondary connecting member **166** may form areas of contact **168, 170** with the growth constraints **158, 160** along at least 40% of the surface area of the upper and lower opposing surface regions **148, 150**, respectively, such as along at least 50% of the surface area of the upper and lower opposing surface regions **148, 150**, respectively. By forming a contact between the at least one connecting member **166** and the growth constraints **158, 160** that makes up a minimum surface area relative to a total surface area of the electrode assembly **106**, proper tension between the growth constraints **158, 160** can be provided. Furthermore, according to one embodiment, the areas of contact **168, 170** can be provided by a single secondary connecting member **166**, or the total area of contact may be the sum of multiple areas of contact **168, 170** provided by a plurality of secondary connecting members **166**, such as one or a plurality of secondary connecting members **166** located at longitudinal ends **117, 119** of the electrode assembly **106**, and/or one or a plurality of interior secondary connecting members **166** that are spaced apart from the longitudinal ends **117, 119** of the electrode assembly **106**.

[0128] Further still, in one embodiment, the primary and secondary growth constraint systems **151, 152**, respectively, (and optionally the tertiary growth constraint system) are capable of restraining growth of the electrode assembly **106** in both the longitudinal direction and the second direction orthogonal to the longitudinal direction, such as the vertical direction (Z axis) (and optionally in the third direction, such as along the X axis), to restrain a volume growth % of the electrode assembly.

[0129] In certain embodiments, one or more of the primary and secondary growth constraint systems **151, 152**, respectively, comprises a member having pores therein, such as a member made of a porous material. For example, referring to FIG. 6A depicting a top view of a secondary growth constraint **158** over an electrode assembly **106**, the secondary growth constraint **158** can comprise pores **176** that permit electrolyte to pass therethrough, so as to access an electrode assembly **106** that is at least partially covered by the secondary growth constraint **158**. In one embodiment, the first and second secondary growth constraints **158, 160**, respectively, have the pores **176** therein. In another embodiment, each of the first and second primary growth constraints **154, 156**, respectively, and the first and second secondary growth constraints **158, 160**, respectively, have the pores **176** therein. In yet another embodiment, only one or only a portion of the first and second secondary growth constraints **158, 160**, respectively, contain the pores therein.

In yet a further embodiment, one or more of the first and second primary connecting members **162, 164**, respectively, and the at least one secondary connecting member **166** contains pores therein. Providing the pores **176** may be advantageous, for example, when the energy storage device **100** or secondary battery **102** contains a plurality of electrode assemblies **106** stacked together in the battery enclosure **104**, to permit electrolyte to flow between the different electrode assemblies **106** in, for example, the secondary battery **102** as shown in the embodiment depicted in FIG. 20. For example, in one embodiment, a porous member making up at least a portion of the primary and secondary growth constraint system **151, 152**, respectively, may have a void fraction of at least 0.25. By way of further example, in some embodiments, a porous member making up at least a portion of the primary and secondary growth constraint systems **151, 152**, respectively, may have a void fraction of at least 0.375. By way of further example, in some embodiments, a porous member making up at least a portion of the primary and secondary growth constraint systems **151, 152**, respectively, may have a void fraction of at least 0.5. By way of further example, in some embodiments, a porous member making up at least a portion of the primary and secondary growth constraint systems **151, 152**, respectively, may have a void fraction of at least 0.625. By way of further example, in some embodiments, a porous member making up at least a portion of the primary and secondary growth constraint systems **151, 152**, respectively, may have a void fraction of at least 0.75.

[0130] In one embodiment, the set of electrode constraints **108** may be assembled and secured to restrain growth of the electrode assembly **106** by at least one of adhering, bonding, and/or gluing components of the primary growth constraint system **151** to components of the secondary growth constraint system **152**. For example, components of the primary growth constraint system **151** may be glued, welded, bonded, or otherwise adhered and secured to components of the secondary growth constraint system **152**. For example, as shown in FIG. 4A, the first and second primary growth constraints **154, 156**, respectively, can be adhered to first and second primary connecting members **162, 164**, respectively, that may also serve as first and second secondary growth constraints **158, 160**, respectively. Conversely, the first and second secondary growth constraints **158, 150**, respectively, can be adhered to at least one secondary connecting member **166** that serves as at least one of the first and second primary growth constraints **154, 156**, respectively, such as growth constraints at the longitudinal ends **117, 119** of the electrode assembly **106**. Referring to FIG. 5, the first and second secondary growth constraints **158, 160**, respectively, can also be adhered to at least one secondary connecting member **166** that is an interior connecting member **166** spaced apart from the longitudinal ends **117, 119**. In one embodiment, by securing portions of the primary and secondary growth constraint systems **151, 152**, respectively, to one another, the cooperative restraint of the electrode assembly **106** growth can be provided.

[0131] FIGS. 6A-6D illustrate embodiments for securing one or more of the first and second secondary growth constraints **158, 160**, respectively, to one or more secondary connecting members **166**. FIGS. 6A-6D provide a top view of an embodiment of the electrode assembly **106** having the first secondary growth constraint **158** over an upper surface region **148** of the lateral surface **142** of the electrode

assembly 106. Also shown are first and second primary growth constraints 154, 156, respectively, spaced apart along a longitudinal axis (Y axis). A secondary connecting member 166 which may correspond to at least a part of an electrode structure 110 and/or counter electrode structure 112 is also shown. In the embodiment as shown, the first secondary growth constraint 158 has pores 176 therein to allow electrolyte and carrier ions to reach the electrode 110 and counter-electrode 112 structures. As described above, in certain embodiments, the first and second primary growth constraints 154, 156, respectively, can serve as the at least one secondary connecting member 166 to connect the first and second secondary growth constraints 158, 160, respectively. Thus, in the version as shown, the first and second secondary growth constraints 158, 160, respectively, can be connected at the periphery of the electrode assembly 106 to the first and second primary growth constraints 154, 156, respectively. However, in one embodiment, the first and second secondary growth constraints 158, 160, respectively, can also be connected via a secondary connecting member 166 that is an interior secondary connecting member 166. In the version as shown, the first secondary growth constraint 158 comprises bonded regions 178 where the growth constraint 158 is bonded to an underlying interior secondary connecting member 166, and further comprises non-bonded regions 180 where the growth constraint 158 is not bonded to an underlying secondary connecting member 166, so as to provide areas of contact 168 between the growth constraint 158 and underlying secondary connecting member 166 in the form of columns of bonded regions 178 that alternate with areas of non-bonded regions 180. In one embodiment, the non-bonded regions 180 further contain open pores 176 where electrolyte and carrier ions can pass. According to one embodiment, the first and second secondary growth constraints 158, 160, respectively, are adhered to a secondary connecting member 166 that comprises at least a portion of an electrode 110 or counter electrode 112 structure, or other interior structure of the electrode assembly 106. The first and second secondary growth constraints 158, 160, respectively, in one embodiment, can be adhered to the top and bottom ends of the counter-electrode structures 112 or other interior structures forming the secondary connecting member 166, to form columns of adhered areas 178 corresponding to where the constraint is adhered to a counter-electrode 112 or other interior structure, and columns of non-adhered areas 180 between the counter-electrode 112 or other interior structures. Furthermore, the first and second secondary growth constraints 158, 160, respectively, may be bonded or adhered to the counter-electrode structure 112 or other structure forming the at least one secondary connecting member 166 such that pores 176 remain open at least in the non-bonded areas 180, and may also be adhered such that pores 176 in the bonded regions 178 can remain relatively open to allow electrolyte and carrier ions to pass therethrough.

[0132] In yet another embodiment as shown in FIG. 6B, the first and second secondary growth constraints 158, 160, respectively, are connected at the periphery of the electrode assembly 106 to the first and second primary growth constraints 154, 156, respectively, and may also be connected via a secondary connecting member 166 that is an interior secondary connecting member 166. In the version as shown, the first secondary growth constraint 158 comprises bonded regions 178 where the growth constraint 158 is bonded to an

underlying interior secondary connecting member 166, and further comprises non-bonded regions 180 where the growth constraint 158 is not bonded to an underlying secondary connecting member 166, so as to provide areas of contact 168 between the growth constraint 158 and underlying secondary connecting member 166 in the form of rows of bonded regions 178 that alternate with areas of non-bonded regions 180. These bonded and non-bonded regions 178, 180, respectively, in this embodiment can extend across a dimension of the secondary connecting member 166, which may be in the transverse direction (X axis) as shown in FIG. 6B, as opposed to in the longitudinal direction (Y axis) as in FIG. 6A. Alternatively, the bonded and non-bonded regions 178, 180, respectively, can extend across both longitudinal and transverse directions in a predetermined pattern. In one embodiment, the non-bonded regions 180 further contain open pores 176 where electrolyte and carrier ions can pass. The first and second secondary growth constraints 158, 160, respectively, can in one embodiment, be adhered to the top and bottom ends of the counter-electrode structures 112 or other interior structures forming the secondary connecting member 166, to form rows of adhered areas 178 corresponding to where the growth constraint is adhered to a counter-electrode 112 or other interior structure, and areas of non-adhered areas 180 between the counter-electrode 112 or other interior structures. Furthermore, the first and second secondary growth constraints 158, 160, respectively, may be bonded or adhered to the counter-electrode structure 112 or other structure forming the at least one secondary connecting member 166 such that pores 176 remain open at least in the non-bonded areas 180, and may also be adhered such that pores 176 in the bonded regions 178 can remain relatively open to allow electrolyte and carrier ions to pass therethrough.

[0133] In yet another embodiment as shown in FIG. 6C, an alternative configuration for connection of the first and second secondary growth constraint members 158, 160, respectively, to the at least one secondary connecting member 166 is shown. More specifically, the bonded and non-bonded regions 178, 180, respectively, of the secondary growth constraints 158, 160 are shown to be symmetric about an axis of adhesion AG located towards the center of the electrode assembly 106 in the longitudinal direction (Y axis). As shown in this embodiment, the first and second secondary growth constraints 158, 160, respectively, are attached to the ends of secondary connecting members 166 that comprise an electrode 110, counter-electrode 112, or other interior electrode assembly structure, but the columns of bonded and non-bonded areas are not of equal size. That is, the growth constraints 158, 160 can be selectively bonded to interior secondary connecting members 166 in an alternating or other sequence, such that the amount of non-bonded area 180 exceeds the amount of bonded area 178, for example, to provide for adequate numbers of pores 176 open for passage of electrolyte therethrough. That is, the first and second secondary growth constraints 158, 160, respectively, may be bonded to every other counter-electrode 112 or other interior structure making up the secondary connecting members 166, or to one of every 1+n structures (e.g., counter-electrodes 112), according to an area of the bonded to non-bonded region to be provided.

[0134] FIG. 6D illustrates yet another embodiment of an alternative configuration for connection of the first and second secondary growth constraint members 158, 160,

respectively, to the at least one secondary connecting member **166**. In this version, the bonded and non-bonded regions **178**, **180**, respectively, of the first and second secondary growth constraints **158**, **160**, respectively, form an asymmetric pattern of columns about the axis of adhesion AG. That is, the first and second secondary growth constraints **158**, **160**, respectively, can be adhered to the secondary connecting member **166** corresponding to the electrode **110** or counter-electrode **112** structure or other internal structure in a pattern that is non-symmetric, such as by skipping adhesion to interior structures according to a random or other non-symmetric pattern. In the pattern in the embodiment as shown, the bonded and non-bonded regions **178**, **180**, respectively, form alternating columns with different widths that are not symmetric about the axis of adhesion AG. Furthermore, while an axis of adhesion AG is shown herein as lying in a longitudinal direction (Y axis), the axis of adhesion AG may also lie along the transverse direction (X axis), or there may be two axes of adhesion along the longitudinal and transverse directions, about which the patterns of the bonded and non-bonded regions **178**, **180**, respectively, can be formed. Similarly, for each pattern described and/or shown with respect to FIGS. 6A-6D, it is understood that a pattern shown along the longitudinal direction (Y axis) could instead be formed along the transverse direction (X axis), or vice versa, or a combination of patterns in both directions can be formed.

[0135] In one embodiment, an area of a bonded region **178** of the first or second secondary growth constraints **158**, **160**, respectively, along any secondary connecting member **166**, and/or along at least one of the first or second primary growth constraints **154**, **156**, respectively, to a total area of the bonded and non-bonded regions along the constraint, is at least 50%, such as at least 75%, and even at least 90%, such as 100%. In another embodiment, the first and second secondary growth constraints **158**, **160**, respectively, can be adhered to a secondary connecting member **166** corresponding to an electrode **110** or counter-electrode **112** structure or other interior structure of the electrode assembly **106** in such a way that the pores **176** in the bonded regions **178** remain open. That is, the first and second secondary growth constraints **158**, **160**, respectively, can be bonded to the secondary connecting member **166** such that the pores **176** in the growth constraints are not occluded by any adhesive or other means used to adhere the growth constraint(s) to the connecting member(s). According to one embodiment, the first and second secondary growth constraints **158**, **160**, respectively, are connected to the at least one secondary connecting members **166** to provide an open area having the pores **176** of at least 5% of the area of the growth constraints **158**, **160**, and even an open area having the pores **176** of at least 10% of the area of the growth constraints **158**, **160**, and even an open area having the pores **176** of at least 25% of the area of the growth constraints **158**, **160**, such as an open area having the pores **176** of at least 50% of the area of the growth constraints **158**, **160**.

[0136] While the embodiments described above may be characterized with the pores **176** aligned as columns along the Y axis, it will be appreciated by those of skill in the art that the pores **176** may be characterized as being oriented in rows along the X axis in FIGS. 6A-6D, as well, and the adhesive or other means of adhesion may be applied horizontally or along the X axis to assemble the set of electrode constraints **108**. Furthermore, the adhesive or other bonding

means may be applied to yield mesh-like air pores **176**. Further, the axis of adhesion AG, as described above, may also be oriented horizontally, or along the X axis, to provide analogous symmetric and asymmetric adhesion and/or bonding patterns.

[0137] Further, while the pores **176** and non-bonded regions **180** have been described above as being aligned in columns along the Y axis and in rows along the X axis (i.e., in a linear fashion), it has been further contemplated that the pores **176** and/or non-bonded regions **180** may be arranged in a non-linear fashion. For example, in certain embodiments, the pores **176** may be distributed throughout the surface of the first and second secondary growth constraints **158**, **160**, respectively, in a non-organized or random fashion. Accordingly, in one embodiment, adhesive or other adhesion means may be applied in any fashion, so long as the resulting structure has adequate pores **176** that are not excessively occluded, and contains the non-bonded regions **180** having the non-occluded pores **176**.

Secondary Constraint System Sub-Architecture

[0138] According to one embodiment, as discussed above, one or more of the first and second secondary growth constraints **158**, **160**, respectively, can be connected together via a secondary connecting member **166** that is a part of an interior structure of the electrode assembly **106**, such as a part of an electrode **110** and/or counter-electrode structure **112**. In one embodiment, by providing connection between the constraints via structures within the electrode assembly **106**, a tightly constrained structure can be realized that adequately compensates for strain produced by growth of the electrode structure **110**. For example, in one embodiment, the first and second secondary growth constraints **158**, **160**, respectively, may constrain growth in a direction orthogonal to the longitudinal direction, such as the vertical direction, by being placed in tension with one another via connection through a connecting member **166** that is a part of an electrode **110** or counter-electrode structure **112**. In yet a further embodiment, growth of an electrode structure **110** (e.g., an anode structure) can be countered by connection of the secondary growth constraints **158**, **160** through a counter-electrode structure **112** (e.g., cathode) that serves as the secondary connecting member **166**.

[0139] In general, in certain embodiments, components of the primary growth constraint system **151** and the secondary growth constraint system **152** may be attached to the electrode **110** and/or counter-electrode structures **112**, respectively, within an electrode assembly **106**, and components of the secondary growth constraint system **152** may also be embodied as the electrode **110** and/or counter-electrode structures **112**, respectively, within an electrode assembly **106**, not only to provide effective restraint but also to more efficiently utilize the volume of the electrode assembly **106** without excessively increasing the size of an energy storage device **110** or a secondary battery **102** having the electrode assembly **106**. For example, in one embodiment, the primary growth constraint system **151** and/or secondary growth constraint system **152** may be attached to one or more electrode structures **110**. By way of further example, in one embodiment, the primary growth constraint system **151** and/or secondary growth constraint system **152** may be attached to one or more counter-electrode structures **112**. By way of further example, in certain embodiments, the at least one secondary connecting member **166** may be embodied as

the population of electrode structures **110**. By way of further example, in certain embodiments, the at least one secondary connecting member **166** may be embodied as the population of counter-electrode structures **112**.

[0140] Referring now to FIG. 7, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIG. 7 shows a cross section, along the line A-A' as in FIG. 1, of a set of electrode constraints **108**, including one embodiment of both a primary growth constraint system **151** and one embodiment of a secondary growth constraint system **152**. Primary growth constraint system **151** includes a first primary growth constraint **154** and a second primary growth constraint **156**, as described above, and a first primary connecting member **162** and a second primary connecting member **164**, as described above. Secondary growth constraint system **152** includes a first secondary growth constraint **158**, a second secondary growth constraint **160**, and at least one secondary connecting member **166** embodied as the population of electrode structures **110** and/or the population of counter-electrode structures **112**; therefore, in this embodiment, the at least one secondary connecting member **166**, electrode structures **110**, and/or counter-electrode structures **112** can be understood to be interchangeable. Furthermore, the separator **130** may also form a portion of a secondary connecting member **166**. Further, in this embodiment, first primary connecting member **162** and first secondary growth constraint **158** are interchangeable, as described above. Further still, in this embodiment, second primary connecting member **164** and second secondary growth constraint **160** are interchangeable, as described above. More specifically, illustrated in FIG. 7 is one embodiment of a flush connection of the secondary connecting member **166** corresponding to the electrode **110** or counter-electrode structure **112** with the first secondary growth constraint **158** and second secondary growth constraint **160**. The flush connection may further include a layer of glue **182** between the first secondary growth constraint **158** and secondary connecting member **166**, and a layer of glue **182** between the second secondary growth constraint **160** and secondary connecting member **166**. The layers of glue **182** affix first secondary growth constraint **158** to secondary connecting members **166**, and affix the second secondary growth constraint **160** to secondary connecting member **166**.

[0141] Also, one or more of the first and second primary growth constraints **154**, **156**, first and second primary connecting members **162**, **164**, first and second secondary growth constraints **158**, **160**, and at least one secondary connecting member **166** may be provided in the form of a plurality of segments **1088** or parts that can be joined together to form a single member. For example, as shown in the embodiment as illustrated in FIG. 7, a first secondary growth constraint **158** is provided in the form of a main middle segment **1088a** and first and second end segments **1088b** located towards the longitudinal ends **117**, **119** of the electrode assembly **106**, with the middle segment **1088a** being connected to each first and second end segment **1088b** by a connecting portion **1089** provided to connect the segments **1088**, such as notches formed in the segments **1088** that can be interconnected to join the segments **1088** to

one another. A second secondary growth constraint **160** may similarly be provided in the form of a plurality of segments **1088** that can be connected together to form the constraint, as shown in FIG. 7. In one embodiment, one or more of the secondary growth constraints **158**, **160**, at least one primary connecting member **162**, and/or at least one secondary connecting member **166** may also be provided in the form of a plurality of segments **1088** that can be connected together via a connecting portions such as notches to form the complete member. According to one embodiment, the connection of the segments **1088** together via the notch or other connecting portion may provide for pre-tensioning of the member formed of the plurality of segments when the segments are connected.

[0142] Further illustrated in FIG. 7, in one embodiment, are members of the electrode population **110** having an electrode active material layer **132**, an ionically porous electrode current collector **136**, and an electrode backbone **134** that supports the electrode active material layer **132** and the electrode current collector **136**. Similarly, in one embodiment, illustrated in FIG. 7 are members of the counter-electrode population **112** having a counter-electrode active material layer **138**, a counter-electrode current collector **140**, and a counter-electrode backbone **141** that supports the counter-electrode active material layer **138** and the counter-electrode current collector **140**.

[0143] Without being bound to any particular theory (e.g., as in FIG. 7), in certain embodiments, members of the electrode population **110** include an electrode active material layer **132**, an electrode current collector **136**, and an electrode backbone **134** that supports the electrode active material layer **132** and the electrode current collector **136**. Similarly, in certain embodiments, members of the counter-electrode population **112** include a counter-electrode active material layer **138**, a counter-electrode current collector **140**, and a counter-electrode backbone **141** that supports the counter-electrode active material layer **138** and the counter-electrode current collector **140**.

[0144] While members of the electrode population **110** have been illustrated and described herein to include the electrode active material layer **132** being directly adjacent to the electrode backbone **134**, and the electrode current collector **136** directly adjacent to and effectively surrounding the electrode backbone **134** and the electrode active material layer **132**, those of skill in the art will appreciate other arrangements of the electrode population **110** have been contemplated. For example, in one embodiment (not shown), the electrode population **110** may include the electrode active material layer **132** being directly adjacent to the electrode current collector **136**, and the electrode current collector **136** being directly adjacent to the electrode backbone **134**. Stated alternatively, the electrode backbone **134** may be effectively surrounded by the electrode current collector **136**, with the electrode active material layer **132** flanking and being directly adjacent to the electrode current collector **136**. As will be appreciated by those of skill in the art, any suitable configuration of the electrode population **110** and/or the counter-electrode population **112** may be applicable to the inventive subject matter described herein, so long as the electrode active material layer **132** is separated from the counter-electrode active material layer **138** via separator **130**. Also, the electrode current collector **136** is required to be ion permeable if it is located between the electrode active material layer **132** and separator **130**; and

the counter-electrode current collector **140** is required to be ion permeable if it is located between the counter-electrode active material layer **138** and separator **130**.

[0145] For ease of illustration, only three members of the electrode population **110** and four members of the counter-electrode population **112** are depicted; in practice, however, an energy storage device **100** or secondary battery **102** using the inventive subject matter herein may include additional members of the electrode **110** and counter-electrode **112** populations depending on the application of the energy storage device **100** or secondary battery **102**, as described above. Further still, illustrated in FIG. 7 is a microporous separator **130** electrically insulating the electrode active material layer **132** from the counter-electrode active material layer **138**.

[0146] As described above, in certain embodiments, each member of the population of electrode structures **110** may expand upon insertion of carrier ions (not shown) within an electrolyte (not shown) into the electrode structures **110**, and contract upon extraction of carrier ions from electrode structures **110**. For example, in one embodiment, the electrode structures **110** may be anodically active. By way of further example, in one embodiment, the electrode structures **110** may be cathodically active.

[0147] Furthermore, to connect the first and second secondary growth constraints **158**, **160**, respectively, the constraints **158**, **160** can be attached to the at least one connecting member **166** by a suitable means, such as by gluing as shown, or alternatively by being welded, such as by being welded to the current collectors **136**, **140**. For example, the first and/or second secondary growth constraints **158**, **160**, respectively, can be attached to a secondary connecting member **166** corresponding to at least one of an electrode structure **110** and/or counter-electrode structure **112**, such as at least one of an electrode and/or counter-electrode backbone **134**, **141**, respectively, an electrode and/or counter-electrode current collector **136**, **140**, respectively, by at least one of adhering, gluing, bonding, welding, and the like. According to one embodiment, the first and/or second secondary growth constraints **158**, **160**, respectively, can be attached to the secondary connecting member **166** by mechanically pressing the first and/or second secondary growth constraint **158**, **160**, respectively, to an end of one or more secondary connecting member **166**, such as ends of the population of electrode **100** and/or counter-electrode structures **112**, while using a glue or other adhesive material to adhere one or more ends of the electrode **110** and/or counter-electrode structures **112** to at least one of the first and/or second secondary growth constraints **158**, **160**, respectively.

[0148] FIGS. 8A-B depict force schematics, according to one embodiment, showing the forces exerted on the electrode assembly **106** by the set of electrode constraints **108**, as well as the forces being exerted by electrode structures **110** upon repeated cycling of a secondary battery **102** containing the electrode assembly **106**. As shown in FIGS. 8A-B, repeated cycling through charge and discharge of the secondary battery **102** can cause growth in electrode structures **110**, such as in electrode active material layers **132** of the electrode structures **110**, due to intercalation and/or alloying of ions (e.g., Li) into the electrode active material layers **132** of the electrode structures **110**. Thus, the electrode structures **110** can exert opposing forces **198a** in the vertical direction, as well as opposing forces **198b** in the longitudinal direction, due to the growth in volume of the

electrode structure **110**. While not specifically shown, the electrode structure **110** may also exert opposing forces in the transverse direction due to the change in volume. To counteract these forces, and to restrain overall growth of the electrode assembly **106**, in one embodiment, the set of electrode constraints **108** includes the primary growth constraint system **151** with the first and second primary growth constraints **154**, **156**, respectively, at the longitudinal ends **117**, **119** of the electrode assembly **106**, which exert forces **200a** in the longitudinal direction to counter the longitudinal forces **198b** exerted by the electrode structure **110**. Similarly, in one embodiment, the set of electrode constraints **108** includes the secondary growth constraint system **152** with the first and second secondary growth constraints **158**, **160**, respectively, at opposing surfaces along the vertical direction of the electrode assembly **106**, which exert forces **200b** in the vertical direction to counter the vertical forces **198a** exerted by the electrode structure **110**. Furthermore, a tertiary growth constraint system **155** (not shown) can also be provided, alternatively or in addition, to one or more of the first and second growth constraint systems **151**, **152**, respectively, to exert counter forces in the transverse direction to counteract transverse forces exerted by volume changes of the electrode structures **110** in the electrode assembly **106**. Accordingly, the set of electrode constraints **108** may be capable of at least partially countering the forces exerted by the electrode structure **110** by volume change of the electrode structure **110** during cycling between charge and discharge, such that an overall macroscopic growth of the electrode assembly **106** can be controlled and restrained.

Population of Electrode Structures

[0149] Referring again to FIG. 7, each member of the population of electrode structures **110** may also include a top **1052** adjacent to the first secondary growth constraint **158**, a bottom **1054** adjacent to the second secondary growth constraint **160**, and a lateral surface (not marked) surrounding a vertical axis AEs (not marked) parallel to the Z axis, the lateral surface connecting the top **1052** and the bottom **1054**. The electrode structures **110** further include a length L_{ES} , a width W_{ES} , and a height H_{ES} . The length L_{ES} being bounded by the lateral surface and measured along the X axis. The width W_{ES} being bounded by the lateral surface and measured along the Y axis, and the height H_{ES} being measured along the vertical axis Aus or the Z axis from the top **1052** to the bottom **1054**.

[0150] The L_{ES} of the members of the electrode population **110** will vary depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s). In general, however, the members of the electrode population **110** will typically have a L_{ES} in the range of about 5 mm to about 500 mm. For example, in one such embodiment, the members of the electrode population **110** have a L_{ES} of about 10 mm to about 250 mm. By way of further example, in one such embodiment, the members of the electrode population **110** have a L_{ES} of about 20 mm to about 100 mm.

[0151] The W_{ES} of the members of the electrode population **110** will also vary depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s). In general, however, each member of the electrode population **110** will typically have a W_{ES} within the range of about 0.01 mm to 2.5 mm. For example, in one embodiment, the W_{ES} of each member of the electrode population **110** will be in the range of about 0.025 mm to about 2 mm. By way

of further example, in one embodiment, the W_{ES} of each member of the electrode population 110 will be in the range of about 0.05 mm to about 1 mm.

[0152] The H_{ES} of the members of the electrode population 110 will also vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, members of the electrode population 110 will typically have a H_{ES} within the range of about 0.05 mm to about 10 mm. For example, in one embodiment, the H_{ES} of each member of the electrode population 110 will be in the range of about 0.05 mm to about 5 mm. By way of further example, in one embodiment, the H_{ES} of each member of the electrode population 110 will be in the range of about 0.1 mm to about 1 mm.

[0153] In another embodiment, each member of the population of electrode structures 110 may include an electrode structure backbone 134 having a vertical axis A_{ESB} parallel to the Z axis. The electrode structure backbone 134 may also include a layer of electrode active material 132 surrounding the electrode structure backbone 134 about the vertical axis A_{ESB} . Stated alternatively, the electrode structure backbone 134 provides mechanical stability for the layer of electrode active material 132, and may provide a point of attachment for the primary growth constraint system 151 and/or secondary constraint system 152. In certain embodiments, the layer of electrode active material 132 expands upon insertion of carrier ions into the layer of electrode active material 132, and contracts upon extraction of carrier ions from the layer of electrode active material 132. For example, in one embodiment, the layer of electrode active material 132 may be anodically active. By way of further example, in one embodiment, the layer of electrode active material 132 may be cathodically active. The electrode structure backbone 134 may also include a top 1056 adjacent to the first secondary growth constraint 158, a bottom 1058 adjacent to the second secondary growth constraint 160, and a lateral surface (not marked) surrounding the vertical axis A_{ESB} and connecting the top 1056 and the bottom 1058. The electrode structure backbone 134 further includes a length $LESB$, a width W_{ESB} , and a height H_{ESB} . The length $LESB$ being bounded by the lateral surface and measured along the X axis. The width W_{ESB} being bounded by the lateral surface and measured along the Y axis, and the height H_{ESB} being measured along the Z axis from the top 1056 to the bottom 1058.

[0154] The $LESB$ of the electrode structure backbone 134 will vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, the electrode structure backbone 134 will typically have a $LESB$ in the range of about 5 mm to about 500 mm. For example, in one such embodiment, the electrode structure backbone 134 will have a $LESB$ of about 10 mm to about 250 mm. By way of further example, in one such embodiment, the electrode structure backbone 134 will have a $LESB$ of about 20 mm to about 100 mm. According to one embodiment, the electrode structure backbone 134 may be the substructure of the electrode structure 110 that acts as the at least one connecting member 166.

[0155] The W_{ESB} of the electrode structure backbone 134 will also vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, each electrode structure backbone 134 will typically have a W_{ESB} of at least 1 micrometer. For example, in one embodiment, the W_{ESB} of each electrode structure backbone 134 may be substantially thicker, but

generally will not have a thickness in excess of 500 micrometers. By way of further example, in one embodiment, the W_{ESB} of each electrode structure backbone 134 will be in the range of about 1 to about 50 micrometers.

[0156] The H_{ESB} of the electrode structure backbone 134 will also vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, the electrode structure backbone 134 will typically have a H_{ESB} of at least about 50 micrometers, more typically at least about 100 micrometers. Further, in general, the electrode structure backbone 134 will typically have a H_{ESB} of no more than about 10,000 micrometers, and more typically no more than about 5,000 micrometers. For example, in one embodiment, the H_{ESB} of each electrode structure backbone 134 will be in the range of about 0.05 mm to about 10 mm. By way of further example, in one embodiment, the H_{ESB} of each electrode structure backbone 134 will be in the range of about 0.05 mm to about 5 mm. By way of further example, in one embodiment, the H_{ESB} of each electrode structure backbone 134 will be in the range of about 0.1 mm to about 1 mm.

[0157] Depending upon the application, electrode structure backbone 134 may be electrically conductive or insulating. For example, in one embodiment, the electrode structure backbone 134 may be electrically conductive and may include electrode current collector 136 for electrode active material 132. In one such embodiment, electrode structure backbone 134 includes an electrode current collector 136 having a conductivity of at least about 10^3 Siemens/cm. By way of further example, in one such embodiment, electrode structure backbone 134 includes an electrode current collector 136 having a conductivity of at least about 10^4 Siemens/cm. By way of further example, in one such embodiment, electrode structure backbone 134 includes an electrode current collector 136 having a conductivity of at least about 10^5 Siemens/cm. In other embodiments, electrode structure backbone 134 is relatively non-conductive. For example, in one embodiment, electrode structure backbone 134 has an electrical conductivity of less than 10 Siemens/cm. By way of further example, in one embodiment, electrode structure backbone 134 has an electrical conductivity of less than 1 Siemens/cm. By way of further example, in one embodiment, electrode structure backbone 134 has an electrical conductivity of less than 10^{-1} Siemens/cm.

[0158] In certain embodiments, electrode structure backbone 134 may include any material that may be shaped, such as metals, semiconductors, organics, ceramics, and glasses. For example, in certain embodiments, materials include semiconductor materials such as silicon and germanium. Alternatively, however, carbon-based organic materials, or metals, such as aluminum, copper, nickel, cobalt, titanium, and tungsten, may also be incorporated into electrode structure backbone 134. In one exemplary embodiment, electrode structure backbone 134 comprises silicon. The silicon, for example, may be single crystal silicon, polycrystalline silicon, amorphous silicon, or a combination thereof.

[0159] In certain embodiments, the electrode active material layer 132 may have a thickness of at least one micrometer. Typically, however, the electrode active material layer 132 thickness will not exceed 200 micrometers. For example, in one embodiment, the electrode active material layer 132 may have a thickness of about 1 to 50 micrometers. By way of further example, in one embodiment, the

electrode active material layer **132** may have a thickness of about 2 to about 75 micrometers. By way of further example, in one embodiment, the electrode active material layer **132** may have a thickness of about 10 to about 100 micrometers. By way of further example, in one embodiment, the electrode active material layer **132** may have a thickness of about 5 to about 50 micrometers.

[0160] In certain embodiments, the electrode current collector **136** includes an ionically permeable conductor material that has sufficient ionic permeability to carrier ions to facilitate the movement of carrier ions from the separator **130** to the electrode active material layer **132**, and sufficient electrical conductivity to enable it to serve as a current collector. Being positioned between the electrode active material layer **132** and the separator **130**, the electrode current collector **136** may facilitate more uniform carrier ion transport by distributing current from the electrode current collector **136** across the surface of the electrode active material layer **132**. This, in turn, may facilitate more uniform insertion and extraction of carrier ions and thereby reduce stress in the electrode active material layer **132** during cycling; since the electrode current collector **136** distributes current to the surface of the electrode active material layer **132** facing the separator **130**, the reactivity of the electrode active material layer **132** for carrier ions will be the greatest where the carrier ion concentration is the greatest.

[0161] The electrode current collector **136** includes an ionically permeable conductor material that is both ionically and electrically conductive. Stated differently, the electrode current collector **136** has a thickness, an electrical conductivity, and an ionic conductivity for carrier ions that facilitates the movement of carrier ions between an immediately adjacent electrode active material layer **132** on one side of the ionically permeable conductor layer and an immediately adjacent separator layer **130** on the other side of the electrode current collector **136** in an electrochemical stack or electrode assembly **106**. On a relative basis, the electrode current collector **136** has an electrical conductance that is greater than its ionic conductance when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. For example, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the electrode current collector **136** will typically be at least 1,000:1, respectively, when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the electrode current collector **136** is at least 5,000:1, respectively, when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the electrode current collector **136** is at least 10,000:1, respectively, when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the electrode current collector **136** is at least 50,000:1, respectively, when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier

ions) of the electrode current collector **136** is at least 100,000:1, respectively, when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**.

[0162] In one embodiment, and when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**, such as when a secondary battery **102** is charging or discharging, the electrode current collector **136** has an ionic conductance that is comparable to the ionic conductance of an adjacent separator layer **130**. For example, in one embodiment, the electrode current collector **136** has an ionic conductance (for carrier ions) that is at least 50% of the ionic conductance of the separator layer **130** (i.e., a ratio of 0.5:1, respectively) when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer **130** is at least 1:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer **130** is at least 1.25:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer **130** is at least 1.5:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer **130** is at least 2:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**.

[0163] In one embodiment, the electrode current collector **136** also has an electrical conductance that is substantially greater than the electrical conductance of the electrode active material layer **132**. For example, in one embodiment, the ratio of the electrical conductance of the electrode current collector **136** to the electrical conductance of the electrode active material layer **132** is at least 100:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the electrical conductance of the electrode current collector **136** to the electrical conductance of the electrode active material layer **132** is at least 500:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the electrical conductance of the electrode current collector **136** to the electrical conductance of the electrode active material layer **132** is at least 1000:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the electrical conductance of the electrode current collector **136** to the electrical conductance of the electrode active material layer **132** is at least 5000:1 when there is an applied current to store energy in the device **100** or an

applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the electrical conductance of the electrode current collector **136** to the electrical conductance of the electrode active material layer **132** is at least 10,000:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**.

[0164] The thickness of the electrode current collector layer **136** (i.e., the shortest distance between the separator **130** and, in one embodiment, the anodically active material layer (e.g., electrode active material layer **132**) between which the electrode current collector layer **136** is sandwiched) in certain embodiments will depend upon the composition of the layer **136** and the performance specifications for the electrochemical stack. In general, when an electrode current collector layer **136** is an ionically permeable conductor layer, it will have a thickness of at least about 300 Angstroms. For example, in some embodiments, it may have a thickness in the range of about 300-800 Angstroms. More typically, however, it will have a thickness greater than about 0.1 micrometers. In general, an ionically permeable conductor layer will have a thickness not greater than about 100 micrometers. Thus, for example, in one embodiment, the electrode current collector layer **136** will have a thickness in the range of about 0.1 to about 10 micrometers. By way of further example, in some embodiments, the electrode current collector layer **136** will have a thickness in the range of about 0.1 to about 5 micrometers. By way of further example, in some embodiments, the electrode current collector layer **136** will have a thickness in the range of about 0.5 to about 3 micrometers. In general, it is preferred that the thickness of the electrode current collector layer **136** be approximately uniform. For example, in one embodiment, it is preferred that the electrode current collector layer **136** have a thickness non-uniformity of less than about 25%. In certain embodiments, the thickness variation is even less. For example, in some embodiments, the electrode current collector layer **136** has a thickness non-uniformity of less than about 20%. By way of further example, in some embodiments, the electrode current collector layer **136** has a thickness non-uniformity of less than about 15%. In some embodiments the ionically permeable conductor layer has a thickness non-uniformity of less than about 10%.

[0165] In one embodiment, the electrode current collector layer **136** is an ionically permeable conductor layer including an electrically conductive component and an ion conductive component that contribute to the ionic permeability and electrical conductivity. Typically, the electrically conductive component will include a continuous electrically conductive material (e.g., a continuous metal or metal alloy) in the form of a mesh or patterned surface, a film, or composite material comprising the continuous electrically conductive material (e.g., a continuous metal or metal alloy). Additionally, the ion conductive component will typically comprise pores, for example, interstices of a mesh, spaces between a patterned metal or metal alloy containing material layer, pores in a metal film, or a solid ion conductor having sufficient diffusivity for carrier ions. In certain embodiments, the ionically permeable conductor layer includes a deposited porous material, an ion-transporting material, an ion-reactive material, a composite material, or a physically porous material. If porous, for example, the ionically permeable conductor layer may have a void fraction of at least about 0.25. In general, however, the void fraction will

typically not exceed about 0.95. More typically, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.25 to about 0.85. In some embodiments, for example, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.35 to about 0.65.

[0166] In the embodiment illustrated in FIG. 7, electrode current collector layer **136** is the sole anode current collector for electrode active material layer **132**. Stated differently, electrode structure backbone **134** may include an anode current collector. In certain other embodiments, however, electrode structure backbone **134** may optionally not include an anode current collector.

Population of Counter-Electrode Structures

[0167] Referring again to FIG. 7, each member of the population of counter-electrode structures **112** may also include a top **1068** adjacent to the first secondary growth constraint **158**, a bottom **1070** adjacent to the second secondary growth constraint **160**, and a lateral surface (not marked) surrounding a vertical axis A_{CES} (not marked) parallel to the Z axis, the lateral surface connecting the top **1068** and the bottom **1070**. The counter-electrode structures **112** further include a length L_{CES} , a width W_{CES} , and a height H_{CES} . The length L_{CES} being bounded by the lateral surface and measured along the X axis. The width W_{CES} being bounded by the lateral surface and measured along the Y axis, and the height H_{CES} being measured along the vertical axis A_{CES} or the Z axis from the top **1068** to the bottom **1070**.

[0168] The L_{CES} of the members of the counter-electrode population **112** will vary depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s). In general, however, the members of the counter-electrode population **112** will typically have a L_{CES} in the range of about 5 mm to about 500 mm. For example, in one such embodiment, the members of the counter-electrode population **112** have a L_{CES} of about 10 mm to about 250 mm. By way of further example, in one such embodiment, the members of the counter-electrode population **112** have a L_{CES} of about 25 mm to about 100 mm.

[0169] The W_{CES} of the members of the counter-electrode population **112** will also vary depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s). In general, however, each member of the counter-electrode population **112** will typically have a W_{CES} within the range of about 0.01 mm to 2.5 mm. For example, in one embodiment, the W_{CES} of each member of the counter-electrode population **112** will be in the range of about 0.025 mm to about 2 mm. By way of further example, in one embodiment, the W_{CES} of each member of the counter-electrode population **112** will be in the range of about 0.05 mm to about 1 mm.

[0170] The H_{CES} of the members of the counter-electrode population **112** will also vary depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s). In general, however, members of the counter-electrode population **112** will typically have a H_{CES} within the range of about 0.05 mm to about 10 mm. For example, in one embodiment, the H_{CES} of each member of the counter-electrode population **112** will be in the range of about 0.05 mm to about 5 mm. By way of further example,

in one embodiment, the H_{CES} of each member of the electrode population 112 will be in the range of about 0.1 mm to about 1 mm.

[0171] In another embodiment, each member of the population of counter-electrode structures 112 may include a counter-electrode structure backbone 141 having a vertical axis A_{CESB} parallel to the Z axis. The counter-electrode structure backbone 141 may also include a layer of counter-electrode active material 138 surrounding the counter-electrode structure backbone 141 about the vertical axis A_{CESB} . Stated alternatively, the counter-electrode structure backbone 141 provides mechanical stability for the layer of counter-electrode active material 138, and may provide a point of attachment for the primary growth constraint system 151 and/or secondary growth constraint system 152. In certain embodiments, the layer of counter-electrode active material 138 expands upon insertion of carrier ions into the layer of counter-electrode active material 138, and contracts upon extraction of carrier ions from the layer of counter-electrode active material 138. For example, in one embodiment, the layer of counter-electrode active material 138 may be anodically active. By way of further example, in one embodiment, the layer of counter-electrode active material 138 may be cathodically active. The counter-electrode structure backbone 141 may also include a top 1072 adjacent to the first secondary growth constraint 158, a bottom 1074 adjacent to the second secondary growth constraint 160, and a lateral surface (not marked) surrounding the vertical axis A_{CESB} and connecting the top 1072 and the bottom 1074. The counter-electrode structure backbone 141 further includes a length L_{CESB} , a width W_{CESB} , and a height H_{CESB} . The length L_{CESB} being bounded by the lateral surface and measured along the X axis. The width W_{CESB} being bounded by the lateral surface and measured along the Y axis, and the height H_{CESB} being measured along the Z axis from the top 1072 to the bottom 1074.

[0172] The L_{CESB} of the counter-electrode structure backbone 141 will vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, the counter-electrode structure backbone 141 will typically have a L_{CESB} in the range of about 5 mm to about 500 mm. For example, in one such embodiment, the counter-electrode structure backbone 141 will have a L_{CESB} of about 10 mm to about 250 mm. By way of further example, in one such embodiment, the counter-electrode structure backbone 141 will have a L_{CESB} of about 20 mm to about 100 mm.

[0173] The W_{CESB} of the counter-electrode structure backbone 141 will also vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, each counter-electrode structure backbone 141 will typically have a W_{CESB} of at least 1 micrometer. For example, in one embodiment, the W_{CESB} of each counter-electrode structure backbone 141 may be substantially thicker, but generally will not have a thickness in excess of 500 micrometers. By way of further example, in one embodiment, the W_{CESB} of each counter-electrode structure backbone 141 will be in the range of about 1 to about 50 micrometers.

[0174] The H_{CESB} of the counter-electrode structure backbone 141 will also vary depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s). In general, however, the counter-electrode structure backbone 141 will typically have a H_{CESB} of at least about

50 micrometers, more typically at least about 100 micrometers. Further, in general, the counter-electrode structure backbone 141 will typically have a H_{CESB} of no more than about 10,000 micrometers, and more typically no more than about 5,000 micrometers. For example, in one embodiment, the H_{CESB} of each counter-electrode structure backbone 141 will be in the range of about 0.05 mm to about 10 mm. By way of further example, in one embodiment, the H_{CESB} of each counter-electrode structure backbone 141 will be in the range of about 0.05 mm to about 5 mm. By way of further example, in one embodiment, the H_{CESB} of each counter-electrode structure backbone 141 will be in the range of about 0.1 mm to about 1 mm.

[0175] Depending upon the application, counter-electrode structure backbone 141 may be electrically conductive or insulating. For example, in one embodiment, the counter-electrode structure backbone 141 may be electrically conductive and may include counter-electrode current collector 140 for counter-electrode active material 138. In one such embodiment, counter-electrode structure backbone 141 includes a counter-electrode current collector 140 having a conductivity of at least about 10^3 Siemens/cm. By way of further example, in one such embodiment, counter-electrode structure backbone 141 includes a counter-electrode current collector 140 having a conductivity of at least about 10^4 Siemens/cm. By way of further example, in one such embodiment, counter-electrode structure backbone 141 includes a counter-electrode current collector 140 having a conductivity of at least about 10^5 Siemens/cm. In other embodiments, counter-electrode structure backbone 141 is relatively nonconductive. For example, in one embodiment, counter-electrode structure backbone 141 has an electrical conductivity of less than 10 Siemens/cm. By way of further example, in one embodiment, counter-electrode structure backbone 141 has an electrical conductivity of less than 1 Siemens/cm. By way of further example, in one embodiment, counter-electrode structure backbone 141 has an electrical conductivity of less than 10^{-1} Siemens/cm.

[0176] In certain embodiments, counter-electrode structure backbone 141 may include any material that may be shaped, such as metals, semiconductors, organics, ceramics, and glasses. For example, in certain embodiments, materials include semiconductor materials such as silicon and germanium. Alternatively, however, carbon-based organic materials, or metals, such as aluminum, copper, nickel, cobalt, titanium, and tungsten, may also be incorporated into counter-electrode structure backbone 141. In one exemplary embodiment, counter-electrode structure backbone 141 comprises silicon. The silicon, for example, may be single crystal silicon, polycrystalline silicon, amorphous silicon, or a combination thereof.

[0177] In certain embodiments, the counter-electrode active material layer 138 may have a thickness of at least one micrometer. Typically, however, the counter-electrode active material layer 138 thickness will not exceed 200 micrometers. For example, in one embodiment, the counter-electrode active material layer 138 may have a thickness of about 1 to 50 micrometers. By way of further example, in one embodiment, the counter-electrode active material layer 138 may have a thickness of about 2 to about 75 micrometers. By way of further example, in one embodiment, the counter-electrode active material layer 138 may have a thickness of about 10 to about 100 micrometers. By way of

further example, in one embodiment, the counter-electrode active material layer 138 may have a thickness of about 5 to about 50 micrometers.

[0178] In certain embodiments, the counter-electrode current collector 140 includes an ionically permeable conductor that has sufficient ionic permeability to carrier ions to facilitate the movement of carrier ions from the separator 130 to the counter-electrode active material layer 138, and sufficient electrical conductivity to enable it to serve as a current collector. Whether or not positioned between the counter-electrode active material layer 138 and the separator 130, the counter-electrode current collector 140 may facilitate more uniform carrier ion transport by distributing current from the counter-electrode current collector 140 across the surface of the counter-electrode active material layer 138. This, in turn, may facilitate more uniform insertion and extraction of carrier ions and thereby reduce stress in the counter-electrode active material layer 138 during cycling; since the counter-electrode current collector 140 distributes current to the surface of the counter-electrode active material layer 138 facing the separator 130, the reactivity of the counter-electrode active material layer 138 for carrier ions will be the greatest where the carrier ion concentration is the greatest.

[0179] The counter-electrode current collector 140 includes an ionically permeable conductor material that is both ionically and electrically conductive. Stated differently, the counter-electrode current collector 140 has a thickness, an electrical conductivity, and an ionic conductivity for carrier ions that facilitates the movement of carrier ions between an immediately adjacent counter-electrode active material layer 138 on one side of the ionically permeable conductor layer and an immediately adjacent separator layer 130 on the other side of the counter-electrode current collector 140 in an electrochemical stack or electrode assembly 106. On a relative basis, the counter-electrode current collector 140 has an electrical conductance that is greater than its ionic conductance when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. For example, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the counter-electrode current collector 140 will typically be at least 1,000:1, respectively, when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the counter-electrode current collector 140 is at least 5,000:1, respectively, when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the counter-electrode current collector 140 is at least 10,000:1, respectively, when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the counter-electrode current collector 140 layer is at least 50,000:1, respectively, when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the counter-electrode

current collector 140 is at least 100,000:1, respectively, when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100.

[0180] In one embodiment, and when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100, such as when an energy storage device 100 or a secondary battery 102 is charging or discharging, the counter-electrode current collector 140 has an ionic conductance that is comparable to the ionic conductance of an adjacent separator layer 130. For example, in one embodiment, the counter-electrode current collector 140 has an ionic conductance (for carrier ions) that is at least 50% of the ionic conductance of the separator layer 130 (i.e., a ratio of 0.5:1, respectively) when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the counter-electrode current collector 140 to the ionic conductance (for carrier ions) of the separator layer 130 is at least 1:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the counter-electrode current collector 140 to the ionic conductance (for carrier ions) of the separator layer 130 is at least 1.25:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the counter-electrode current collector 140 to the ionic conductance (for carrier ions) of the separator layer 130 is at least 1.5:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the ionic conductance (for carrier ions) of the counter-electrode current collector 140 to the ionic conductance (for anode current collector layer) carrier ions) of the separator layer 130 is at least 2:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100.

[0181] In one embodiment, the counter-electrode current collector 140 also has an electrical conductance that is substantially greater than the electrical conductance of the counter-electrode active material layer 138. For example, in one embodiment, the ratio of the electrical conductance of the counter-electrode current collector 140 to the electrical conductance of the counter-electrode active material layer 138 is at least 100:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the electrical conductance of the counter-electrode current collector 140 to the electrical conductance of the counter-electrode active material layer 138 is at least 500:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the electrical conductance of the counter-electrode current collector 140 to the electrical conductance of the counter-electrode active material layer 138 is at least 1000:1 when there is an applied current to store energy in the device 100 or an applied load to discharge the device 100. By way of further example, in some embodiments, the ratio of the electrical conductance of the counter-electrode current collector 140 to the electrical conductance of the counter-

electrode active material layer **138** is at least 5000:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**. By way of further example, in some embodiments, the ratio of the electrical conductance of the counter-electrode current collector **140** to the electrical conductance of the counter-electrode active material layer **138** is at least 10,000:1 when there is an applied current to store energy in the device **100** or an applied load to discharge the device **100**.

[0182] The thickness of the counter-electrode current collector layer **140** (i.e., the shortest distance between the separator **130** and, in one embodiment, the cathodically active material layer (e.g., counter-electrode active material layer **138**) between which the counter-electrode current collector layer **140** is sandwiched) in certain embodiments will depend upon the composition of the layer **140** and the performance specifications for the electrochemical stack. In general, when a counter-electrode current collector layer **140** is an ionically permeable conductor layer, it will have a thickness of at least about 300 Angstroms. For example, in some embodiments, it may have a thickness in the range of about 300-800 Angstroms. More typically, however, it will have a thickness greater than about 0.1 micrometers. In general, an ionically permeable conductor layer will have a thickness not greater than about 100 micrometers. Thus, for example, in one embodiment, the counter-electrode current collector layer **140** will have a thickness in the range of about 0.1 to about 10 micrometers. By way of further example, in some embodiments, the counter-electrode current collector layer **140** will have a thickness in the range of about 0.1 to about 5 micrometers. By way of further example, in some embodiments, the counter-electrode current collector layer **140** will have a thickness in the range of about 0.5 to about 3 micrometers. In general, it is preferred that the thickness of the counter-electrode current collector layer **140** be approximately uniform. For example, in one embodiment, it is preferred that the counter-electrode current collector layer **140** have a thickness non-uniformity of less than about 25%. In certain embodiments, the thickness variation is even less. For example, in some embodiments, the counter-electrode current collector layer **140** has a thickness non-uniformity of less than about 20%. By way of further example, in some embodiments, the counter-electrode current collector layer **140** has a thickness non-uniformity of less than about 15%. In some embodiments, the counter-electrode current collector layer **140** has a thickness non-uniformity of less than about 10%.

[0183] In one embodiment, the counter-electrode current collector layer **140** is an ionically permeable conductor layer including an electrically conductive component and an ion conductive component that contributes to the ionic permeability and electrical conductivity. Typically, the electrically conductive component will include a continuous electrically conductive material (e.g., a continuous metal or metal alloy) in the form of a mesh or patterned surface, a film, or composite material comprising the continuous electrically conductive material (e.g., a continuous metal or metal alloy). Additionally, the ion conductive component will typically comprise pores, for example, interstices of a mesh, spaces between a patterned metal or metal alloy containing material layer, pores in a metal film, or a solid ion conductor having sufficient diffusivity for carrier ions. In certain embodiments, the ionically permeable conductor layer includes a deposited porous material, an ion-transporting material, an

ion-reactive material, a composite material, or a physically porous material. If porous, for example, the ionically permeable conductor layer may have a void fraction of at least about 0.25. In general, however, the void fraction will typically not exceed about 0.95. More typically, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.25 to about 0.85. In some embodiments, for example, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.35 to about 0.65.

[0184] In the embodiment illustrated in FIG. 7, counter-electrode current collector layer **140** is the sole cathode current collector for counter-electrode active material layer **138**. Stated differently, counter-electrode structure backbone **141** may include a cathode current collector **140**. In certain other embodiments, however, counter-electrode structure backbone **141** may optionally not include a cathode current collector **140**.

[0185] In one embodiment, first secondary growth constraint **158** and second secondary growth constraint **160** each may include an inner surface **1060** and **1062**, respectively, and an opposing outer surface **1064** and **1066**, respectively, separated along the z-axis thereby defining a first secondary growth constraint **158** height H_{158} and a second secondary growth constraint **160** height H_{160} . According to aspects of the disclosure, increasing the heights of either the first and/or second secondary growth constraints **158**, **160**, respectively, can increase the stiffness of the constraints, but can also require increased volume, thus causing a reduction in energy density for an energy storage device **100** or a secondary battery **102** containing the electrode assembly **106** and set of constraints **108**. Accordingly, the thickness of the constraints **158**, **160** can be selected in accordance with the constraint material properties, the strength of the constraint required to offset pressure from a predetermined expansion of an electrode **100**, and other factors. For example, in one embodiment, the first and second secondary growth constraint heights H_{158} and H_{160} , respectively, may be less than 50% of the height H_{ES} . By way of further example, in one embodiment, the first and second secondary growth constraint heights H_{158} and H_{160} , respectively, may be less than 25% of the height H_{ES} . By way of further example, in one embodiment, the first and second secondary growth constraint heights H_{158} and H_{160} , respectively, may be less than 10% of the height H_{ES} . By way of further example, in one embodiment, the first and second secondary growth constraint heights H_{158} and H_{160} may be less than about 5% of the height H_{ES} . In some embodiments, the first secondary growth constraint height H_{158} and the second secondary growth constraint height H_{160} may be different, and the materials used for each of the first and second secondary growth constraints **158**, **160** may also be different.

[0186] In certain embodiments, the inner surfaces **1060** and **1062** may include surface features amenable to affixing the population of electrode structures **110** and/or the population of counter-electrode structures **112** thereto, and the outer surfaces **1064** and **1066** may include surface features amenable to the stacking of a plurality of constrained electrode assemblies **106** (i.e., inferred within FIG. 7, but not shown for clarity). For example, in one embodiment, the inner surfaces **1060** and **1062** or the outer surfaces **1064** and **1066** may be planar. By way of further example, in one embodiment, the inner surfaces **1060** and **1062** or the outer surfaces **1064** and **1066** may be non-planar. By way of

further example, in one embodiment, the inner surfaces **1060** and **1062** and the outer surfaces **1064** and **1066** may be planar. By way of further example, in one embodiment, the inner surfaces **1060** and **1062** and the outer surfaces **1064** and **1066** may be non-planar. By way of further example, in one embodiment, the inner surfaces **1060** and **1062** and the outer surfaces **1064** and **1066** may be substantially planar.

[0187] As described elsewhere herein, modes for affixing the at least one secondary connecting member **166** embodied as electrode structures **110** and/or counter-electrodes **112** to the inner surfaces **1060** and **1062** may vary depending upon the energy storage device **100** or secondary battery **102** and their intended use(s). As one exemplary embodiment shown in FIG. 7, the top **1052** and the bottom **1054** of the population of electrode structures **110** (i.e., electrode current collector **136**, as shown) and the top **1068** and bottom **1070** of the population of counter-electrode structures **112** (i.e., counter-electrode current collector **140**, as shown) may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182**. Similarly, a top **1076** and a bottom **1078** of the first primary growth constraint **154**, and a top **1080** and a bottom **1082** of the second primary growth constraint **156** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182**.

[0188] Stated alternatively, in the embodiment shown in FIG. 7, the top **1052** and the bottom **1054** of the population of electrode structures **110** include a height H_{ES} that effectively meets both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**, and may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** in a flush embodiment. In addition, the top **1068** and the bottom **1070** of the population of counter-electrode structures **112** include a height H_{CES} that effectively meets both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**, and may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** in a flush embodiment.

[0189] Further, in another exemplary embodiment, a top **1056** and a bottom **1058** of the electrode backbones **134**, and a top **1072** and a bottom **1074** of the counter-electrode backbones **141** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** (not illustrated). Similarly, a top **1076** and a bottom **1078** of the first primary growth constraint **154**, and a top **1080** and a bottom **1082** of the second primary growth constraint **156** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** (not illustrated with respect to the embodiment described in this paragraph). Stated alternatively, the top **1056** and the bottom **1058** of the electrode backbones **134** include a height H_{ESB} that effectively meets both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the

second secondary growth constraint **160**, and may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** in a flush embodiment. In addition, the top **1072** and the bottom **1074** of the counter-electrode backbones **141** include a height H_{CESB} that effectively meets both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**, and may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182** in a flush embodiment.

[0190] Accordingly, in one embodiment, at least a portion of the population of electrode **110** and/or counter electrode structures **112**, and/or the separator **130** may serve as one or more secondary connecting members **166** to connect the first and second secondary growth constraints **158**, **160**, respectively, to one another in a secondary growth constraint system **152**, thereby providing a compact and space-efficient constraint system to restrain growth of the electrode assembly **106** during cycling thereof. According to one embodiment, any portion of the electrode **110** and/or counter-electrode structures **112**, and/or separator **130** may serve as the one or more secondary connecting members **166**, with the exception of any portion of the electrode **110** and/or counter-electrode structure **112** that swells in volume with charge and discharge cycles. That is, that portion of the electrode **110** and/or counter-electrode structure **112**, such as the electrode active material **132**, that is the cause of the volume change in the electrode assembly **106**, typically will not serve as a part of the set of electrode constraints **108**. In one embodiment, first and second primary growth constraints **154**, **156**, respectively, provided as a part of the primary growth constraint system **151** further inhibit growth in a longitudinal direction, and may also serve as secondary connecting members **166** to connect the first and second secondary growth constraints **158**, **160**, respectively, of the secondary growth constraint system **152**, thereby providing a cooperative, synergistic constraint system (i.e., set of electrode constraints **108**) for restraint of electrode growth/swelling.

Connections Via Counter-Electrode Structures

[0191] Referring now to FIGS. 9A-9C, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page); a separator **130**, and a designation of the stacking direction D, as described above, and co-parallel with the Y axis. More specifically, FIGS. 9A-9C each show a cross section, along the line A-A' as in FIG. 1, where each first primary growth constraint **154** and each second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in each of FIGS. 9A-9C, non-affixed electrode structures **110** may include electrode gaps **1084** between their tops and the first secondary growth constraint **158**, and their bottoms and the second secondary growth constraint **160**. Stated alternatively, in certain embodiments, the top and the bottom **1052**, **1054**, respectively, of each electrode structure **110** may have a gap between the first and second secondary growth constraints

158, 160, respectively. Further, in certain embodiments as shown in FIG. 9C, the top **1052** of the electrode structure **110** may be in contact with, but not affixed to, the first secondary growth constraint **158**, the bottom **1054** of the electrode structure **110** may be in contact with, but not affixed to, the second secondary growth constraint **160**, or the top **1052** of the electrode structure **110** may be in contact with, but not affixed to, the first secondary growth constraint **158** and the bottom **1054** of the electrode structure **110** may be in contact with, but not affixed to, the second secondary growth constraint **160** (not illustrated).

[0192] More specifically, in one embodiment, as shown in FIG. 9A, a plurality of counter-electrode backbones **141** may be affixed to the inner surface **1160** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182**. In certain embodiments, the plurality of counter-electrode backbones **112** affixed to the first and second secondary growth constraints **158, 160**, respectively, may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**. In certain embodiments, the plurality of counter-electrode backbones **141** affixed to the first and second secondary growth constraints **158, 160**, respectively, may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**.

[0193] In one exemplary embodiment, a first symmetric attachment pattern unit may include two counter-electrode backbones **141** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two affixed counter-electrode backbones **141** flank one electrode structure **110**. Accordingly, the first symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and the intended use(s) thereof. In another exemplary embodiment, a second symmetric attachment pattern unit may include two counter-electrode backbones **141** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, the two affixed counter-electrode backbones **141** flanking two or more electrode structures **110** and one or more non-affixed counter-electrode backbones **141**. Accordingly, the second symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and the intended use(s) thereof. Other exemplary symmetric attachment pattern units have been contemplated, as would be appreciated by a person having skill in the art.

[0194] In one exemplary embodiment, a first asymmetric or random attachment pattern may include two or more counter-electrode backbones **141** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two or more affixed counter-electrode backbones **141** may be individually designated as affixed counter-electrode backbone **141A**, affixed counter-electrode backbone **141B**, affixed counter-electrode backbone **141C**, and affixed counter-electrode backbone **141D**. Affixed counter-electrode backbone **141A** and affixed counter-electrode backbone **141B** may flank (1+x) electrode structures **110**, affixed counter-electrode backbone **141B** and affixed counter-electrode backbone **141C** may flank (1+y) electrode structures **110**, and affixed counter-electrode backbone **141C** and affixed counter-electrode backbone **141D**

may flank (1+z) electrode structures **110**, wherein the total amount of electrode structures **110** (i.e., x, y, or z) between any two affixed counter-electrode backbones **141A-141D** are non-equal (i.e., x y z) and may be further separated by non-affixed counter-electrode backbones **141**. Stated alternatively, any number of counter-electrode backbones **141** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, whereby between any two affixed counter-electrode backbones **141** may include any non-equivalent number of electrode structures **110** separated by non-affixed counter-electrode backbones **141**. Other exemplary asymmetric or random attachment patterns have been contemplated, as would be appreciated by a person having skill in the art.

[0195] More specifically, in one embodiment, as shown in FIG. 9B, a plurality of counter-electrode current collectors **140** may be affixed to the inner surface **1160** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182**. In certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158** and **160** may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**. In certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158** and **160**, respectively, may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**.

[0196] In one exemplary embodiment, a first symmetric attachment pattern unit may include two counter-electrode current collectors **140** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two affixed counter-electrode current collectors **140** flank one electrode structure **110**. Accordingly, the first symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and the intended use(s) thereof. In another exemplary embodiment, a second symmetric attachment pattern unit may include two counter-electrode current collectors **140** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, the two affixed counter-electrode current collectors **140** flanking two or more electrode structures **110** and one or more non-affixed counter-electrode current collectors **140**. Accordingly, the second symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and the intended use(s) thereof. Other exemplary symmetric attachment pattern units have been contemplated, as would be appreciated by a person having skill in the art.

[0197] In one exemplary embodiment, a first asymmetric or random attachment pattern may include two or more counter-electrode current collectors **140** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two or more affixed counter-electrode current collectors **140** may be individually designated as affixed counter-electrode current collector **140A**, affixed counter-electrode current collector **140B**, affixed counter-electrode current collector **140C**, and affixed counter-electrode current collector **140D**. Affixed

counter-electrode current collector **140A** and affixed counter-electrode structure current collector **140B** may flank $(1+x)$ electrode structures **110**, affixed counter-electrode current collector **140B** and affixed counter-electrode current collector **140C** may flank $(1+y)$ electrode structures **110**, and affixed counter-electrode current collector **140C** and affixed counter-electrode current collector **140D** may flank $(1+z)$ electrode structures **110**, wherein the total amount of electrode structures **110** (i.e., x , y , or z) between any two affixed counter-electrode current collectors **140A-140D** are non-equal (i.e., $x+y+z$) and may be further separated by non-affixed counter-electrode current collectors **140**. Stated alternatively, any number of counter-electrode current collectors **140** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, whereby between any two affixed counter-electrode current collectors **140** may include any non-equivalent number of electrode structures **110** separated by non-affixed counter-electrode current collectors **140**. Other exemplary asymmetric or random attachment patterns have been contemplated, as would be appreciated by a person having skill in the art.

[0198] Referring now to FIG. 10, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page); and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIG. 10 shows a cross section, along the line A-A' as in FIG. 1, having the first and second primary growth constraints **154**, **166**, respectively, affixed to the first and second secondary growth constraints **158**, **160**, respectively, via glue **182**, as described above. Further, in one embodiment, illustrated is a plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via glue **182**. More specifically, the plurality of counter-electrode current collectors **140** may include a bulbous or dogbone shaped cross section. Stated alternatively, the counter-electrode current collectors **140** may have increased current collector **140** width near the top **1072** and the bottom **1074** of the counter-electrode backbone **141** with respect to a width of the current collector **140** near a midpoint between the top **1072** and the bottom **1074** of the counter-electrode backbone **141**. That is, the bulbous cross-section of the counter-electrode current collector **140** width towards the top of the current collector **140** may taper towards the middle of the counter-electrode current collector **140**, and increase again to provide a bulbous cross-section towards the bottom of the counter-electrode current collector **140**. Accordingly, the application of glue **182** may surround the bulbous or dogbone portions of counter-electrode current collector **140** and affix counter-electrode current collector **140** to first and second secondary growth constraints **158**, **160**, respectively, as described above. In this embodiment, the bulbous or dogbone shaped counter-electrode current collector **140** may provide an increased strength of attachment to the first and second secondary growth constraints **158**, **160**, respectively, compared to other embodiments described herein. Also illustrated in FIG. 10 are electrode structures **110** with corresponding electrode gaps **1084**, each as described above, and separators **130**. Further, in this embodiment, the plurality of counter-electrode current collectors **140** may be affixed in a symmetric or asymmetric pattern as described

above. Further still, in this embodiment, electrode structures **110** may be in contact with, but not affixed to, the first and second secondary growth constraints **158**, **160**, respectively, as described above.

[0199] Another mode for affixing the counter-electrode structures **112** to the first and second secondary growth constraints **158**, **160**, respectively, via glue **182** includes the use of notches within the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**. Referring now to FIGS. 11A-11C, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page); a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. 11A-11C each show a cross section, along the line A-A' as in FIG. 1, where each first primary growth constraint **154** and each second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in each of FIGS. 11A-11C, non-affixed electrode structures **110** may include electrode gaps **1084** between their tops and the first secondary growth constraint **158**, and their bottoms and the second secondary growth constraint **160**, as described in more detail above.

[0200] More specifically, in one embodiment, as shown in FIG. 11A, a plurality of counter-electrode backbones **141** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a notch **1060a** and **1062a**, and a layer of glue **182**. Accordingly, in certain embodiments, the plurality of counter-electrode backbones **141** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via notches **1060a**, **1062a** may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**, as described above. In certain embodiments, the plurality of counter-electrode backbones **141** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via notches **1060a**, **1062a** may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**, as described above.

[0201] In certain embodiments, notches **1060a**, **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively. For example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 25% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 50% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a** or **1060b** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 75% of

the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 90% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). Alternatively stated, each member of the plurality of the counter-electrode backbones **141** may include a height H_{CESB} that effectively meets and extends into both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**, and may be affixed into the notch **1060a** of the first secondary growth constraint **158** and into the notch **1062a** of the second secondary growth constraint **160** via glue **182** in a notched embodiment.

[0202] Further, FIGS. 11A-11C also depict different embodiments for gluing the plurality of the counter-electrode backbones **141** in a notched embodiment. For example, in one embodiment depicted in FIG. 11A, the plurality of counter-electrode backbones **141** may be glued **182** via a counter-electrode backbone top **1072** and a counter-electrode backbone bottom **1074**. By way of further example, in one embodiment depicted in FIG. 11B, the plurality of counter-electrode backbones **141** may be glued **182** via the lateral surfaces of the counter-electrode backbones **141**. By way of further example, in one embodiment depicted in FIG. 11C, the plurality of counter-electrode backbones **141** may be glued **182** via the top **1072**, the bottom **1074**, and the lateral surfaces of the counter-electrode backbones **141**.

[0203] Further, another mode for affixing the counter-electrode structures **112** to the first and second secondary growth constraints **158**, **160**, respectively, via glue **182** includes, again, the use of notches **1060a** and **1062a** within the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160**. Referring now to FIGS. 12A-12C, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page); a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. 12A-12C each show a cross section, along the line A-A' as in FIG. 1, where each first primary growth constraint **154** and each second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in each of FIGS. 12A-12C, non-affixed electrode structures **110** may include electrode gaps **1084** between their tops **1052** and the first secondary growth constraint **158**, and their bottoms **1054** and the second secondary growth constraint **160**, as described in more detail above.

[0204] More specifically, in one embodiment, as shown in FIG. 12A, a plurality of counter-electrode current collectors **140** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062**

of the second secondary growth constraint **160** via a notch **1060a** and **1062a**, and a layer of glue **182**. Accordingly, in certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via notches **1060a**, **1062a** may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**, as described above. In certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via notches **1060a**, **1062a** may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**, as described above.

[0205] In certain embodiments, notches **1060a**, **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively. For example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 25% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 50% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a** or **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 75% of the height of the first and the second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). Alternatively stated, each member of the plurality of the counter-electrode current collectors **140** may effectively meet and extend into both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** (akin to the height H_{CESB} , as described above), and may be affixed into the notch **1060a** of the first secondary growth constraint **158** and into the notch **1062a** of the second secondary growth constraint **160** via glue **182** in a notched embodiment.

[0206] Further, FIGS. 12A-12C also depict different embodiments for gluing the plurality of the counter-electrode current collectors **140** in a notched embodiment. For example, in one embodiment depicted in FIG. 12A, the plurality of counter-electrode current collectors **140** may be glued **182** via a counter-electrode current collector top **1486** and a counter-electrode current collector bottom **1488**. By way of further example, in one embodiment depicted in FIG. 12B, the plurality of counter-electrode current collectors **140**

may be glued **182** via the lateral surfaces of the counter-electrode current collectors **140** (akin to the lateral surfaces of the counter-electrode backbones **141**, as described above). By way of further example, in one embodiment depicted in FIG. **12C**, the plurality of counter-electrode current collectors **140** may be glued **182** via the top **1486**, the bottom **1488**, and the lateral surfaces of the counter-electrode current collectors **140**.

[0207] In certain embodiments, a plurality of counter-electrode backbones **141** or a plurality of counter-electrode current collectors **140** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160** via a slot in each of the first secondary growth constraint **158** and the second secondary growth constraint **160**, via an interlocking connection embodiment. Referring now to FIGS. **13A-13C** and **14**, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. **13A-13C** and **14** each show a cross section, along the line A-A' as in FIG. **1**, where each first primary growth constraint **154** and each second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in each of FIGS. **13A-13C** and **14**, non-affixed electrode structures **110** may include electrode gaps **1084** between their tops **1052** and the first secondary growth constraint **158**, and their bottoms **1054** and the second secondary growth constraint **160**, as described in more detail above.

[0208] More specifically, in one embodiment shown in FIG. **13A**, a plurality of counter-electrode backbones **141** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160** via a slot **1060b** and **1062b**, and a layer of glue **182**. Accordingly, in certain embodiments, the plurality of counter-electrode backbones **141** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b** and **1062b** may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**, as described above. In certain embodiments, the plurality of counter-electrode backbones **141** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b** and **1062b** may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode backbones **141**, as described above.

[0209] In certain embodiments, slots **1060b** and **1062b** in each of the first secondary growth constraint **158** and the second secondary growth constraint **160** may extend through the first secondary growth constraint **158** and the second secondary growth constraint **160**, respectively, in order to receive the plurality of counter-electrode backbones **141** in an interlocked embodiment. Stated alternatively, the plurality of counter-electrode backbones **141** include a height H_{CESB} that meets and extends entirely through both the first secondary growth constraint height H_{158} , as described above, via slot **1060b** and the second secondary growth constraint height H_{160} , as described above via slot **1062b**, thereby interlocking with both the first secondary growth constraint **158** and the second secondary growth constraint

160 in an interlocked embodiment. In certain embodiments, glue **182** may be used to affix or reinforce the interlocking connection between the lateral surfaces of the plurality of counter-electrode backbones **141** and the slots **1060b**, **1062b**, respectively.

[0210] More specifically, as illustrated by FIGS. **13B-13C**, slots **1060b** and **1062b** may be characterized by an aspect ratio. For example, in certain embodiments as illustrated in FIG. **13B**, slot **1060b** may include a first dimension S_1 defined as the distance between the top **1072** of the counter-electrode backbone **141** and the outer surface **1064** of the first secondary growth constraint **158**, and a second dimension S_2 defined as the distance between two lateral surfaces of the counter-electrode backbone **141**, as described above. Accordingly, for example, in one embodiment S_1 may be the same and/or similar dimension as the secondary growth constraint heights H_{158} and H_{160} described above, which in turn may have a height selected in relation to a counter-electrode structure height H_{CES} . For example, in one embodiment, S_1 may be less than 50% of a counter-electrode height H_{CES} . By way of further example, in one embodiment, S_1 may be less than 25% of a counter-electrode height H_{CES} . By way of further example, in one embodiment, S_1 may be less than 10% of a counter-electrode height H_{CES} , such as less than 5% of a counter-electrode height H_{CES} . Accordingly, for a counter-electrode height H_{CES} in the range of 0.05 mm to 10 mm, S_1 may have a value in the range of 0.025 mm to 0.5 mm. Furthermore, in one embodiment, S_2 may be at least 1 micrometer. By way of further example, in one embodiment, S_2 may generally not exceed 500 micrometers. By way of further example, in one embodiment, S_2 may be in the range of 1 to about 50 micrometers. As such, for example, in one embodiment, the aspect ratio $S_1:S_2$ may be in a range of from 0.05 to 500. By way of further example, in one embodiment, the aspect ratio $S_1:S_2$ may be in a range of from 0.5 to 100.

[0211] Further, as illustrated in FIG. **13C**, slot **1062b** may include a first dimension S_3 defined as the distance between the bottom **1074** of the counter-electrode backbone **141** and the outer surface **1066** of the second secondary growth constraint **160**, and a second dimension S_4 defined as the distance between two lateral surfaces of the counter-electrode backbone **141**, as described above. In one embodiment, S_3 may be the same and/or similar dimension as the secondary growth constraint heights H_{158} and H_{160} described above, which in turn may have a height selected in relation to a counter-electrode height H_{CES} . For example, in one embodiment, S_3 may be less than 50% of a counter-electrode height H_{CES} . By way of further example, in one embodiment, S_3 may be less than 25% of a counter-electrode height H_{CES} . By way of further example, in one embodiment, S_3 may be less than 10% of a counter-electrode height H_{CES} , such as less than 5% of a counter-electrode height H_{CES} . Furthermore, in one embodiment S_2 may be at least 1 micrometer. By way of further example, in one embodiment, S_2 may generally not exceed 500 micrometers. By way of further example, in one embodiment, S_2 may be in the range of 1 to about 50 micrometers. As such, for example, in one embodiment, the aspect ratio $S_3:S_4$ may be in a range of from 0.05 to 500. By way of further example, in one embodiment, the aspect ratio $S_3:S_4$ may be in a range of from 0.5 to 100.

[0212] Referring now to FIG. **14**, in another embodiment, a plurality of counter-electrode current collectors **140** may

be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160** via a slot **1060b** and **1062b**, and a layer of glue **182**. Accordingly, in certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b**, **1062b** may include a symmetrical pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**, as described above. In certain embodiments, the plurality of counter-electrode current collectors **140** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b**, **1062b** may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed counter-electrode current collectors **140**, as described above.

[0213] In certain embodiments, slots **1060b**, **1062b** in each of the first secondary growth constraint **158** and the second secondary growth constraint **160** may extend through the first secondary growth constraint **158** and the second secondary growth constraint **160**, respectively, in order to receive the plurality of counter-electrode current collectors **140** in another interlocked embodiment. Stated alternatively, the plurality of counter-electrode current collectors **140** may effectively meet and extend entirely through both the first secondary growth constraint **158** and the second secondary growth constraint **160** (akin to the height H_{CESB} , as described above), and may be affixed into slots **1060b** and **1062b** via glue **182** in another interlocked embodiment.

Connections via Electrode Structures

[0214] In alternative embodiments described below, the electrode structures **110** may also be independently affixed to the first and second secondary growth constraints **158**, **160**, respectively. Referring now to FIGS. **15A-15B**, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. **15A-15B** each show a cross section, along the line A-A' as in FIG. **1**, where each first primary growth constraint **154** and each second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in each of FIGS. **15A-15B**, non-affixed counter-electrode structures **112** may include counter-electrode gaps **1086** between their tops **1068** and the first secondary growth constraint **158**, and their bottoms **1070** and the second secondary growth constraint **160**. Stated alternatively, in certain embodiments, the top **1068** and the bottom **1070** of each counter-electrode structure **112** may have a gap **1086** between the first and second secondary constraints **158**, **160**, respectively. Further, in certain embodiments, also shown in FIGS. **15A-15B**, the top **1068** of the counter-electrode structure **112** may be in contact with, but not affixed to, the first secondary growth constraint **158**, the bottom **1070** of the counter-electrode structure **112** may be in contact with, but not affixed to, the second secondary growth constraint **160**, or the top **1068** of the counter-electrode structure **112** may be in contact with, but not affixed to, the first secondary growth constraint **158** and the bottom **1070** of the counter-

electrode structure **112** may be in contact with, but not affixed to, the second secondary growth constraint **160** (not illustrated).

[0215] More specifically, in one embodiment, as shown in FIG. **15A**, a plurality of electrode backbones **134** may be affixed to the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** via a layer of glue **182**. In certain embodiments, the plurality of electrode backbones **134** affixed to the first and second secondary growth constraints **158**, **160**, respectively, may include a symmetrical pattern about a gluing axis A_G with respect to affixed electrode backbones **134**. In certain embodiments, the plurality of electrode backbones **134** affixed to the first and second secondary growth constraints **158**, **160**, respectively, may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed electrode backbones **134**.

[0216] In one exemplary embodiment, a first symmetric attachment pattern unit may include two electrode backbones **134** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two affixed electrode backbones **134** flank one counter-electrode structure **112**. Accordingly, the first symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s) thereof. In another exemplary embodiment, a second symmetric attachment pattern unit may include two electrode backbones **134** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, the two affixed electrode backbones **134** flanking two or more counter-electrode structures **112** and one or more non-affixed electrode backbones **134**. Accordingly, the second symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device **100** or the secondary battery **102** and their intended use(s) thereof. Other exemplary symmetric attachment pattern units have been contemplated, as would be appreciated by a person having skill in the art.

[0217] In one exemplary embodiment, a first asymmetric or random attachment pattern may include two or more electrode backbones **134** affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, where the two or more affixed electrode backbones **134** may be individually designated as affixed electrode backbone **134A**, affixed electrode backbone **134B**, affixed electrode backbone **134C**, and affixed electrode backbone **134D**. Affixed electrode backbone **134A** and affixed electrode backbone **134B** may flank (1+x) counter-electrode structures **112**, affixed electrode backbone **134B** and affixed electrode backbone **134C** may flank (1+y) counter-electrode structures **112**, and affixed electrode backbone **134C** and affixed electrode backbone **134D** may flank (1+z) counter-electrode structures **112**, wherein the total amount of counter-electrode structures **112** (i.e., x, y, or z) between any two affixed electrode backbones **134A-134D** are non-equal (i.e., $x \neq y \neq z$) and may be further separated by non-affixed electrode backbones **134**. Stated alternatively, any number of electrode backbones **134** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160**, as above, whereby between any two affixed electrode backbones **134** may

include any non-equivalent number of counter-electrode structures 112 separated by non-affixed electrode backbones 134. Other exemplary asymmetric or random attachment patterns have been contemplated, as would be appreciated by a person having skill in the art.

[0218] More specifically, in one embodiment, as shown in FIG. 15B, a plurality of electrode current collectors 136 may be affixed to the inner surface 1060 of the first secondary growth constraint 158 and the inner surface 1062 of the second secondary growth constraint 160 via a layer of glue 182. In certain embodiments, the plurality of electrode current collectors 136 affixed to the first and second secondary growth constraints 158, 160, respectively, may include a symmetrical pattern about a gluing axis A_G with respect to affixed electrode current collectors 136. In certain embodiments, the plurality of electrode current collectors 136 affixed to the first and second secondary growth constraints 158, 160, respectively, may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed electrode current collectors 136.

[0219] In one exemplary embodiment, a first symmetric attachment pattern unit may include two electrode current collectors 136 affixed to the first secondary growth constraint 158 and the second secondary growth constraint 160, as above, where the two affixed electrode current collectors 136 flank one counter-electrode structure 112. Accordingly, the first symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s) thereof. In another exemplary embodiment, a second symmetric attachment pattern unit may include two electrode current collectors 136 affixed to the first secondary growth constraint 158 and the second secondary growth constraint 160, as above, the two affixed electrode current collectors 136 flanking two or more counter-electrode structures 112 and one or more non-affixed electrode current collectors 136. Accordingly, the second symmetric attachment pattern unit may repeat, as needed, along the stacking direction D depending upon the energy storage device 100 or the secondary battery 102 and their intended use(s) thereof. Other exemplary symmetric attachment pattern units have been contemplated, as would be appreciated by a person having skill in the art.

[0220] In one exemplary embodiment, a first asymmetric or random attachment pattern may include two or more electrode current collectors 136 affixed to the first secondary growth constraint 158 and the second secondary growth constraint 160, as above, where the two or more affixed electrode current collectors 136 may be individually designated as affixed electrode current collector 136A, affixed electrode current collector 136B, affixed electrode current collector 136C, and affixed electrode current collector 136D. Affixed electrode current collector 136A and affixed electrode current collector 136B may flank (1+x) counter-electrode structures 112, affixed electrode current collector 136B and affixed electrode current collector 136C may flank (1+y) counter-electrode structures 112, and affixed electrode current collector 136C and affixed electrode current collector 136D may flank (1+z) counter-electrode structures 112, wherein the total amount of counter-electrode structures 112 (i.e., x, y, or z) between any two affixed electrode current collectors 136A-136D are non-equal (i.e., $x \neq y \neq z$) and may be further separated by non-affixed electrode current collectors 136. Stated alternatively, any number of electrode

current collectors 136 may be affixed to the first secondary growth constraint 158 and the second secondary growth constraint 160, as above, whereby between any two affixed electrode current collectors 136 may include any non-equivalent number of counter-electrode structures 112 separated by non-affixed electrode current collectors 136. Other exemplary asymmetric or random attachment patterns have been contemplated, as would be appreciated by a person having skill in the art.

[0221] Another mode for affixing the electrode structures 110 to the first and second secondary growth constraints 158, 160, respectively, via glue 182 includes the use of notches 1060a, 1062a within the inner surface 1060 of the first secondary growth constraint 158 and the inner surface 1062 of the second secondary growth constraint 160. Referring now to FIGS. 16A-16C, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page); a separator 130, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. 16A-16C each show a cross section, along the line A-A' as in FIG. 1, where each first primary growth constraint 154 and each second primary growth constraint 156 may be attached via a layer of glue 182 to the first secondary growth constraint 158 and second secondary growth constraint 160, as described above. In certain embodiments, as shown in each of FIGS. 16A-16C, non-affixed counter-electrode structures 112 may include counter-electrode gaps 1086 between their tops 1068 and the first secondary growth constraint 158, and their bottoms 1070 and the second secondary growth constraint 160, as described in more detail above.

[0222] More specifically, in one embodiment, as shown in FIG. 16A, a plurality of electrode current collectors 136 may be affixed to the inner surface 1060 of the first secondary growth constraint 158 and the inner surface 1062 of the second secondary growth constraint 160 via a notch 1060a and 1062a, and a layer of glue 182. Accordingly, in certain embodiments, the plurality of electrode current collectors 136 affixed to the first and second secondary growth constraints 158, 160, respectively, via notches 1060a, 1062a may include a symmetrical pattern about a gluing axis A_G with respect to affixed electrode current collectors 136, as described above. In certain embodiments, the plurality of electrode current collectors 136 affixed to the first and second secondary growth constraints 158, 160, respectively, via notches 1060a, 1062a may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed electrode current collectors 136, as described above.

[0223] In certain embodiments, notches 1060a, 1062a may have a depth within the first and second secondary growth constraints 158, 160, respectively. For example, in one embodiment, a notch 1060a, 1062a may have a depth within the first and second secondary growth constraints 158, 160, respectively, of 25% of the height of the first and second secondary growth constraints 158, 160, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch 1060a, 1062a may have a depth within the first and second secondary growth constraints 158, 160, respectively, of 50% of the height of the first and second secondary growth constraints 158, 160, respectively

(i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a**, **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 75% of the height of the first and second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). By way of further example, in one embodiment, a notch **1060a**, **1062a** may have a depth within the first and second secondary growth constraints **158**, **160**, respectively, of 90% of the height of the first and second secondary growth constraints **158**, **160**, respectively (i.e., the heights of the first and second secondary growth constraints in this embodiment may be analogous to H_{158} and H_{160} , as described above). Alternatively stated, each member of the plurality of the electrode current collectors **136** may effectively meet and extend into both the inner surface **1060** of the first secondary growth constraint **158** and the inner surface **1062** of the second secondary growth constraint **160** (akin to the height H_{CESB} , as described above), and may be affixed into the notch **1060a** of the first secondary growth constraint **158** and into the notch **1062a** of the second secondary growth constraint **160** via glue **182** in a notched embodiment.

[0224] Further, FIGS. **16A-16C** also depict different embodiments for gluing the plurality of the electrode current collectors **136** in a notched embodiment. For example, in one embodiment depicted in FIG. **16A**, the plurality of electrode current collectors **136** may be glued **182** via an electrode current collector top **1892** and an electrode current collector bottom **1894**. By way of further example, in one embodiment depicted in FIG. **16B**, the plurality of electrode current collectors **136** may be glued **182** via the lateral surfaces of the electrode current collectors **136** (akin to the lateral surfaces of the electrode backbones **134**, as described above). By way of further example, in one embodiment depicted in FIG. **16C**, the plurality of electrode current collectors **136** may be glued **182** via the top **1892**, the bottom **1894**, and the lateral surfaces of the electrode current collectors **136**.

[0225] In certain embodiments, a plurality of electrode current collectors **136** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160** via a slot **1060b**, **1062b** in each of the first secondary growth constraint **158** and the second secondary growth constraint **160**, via an interlocking connection embodiment. Referring now to FIG. **17**, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIG. **17** shows a cross section, along the line A-A' as in FIG. **1**, where first primary growth constraint **154** and second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. In certain embodiments, as shown in FIG. **17**, non-affixed counter-electrode structures **112** may include counter-electrode gaps **1086** between their tops **1068** and the first secondary growth constraint **158**, and

their bottoms **1070** and the second secondary growth constraint **160**, as described in more detail above.

[0226] More specifically, in one embodiment shown in FIG. **17**, a plurality of electrode current collectors **136** may be affixed to the first secondary growth constraint **158** and the second secondary growth constraint **160** via a slot **1060b** and **1062b** and a layer of glue **182**. Accordingly, in certain embodiments, the plurality of electrode current collectors **136** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b**, **1062b** may include a symmetrical pattern about a gluing axis A_G with respect to affixed electrode current collectors **136**, as described above. In certain embodiments, the plurality of electrode current collectors **136** affixed to the first and second secondary growth constraints **158**, **160**, respectively, via slots **1060b**, **1062b** may include an asymmetric or random pattern about a gluing axis A_G with respect to affixed electrode current collectors **136**, as described above.

[0227] In certain embodiments, slots **1060b**, **1062b** in each of the first secondary growth constraint **158** and the second secondary growth constraint **160** may extend through the first secondary growth constraint **158** and the second secondary growth constraint **160**, respectively, in order to receive the plurality of electrode current collectors **136** in an interlocked embodiment. Stated alternatively, the plurality of electrode current collectors **136** may effectively meet and extend entirely through both the first secondary growth constraint **158** and the second secondary growth constraint **160** (akin to the height H_{CESB} , as described above), and may be affixed into slots **1060b** and **1062b** via glue **182** in another interlocked embodiment.

Connections Via Primary Growth Constraints

[0228] In another embodiment, a constrained electrode assembly **106** may include a set of electrode constraints **108** wherein the secondary connecting member **166** includes the first and second primary growth constraints **154**, **156** respectively, and yet still restrains growth of an electrode assembly **106** in both the longitudinal direction (i.e., along the Y axis) and/or the stacking direction D, and the vertical direction (i.e., along the Z axis) simultaneously, as described above. Referring now to FIGS. **18A-18B**, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIGS. **18A-18B** each show a cross section, along the line A-A' as in FIG. **1**, of a set of electrode constraints **108**, including one embodiment of both a primary growth constraint system **151** and one embodiment of a secondary growth constraint system **152**. Primary growth constraint system **151** includes a first primary growth constraint **154** and a second primary growth constraint **156**, as described above, and a first primary connecting member **162** and a second primary connecting member **164**, as described above. Secondary growth constraint system **152** includes a first secondary growth constraint **158**, a second secondary growth constraint **160**, and a secondary connecting member **166** embodied as first primary growth constraint **154** and/or second primary growth constraint **156**; therefore, in this embodiment, secondary connecting member **166**, first primary growth constraint **154**, and second primary growth constraint **156** are interchangeable. Further, in this embodi-

ment, first primary connecting member **162** and first secondary growth constraint **158** are interchangeable, as described above. Further still, in this embodiment, second primary connecting member **164** and second secondary growth constraint **160** are interchangeable, as described above.

[0229] First primary growth constraint **154** and second primary growth constraint **156** may be attached via a layer of glue **182** to the first secondary growth constraint **158** and second secondary growth constraint **160**, as described above. Stated alternatively, in the embodiments shown in FIGS. **18A-18B**, the set of electrode constraints **108** include a first primary connecting member **162** that may be the first secondary growth constraint **158** in a hybridized embodiment, and a second primary connecting member **164** that may be the second secondary growth constraint **160** in a hybridized embodiment. As such, the first and second primary connecting members **162**, **164**, respectively, may be under tension when restraining growth in the longitudinal direction, and may also function as first and second secondary growth constraints **158**, **160**, respectively (i.e., compression members) when restraining growth in the vertical direction.

[0230] More specifically, in one embodiment as shown in FIG. **18A**, non-affixed electrode structures **110** and non-affixed counter-electrode structures **1128** may include corresponding electrode gaps **1084** and corresponding counter-electrode gaps **1086** between each of their tops, respectively (i.e., **1052** and **1068**), and the first secondary growth constraint **158**, and each of their bottoms, respectively (i.e., **1054** and **1070**), and the second secondary growth constraint **160**, as described in more detail above.

[0231] More specifically, in one embodiment as shown in FIG. **18B**, the set of electrode constraints **108** further includes a second separator **130a** adjacent to both the hybridized first secondary growth constraint **158**/first primary connecting member **162** and the hybridized second secondary growth constraint **160**/second primary connecting member **164**.

Fused Constraint System

[0232] In some embodiments, a set of electrode constraints **108** may be fused together. For example, in one embodiment, the primary growth constraint system **151** may be fused with the secondary growth constraint system **152**. By way of further example, in one embodiment, the secondary growth constraint system **152** may be fused with the primary growth constraint system **151**. Stated alternatively, aspects of the primary growth constraint system **151** (e.g., the first and second primary growth constraints **154**, **156**, respectively) may coexist (i.e., may be fused with) aspects of the secondary growth constraint system **152** (e.g., the first and second secondary growth constraints **158**, **160**, respectively) in a unibody-type system. Referring now to FIG. **19**, a Cartesian coordinate system is shown for reference having a vertical axis (Z axis), a longitudinal axis (Y axis), and a transverse axis (X axis); wherein the X axis is oriented as coming out of the plane of the page; a separator **130**, and a designation of the stacking direction D, as described above, co-parallel with the Y axis. More specifically, FIG. **19** shows a cross section, along the line A-A' as in FIG. **1**, of a fused electrode constraint **108**, including one embodiment of a primary growth constraint system **151** fused with one embodiment of a secondary growth constraint system **152**.

[0233] Further illustrated in FIG. **19**, in one embodiment, are members of the electrode population **110** having an electrode active material layer **132**, and an electrode current collector **136**. Similarly, in one embodiment, illustrated in FIG. **19** are members of the counter-electrode population **112** having a counter-electrode active material layer **138**, and a counter-electrode current collector **140**. For case of illustration, only two members of the electrode population **110** and three members of the counter-electrode population **112** are depicted; in practice, however, an energy storage device **100** or a secondary battery **102** using the inventive subject matter herein may include additional members of the electrode **110** and counter-electrode **112** populations depending on the application of the energy storage device **100** or secondary battery **102**, as described above. More specifically, illustrated in the fused embodiment of FIG. **19**, the secondary connecting member **166** may be embodied as the electrode and/or counter-electrode backbones **134**, **141**, respectively, as described above, but each may be fused to each of the first and second secondary growth constraints **158**, **160**, respectively, as described above. Similarly, the first primary growth constraint **154** and the second primary growth constraint **156** may be fused to the first and second secondary growth constraints **158**, **160**, respectively, thereby ultimately forming a fused or unibody constraint **108**.

Secondary Battery

[0234] Referring now to FIG. **20**, illustrated is an exploded view of one embodiment of a secondary battery **102** having a plurality of sets of electrode constraints **108a** of the present disclosure. The secondary battery **102** includes battery enclosure **104** and a set of electrode assemblies **106a** within the battery enclosure **104**, each of the electrode assemblies **106** having a first longitudinal end surface **116**, an opposing second longitudinal end surface **118** (i.e., separated from first longitudinal end surface **116** along the Y axis the Cartesian coordinate system shown), as described above. Each electrode assembly **106** includes a population of electrode structures **110** and a population of counter-electrode structures **112**, stacked relative to each other within each of the electrode assemblies **106** in a stacking direction D; stated differently, the populations of electrode **110** and counter-electrode **112** structures are arranged in an alternating series of electrodes **110** and counter-electrodes **112** with the series progressing in the stacking direction D between first and second longitudinal end surfaces **116**, **118**, respectively (see, e.g., FIG. **2A**; as illustrated in FIG. **2A** and FIG. **20**, stacking direction D parallels the Y axis of the Cartesian coordinate system(s) shown), as described above. In addition, the stacking direction D within an individual electrode assembly **106** is perpendicular to the direction of stacking of a collection of electrode assemblies **106** within a set **106a** (i.e., an electrode assembly stacking direction); stated differently, the electrode assemblies **106** are disposed relative to each other in a direction within a set **106a** that is perpendicular to the stacking direction D within an individual electrode assembly **106** (e.g., the electrode assembly stacking direction is in a direction corresponding to the Z axis of the Cartesian coordinate system shown, whereas the stacking direction D within individual electrode assemblies **106** is in a direction corresponding to the Y axis of the Cartesian coordinate system shown).

[0235] While the set of electrode assemblies **106a** depicted in the embodiment shown in FIG. **20** contains

individual electrode assemblies **106** having the same general size, one or more of the individual electrode assemblies **106** may also and/or alternatively have different sizes in at least one dimension thereof, than the other electrode assemblies **106** in the set **106a**. For example, according to one embodiment, the electrode assemblies **106** that are stacked together to form the set **106a** provided in the secondary battery **102** may have different maximum widths W_{EA} in the longitudinal direction (i.e., stacking direction D) of each assembly **106**. According to another embodiment, the electrode assemblies **106** making up the stacked set **106a** provided in the secondary battery **102** may have different maximum lengths L_{EA} along the transverse axis that is orthogonal to the longitudinal axis. By way of further example, in one embodiment, each electrode assembly **106** that is stacked together to form the set of electrode assemblies **106a** in the secondary battery **102** has a maximum width W_{EA} along the longitudinal axis and a maximum length L_{EA} along the transverse axis that is selected to provide an area of $L_{EA} \times W_{EA}$ that decreases along a direction in which the electrode assemblies **106** are stacked together to form the set of electrode assemblies **106a**. For example, the maximum width W_{EA} and maximum length L_{EA} of each electrode assembly **106** may be selected to be less than that of an electrode assembly **106** adjacent thereto in a first direction in which the assemblies **106** are stacked, and to be greater than that of an electrode assembly **106** adjacent thereto in a second direction that is opposite thereto, such that the electrode assemblies **106** are stacked together to form a secondary battery **102** having a set of electrode assemblies **106a** in a pyramidal shape. Alternatively, the maximum lengths L_{EA} and maximum widths W_{EA} for each electrode assembly **106** can be selected to provide different shapes and/or configurations for the stacked electrode assembly set **106a**. The maximum vertical height H_{EA} for one or more of the electrode assemblies **106** can also and/or alternatively be selected to be different from other assemblies **106** in the set **106a** and/or to provide a stacked set **106a** having a predetermined shape and/or configuration.

[0236] Tabs **190**, **192** project out of the battery enclosure **104** and provide an electrical connection between the electrode assemblies **106** of set **106a** and an energy supply or consumer (not shown). More specifically, in this embodiment tab **190** is electrically connected to tab extension **191** (e.g., using an electrically conductive glue), and tab extension **191** is electrically connected to the electrodes **110** comprised by each of the electrode assemblies **106**. Similarly, tab **192** is electrically connected to tab extension **193** (e.g., using an electrically conductive glue), and tab extension **193** is electrically connected to the counter-electrodes **112** comprised by each of electrode assemblies **106**.

[0237] Each electrode assembly **106** in the embodiment illustrated in FIG. 20 has an associated primary growth constraint system **151** to restrain growth in the longitudinal direction (i.e., stacking direction D). Alternatively, in one embodiment, a plurality of electrode assemblies **106** making up a set **106a** may share at least a portion of the primary growth constraint system **151**. In the embodiment as shown, each primary growth constraint system **151** includes first and second primary growth constraints **154**, **156**, respectively, that may overlie first and second longitudinal end surfaces **116**, **118**, respectively, as described above; and first and second opposing primary connecting members **162**, **164**, respectively, that may overlie lateral surfaces **142**, as

described above. First and second opposing primary connecting members **162**, **164**, respectively, may pull first and second primary growth constraints **154**, **156**, respectively, towards each other, or alternatively stated, assist in restraining growth of the electrode assembly **106** in the longitudinal direction, and primary growth constraints **154**, **156** may apply a compressive or restraint force to the opposing first and second longitudinal end surfaces **116**, **118**, respectively. As a result, expansion of the electrode assembly **106** in the longitudinal direction is inhibited during formation and/or cycling of the battery **102** between charged and discharged states. Additionally, primary growth constraint system **151** exerts a pressure on the electrode assembly **106** in the longitudinal direction (i.e., stacking direction D) that exceeds the pressure maintained on the electrode assembly **106** in either of the two directions that are mutually perpendicular to each other and are perpendicular to the longitudinal direction (e.g., as illustrated, the longitudinal direction corresponds to the direction of the Y axis, and the two directions that are mutually perpendicular to each other and to the longitudinal direction correspond to the directions of the X axis and the Z axis, respectively, of the illustrated Cartesian coordinate system).

[0238] Further, each electrode assembly **106** in the embodiment illustrated in FIG. 20 has an associated secondary growth constraint system **152** to restrain growth in the vertical direction (i.e., expansion of the electrode assembly **106**, electrodes **110**, and/or counter-electrodes **112** in the vertical direction (i.e., along the Z axis of the Cartesian coordinate system)). Alternatively, in one embodiment, a plurality of electrode assemblies **106** making up a set **106a** share at least a portion of the secondary growth constraint system **152**. Each secondary growth constraint system **152** includes first and second secondary growth constraints **158**, **160**, respectively, that may overlie corresponding lateral surfaces **142**, respectively, and at least one secondary connecting member **166**, each as described in more detail above. Secondary connecting members **166** may pull first and second secondary growth constraints **158**, **160**, respectively, towards each other, or alternatively stated, assist in restraining growth of the electrode assembly **106** in the vertical direction, and first and second secondary growth constraints **158**, **160**, respectively, may apply a compressive or restraint force to the lateral surfaces **142**, each as described above in more detail. As a result, expansion of the electrode assembly **106** in the vertical direction is inhibited during formation and/or cycling of the battery **102** between charged and discharged states. Additionally, secondary growth constraint system **152** exerts a pressure on the electrode assembly **106** in the vertical direction (i.e., parallel to the Z axis of the Cartesian coordinate system) that exceeds the pressure maintained on the electrode assembly **106** in either of the two directions that are mutually perpendicular to each other and are perpendicular to the vertical direction (e.g., as illustrated, the vertical direction corresponds to the direction of the Z axis, and the two directions that are mutually perpendicular to each other and to the vertical direction correspond to the directions of the X axis and the Y axis, respectively, of the illustrated Cartesian coordinate system).

[0239] Further still, each electrode assembly **106** in the embodiment illustrated in FIG. 20 has an associated primary growth constraint system **151**—and an associated secondary growth constraint system **152**—to restrain growth in the longitudinal direction and the vertical direction, as described

in more detail above. Furthermore, according to certain embodiments, the electrode and/or counter-electrode tabs **190**, **192**, respectively, and tab extensions **191**, **193** can serve as a part of the tertiary growth constraint system **155**. For example, in certain embodiments, the tab extensions **191**, **193** may extend along the opposing transverse surface regions **144**, **146** to act as a part of the tertiary constraint system **155**, such as the first and second tertiary growth constraints **157**, **159**. The tab extensions **191**, **193** can be connected to the primary growth constraints **154**, **156** at the longitudinal ends **117**, **119** of the electrode assembly **106**, such that the primary growth constraints **154**, **156** serve as the at least one tertiary connecting member **165** that places the tab extensions **191**, **193** in tension with one another to compress the electrode assembly **106** along the transverse direction, and act as first and second tertiary growth constraints **157**, **159**, respectively. Conversely, the tabs **190**, **192** and/or tab extensions **191**, **193** can also serve as the first and second primary connecting members **162**, **164**, respectively, for the first and second primary growth constraints **154**, **156**, respectively, according to one embodiment. In yet another embodiment, the tabs **190**, **192** and/or tab extensions **191**, **193** can serve as a part of the secondary growth constraint system **152**, such as by forming a part of the at least one secondary connecting member **166** connecting the secondary growth constraints **158**, **160**. Accordingly, the tabs **190**, **192** and/or tab extensions **191**, **193** can assist in restraining overall macroscopic growth of the electrode assembly **106** by either serving as a part of one or more of the primary and secondary constraint systems **151**, **152**, respectively, and/or by forming a part of a tertiary growth constraint system **155** to constrain the electrode assembly **106** in a direction orthogonal to the direction being constrained by one or more of the primary and secondary growth constraint systems **151**, **152**, respectively.

[0240] To complete the assembly of the secondary battery **102**, battery enclosure **104** is filled with a non-aqueous electrolyte (not shown) and lid **104a** is folded over (along fold line, FL) and sealed to upper surface **104b**. When fully assembled, the sealed secondary battery **102** occupies a volume bounded by its exterior surfaces (i.e., the displacement volume), the secondary battery enclosure **104** occupies a volume corresponding to the displacement volume of the battery (including lid **104a**) less its interior volume (i.e., the prismatic volume bounded by interior surfaces **104c**, **104d**, **104e**, **104f**, **104g** and lid **104a**) and each growth constraint **151**, **152** of set **106a** occupies a volume corresponding to its respective displacement volume. In combination, therefore, the battery enclosure **104** and growth constraints **151**, **152** occupy no more than 75% of the volume bounded by the outer surface of the battery enclosure **104** (i.e., the displacement volume of the battery). For example, in one such embodiment, the growth constraints **151**, **152** and battery enclosure **104**, in combination, occupy no more than 60% of the volume bounded by the outer surface of the battery enclosure **104**. By way of further example, in one such embodiment, the constraints **151**, **152** and battery enclosure **104**, in combination, occupy no more than 45% of the volume bounded by the outer surface of the battery enclosure **104**. By way of further example, in one such embodiment, the constraints **151**, **152** and battery enclosure **104**, in combination, occupy no more than 30% of the volume bounded by the outer surface of the battery enclosure **104**. By way of further example, in one such embodiment, the

constraints **151**, **152** and battery enclosure **104**, in combination, occupy no more than 20% of the volume bounded by the outer surface of the battery enclosure.

[0241] For case of illustration in FIG. 20, secondary battery **102** includes only one set **106a** of electrode assemblies **106** and the set **106a** includes only six electrode assemblies **106**. In practice, the secondary battery **102** may include more than one set of electrode assemblies **106a**, with each of the sets **106a** being disposed laterally relative to each other (e.g., in a relative direction lying within the X-Y plane of the Cartesian coordinate system of FIG. 20) or vertically relative to each other (e.g., in a direction substantially parallel to the Z axis of the Cartesian coordinate system of FIG. 20). Additionally, in each of these embodiments, each of the sets of electrode assemblies **106a** may include one or more electrode assemblies **106**. For example, in certain embodiments, the secondary battery **102** may comprise one, two, or more sets of electrode assemblies **106a**, with each such set **106a** including one or more electrode assemblies **106** (e.g., 1, 2, 3, 4, 5, 6, 10, 15, or more electrode assemblies **106** within each such set **106a**) and, when the battery **102** includes two or more such sets **106a**, the sets **106a** may be laterally or vertically disposed relative to other sets of electrode assemblies **106a** included in the secondary battery **102**. In each of these various embodiments, each individual electrode assembly **106** may have its own growth constraint(s), as described above (i.e., a 1:1 relationship between electrode assemblies **106** and constraints **151**, **152**), two more electrode assemblies **106** may have a common growth constraint(s) **151**, **152**, as described above (i.e., a set of constraints **108** for two or more electrode assemblies **106**), or two or more electrode assemblies **106** may share components of a growth constraint(s) **151**, **152** (i.e., two or more electrode assemblies **106** may have a common compression member (e.g., second secondary growth constraint **158**) and/or tension members **166**, for example, as in the fused embodiment, as described above).

Other Battery Components

[0242] In certain embodiments, the set of electrode constraints **108**, including a primary growth constraint system **151** and a secondary growth constraint system **152**, as described above, may be derived from a sheet **2000** having a length L_1 , width W_1 , and thickness t_1 , as shown for example in FIG. 20. More specifically, to form a primary growth constraint system **151**, a sheet **2000** may be wrapped around an electrode assembly **106** and folded at folded at edges **2001** to enclose the electrode assembly **106**. Alternatively, in one embodiment, the sheet **2000** may be wrapped around a plurality of electrode assemblies **106** that are stacked to form an electrode assembly set **106a**. The edges of the sheet may overlap each other, and are welded, glued, or otherwise secured to each other to form a primary growth constraint system **151** including first primary growth constraint **154** and second primary growth constraint **156**, and first primary connecting member **162** and second primary connecting member **164**. In this embodiment, the primary growth constraint system **151** has a volume corresponding to the displacement volume of sheet **2000** (i.e., the multiplication product of L_1 , W_1 and t_1). In one embodiment, the at least one primary connecting member is stretched in the stacking direction D to place the member in tension, which causes a compressive force to be exerted by the first and second primary growth constraints. Alternatively, the at least

one secondary connecting member can be stretched in the second direction to place the member in tension, which causes a compressive force to be exerted by the first and second secondary growth constraints. In an alternative embodiment, instead of stretching the connecting members to place them in tension, the connecting members and/or growth constraints or other portion of one or more of the primary and secondary growth constraint systems may be pre-tensioned prior to installation over and/or in the electrode assembly. In another alternative embodiment, the connecting members and/or growth constraints and/or other portions of one or more of the primary and secondary growth constraint systems are not initially under tension at the time of installation into and/or over the electrode assembly, but rather, formation of the battery causes the electrode assembly to expand and induce tension in portions of the primary and/or secondary growth constraint systems such as the connecting members and/or growth constraints. (i.e., self-tensioning).

[0243] Sheet **2000** may comprise any of a wide range of compatible materials capable of applying the desired force to the electrode assembly **106**. In general, the primary growth constraint system **151** will typically comprise a material that has an ultimate tensile strength of at least 10,000 psi (>70 MPa), that is compatible with the battery electrolyte, does not significantly corrode at the floating or anode potential for the battery **102**, and does not significantly react or lose mechanical strength at 45° C., and even up to 70° C. For example, the primary growth constraint system **151** may comprise any of a wide range of metals, alloys, ceramics, glass, plastics, or a combination thereof (i.e., a composite). In one exemplary embodiment, primary growth constraint system **151** comprises a metal such as stainless steel (e.g., SS 316, 440C or 440C hard), aluminum (e.g., aluminum 7075-T6, hard H18), titanium (e.g., 6Al-4V), beryllium, beryllium copper (hard), copper (O₂ free, hard), nickel; in general, however, when the primary growth constraint system **151** comprises metal it is generally preferred that it be incorporated in a manner that limits corrosion and limits creating an electrical short between the electrodes **110** and counter-electrodes **112**. In another exemplary embodiment, the primary growth constraint system **151** comprises a ceramic such as alumina (e.g., sintered or Coorstek AD96), zirconia (e.g., Coorstek YZTP), yttria-stabilized zirconia (e.g., ENrG E-Strate®). In another exemplary embodiment, the primary growth constraint system **151** comprises a glass such as Schott D263 tempered glass. In another exemplary embodiment, the primary growth constraint system **151** comprises a plastic such as polyetheretherketone (PEEK) (e.g., Aptiv 1102), PEEK with carbon (e.g., Victrex 90HMF40 or Xycomp 1000-04), polyphenylene sulfide (PPS) with carbon (e.g., Tepex Dynalite 207), polyetheretherketone (PEEK) with 30% glass, (e.g., Victrex 90HMF40 or Xycomp 1000-04), polyimide (e.g., Kapton®). In another exemplary embodiment, the primary growth constraint system **151** comprises a composite such as E Glass Std Fabric/Epoxy, 0 deg, E Glass UD/Epoxy, 0 deg, Kevlar Std Fabric/Epoxy, 0 deg, Kevlar UD/Epoxy, 0 deg, Carbon Std Fabric/Epoxy, 0 deg, Carbon UD/Epoxy, 0 deg, Toyobo Zylon® HM Fiber/Epoxy. In another exemplary embodiment, the primary growth constraint system **151** comprises fibers such as Kevlar **49** Aramid Fiber, S Glass Fibers, Carbon Fibers, Vectran UM LCP Fibers, Dyneema, Zylon.

[0244] Thickness (t_1) of the primary growth constraint system **151** will depend upon a range of factors including, for example, the material(s) of construction of the primary growth constraint system **151**, the overall dimensions of the electrode assembly **106**, and the composition of a battery anode and cathode. In some embodiments, for example, the primary growth constraint system **151** will comprise a sheet having a thickness in the range of about 10 to about 100 micrometers. For example, in one such embodiment, the primary growth constraint system **151** comprises a stainless steel sheet (e.g., SS316) having a thickness of about 30 μ m. By way of further example, in another such embodiment, the primary growth constraint system **151** comprises an aluminum sheet (e.g., 7075-T6) having a thickness of about 40 μ m. By way of further example, in another such embodiment, the primary growth constraint system **151** comprises a zirconia sheet (e.g., Coorstek YZTP) having a thickness of about 30 μ m. By way of further example, in another such embodiment, the primary growth constraint system **151** comprises an E Glass UD/Epoxy 0 deg sheet having a thickness of about 75 μ m. By way of further example, in another such embodiment, the primary growth constraint system **151** comprises 12 μ m carbon fibers at >50% packing density.

[0245] Without being bound to any particular theory, methods for gluing, as described herein, may include gluing, soldering, bonding, sintering, press contacting, brazing, thermal spraying joining, clamping, or combinations thereof. Gluing may include joining the materials with conductive materials such as conducting epoxies, conducting elastomers, mixtures of insulating organic glue filled with conducting metals, such as nickel filled epoxy, carbon filled epoxy etc. Conductive pastes may be used to join the materials together and the joining strength could be tailored by temperature (sintering), light (UV curing, cross-linking), chemical curing (catalyst based cross linking). Bonding processes may include wire bonding, ribbon bonding, ultrasonic bonding. Welding processes may include ultrasonic welding, resistance welding, laser beam welding, electron beam welding, induction welding, and cold welding. Joining of these materials can also be performed by using a coating process such as a thermal spray coating such as plasma spraying, flame spraying, arc spraying, to join materials together. For example, a nickel or copper mesh can be joined onto a nickel bus using a thermal spray of nickel as a glue.

[0246] Members of the electrode **110** and counter-electrode **112** populations include an electroactive material capable of absorbing and releasing a carrier ion such as lithium, sodium, potassium, calcium, magnesium or aluminum ions. In some embodiments, members of the electrode structure **110** population include an anodically active electroactive material (sometimes referred to as a negative electrode) and members of the counter-electrode structure **112** population include a cathodically active electroactive material (sometimes referred to as a positive electrode). In other embodiments, members of the electrode structure **110** population include a cathodically active electroactive material and members of the counter-electrode structure **112** population comprise an anodically active electroactive material. In each of the embodiments and examples recited in this paragraph, negative electrode active material may be a particulate agglomerate electrode or a monolithic electrode.

[0247] Exemplary anodically active electroactive materials include carbon materials such as graphite and soft or hard carbons, or any of a range of metals, semi-metals, alloys, oxides and compounds capable of forming an alloy with lithium. Specific examples of the metals or semi-metals capable of constituting the anode material include tin, lead, magnesium, aluminum, boron, gallium, silicon, indium, zirconium, germanium, bismuth, cadmium, antimony, silver, zinc, arsenic, hafnium, yttrium, and palladium. In one exemplary embodiment, the anodically active material comprises aluminum, tin, or silicon, or an oxide thereof, a nitride thereof, a fluoride thereof, or other alloy thereof. In another exemplary embodiment, the anodically active material comprises silicon or an alloy thereof.

[0248] Exemplary cathodically active materials include any of a wide range of cathode active materials. For example, for a lithium-ion battery, the cathodically active material may comprise a cathode material selected from transition metal oxides, transition metal sulfides, transition metal nitrides, lithium-transition metal oxides, lithium-transition metal sulfides, and lithium-transition metal nitrides may be selectively used. The transition metal elements of these transition metal oxides, transition metal sulfides, and transition metal nitrides can include metal elements having a d-shell or f-shell. Specific examples of such metal element are Sc, Y, lanthanoids, actinoids, Ti, Zr, Hf, V, Nb, Ta, Cr, Mo, W, Mn, Tc, Re, Fe, Ru, Os, Co, Rh, Ir, Ni, Pb, Pt, Cu, Ag, and Au. Additional cathode active materials include LiCoO_2 , $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$, $\text{Li}(\text{Ni}_x\text{Co}_y\text{Al}_2)\text{O}_2$, LiFePO_4 , Li_2MnO_4 , V_2O_5 , molybdenum oxysulfides, phosphates, silicates, vanadates and combinations thereof.

[0249] In one embodiment, the anodically active material is microstructured to provide a significant void volume fraction to accommodate volume expansion and contraction as lithium ions (or other carrier ions) are incorporated into or leave the negative electrode active material during charging and discharging processes. In general, the void volume fraction of the negative electrode active material is at least 0.1. Typically, however, the void volume fraction of the negative electrode active material is not greater than 0.8. For example, in one embodiment, the void volume fraction of the negative electrode active material is about 0.15 to about 0.75. By way of the further example, in one embodiment, the void volume fraction of the negative electrode active material is about 0.2 to about 0.7. By way of the further example, in one embodiment, the void volume fraction of the negative electrode active material is about 0.25 to about 0.6.

[0250] Depending upon the composition of the microstructured negative electrode active material and the method of its formation, the microstructured negative electrode active material may comprise macroporous, microporous, or mesoporous material layers or a combination thereof, such as a combination of microporous and mesoporous, or a combination of mesoporous and macroporous. Microporous material is typically characterized by a pore dimension of less than 10 nm, a wall dimension of less than 10 nm, a pore depth of 1-50 micrometers, and a pore morphology that is generally characterized by a "spongy" and irregular appearance, walls that are not smooth, and branched pores. Mesoporous material is typically characterized by a pore dimension of 10-50 nm, a wall dimension of 10-50 nm, a pore depth of 1-100 micrometers, and a pore morphology that is generally characterized by branched pores that are somewhat well defined or dendritic pores. Macroporous material

is typically characterized by a pore dimension of greater than 50 nm, a wall dimension of greater than 50 nm, a pore depth of 1-500 micrometers, and a pore morphology that may be varied, straight, branched, or dendritic, and smooth or rough-walled. Additionally, the void volume may comprise open or closed voids, or a combination thereof. In one embodiment, the void volume comprises open voids, that is, the negative electrode active material contains voids having openings at the lateral surface of the negative electrode active material through which lithium ions (or other carrier ions) can enter or leave the negative electrode active material; for example, lithium ions may enter the negative electrode active material through the void openings after leaving the positive electrode active material. In another embodiment, the void volume comprises closed voids, that is, the negative electrode active material contains voids that are enclosed by negative electrode active material. In general, open voids can provide greater interfacial surface area for the carrier ions whereas closed voids tend to be less susceptible to solid electrolyte interface while each provides room for expansion of the negative electrode active material upon the entry of carrier ions. In certain embodiments, therefore, it is preferred that the negative electrode active material comprise a combination of open and closed voids.

[0251] In one embodiment, negative electrode active material comprises porous aluminum, tin or silicon or an alloy thereof. Porous silicon layers may be formed, for example, by anodization, by etching (e.g., by depositing precious metals such as gold, platinum, silver or gold/palladium on the surface of single crystal silicon and etching the surface with a mixture of hydrofluoric acid and hydrogen peroxide), or by other methods known in the art such as patterned chemical etching. Additionally, the porous negative electrode active material will generally have a porosity fraction of at least about 0.1, but less than 0.8 and have a thickness of about 1 to about 100 micrometers. For example, in one embodiment, negative electrode active material comprises porous silicon, has a thickness of about 5 to about 100 micrometers, and has a porosity fraction of about 0.15 to about 0.75. By way of further example, in one embodiment, negative electrode active material comprises porous silicon, has a thickness of about 10 to about 80 micrometers, and has a porosity fraction of about 0.15 to about 0.7. By way of further example, in one such embodiment, negative electrode active material comprises porous silicon, has a thickness of about 20 to about 50 micrometers, and has a porosity fraction of about 0.25 to about 0.6. By way of further example, in one embodiment, negative electrode active material comprises a porous silicon alloy (such as nickel silicide), has a thickness of about 5 to about 100 micrometers, and has a porosity fraction of about 0.15 to about 0.75.

[0252] In another embodiment, negative electrode active material comprises fibers of aluminum, tin or silicon, or an alloy thereof. Individual fibers may have a diameter (thickness dimension) of about 5 nm to about 10,000 nm and a length generally corresponding to the thickness of the negative electrode active material. Fibers (nanowires) of silicon may be formed, for example, by chemical vapor deposition or other techniques known in the art such as vapor liquid solid (VLS) growth and solid liquid solid (SLS) growth. Additionally, the negative electrode active material will generally have a porosity fraction of at least about 0.1, but less than 0.8 and have a thickness of about 1 to about 200 micrometers. For example, in one embodiment, negative

electrode active material comprises silicon nanowires, has a thickness of about 5 to about 100 micrometers, and has a porosity fraction of about 0.15 to about 0.75. By way of further example, in one embodiment, negative electrode active material comprises silicon nanowires, has a thickness of about 10 to about 80 micrometers, and has a porosity fraction of about 0.15 to about 0.7. By way of further example, in one such embodiment, negative electrode active material comprises silicon nanowires, has a thickness of about 20 to about 50 micrometers, and has a porosity fraction of about 0.25 to about 0.6. By way of further example, in one embodiment, negative electrode active material comprises nanowires of a silicon alloy (such as nickel silicide), has a thickness of about 5 to about 100 micrometers, and has a porosity fraction of about 0.15 to about 0.75.

[0253] In one embodiment, each member of the electrode **110** population has a bottom, a top, and a longitudinal axis (A_E) extending from the bottom to the top thereof and in a direction generally perpendicular to the direction in which the alternating sequence of electrode structures **110** and counter-electrode structures **112** progresses. Additionally, each member of the electrode **110** population has a length (L_E) measured along the longitudinal axis (A_E) of the electrode, a width (W_E) measured in the direction in which the alternating sequence of electrode structures and counter-electrode structures progresses, and a height (H_E) measured in a direction that is perpendicular to each of the directions of measurement of the length (L_E) and the width (W_E). Each member of the electrode population also has a perimeter (P_E) that corresponds to the sum of the length(s) of the side(s) of a projection of the electrode in a plane that is normal to its longitudinal axis.

[0254] The length (L_E) of the members of the electrode population will vary depending upon the energy storage device and its intended use. In general, however, the members of the electrode population will typically have a length (L_E) in the range of about 5 mm to about 500 mm. For example, in one such embodiment, the members of the electrode population have a length (L_E) of about 10 mm to about 250 mm. By way of further example, in one such embodiment the members of the electrode population have a length (L_E) of about 25 mm to about 100 mm.

[0255] The width (W_E) of the members of the electrode population will also vary depending upon the energy storage device and its intended use. In general, however, each member of the electrode population will typically have a width (W_E) within the range of about 0.01 mm to 2.5 mm. For example, in one embodiment, the width (W_E) of each member of the electrode population will be in the range of about 0.025 mm to about 2 mm. By way of further example, in one embodiment, the width (W_E) of each member of the electrode population will be in the range of about 0.05 mm to about 1 mm.

[0256] The height (H_E) of the members of the electrode population will also vary depending upon the energy storage device and its intended use. In general, however, members of the electrode population will typically have a height (H_E) within the range of about 0.05 mm to about 10 mm. For example, in one embodiment, the height (H_E) of each member of the electrode population will be in the range of about 0.05 mm to about 5 mm. By way of further example, in one embodiment, the height (H_E) of each member of the electrode population will be in the range of about 0.1 mm to

about 1 mm. According to one embodiment, the members of the electrode population include one or more first electrode members having a first height, and one or more second electrode members having a second height that is other than the first. For example, in one embodiment, the one or more first electrode members may have a height selected to allow the electrode members to contact a portion of the secondary constraint system in the vertical direction (Z axis). For example, the height of the one or more first electrode members may be sufficient such that the first electrode members extend between and contact both the first and second secondary growth constraints **158**, **160** along the vertical axis, such as when at least one of the first electrode members or a substructure thereof serves as a secondary connecting member **166**. Furthermore, according to one embodiment, one or more second electrode members may have a height that is less than the one or more first electrode members, such that for example the one or more second electrode members do not fully extend to contact both of the first and second secondary growth constraints **158**, **160**. In yet another embodiment, the different heights for the one or more first electrode members and one or more second electrode members may be selected to accommodate a predetermined shape for the electrode assembly **106**, such as an electrode assembly shape having a different heights along one or more of the longitudinal and/or transverse axis, and/or to provide predetermined performance characteristics for the secondary battery.

[0257] The perimeter (P_E) of the members of the electrode population will similarly vary depending upon the energy storage device and its intended use. In general, however, members of the electrode population will typically have a perimeter (P_E) within the range of about 0.025 mm to about 25 mm. For example, in one embodiment, the perimeter (P_E) of each member of the electrode population will be in the range of about 0.1 mm to about 15 mm. By way of further example, in one embodiment, the perimeter (P_E) of each member of the electrode population will be in the range of about 0.5 mm to about 10 mm.

[0258] In general, members of the electrode population have a length (L_E) that is substantially greater than each of its width (W_E) and its height (H_E). For example, in one embodiment, the ratio of L_E to each of W_E and H_E is at least 5:1, respectively (that is, the ratio of L_E to W_E is at least 5:1, respectively and the ratio of L_E to H_E is at least 5:1, respectively), for each member of the electrode population. By way of further example, in one embodiment the ratio of L_E to each of W_E and H_E is at least 10:1. By way of further example, in one embodiment, the ratio of L_E to each of W_E and H_E is at least 15:1. By way of further example, in one embodiment, the ratio of L_E to each of W_E and H_E is at least 20:1, for each member of the electrode population.

[0259] Additionally, it is generally preferred that members of the electrode population have a length (L_E) that is substantially greater than its perimeter (P_E); for example, in one embodiment, the ratio of L_E to P_E is at least 1.25:1, respectively, for each member of the electrode population. By way of further example, in one embodiment the ratio of L_E to P_E is at least 2.5:1, respectively, for each member of the electrode population. By way of further example, in one embodiment, the ratio of L_E to P_E is at least 3.75:1, respectively, for each member of the electrode population.

[0260] In one embodiment, the ratio of the height (H_E) to the width (W_E) of the members of the electrode population

is at least 0.4:1, respectively. For example, in one embodiment, the ratio of H_E to W_E will be at least 2:1, respectively, for each member of the electrode population. By way of further example, in one embodiment the ratio of H_E to W_E will be at least 10:1, respectively. By way of further example, in one embodiment the ratio of H_E to W_E will be at least 20:1, respectively. Typically, however, the ratio of H_E to W_E will generally be less than 1,000:1, respectively. For example, in one embodiment the ratio of H_E to W_E will be less than 500:1, respectively. By way of further example, in one embodiment the ratio of H_E to W_E will be less than 100:1, respectively. By way of further example, in one embodiment the ratio of H_E to W_E will be less than 10:1, respectively. By way of further example, in one embodiment the ratio of H_E to W_E will be in the range of about 2:1 to about 100:1, respectively, for each member of the electrode population.

[0261] Each member of the counter-electrode population has a bottom, a top, and a longitudinal axis (A_{CE}) extending from the bottom to the top thereof and in a direction generally perpendicular to the direction in which the alternating sequence of electrode structures and counter-electrode structures progresses. Additionally, each member of the counter-electrode population has a length (L_{CE}) measured along the longitudinal axis (A_{CE}), a width (W_{CE}) measured in the direction in which the alternating sequence of electrode structures and counter-electrode structures progresses, and a height (H_{CE}) measured in a direction that is perpendicular to each of the directions of measurement of the length (L_{CE}) and the width (W_{CE}). Each member of the counter-electrode population also has a perimeter (P_{CE}) that corresponds to the sum of the length(s) of the side(s) of a projection of the counter-electrode in a plane that is normal to its longitudinal axis.

[0262] The length (L_{CE}) of the members of the counter-electrode population will vary depending upon the energy storage device and its intended use. In general, however, each member of the counter-electrode population will typically have a length (L_{CE}) in the range of about 5 mm to about 500 mm. For example, in one such embodiment, each member of the counter-electrode population has a length (L_{CE}) of about 10 mm to about 250 mm. By way of further example, in one such embodiment each member of the counter-electrode population has a length (L_{CE}) of about 25 mm to about 100 mm.

[0263] The width (W_{CE}) of the members of the counter-electrode population will also vary depending upon the energy storage device and its intended use. In general, however, members of the counter-electrode population will typically have a width (W_{CE}) within the range of about 0.01 mm to 2.5 mm. For example, in one embodiment, the width (W_{CE}) of each member of the counter-electrode population will be in the range of about 0.025 mm to about 2 mm. By way of further example, in one embodiment, the width (W_{CE}) of each member of the counter-electrode population will be in the range of about 0.05 mm to about 1 mm.

[0264] The height (H_{CE}) of the members of the counter-electrode population will also vary depending upon the energy storage device and its intended use. In general, however, members of the counter-electrode population will typically have a height (H_{CE}) within the range of about 0.05 mm to about 10 mm. For example, in one embodiment, the height (H_{CE}) of each member of the counter-electrode population will be in the range of about 0.05 mm to about

5 mm. By way of further example, in one embodiment, the height (H_{CE}) of each member of the counter-electrode population will be in the range of about 0.1 mm to about 1 mm. According to one embodiment, the members of the counter-electrode population include one or more first counter-electrode members having a first height, and one or more second counter-electrode members having a second height that is other than the first. For example, in one embodiment, the one or more first counter-electrode members may have a height selected to allow the counter-electrode members to contact a portion of the secondary constraint system in the vertical direction (Z axis). For example, the height of the one or more first counter-electrode members may be sufficient such that the first counter-electrode members extend between and contact both the first and second secondary growth constraints **158**, **160** along the vertical axis, such as when at least one of the first counter-electrode members or a substructure thereof serves as a secondary connecting member **166**. Furthermore, according to one embodiment, one or more second counter-electrode members may have a height that is less than the one or more first counter-electrode members, such that for example the one or more second counter-electrode members do not fully extend to contact both of the first and second secondary growth constraints **158**, **160**. In yet another embodiment, the different heights for the one or more first counter-electrode members and one or more second counter-electrode members may be selected to accommodate a predetermined shape for the electrode assembly **106**, such as an electrode assembly shape having a different heights along one or more of the longitudinal and/or transverse axis, and/or to provide predetermined performance characteristics for the secondary battery.

[0265] The perimeter (P_{CE}) of the members of the counter-electrode population will also vary depending upon the energy storage device and its intended use. In general, however, members of the counter-electrode population will typically have a perimeter (P_{CE}) within the range of about 0.025 mm to about 25 mm. For example, in one embodiment, the perimeter (P_{CE}) of each member of the counter-electrode population will be in the range of about 0.1 mm to about 15 mm. By way of further example, in one embodiment, the perimeter (P_{CE}) of each member of the counter-electrode population will be in the range of about 0.5 mm to about 10 mm.

[0266] In general, each member of the counter-electrode population has a length (L_{CE}) that is substantially greater than width (W_{CE}) and substantially greater than its height (H_{CE}). For example, in one embodiment, the ratio of L_{CE} to each of W_{CE} and H_{CE} is at least 5:1, respectively (that is, the ratio of L_{CE} to W_{CE} is at least 5:1, respectively and the ratio of L_{CE} to H_{CE} is at least 5:1, respectively), for each member of the counter-electrode population. By way of further example, in one embodiment the ratio of L_{CE} to each of W_{CE} and H_{CE} is at least 10:1 for each member of the counter-electrode population. By way of further example, in one embodiment, the ratio of L_{CE} to each of W_{CE} and H_{CE} is at least 15:1 for each member of the counter-electrode population. By way of further example, in one embodiment, the ratio of L_{CE} to each of W_{CE} and H_{CE} is at least 20:1 for each member of the counter-electrode population.

[0267] Additionally, it is generally preferred that members of the counter-electrode population have a length (L_{CE}) that is substantially greater than its perimeter (P_{CE}); for example, in one embodiment, the ratio of L_{CE} to P_{CE} is at least 1.25:1,

respectively, for each member of the counter-electrode population. By way of further example, in one embodiment the ratio of L_{CE} to P_{CE} is at least 2.5:1, respectively, for each member of the counter-electrode population. By way of further example, in one embodiment, the ratio of L_{CE} to P_{CE} is at least 3.75:1, respectively, for each member of the counter-electrode population.

[0268] In one embodiment, the ratio of the height (H_{CE}) to the width (W_{CE}) of the members of the counter-electrode population is at least 0.4:1, respectively. For example, in one embodiment, the ratio of H_{CE} to W_{CE} will be at least 2:1, respectively, for each member of the counter-electrode population. By way of further example, in one embodiment the ratio of H_{CE} to W_{CE} will be at least 10:1, respectively, for each member of the counter-electrode population. By way of further example, in one embodiment the ratio of H_{CE} to W_{CE} will be at least 20:1, respectively, for each member of the counter-electrode population. Typically, however, the ratio of H_{CE} to W_{CE} will generally be less than 1,000:1, respectively, for each member of the electrode population. For example, in one embodiment the ratio of H_{CE} to W_{CE} will be less than 500:1, respectively, for each member of the counter-electrode population. By way of further example, in one embodiment the ratio of H_{CE} to W_{CE} will be less than 100:1, respectively. By way of further example, in one embodiment the ratio of H_{CE} to W_{CE} will be less than 10:1, respectively. By way of further example, in one embodiment the ratio of H_{CE} to W_{CE} will be in the range of about 2:1 to about 100:1, respectively, for each member of the counter-electrode population.

[0269] In one embodiment the negative electrode current conductor layer 136 comprised by each member of the negative electrode population has a length L_{NC} that is at least 50% of the length L_{NE} of the member comprising such negative electrode current collector. By way of further example, in one embodiment the negative electrode current conductor layer 136 comprised by each member of the negative electrode population has a length L_{NC} that is at least 60% of the length L_{NE} of the member comprising such negative electrode current collector. By way of further example, in one embodiment the negative electrode current conductor layer 136 comprised by each member of the negative electrode population has a length L_{NC} that is at least 70% of the length L_{NE} of the member comprising such negative electrode current collector. By way of further example, in one embodiment the negative electrode current conductor layer 136 comprised by each member of the negative electrode population has a length L_{NC} that is at least 80% of the length L_{NE} of the member comprising such negative electrode current collector. By way of further example, in one embodiment the negative electrode current conductor layer 136 comprised by each member of the negative electrode population has a length L_{NC} that is at least 90% of the length L_{NE} of the member comprising such negative electrode current collector.

[0270] In one embodiment, the positive electrode current conductor 140 comprised by each member of the positive electrode population has a length L_{PC} that is at least 50% of the length L_{PE} of the member comprising such positive electrode current collector. By way of further example, in one embodiment the positive electrode current conductor 140 comprised by each member of the positive electrode population has a length L_{PC} that is at least 60% of the length L_{PE} of the member comprising such positive electrode

current collector. By way of further example, in one embodiment the positive electrode current conductor 140 comprised by each member of the positive electrode population has a length L_{PC} that is at least 70% of the length L_{PE} of the member comprising such positive electrode current collector. By way of further example, in one embodiment the positive electrode current conductor 140 comprised by each member of the positive electrode population has a length L_{PC} that is at least 80% of the length L_{PE} of the member comprising such positive electrode current collector. By way of further example, in one embodiment the positive electrode current conductor 140 comprised by each member of the positive electrode population has a length L_{PC} that is at least 90% of the length L_{PE} of the member comprising such positive electrode current collector.

[0271] In one embodiment negative electrode current collector layer 136 comprises an ionically permeable conductor material that is both ionically and electrically conductive. Stated differently, the negative electrode current collector layer has a thickness, an electrical conductivity, and an ionic conductivity for carrier ions that facilitates the movement of carrier ions between an immediately adjacent active electrode material layer one side of the ionically permeable conductor layer and an immediately adjacent separator layer on the other side of the negative electrode current collector layer in an electrochemical stack. On a relative basis, the negative electrode current collector layer has an electrical conductance that is greater than its ionic conductance when there is an applied current to store energy in the device or an applied load to discharge the device. For example, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the negative electrode current collector layer will typically be at least 1,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the negative electrode current collector layer is at least 5,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the negative electrode current collector layer is at least 10,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the negative electrode current collector layer is at least 50,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the negative electrode current collector layer is at least 100,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device.

[0272] In those embodiments in which negative electrode current collector 136 comprises an ionically permeable conductor material that is both ionically and electrically conductive, negative electrode current collector 136 may have an ionic conductance that is comparable to the ionic conductance of an adjacent separator layer when a current is applied to store energy in the device or a load is applied to

discharge the device, such as when a secondary battery is charging or discharging. For example, in one embodiment negative electrode current collector **136** has an ionic conductance (for carrier ions) that is at least 50% of the ionic conductance of the separator layer (i.e., a ratio of 0.5:1, respectively) when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the ionic conductance (for carrier ions) of negative electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer is at least 1:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the ionic conductance (for carrier ions) of negative electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer is at least 1.25:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the ionic conductance (for carrier ions) of negative electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer is at least 1.5:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the ionic conductance (for carrier ions) of negative electrode current collector **136** to the ionic conductance (for carrier ions) of the separator layer is at least 2:1 when there is an applied current to store energy in the device or an applied load to discharge the device.

[0273] In one embodiment, negative electrode current collector **136** also has an electrical conductance that is substantially greater than the electrical conductance of the negative electrode active material layer. For example, in one embodiment the ratio of the electrical conductance of negative electrode current collector **136** to the electrical conductance of the negative electrode active material layer is at least 100:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the electrical conductance of negative electrode current collector **136** to the electrical conductance of the negative electrode active material layer is at least 500:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the electrical conductance of negative electrode current collector **136** to the electrical conductance of the negative electrode active material layer is at least 1000:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the electrical conductance of negative electrode current collector **136** to the electrical conductance of the negative electrode active material layer is at least 5000:1 when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in some embodiments the ratio of the electrical conductance of negative electrode current collector **136** to the electrical conductance of the negative electrode active material layer is at least 10,000:1 when there is an applied current to store energy in the device or an applied load to discharge the device.

[0274] The thickness of negative electrode current collector **136** (i.e., the shortest distance between the separator and

the negative electrode active material layer between which negative electrode current collector layer **136** is sandwiched) in this embodiment will depend upon the composition of the layer and the performance specifications for the electrochemical stack. In general, when a negative electrode current collector layer is an ionically permeable conductor layer, it will have a thickness of at least about 300 Angstroms. For example, in some embodiments it may have a thickness in the range of about 300-800 Angstroms. More typically, however, it will have a thickness greater than about 0.1 micrometers. In general, an ionically permeable conductor layer will have a thickness not greater than about 100 micrometers. Thus, for example, in one embodiment, negative electrode current collector **136** will have a thickness in the range of about 0.1 to about 10 micrometers. By way of further example, in some embodiments, negative electrode current collector **136** will have a thickness in the range of about 0.1 to about 5 micrometers. By way of further example, in some embodiments, negative electrode current collector **136** will have a thickness in the range of about 0.5 to about 3 micrometers. In general, it is preferred that the thickness of negative electrode current collector **136** be approximately uniform. For example, in one embodiment it is preferred that negative electrode current collector **136** have a thickness non-uniformity of less than about 25% wherein thickness non-uniformity is defined as the quantity of the maximum thickness of the layer minus the minimum thickness of the layer, divided by the average layer thickness. In certain embodiments, the thickness variation is even less. For example, in some embodiments negative electrode current collector **136** has a thickness non-uniformity of less than about 20%. By way of further example, in some embodiments negative electrode current collector **136** has a thickness non-uniformity of less than about 15%. In some embodiments the ionically permeable conductor layer has a thickness non-uniformity of less than about 10%.

[0275] In one preferred embodiment, negative electrode current collector **136** is an ionically permeable conductor layer comprising an electrically conductive component and an ion conductive component that contribute to the ionic permeability and electrical conductivity. Typically, the electrically conductive component will comprise a continuous electrically conductive material (such as a continuous metal or metal alloy) in the form of a mesh or patterned surface, a film, or composite material comprising the continuous electrically conductive material (such as a continuous metal or metal alloy). Additionally, the ion conductive component will typically comprise pores, e.g., interstices of a mesh, spaces between a patterned metal or metal alloy containing material layer, pores in a metal film, or a solid ion conductor having sufficient diffusivity for carrier ions. In certain embodiments, the ionically permeable conductor layer comprises a deposited porous material, an ion-transporting material, an ion-reactive material, a composite material, or a physically porous material. If porous, for example, the ionically permeable conductor layer may have a void fraction of at least about 0.25. In general, however, the void fraction will typically not exceed about 0.95. More typically, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.25 to about 0.85. In some embodiments, for example, when the ionically permeable conductor layer is porous the void fraction may be in the range of about 0.35 to about 0.65.

[0276] Being positioned between the negative electrode active material layer and the separator, negative electrode current collector **136** may facilitate more uniform carrier ion transport by distributing current from the negative electrode current collector across the surface of the negative electrode active material layer. This, in turn, may facilitate more uniform insertion and extraction of carrier ions and thereby reduce stress in the negative electrode active material during cycling; since negative electrode current collector **136** distributes current to the surface of the negative electrode active material layer facing the separator, the reactivity of the negative electrode active material layer for carrier ions will be the greatest where the carrier ion concentration is the greatest. In yet another embodiment, the positions of the negative electrode current collector **136** and the negative electrode active material layer may be reversed.

[0277] According to one embodiment, each member of the positive electrodes has a positive electrode current collector **140** that may be disposed, for example, between the positive electrode backbone and the positive electrode active material layer. Furthermore, one or more of the negative electrode current collector **136** and positive electrode current collector **140** may comprise a metal such as aluminum, carbon, chromium, gold, nickel, NiP, palladium, platinum, rhodium, ruthenium, an alloy of silicon and nickel, titanium, or a combination thereof (see “Current collectors for positive electrodes of lithium-based batteries” by A. H. Whitehead and M. Schreiber, *Journal of the Electrochemical Society*, 152 (11) A2105-A2113 (2005)). By way of further example, in one embodiment, positive electrode current collector **140** comprises gold or an alloy thereof such as gold silicide. By way of further example, in one embodiment, positive electrode current collector **140** comprises nickel or an alloy thereof such as nickel silicide.

[0278] In an alternative embodiment, the positions of the positive electrode current collector layer and the positive electrode active material layer may be reversed, for example such that the positive electrode current collector layer is positioned between the separator layer and the positive electrode active material layer. In such embodiments, the positive electrode current collector **140** for the immediately adjacent positive electrode active material layer comprises an ionically permeable conductor having a composition and construction as described in connection with the negative electrode current collector layer; that is, the positive electrode current collector layer comprises a layer of an ionically permeable conductor material that is both ionically and electrically conductive. In this embodiment, the positive electrode current collector layer has a thickness, an electrical conductivity, and an ionic conductivity for carrier ions that facilitates the movement of carrier ions between an immediately adjacent positive electrode active material layer on one side of the positive electrode current collector layer and an immediately adjacent separator layer on the other side of the positive electrode current collector layer in an electrochemical stack. On a relative basis in this embodiment, the positive electrode current collector layer has an electrical conductance that is greater than its ionic conductance when there is an applied current to store energy in the device or an applied load to discharge the device. For example, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the positive electrode current collector layer will typically be at least 1,000:1, respectively, when there is an applied current to store energy in the device or an applied

load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the positive electrode current collector layer is at least 5,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the positive electrode current collector layer is at least 10,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the positive electrode current collector layer is at least 50,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device. By way of further example, in one such embodiment, the ratio of the electrical conductance to the ionic conductance (for carrier ions) of the positive electrode current collector layer is at least 100,000:1, respectively, when there is an applied current to store energy in the device or an applied load to discharge the device.

[0279] Electrically insulating separator layers **130** may surround and electrically isolate each member of the electrode structure **110** population from each member of the counter-electrode structure **112** population. Electrically insulating separator layers **130** will typically include a microporous separator material that can be permeated with a non-aqueous electrolyte; for example, in one embodiment, the microporous separator material includes pores having a diameter of at least 50 Å, more typically in the range of about 2,500 Å, and a porosity in the range of about 25% to about 75%, more typically in the range of about 35-55%. Additionally, the microporous separator material may be permeated with a non-aqueous electrolyte to permit conduction of carrier ions between adjacent members of the electrode and counter-electrode populations. In certain embodiments, for example, and ignoring the porosity of the microporous separator material, at least 70 vol % of electrically insulating separator material between a member of the electrode structure **110** population and the nearest member(s) of the counter-electrode structure **112** population (i.e., an “adjacent pair”) for ion exchange during a charging or discharging cycle is a microporous separator material; stated differently, microporous separator material constitutes at least 70 vol % of the electrically insulating material between a member of the electrode structure **110** population and the nearest member of the counter-electrode **112** structure population. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator material, microporous separator material constitutes at least 75 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and members of the counter-electrode structure **112** population, respectively. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator material, the microporous separator material constitutes at least 80 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and members of the counter-electrode structure **112** population, respectively. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator

material, the microporous separator material constitutes at least 85 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and members of the counter-electrode structure **112** population, respectively. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator material, the microporous separator material constitutes at least 90 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and member of the counter-electrode structure **112** population, respectively. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator material, the microporous separator material constitutes at least 95 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and members of the counter-electrode structure **112** population, respectively. By way of further example, in one embodiment, and ignoring the porosity of the microporous separator material, the microporous separator material constitutes at least 99 vol % of the electrically insulating separator material layer between adjacent pairs of members of the electrode structure **110** population and members of the counter-electrode structure **112** population, respectively.

[0280] In one embodiment, the microporous separator material comprises a particulate material and a binder, and has a porosity (void fraction) of at least about 20 vol. % The pores of the microporous separator material will have a diameter of at least 50 Å and will typically fall within the range of about 250 to 2,500 Å. The microporous separator material will typically have a porosity of less than about 75%. In one embodiment, the microporous separator material has a porosity (void fraction) of at least about 25 vol %. In one embodiment, the microporous separator material will have a porosity of about 35-55%.

[0281] The binder for the microporous separator material may be selected from a wide range of inorganic or polymeric materials. For example, in one embodiment, the binder is an organic material selected from the group consisting of silicates, phosphates, aluminates, aluminosilicates, and hydroxides such as magnesium hydroxide, calcium hydroxide, etc. For example, in one embodiment, the binder is a fluoropolymer derived from monomers containing vinylidene fluoride, hexafluoropropylene, tetrafluoropropene, and the like. In another embodiment, the binder is a polyolefin such as polyethylene, polypropylene, or polybutene, having any of a range of varying molecular weights and densities. In another embodiment, the binder is selected from the group consisting of ethylene-diene-propene terpolymer, polystyrene, polymethyl methacrylate, polyethylene glycol, polyvinyl acetate, polyvinyl butyral, polyacetal, and polyethyleneglycol diacrylate. In another embodiment, the binder is selected from the group consisting of methyl cellulose, carboxymethyl cellulose, styrene rubber, butadiene rubber, styrene-butadiene rubber, isoprene rubber, polyacrylamide, polyvinyl ether, polyacrylic acid, polymethacrylic acid, and polyethylene oxide. In another embodiment, the binder is selected from the group consisting of acrylates, styrenes, epoxies, and silicones. In another embodiment, the binder is a copolymer or blend of two or more of the aforementioned polymers.

[0282] The particulate material comprised by the microporous separator material may also be selected from a wide

range of materials. In general, such materials have a relatively low electronic and ionic conductivity at operating temperatures and do not corrode under the operating voltages of the battery electrode or current collector contacting the microporous separator material. For example, in one embodiment, the particulate material has a conductivity for carrier ions (e.g., lithium) of less than 1×10^{-4} S/cm. By way of further example, in one embodiment, the particulate material has a conductivity for carrier ions of less than 1×10^{-5} S/cm. By way of further example, in one embodiment, the particulate material has a conductivity for carrier ions of less than 1×10^{-6} S/cm. Exemplary particulate materials include particulate polyethylene, polypropylene, a TiO_2 -polymer composite, silica aerogel, fumed silica, silica gel, silica hydrogel, silica xerogel, silica sol, colloidal silica, alumina, titania, magnesia, kaolin, talc, diatomaceous earth, calcium silicate, aluminum silicate, calcium carbonate, magnesium carbonate, or a combination thereof. For example, in one embodiment, the particulate material comprises a particulate oxide or nitride such as TiO_2 , SiO_2 , Al_2O_3 , GeO_2 , B_2O_3 , Bi_2O_3 , BaO , ZnO , ZrO_2 , BN , Si_3N_4 , Ge_3N_4 . See, for example, P. Arora and J. Zhang, "Battery Separators" Chemical Reviews 2004, 104, 4419-4462). In one embodiment, the particulate material will have an average particle size of about 20 nm to 2 micrometers, more typically 200 nm to 1.5 micrometers. In one embodiment, the particulate material will have an average particle size of about 500 nm to 1 micrometer.

[0283] In an alternative embodiment, the particulate material comprised by the microporous separator material may be bound by techniques such as sintering, binding, curing, etc. while maintaining the void fraction desired for electrolyte ingress to provide the ionic conductivity for the functioning of the battery.

[0284] Microporous separator materials may be deposited, for example, by electrophoretic deposition of a particulate separator material in which particles are coalesced by surface energy such as electrostatic attraction or van der Waals forces, slurry deposition (including spin or spray coating) of a particulate separator material, screen printing, dip coating, and electrostatic spray deposition. Binders may be included in the deposition process; for example, the particulate material may be slurry deposited with a dissolved binder that precipitates upon solvent evaporation, electrophoretically deposited in the presence of a dissolved binder material, or co-electrophoretically deposited with a binder and insulating particles etc. Alternatively, or additionally, binders may be added after the particles are deposited into or onto the electrode structure; for example, the particulate material may be dispersed in an organic binder solution and dip coated or spray-coated, followed by drying, melting, or cross-linking the binder material to provide adhesion strength.

[0285] In an assembled energy storage device, the microporous separator material is permeated with a non-aqueous electrolyte suitable for use as a secondary battery electrolyte. Typically, the non-aqueous electrolyte comprises a lithium salt dissolved in an organic solvent. Exemplary lithium salts include inorganic lithium salts such as LiClO_4 , LiBF_4 , LiPF_6 , LiAsF_6 , LiCl , and LiBr ; and organic lithium salts such as $\text{LiB}(\text{C}_6\text{H}_5)_4$, $\text{LiN}(\text{SO}_2\text{CF}_3)_2$, $\text{LiN}(\text{SO}_2\text{CF}_3)_3$, $\text{LiNSO}_2\text{CF}_3$, $\text{LiNSO}_2\text{CF}_5$, $\text{LiNSO}_2\text{C}_4\text{F}_9$, $\text{LiNSO}_2\text{C}_5\text{F}_{11}$, $\text{LiNSO}_2\text{C}_6\text{F}_{13}$, and $\text{LiNSO}_2\text{C}_7\text{F}_{15}$. Exemplary organic solvents to dissolve the lithium salt include cyclic esters, chain

esters, cyclic ethers, and chain ethers. Specific examples of the cyclic esters include propylene carbonate, butylene carbonate, γ -butyrolactone, vinylene carbonate, 2-methyl- γ -butyrolactone, acetyl- γ -butyrolactone, and γ -valerolactone. Specific examples of the chain esters include dimethyl carbonate, diethyl carbonate, dibutyl carbonate, dipropyl carbonate, methyl ethyl carbonate, methyl butyl carbonate, methyl propyl carbonate, ethyl butyl carbonate, ethyl propyl carbonate, butyl propyl carbonate, alkyl propionates, dialkyl malonates, and alkyl acetates. Specific examples of the cyclic ethers include tetrahydrofuran, alkyltetrahydrofurans, dialkyltetrahydrofurans, alkoxytetrahydrofurans, dialkoxymethyltetrahydrofurans, 1,3-dioxolane, alkyl-1,3-dioxolanes, and 1,4-dioxolane. Specific examples of the chain ethers include 1,2-dimethoxyethane, 1,2-diethoxyethane, diethyl ether, ethylene glycol dialkyl ethers, diethylene glycol dialkyl ethers, trichylene glycol dialkyl ethers, and tetraethylene glycol dialkyl ethers.

[0286] Furthermore, according to one embodiment, components of the secondary battery 102 including the microporous separator 130 and other electrode 110 and/or counter-electrode 112 structures comprise a configuration and composition that allow the components to function, even in a case where expansion of electrode active material 132 occurs during charge and discharge of the secondary battery 102. That is, the components may be structured such that failure of the components due to expansion of the electrode active material 132 during charge/discharge thereof is within acceptable limits.

Electrode Constraint Parameters

[0287] According to one embodiment, the design of the set of electrode constraints 108 depends on parameters including: (i) the force exerted on components of the set of electrode constraints 108 due to the expansion of the electrode active material layers 132; and (ii) the strength of the set of electrode constraints 108 that is required to counteract force exerted by the expansion of the electrode active material layers 132. For example, according to one embodiment, the forces exerted on the system by the expansion of the electrode active material are dependent on the cross-sectional electrode area along a particular direction. For example, the force exerted in the longitudinal direction will be proportional to the length of the electrode (L_E) multiplied by the height of the electrode (H_E); in the vertical direction, the force would be proportional to the length of the electrode (L_E) multiplied by the width of the electrode (W_E), and the force in the transverse direction would be proportional to the width of the electrode (W_E) multiplied by the height of the electrode (H_E).

[0288] The design of the primary growth constraints 154, 156 may be dependent on a number of variables. The primary growth constraints 154, 156 restrain macroscopic growth of the electrode assembly 106 that is due to expansion of the electrode active material layers 132 in the longitudinal direction. In the embodiment as shown in FIG. 8A, the primary growth constraints 154, 156 act in concert with the at least one primary connecting member 158 (e.g., first and second primary connecting members 158 and 160), to restrain growth of the electrode structures 110 having the electrode active material layers 132. In restraining the growth, the at least one connecting member 158 places the primary growth constraints 154, 156 in tension with one another, such that they exert a compressive force to coun-

teract the forces exerted by growth of the electrode active material layers 132. According to one embodiment, when a force is exerted on the primary growth constraints 154, 156, depending on the tensile strength of the primary connecting members 158, the primary growth constraints 154, 156 can do at least one of: (i) translate away from each other (move apart in the longitudinal direction); (ii) compress in thickness; and (iii) bend and/or deflect along the longitudinal direction, to accommodate the force. The extent of translation of the primary growth constraints 154, 156 away from each other may depend on the design of the primary connecting members 158, 160. The amount the primary growth constraints 154, 156 can compress is a function of the primary growth constraint material properties, e.g., the compressive strength of the material that forms the primary growth constraints 154, 156. According to one embodiment, the amount that the primary growth constraints 154, 156 can bend may depend on the following: (i) the force exerted by the growth of the electrode structures 110 in the longitudinal direction, (ii) the elastic modulus of the primary growth constraints 154, 156; (iii) the distance between primary connecting members 158, 160 in the vertical direction; and (iv) the thickness (width) of the primary growth constraints 154, 156. In one embodiment, a maximum deflection of the primary growth constraints 154, 156 may occur at the midpoint of the growth constraints 154, 156 in a vertical direction between the primary connecting members 158, 160. The deflection increases with the fourth power of the distance between the primary connecting members 158, 160 along the vertical direction, decreases linearly with the constraint material modulus, and decreases with the 3rd power of the primary growth constraint thickness (width). The equation governing the deflection due to bending of the primary growth constraints 154, 156 can be written as:

$$\delta = 60wL^4/Eh^3$$

[0289] where w =total distributed load applied on the primary growth constraint 154, 156 due to the electrode expansion; L =distance between the primary connecting members 158, 160 along the vertical direction; E =elastic modulus of the primary growth constraints 154, 156, and h =thickness (width) of the primary growth constraints 154, 156.

[0290] In one embodiment, the stress on the primary growth constraints 154, 156 due to the expansion of the electrode active material 132 can be calculated using the following equation:

$$\sigma = 3wL^2/4h^2$$

[0291] where w =total distributed load applied on the primary growth constraints 154, 156 due to the expansion of the electrode active material layers 132; L =distance between primary connecting members 158, 160 along the vertical direction; and h =thickness (width) of the primary growth constraints 154, 156. In one embodiment, the highest stress on the primary growth constraints 154, 156 is at the point of attachment of the primary growth constraints 154, 156 to the primary connecting members 158, 160. In one embodiment, the stress increases with the square of the distance between the primary connecting members 158, 160, and decreases with the square of the thickness of the primary growth constraints 154, 156.

Variables Affecting Primary Connecting Member Design

[0292] A number of variables may affect the design of the at least one primary connecting member 158, such as the first and second primary connecting members 158, 160 as shown in the embodiment depicted in FIG. 8A. In one embodiment, the primary connecting members 158, 160 may provide sufficient resistance to counteract forces that could otherwise result in the primary growth constraints 154, 156 translating away from each other (moving apart). In one embodiment, the equation that governs the tensile stress on the primary connecting members 158, 160 can be written as follows:

$$\sigma = PL/2t$$

[0293] where P=pressure applied due to expansion of the electrode active material layers 132 on the primary growth constraints; L=distance between the primary connecting members 158, 160 along the vertical direction, and t=thickness of the connecting members 158, 160 in the vertical direction.

Variables Affecting Secondary Growth Constraint Design

[0294] A number of variables may affect the design of the first and second secondary growth constraints 158, 160, as shown in the embodiment depicted in FIG. 8B. In one embodiment, the variables affecting the design of the secondary growth constraints 158, 160 are similar to the variables affecting the design of the primary growth constraints 154, 156, but translated into the orthogonal direction. For example, in one embodiment, the equation governing the deflection due to bending of the secondary growth constraints 158, 160 can be written as:

$$\delta = 60wy^4/Et^3$$

[0295] where w=total distributed load applied on the secondary growth constraints 158, 160 due to the expansion of the electrode active material layers 132; y=distance between the secondary connecting members 166 (such as first and second primary growth constraints 154, 156 acting as secondary connecting members 166) in the longitudinal direction; E=elastic modulus of the secondary growth constraints 158, 160, and t=thickness of the secondary growth constraints 158, 160. In another embodiment, the stress on the secondary growth constraints 158, 160 can be written as:

$$\sigma = 3wy^2/4t^2$$

[0296] where w=total distributed load applied on the secondary growth constraints 158, 160 due to the expansion of the electrode active material layers 132; y=distance between the secondary connecting members 154, 156 along the longitudinal direction; and t=thickness of the secondary growth constraints 158, 160.

Variables Affecting Secondary Connecting Member Design

[0297] A number of variables may affect the design of the at least one secondary connecting member 166, such as first and second secondary connecting members 154, 156, as shown in the embodiment depicted in FIG. 8B. In one embodiment, the tensile stress on secondary connecting members 154, 156 can be written similarly to that for the primary connecting members 158, 160 as follows:

$$\sigma = Py/2h,$$

[0298] where P=pressure applied due to the expansion of the electrode active material layers 132 on the secondary growth constraints 158, 160; y=distance between the connecting members 154, 156 along the longitudinal direction, and h=thickness of the secondary connecting members 154, 156 in the longitudinal direction.

[0299] In one embodiment, the at least one connecting member 166 for the secondary growth constraints 158, 160 are not located at the longitudinal ends 117, 119 of the electrode assembly 106, but may instead be located internally within the electrode assembly 106. For example, a portion of the counter electrode structures 112 may act as secondary connecting members 166 that connect the secondary growth constraints 158, 160 to one another. In such a case where the at least one secondary connecting member 166 is an internal member, and where the expansion of the electrode active material layers 132 occurs on either side of the secondary connecting member 166, the tensile stress on the internal secondary connecting members 166 can be calculated as follows:

$$\sigma = Py/h$$

[0300] where P=pressure applied due to expansion of the electrode active material on regions of the secondary growth constraints 158, 160 that are in between the internal first and second secondary connecting members 166 (e.g., counter electrode structures 112 separated from each other in the longitudinal direction); y=distance between the internal secondary connecting members 166 along the longitudinal direction, and h=thickness of the internal secondary connecting members 166 in the longitudinal direction. According to this embodiment, only one half of the thickness of the internal secondary connecting member 166 (e.g., counter-electrode structure 112) contributes towards restraining the expansion due to the electrode active material on one side, with the other half of the thickness of the internal secondary connecting member 166 contributing to the restraining of the expansion due to the electrode active material on the other side.

EXAMPLES

Example 1

[0301] The present example demonstrates an example of a method of fabricating an electrode assembly 106 having the set of constraints 108. FIG. 21 is referred to herein as an embodiment of the method.

[0302] In the example, in Step S1, the anode and cathode backbones 134, 141, respectively, are fabricated. The anodes 110 are then fabricated by forming anodically active material 132, such as a silicon-containing material, on the anode backbones 134, as shown in Step S2. In Step S3, the anode and cathode current collectors 136, 140, respectively, are fabricated. The set of constraints 108 is fabricated, as in Step S4. In Step S5, the material for forming the separators 130 is deposited on the sides of the anode current collectors 136. In Step S6, the cathodically active material 138 is formed by depositing the material between the anode structures 110 and the cathode backbone 141. Finally, in Step S7, the electrode assembly 106 is connected to tabs, packaged, filled with electrolyte and sealed. According to one embodiment, the

fabrication of the set of constraints **108** in Step S4 before the separator **130** and/or cathode active material **138** are deposited in Steps S5 and S6 may provide for improved adhesion of the anode structure **110** to the set of constraints **108**, without excess residue that might otherwise remain if the set of constraints **108** were formed after the separator **130** and/or cathodically active material **138** was provided to the electrode assembly **106**.

[0303] The following embodiments are provided to illustrate aspects of the disclosure, although the embodiments are not intended to be limiting and other aspects and/or embodiments may also be provided.

[0304] Embodiment 1. A secondary battery for cycling between a charged and a discharged state, the secondary battery comprising a battery enclosure, an electrode assembly, carrier ions, a non-aqueous liquid electrolyte within the battery enclosure, and a set of electrode constraints, wherein

[0305] the electrode assembly has mutually perpendicular longitudinal, transverse, and vertical axes, a first longitudinal end surface and a second longitudinal end surface separated from each other in the longitudinal direction, and a lateral surface surrounding an electrode assembly longitudinal axis A_{EA} and connecting the first and second longitudinal end surfaces, the lateral surface having opposing first and second regions on opposite sides of the longitudinal axis and separated in a first direction that is orthogonal to the longitudinal axis, the electrode assembly having a maximum width W_{EA} measured in the longitudinal direction, a maximum length L_{EA} bounded by the lateral surface and measured in the transverse direction, and a maximum height H_{EA} bounded by the lateral surface and measured in the vertical direction, the ratio of each of L_{EA} and W_{EA} to H_{EA} being at least 2:1, respectively,

[0306] the electrode assembly further comprises a population of electrode structures, a population of counter-electrode structures, and an electrically insulating microporous separator material electrically separating members of the electrode and counter-electrode populations, members of the electrode and counter-electrode structure populations being arranged in an alternating sequence in the longitudinal direction,

[0307] each member of the population of electrode structures comprises a layer of an electrode active material and each member of the population of counter-electrode structures comprises a layer of a counter-electrode active material, wherein the electrode active material has the capacity to accept more than one mole of carrier ion per mole of electrode active material when the secondary battery is charged from a discharged state to a charged state,

[0308] the set of electrode constraints comprises a primary constraint system comprising first and second primary growth constraints and at least one primary connecting member, the first and second primary growth constraints separated from each other in the longitudinal direction, and the at least one primary connecting member connecting the first and second primary growth constraints, wherein the primary constraint system restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20%,

[0309] the set of electrode constraints further comprising a secondary constraint system comprising first and second secondary growth constraints separated in a second direction and connected by at least one secondary connecting member, wherein the secondary constraint system at least partially restrains growth of the electrode assembly in the second direction upon cycling of the secondary battery, the second direction being orthogonal to the longitudinal direction,

[0310] the charged state is at least 75% of a rated capacity of the secondary battery, and the discharged state is less than 25% of the rated capacity of the secondary battery.

[0311] Embodiment 2. The secondary battery of Embodiment 1, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30 consecutive cycles of the secondary battery is less than 20%.

[0312] Embodiment 3. The secondary battery of Embodiment 1, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 50 consecutive cycles of the secondary battery is less than 20%.

[0313] Embodiment 4. The secondary battery of Embodiment 1, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 80 consecutive cycles of the secondary battery is less than 20%.

[0314] Embodiment 5. The secondary battery of Embodiment 1, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction to less than 20% over 100 consecutive cycles of the secondary battery.

[0315] Embodiment 6. The secondary battery of Embodiment 1, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 1000 consecutive cycles of the secondary battery is less than 20%.

[0316] Embodiment 7. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10 consecutive cycles of the secondary battery is less than 10%.

[0317] Embodiment 8. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 10%.

[0318] Embodiment 9. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30 consecutive cycles of the secondary battery is less than 10%.

[0319] Embodiment 10. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 50 consecutive cycles of the secondary battery is less than 10%.

[0320] Embodiment 11. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 80 consecutive cycles of the secondary battery is less than 10%.

[0321] Embodiment 12. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 100 consecutive cycles of the secondary battery is less than 10%.

[0322] Embodiment 13. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 5 consecutive cycles of the secondary battery is less than 5%.

[0323] Embodiment 14. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 10 consecutive cycles of the secondary battery is less than 5%.

[0324] Embodiment 15. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 5%.

[0325] Embodiment 16. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 30 consecutive cycles of the secondary battery is less than 5%.

[0326] Embodiment 17. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 50 consecutive cycles of the secondary battery is less than 5%.

[0327] Embodiment 18. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 80 consecutive cycles of the secondary battery is less than 5%.

[0328] Embodiment 19. The secondary battery as in any preceding Embodiment, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction per cycle of the secondary battery is less than 1%.

[0329] Embodiment 20. The secondary battery as in any preceding Embodiment, wherein the secondary growth constraint system restrains growth of the electrode assembly in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 20 consecutive cycles upon repeated cycling of the secondary battery is less than 20%.

[0330] Embodiment 21. The secondary battery as in any preceding Embodiment, wherein the secondary growth constraint system restrains growth of the electrode assembly in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 10 consecutive cycles of the secondary battery is less than 10%.

[0331] Embodiment 22. The secondary battery as in any preceding Embodiment, wherein the secondary growth constraint system restrains growth of the electrode assembly in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction over 5 consecutive cycles of the secondary battery is less than 5%.

[0332] Embodiment 23. The secondary battery as in any preceding Embodiment, wherein the secondary growth constraint system restrains growth of the electrode assembly in the second direction such that any increase in the Feret diameter of the electrode assembly in the second direction per cycle of the secondary battery is less than 1%.

[0333] Embodiment 24. The secondary battery as in any preceding Embodiment, wherein the first primary growth constraint at least partially covers the first longitudinal end surface of the electrode assembly, and the second primary growth constraint at least partially covers the second longitudinal end surface of the electrode assembly.

[0334] Embodiment 25. The secondary battery as in any preceding Embodiment, wherein a surface area of a projection of the electrode assembly in a plane orthogonal to the stacking direction, is smaller than the surface areas of projections of the electrode assembly onto other orthogonal planes.

[0335] Embodiment 26. The secondary battery as in any preceding Embodiment, wherein a surface area of a projection of an electrode structure in a plane orthogonal to the stacking direction, is larger than the surface areas of projections of the electrode structure onto other orthogonal planes.

[0336] Embodiment 27. The secondary battery as in any preceding Embodiment, wherein at least a portion of the primary growth constraint system is pre-tensioned to exert a compressive force on at least a portion of the electrode assembly in the longitudinal direction, prior to cycling of the secondary battery between charged and discharged states.

[0337] Embodiment 28. The secondary battery as in any preceding Embodiment, wherein the primary constraint system comprises first and second primary connecting members that are separated from each other in the first direction and connect the first and second primary growth constraints.

[0338] Embodiment 29. The secondary battery as in any preceding Embodiment, wherein the first primary connect-

ing member is the first secondary growth constraint, the second primary connecting member is the second secondary growth constraint, and the first primary growth constraint or the second primary growth constraint is the first secondary connecting member.

[0339] Embodiment 30. The secondary battery as in any preceding Embodiment, wherein the at least one secondary connecting member comprises a member that is interior to longitudinal first and second ends of the electrode assembly along the longitudinal axis.

[0340] Embodiment 31. The secondary battery as in any preceding Embodiment, wherein the at least one secondary connecting member comprises at least a portion of one or more of the electrode and counter electrode structures.

[0341] Embodiment 32. The secondary battery as in any preceding Embodiment, wherein the at least one secondary connecting member comprises a portion of at least one of an electrode backbone structure and a counter-electrode backbone structure.

[0342] Embodiment 33. The secondary battery as in any preceding Embodiment, wherein the at least one secondary connecting member comprises a portion of one or more of an electrode current collector and a counter-electrode current collector.

[0343] Embodiment 34. The secondary battery as in any preceding Embodiment, wherein at least one of the first and second primary growth constraints is interior to longitudinal first and second ends of the electrode assembly along the longitudinal axis.

[0344] Embodiment 35. The secondary battery as in any preceding claim, wherein at least one of the first and second primary growth constraints comprises at least a portion of one or more of the electrode and counter electrode structures.

[0345] Embodiment 36. The secondary battery as in any preceding Embodiment, wherein at least one of the first and second primary growth constraints comprises a portion of at least one of an electrode backbone structure and a counter-electrode backbone structure.

[0346] Embodiment 37. The secondary battery as in any preceding Embodiment, wherein at least one of the first and second primary growth constraints comprises a portion of one or more of an electrode current collector and a counter-electrode current collector.

[0347] Embodiment 38. The secondary battery as in any preceding Embodiment, further comprising a tertiary constraint system comprising first and second tertiary growth constraints separated in a third direction and connected by at least one tertiary connecting member wherein the tertiary constraint system restrains growth of the electrode assembly in the third direction in charging of the secondary battery from the discharged state to the charged state, the third direction being orthogonal to the longitudinal direction and second direction.

[0348] Embodiment 39. The secondary battery as in any preceding Embodiment wherein the electrode active material is anodically active and the counter-electrode active material is cathodically active.

[0349] Embodiment 40. The secondary battery as in any preceding Embodiment wherein each member of the population of electrode structures comprises a backbone.

[0350] Embodiment 41. The secondary battery as in any preceding Embodiment wherein each member of the population of counter-electrode structures comprises a backbone.

[0351] Embodiment 42. The secondary battery as in any preceding Embodiment wherein the secondary constraint system restrains growth of the electrode assembly in the vertical direction with a restraining force of greater than 1000 psi and a skew of less than 0.2 mm/m.

[0352] Embodiment 43. The secondary battery as in any preceding Embodiment wherein the secondary growth constraint restrains growth of the electrode assembly in the vertical direction with less than 5% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m.

[0353] Embodiment 44. The secondary battery as in any preceding Embodiment wherein the secondary growth constraint restrains growth of the electrode assembly in the vertical direction with less than 3% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m.

[0354] Embodiment 45. The secondary battery as in any preceding Embodiment wherein the secondary growth constraint restrains growth of the electrode assembly in the vertical direction with less than 1% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m.

[0355] Embodiment 46. The secondary battery as in any preceding Embodiment wherein the secondary growth constraint restrains growth of the electrode assembly in the vertical direction with less than 15% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m after 50 battery cycles.

[0356] Embodiment 47. The secondary battery as in any preceding Embodiment wherein the secondary growth constraint restrains growth of the electrode assembly in the vertical direction with less than 5% displacement at less than or equal to 10,000 psi and a skew of less than 0.2 mm/m after 150 battery cycles.

[0357] Embodiment 48. The secondary battery as in any preceding Embodiment wherein members of the population of counter-electrode structures comprise a top adjacent to the first secondary growth constraint, a bottom adjacent to the second secondary growth constraint, a vertical axis A_{CES} parallel to and in the vertical direction extending from the top to the bottom, a lateral electrode surface surrounding the vertical axis A_{CES} and connecting the top and the bottom, the lateral electrode surface having opposing first and second regions on opposite sides of the vertical axis and separated in a first direction that is orthogonal to the vertical axis, a length L_{CES} , a width W_{CES} , and a height H_{CES} , the length L_{CES} being bounded by the lateral electrode surface and measured in the transverse direction, the width W_{CES} being bounded by the lateral electrode surface and measured in the longitudinal direction, and the height H_{CES} being measured in the direction of the vertical axis A_{CES} from the top to the bottom, wherein

[0358] the first and second secondary growth constraints each comprise an inner surface and an opposing outer surface, the inner surface and the outer surface of each are substantially co-planar and the distance between the inner surface and the opposing outer surface of each of the first and second secondary growth constraints defines a height of each that is measured in the vertical direction from the inner surface to the outer surface of each, the inner surfaces of each being affixed to the top and bottom of the population of electrode structures.

[0359] Embodiment 49. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise

a notch, and the population of counter-electrode structures height H_{CES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 25% of the first and second secondary growth constraint heights.

[0360] Embodiment 50. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of counter-electrode structures height H_{CES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 50% of the first and second secondary growth constraint heights.

[0361] Embodiment 51. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of counter-electrode structures height H_{CES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 75% of the first and second secondary growth constraint heights.

[0362] Embodiment 52. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of counter-electrode structures height H_{CES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 90% of the first and second secondary growth constraint heights.

[0363] Embodiment 53. The secondary battery as in any preceding Embodiment wherein each of the first and second secondary growth constraints comprise a slot, and the population of counter-electrode structures height extends through and is affixed within the slot forming an interlocking connection between the population of electrode structures and each of the first and second secondary growth constraints.

[0364] Embodiment 54. The secondary battery as in any preceding Embodiment wherein members of the population of electrode structures comprise a top adjacent to the first secondary growth constraint, a bottom adjacent to the second secondary growth constraint, a vertical axis A_{ES} parallel to and in the vertical direction extending from the top to the bottom, a lateral electrode surface surrounding the vertical axis A_{ES} and connecting the top and the bottom, the lateral electrode surface having opposing first and second regions on opposite sides of the vertical axis and separated in a first direction that is orthogonal to the vertical axis, a length L_{ES} , a width W_{ES} , and a height H_{ES} , the length L_{ES} being bounded by the lateral electrode surface and measured in the transverse direction, the width W_{ES} being bounded by the lateral electrode surface and measured in the longitudinal direction, and the height H_{ES} being measured in the direction of the vertical axis A_{ES} from the top to the bottom, wherein

[0365] the first and second secondary growth constraints each comprise an inner surface and an opposing outer surface, the inner surface and the outer surface of each are substantially co-planar and the distance between the inner surface and the opposing outer surface of each of the first and second secondary growth constraints defines a height of each that is measured in the vertical direction from the inner surface to the outer surface of each, the inner surfaces of each being affixed to the top and bottom of the population of electrode structures.

[0366] Embodiment 55. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of electrode structures height H_{ES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 25% of the first and second secondary growth constraint heights.

[0367] Embodiment 56. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of electrode structures height H_{ES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 50% of the first and second secondary growth constraint heights.

[0368] Embodiment 57. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of electrode structures height H_{ES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 75% of the first and second secondary growth constraint heights.

[0369] Embodiment 58. The secondary battery as in any preceding Embodiment wherein the inner surfaces of each of the first and second secondary growth constraints comprise a notch, and the population of electrode structures height H_{ES} extends into and is affixed within the notch, the notch having a depth defined along the vertical direction of 90% of the first and second secondary growth constraint heights.

[0370] Embodiment 59. The secondary battery as in any preceding Embodiment wherein each of the first and second secondary growth constraints comprise a slot, and the population of electrode structures height extends through and is affixed within the slot forming an interlocking connection between the population of electrode structures and each of the first and second secondary growth constraints.

[0371] Embodiment 60. A secondary battery as in any preceding Embodiment, wherein the set of electrode constraints further comprising a fused secondary constraint system comprising first and second secondary growth constraints separated in a second direction and fused with at least one first secondary connecting member.

[0372] Embodiment 61. The secondary battery as in any preceding Embodiment wherein members of the population of counter-electrode structures comprise a top adjacent to the first secondary growth constraint, a bottom adjacent to the second secondary growth constraint, a vertical axis A_{CES} parallel to and in the vertical direction extending from the top to the bottom, a lateral electrode surface surrounding the vertical axis A_{CES} and connecting the top and the bottom, the lateral electrode surface having opposing first and second regions on opposite sides of the vertical axis and separated in a first direction that is orthogonal to the vertical axis, a length L_{CES} , a width W_{CES} , and a height H_{CES} , the length L_{CES} being bounded by the lateral electrode surface and measured in the transverse direction, the width W_{CES} being bounded by the lateral electrode surface and measured in the longitudinal direction, and the height H_{CES} being measured in the direction of the vertical axis A_{CES} from the top to the bottom, wherein

[0373] the first and second secondary growth constraints each comprise an inner surface and an opposing outer surface, the inner surface and the outer surface of each are substantially co-planar and the distance between the inner surface and the opposing outer surface of each of the first

and second secondary growth constraints defines a height of each that is measured in the vertical direction from the inner surface to the outer surface of each, the inner surfaces of each being fused to the top and bottom of the population of counter-electrode structures.

[0374] Embodiment 62. The secondary battery as in any preceding Embodiment wherein members of the population of electrode structures comprise a top adjacent to the first secondary growth constraint, a bottom adjacent to the second secondary growth constraint, a vertical axis A_{ES} parallel to and in the vertical direction extending from the top to the bottom, a lateral electrode surface surrounding the vertical axis A_{ES} and connecting the top and the bottom, the lateral electrode surface having opposing first and second regions on opposite sides of the vertical axis and separated in a first direction that is orthogonal to the vertical axis, a length L_{ES} , a width W_{ES} , and a height H_{ES} , the length L_{ES} being bounded by the lateral electrode surface and measured in the transverse direction, the width W_{ES} being bounded by the lateral electrode surface and measured in the longitudinal direction, and the height H_{ES} being measured in the direction of the vertical axis A_{ES} from the top to the bottom, wherein

[0375] the first and second secondary growth constraints each comprise an inner surface and an opposing outer surface, the inner surface and the outer surface of each are substantially co-planar and the distance between the inner surface and the opposing outer surface of each of the first and second secondary growth constraints defines a height of each that is measured in the vertical direction from the inner surface to the outer surface of each, the inner surfaces of each being fused to the top and bottom of the population of electrode structures.

[0376] Embodiment 63. The secondary battery as in any preceding Embodiment wherein at least one of an electrode structure and counter-electrode structure comprise a top adjacent to the first secondary growth constraint, a bottom adjacent to the second secondary growth constraint, a vertical axis A_{ES} parallel to and in the vertical direction extending from top to bottom, a lateral electrode surface surrounding the vertical axis and connecting top and bottom, the lateral electrode surface having a width W_{ES} bounded by the lateral surface and measured in the longitudinal direction, wherein

[0377] the width W_{ES} tapers from a first width adjacent the top to a second width that is smaller than the first width at a region along the vertical axis between the top and bottom.

[0378] Embodiment 64. The secondary battery as in any preceding Embodiment, wherein the at least one secondary connecting member corresponds to at least one of the first and second primary growth constraints at the longitudinal ends of the electrode assembly.

[0379] Embodiment 65. The secondary battery as in any preceding Embodiment wherein the electrically insulating microporous separator material comprises a particulate material and a binder, has a void fraction of at least 20 vol. %, and is permeated by the non-aqueous liquid electrolyte.

[0380] Embodiment 66. The secondary battery as in any preceding Embodiment wherein the carrier ions are selected from the group consisting of lithium, potassium, sodium, calcium, and magnesium.

[0381] Embodiment 67. The secondary battery as in any preceding Embodiment wherein the non-aqueous liquid electrolyte comprises a lithium salt dissolved in an organic solvent.

[0382] Embodiment 68. The secondary battery as in any preceding Embodiment wherein the first and second secondary growth constraints each comprise a thickness that is less than 50% of the electrode or counter-electrode height.

[0383] Embodiment 69. The secondary battery as in any preceding Embodiment wherein the first and second secondary growth constraints each comprise a thickness that is less than 20% of the electrode or counter-electrode height.

[0384] Embodiment 70. The secondary battery as in any preceding Embodiment wherein the first and second secondary growth constraints each comprise a thickness that is less than 10% of the electrode or counter-electrode height.

[0385] Embodiment 71. The secondary battery as in any preceding Embodiment wherein the set of electrode constraints inhibits expansion of the electrode active material layers in the vertical direction upon insertion of the carrier ions into the electrode active material as measured by scanning electron microscopy (SEM).

[0386] Embodiment 72. The secondary battery as in any preceding Embodiment wherein the first and second primary growth constraints impose an average compressive force to each of the first and second longitudinal ends of at least 0.7 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0387] Embodiment 73. The secondary battery as in any preceding Embodiment wherein the first and second primary growth constraints impose an average compressive force to each of the first and second longitudinal ends of at least 1.75 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0388] Embodiment 74. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 2.8 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0389] Embodiment 75. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 3.5 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0390] Embodiment 76. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 5.25 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0391] Embodiment 77. The secondary battery according to any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 7 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0392] Embodiment 78. The secondary battery according to any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 8.75 kPa, averaged over the surface area of the first and second projected longitudinal ends, respectively.

[0393] Embodiment 79. The secondary battery according to any preceding Embodiment wherein the first and second primary growth constraints imposes an average compressive force to each of the first and second longitudinal ends of at least 10 kPa, averaged over the surface area of the first and second longitudinal ends, respectively.

[0394] Embodiment 80. The secondary battery of any preceding Embodiment wherein the surface area of the first and second longitudinal end surfaces is less than 25% of the surface area of the electrode assembly.

[0395] Embodiment 81. The secondary battery of any preceding Embodiment wherein the surface area of the first and second longitudinal end surfaces is less than 20% of the surface area of the electrode assembly.

[0396] Embodiment 82. The secondary battery of any preceding Embodiment wherein the surface area of the first and second longitudinal end surfaces is less than 15% of the surface area of the electrode assembly.

[0397] Embodiment 83. The secondary battery of any preceding Embodiment wherein the surface area of the first and second longitudinal end surfaces is less than 10% of the surface area of the electrode assembly.

[0398] Embodiment 84. The secondary battery of any preceding Embodiment wherein the constraint and enclosure have a combined volume that is less than 60% of the volume enclosed by the battery enclosure.

[0399] Embodiment 85. The secondary battery of any preceding Embodiment wherein the constraint and enclosure have a combined volume that is less than 45% of the volume enclosed by the battery enclosure.

[0400] Embodiment 86. The secondary battery of any preceding Embodiment wherein the constraint and enclosure have a combined volume that is less than 30% of the volume enclosed by the battery enclosure.

[0401] Embodiment 87. The secondary battery of any preceding Embodiment wherein the constraint and enclosure have a combined volume that is less than 20% of the volume enclosed by the battery enclosure.

[0402] Embodiment 88. The secondary battery of any preceding Embodiment wherein the first and second longitudinal end surfaces are under a compressive load when the secondary battery is charged to at least 80% of its rated capacity.

[0403] Embodiment 89. The secondary battery of any preceding Embodiment wherein the secondary battery comprises a set of electrode assemblies, the set comprising at least two electrode assemblies.

[0404] Embodiment 90. The secondary battery of any preceding Embodiment claim wherein the electrode assembly comprises at least 5 electrode structures and at least 5 counter-electrode structures.

[0405] Embodiment 91. The secondary battery of any preceding Embodiment wherein the electrode assembly comprises at least 10 electrode structures and at least 10 counter-electrode structures.

[0406] Embodiment 92. The secondary battery of any preceding Embodiment wherein the electrode assembly comprises at least 50 electrode structures and at least 50 counter-electrode structures.

[0407] Embodiment 93. The secondary battery of any preceding Embodiment wherein the electrode assembly comprises at least 100 electrode structures and at least 100 counter-electrode structures.

[0408] Embodiment 94. The secondary battery of any preceding Embodiment wherein the electrode assembly comprises at least 500 electrode structures and at least 500 counter-electrode structures.

[0409] Embodiment 95. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a material having an ultimate tensile strength of at least 10,000 psi (>70 MPa).

[0410] Embodiment 96. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a material that is compatible with the battery electrolyte.

[0411] Embodiment 97. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a material that does not significantly corrode at the floating or anode potential for the battery.

[0412] Embodiment 98. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a material that does not significantly react or lose mechanical strength at 45° C.

[0413] Embodiment 99. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a material that does not significantly react or lose mechanical strength at 70° C.

[0414] Embodiment 100. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises metal, metal alloy, ceramic, glass, plastic, or a combination thereof.

[0415] Embodiment 101. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a sheet of material having a thickness in the range of about 10 to about 100 micrometers.

[0416] Embodiment 102. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises a sheet of material having a thickness in the range of about 30 to about 75 micrometers.

[0417] Embodiment 103. The secondary battery of any preceding Embodiment wherein at least one of the primary and secondary constraint systems comprises carbon fibers at >50% packing density.

[0418] Embodiment 104. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints exert a pressure on the first and second longitudinal end surfaces that exceeds the pressure maintained on the electrode assembly in each of two directions that are mutually perpendicular and perpendicular to the stacking direction by factor of at least 3.

[0419] Embodiment 105. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints exert a pressure on the first and second longitudinal end surfaces that exceeds the pressure maintained on the electrode assembly in each of two directions that are mutually perpendicular and perpendicular to the stacking direction by factor of at least 3.

[0420] Embodiment 106. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints exert a pressure on the first and second longitudinal end surfaces that exceeds the pressure main-

tained on the electrode assembly in each of two directions that are mutually perpendicular and perpendicular to the stacking direction by factor of at least 4.

[0421] Embodiment 107. The secondary battery of any preceding Embodiment wherein the first and second primary growth constraints exert a pressure on the first and second longitudinal end surfaces that exceeds the pressure maintained on the electrode assembly in each of two directions that are mutually perpendicular and perpendicular to the stacking direction by factor of at least 5.

[0422] Embodiment 108. The secondary battery of any preceding Embodiment, wherein portions of the set of electrode constraints that are external to the electrode assembly occupy no more than 80% of the total combined volume of the electrode assembly and the external portions of the electrode constraints.

[0423] Embodiment 109. The secondary battery of any preceding Embodiment, wherein portions of the primary growth constraint system that are external to the electrode assembly occupy no more than 40% of the total combined volume of the electrode assembly and external portions of the primary growth constraint system.

[0424] Embodiment 110. The secondary battery of any preceding Embodiment, wherein portions of the secondary growth constraint system that are external to the electrode assembly occupy no more than 40% of the total combined volume of the electrode assembly and external portions of the secondary growth constraint system.

[0425] Embodiment 111. The secondary battery of any preceding Embodiment, wherein a projection of the members of the electrode population and the counter-electrode populations onto the first longitudinal end surface circumscribes a first projected area, and a projection of the members of the electrode population and the counter-electrode populations onto the second longitudinal end surface circumscribes a second projected area, and wherein the first and second projected areas each comprise at least 50% of the surface area of the first and second longitudinal end surfaces, respectively.

[0426] Embodiment 112. The secondary battery of any preceding Embodiment, wherein the first and second primary growth constraints deflect upon repeated cycling of the secondary battery between charged and discharged states according to the following formula:

$$\delta = 60wL^4/Eh^3,$$

[0427] wherein w is total distributed load applied to the first and second primary growth constraints upon repeated cycling of the secondary battery between charged and discharged states, L is the distance between first and second primary connecting members in the vertical direction, E is the elastic modulus of the first and second primary growth constraints, and h is the thickness of the first and second primary growth constraints.

[0428] Embodiment 113. The secondary battery of any preceding Embodiment, wherein the stress on the first and second primary growth constraints upon repeated cycling of the secondary battery between charged and discharged states is as follows:

$$\sigma = 3wL^2/4h^2$$

[0429] wherein w is total distributed load applied on the first and second primary growth constraints upon repeated cycling of the secondary battery between

charged and discharged states, L is the distance between first and second primary connecting members in the vertical direction, and h is the thickness of the first and second primary growth constraints.

[0430] Embodiment 114. The secondary battery of any preceding Embodiment, wherein the tensile stress on the first and second primary connecting members is as follows:

$$\sigma = PL/2t$$

[0431] wherein P is pressure applied due to the first and second primary growth constraints upon repeated cycling of the secondary battery between charged and discharged states, L is the distance between the first and second primary connecting members along the vertical direction, and t is the thickness of the first and second primary connecting members in the vertical direction.

[0432] Embodiment 115. The secondary battery of any preceding Embodiment, wherein the first and second secondary growth constraints deflect upon repeated cycling of the secondary battery between charged and discharged states according to the following formula

$$\delta = 60wy_A/Et^3,$$

[0433] wherein w is the total distributed load applied on the first and second secondary growth constraints upon repeated cycling of the secondary battery between charged and discharged states, y is the distance between the first and second secondary connecting members in the longitudinal direction, E is the elastic modulus of the first and second secondary growth constraints, and t is the thickness of the first and second secondary growth constraints.

[0434] Embodiment 116. The secondary battery of any preceding Embodiment, wherein the stress on the first and second secondary growth constraints is as follows:

$$\sigma = 3wy^2/4t^2$$

[0435] wherein w is the total distributed load applied on the first and second secondary growth constraints upon repeated cycling of the secondary battery between charged and discharged states, y is the distance between the first and second secondary connecting members along the longitudinal direction, and t is the thickness of the first and second secondary growth constraints.

[0436] Embodiment 117. The secondary battery of any preceding Embodiment, wherein the tensile stress on the first and second secondary connecting members is as follows:

$$\sigma = Py/2h,$$

[0437] wherein P is the pressure applied on the first and second secondary growth constraints upon repeated cycling of the secondary battery, y is the distance between the first and second secondary connecting members along the longitudinal direction, and h is the thickness of the first and second secondary connecting members in the longitudinal direction.

[0438] Embodiment 118. The secondary battery of any preceding Embodiment, wherein the tensile stress on internal secondary connecting members is as follows:

$$\sigma = Py/h$$

[0439] wherein P is the pressure applied to the first and second secondary growth constraints upon cycling of the of the secondary battery between charged and

discharge states, due to expansion of the electrode active material on regions that are in between internal first and second secondary connecting members, y is the distance between the internal first and second secondary connecting members along the longitudinal direction, and h is the thickness of the internal first and second secondary connecting members in the longitudinal direction.

[0440] Embodiment 119. A secondary battery for cycling between a charged and a discharged state, the secondary battery comprising a battery enclosure, an electrode assembly, carrier ions, a non-aqueous liquid electrolyte within the battery enclosure, and a set of electrode constraints, wherein

[0441] the electrode assembly has mutually perpendicular longitudinal, transverse, and vertical axes, a first longitudinal end surface and a second longitudinal end surface separated from each other in the longitudinal direction, and a lateral surface surrounding an electrode assembly longitudinal axis AHA and connecting the first and second longitudinal end surfaces, the lateral surface having opposing first and second regions on opposite sides of the longitudinal axis and separated in a first direction that is orthogonal to the longitudinal axis, the electrode assembly having a maximum width W_{EA} measured in the longitudinal direction, a maximum length L_{EA} bounded by the lateral surface and measured in the transverse direction, and a maximum height H_{EA} bounded by the lateral surface and measured in the vertical direction, the ratio of each of L_{EA} and W_{EA} to H_{EA} being at least 2:1, respectively,

[0442] the electrode assembly further comprises a population of electrode structures, a population of counter-electrode structures, and an electrically insulating microporous separator material electrically separating members of the electrode and counter-electrode populations, members of the electrode and counter-electrode structure populations being arranged in an alternating sequence in the longitudinal direction,

[0443] each member of the population of electrode structures comprises a layer of an electrode active material and each member of the population of counter-electrode structures comprises a layer of a counter-electrode active material, wherein the electrode active material has the capacity to accept more than one mole of carrier ion per mole of electrode active material when the secondary battery is charged from a discharged state to a charged state,

[0444] the set of electrode constraints comprises a primary constraint system comprising first and second primary growth constraints and at least one primary connecting member, the first and second primary growth constraints separated from each other in the longitudinal direction, and the at least one primary connecting member connecting the first and second primary growth constraints, wherein the primary constraint array restrains growth of the electrode assembly in the longitudinal direction such that any increase in the Feret diameter of the electrode assembly in the longitudinal direction over 20 consecutive cycles of the secondary battery is less than 20%,

[0445] the charged state is at least 75% of a rated capacity of the secondary battery, and the discharged state is less than 25% of the rated capacity of the secondary battery.

INCORPORATION BY REFERENCE

[0446] All publications and patents mentioned herein, including those items listed below, are hereby incorporated by reference in their entirety for all purposes as if each individual publication or patent was specifically and individually incorporated by reference. In case of conflict, the present application, including any definitions herein, will control.

EQUIVALENTS

[0447] While specific embodiments have been discussed, the above specification is illustrative, and not restrictive. Many variations will become apparent to those skilled in the art upon review of this specification. The full scope of the embodiments should be determined by reference to the claims, along with their full scope of equivalents, and the specification, along with such variations.

[0448] Unless otherwise indicated, all numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in this specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained.

What is claimed is:

1. A device for electrical charging and discharging, the device comprising:

electrolytic cells stacked along an axis to form an electrode assembly, the electrode assembly being configured for repeated electrical charging and discharging in cycles at least in part by utilizing charge carriers comprising alkaline ions, alkaline earth ions, or aluminum ions, each of the electrolytic cells having an anode active material including an allotrope of elemental carbon and/or silicon; and

an enclosure configured to enclose the electrode assembly that undergoes an alteration of a volume of the electrode assembly, the alteration of the volume resulting in application of pressure on the enclosure, the alteration of the volume occurring during the cycles, the enclosure being configured to restrain a volumetric growth of the electrode assembly in a direction, the direction being along the axis or orthogonal to the axis, the device being configured to restrain a volumetric growth of the electrode assembly in the direction such that an increase in a Feret diameter of the electrode assembly in the direction over 20 cycles of the device is less than 20%, the cycles being consecutive.

2. The device of claim 1, wherein the device restrains growth of the electrode assembly in the direction such that an increase in a Feret diameter of the electrode assembly in the direction over 1000 consecutive cycles of the device is less than 20%.

3. The device of claim 1, wherein the charge carriers comprise lithium ions.

4. The device of claim 1, wherein the electrical charging and discharging is between a charged state that is charged to at least 75% of a rated capacity of the electrode assembly, and a discharged state that is discharged less than 25% of the rated capacity.

5. The device of claim 1, wherein each of the electrolytic cells comprises an electrode active material including silicon.

6. The device of claim 1, wherein the enclosure is configured to restrain a growth of the volume of the electrode assembly along the axis.

7. The device of claim 1, wherein the enclosure is configured to restrain a growth of the volume of the electrode assembly in the direction orthogonal to the axis.

8. The device of claim 1, wherein the device comprises an electrolyte that is non-aqueous.

9. The device of claim 8, wherein the electrolyte is a liquid.

10. The device of claim 1, wherein each of the electrolytic cells comprises an anode, a cathode, and a separator configured for traversal of the charge carriers therethrough during the cycles.

11. The device of claim 1, wherein the electrode assembly includes at least 10 of the electrolytic cells.

12. The device of claim 1, wherein the electrode assembly includes at least 10 electrodes.

13. The device of claim 1, wherein each of the electrolytic cells comprises an electrode, the electrode having a rectangular cross section having a width perpendicular to a height, the width to height of each of the electrodes is at least about 1:5.

14. The device of claim 1, wherein the enclosure is a rectangular prism.

15. The device of claim 14, wherein the rectangular prism is elongated.

16. The device of claim 1, wherein the electrolytic cells includes at least 10 cells, each of the electrolytic cells comprises an electrode, the electrode having a rectangular cross section having a width perpendicular to a height, the width to height of each of the electrodes is at least about 1:5.

17. The device of claim 1, wherein the enclosure is configured such that two terminal tabs project out of a side of the enclosure having a shape of a rectangular prism, to facilitate the electrical charging and discharging.

18. The device of claim 1, wherein the device is configured to power consumer electronics.

19. The device of claim 1, wherein each of the electrolytic cells comprises a separator, each of the electrolytic cells comprising a polymer in a component other than the separator, the component being of each of the electrolytic cells.

20. The device of claim 1, wherein the allotrope of elemental carbon comprises graphite, soft carbon, or hard carbon.

21. The device of claim 1, wherein the enclosure seals the electrode assembly within the enclosure.

22. The device of claim 1, wherein the enclosure has an external surface exposed to a consumer.

23. The device of claim 1, wherein the pressure causes buckling of the electrode assembly.

24. Method of the electrical charging and discharging, the method comprising: (a) forming and/or providing, the device in claim 1; and (b) using the device for the electrical charging and discharging.

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